# Hit Location & Wound Tables

Historically weighted d100 hit locations with system-agnostic wound effects. Roll d100; 00 = 100.

## Procedure

1. Roll d100 on the **location table** for the fight, in the column for the attacking weapon. If a **situation** applies (shield side, target down, fleeing...), use that column on the weapon's situation page instead.

2. **Called shot:** Roll the location twice; keep whichever result lands in the called zone. If both or neither do, keep the first. Your system sets any to-hit penalty.

3. Roll d100 on the **mechanism table** for the weapon: cut, pierce or crush.

4. Set **severity** from your own system's damage (see guide), or roll d100 on the severity table (ballistic hits use their own table, by hit zone).

5. **Armour** (adjusted tables only): lower the severity by the armour's steps; see the Armour pages.

6. Find the wound on the **wound effects** page for that mechanism: location row, severity column.

7. Track it: Bleed drains Blood, Pain stacks as a penalty, the Lethal timer runs until treated.

## Evidence vs adjusted tables

**Evidence** tables show where blows marked bone, so armour and soft-tissue blind spots are already baked in: don't subtract armour again. **Adjusted** tables show where blows land; resolve armour separately by the location's armour slot.

## Confidence

**Historical** - taken directly from a source

**Fitted** - estimated to match published summary figures

**Extrapolated** - design estimate from analogous evidence

# Crusades



*Sidon, 1253 or 1260*

From 1099 to 1291 Frankish crusader states held the coast of the Levant. Sidon, a walled port in today's Lebanon, was sacked twice in the 13th century: in 1253 by an army from Damascus while Louis IX of France was rebuilding its walls (Joinville describes the king helping to bury the rotting dead), and in 1260 by the Mongols under Kitbuqa. Two mass graves in the town ditch near St Louis' castle, dated to the mid-13th century, hold at least 25 men, Europeans and locals, killed in one of these attacks; their bones show sword, axe and mace wounds, many from behind.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Defenders of Sidon** | Frankish knights and sergeants, crossbowmen and townsmen of mixed European and Near Eastern descent. | Knights and many sergeants in mail hauberks with helmets and shields; others in padded coats or nothing. |
| **Attackers** | The army from Damascus that sacked the town in 1253, or the Mongols in 1260; which raid filled the graves is not known. | Horsemen in mail or lamellar with round shields, lance, sword, mace and bow; Mongol horse archers in lamellar or padded coats. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Frankish knight** (picture) | Defenders of Sidon | Mail hauberk and helm (c.1100-1300) | Sword (arming sword) | Mail hauberk, great helm and shield. |
|  | **Frankish sergeant** (picture) | Defenders of Sidon | Mail hauberk and helm (c.1100-1300) | Axe (hand or long-hafted) | Heavy blade wounds at Sidon point to swords and axes. |
|  | **Crossbowman** | Defenders of Sidon | Gambeson and cap | Crossbow (bolts) | Padded coat; the town's walls were still being rebuilt in 1253. |
|  | **Townsman** | Defenders of Sidon | Unarmoured | Spear / lance | Caught in the sack. |
|  | **Syrian horseman** (picture) | Attackers | Mail hauberk and helm (c.1100-1300) | Spiked club / morning star (knout) | Mace blows from the saddle: the Sidon wounds fall on the head, neck and shoulders. |
|  | **Mongol horse archer** (picture) | Attackers | Hardened leather | Bow (war arrows) | Kitbuqa's raid of 1260. |

**Sources.** Mikulski R.N.R. et al., 'Weapon injuries in the crusader mass graves from a 13th century attack on the port city of Sidon (Lebanon)', PLoS ONE 16(8): e0256517 (2021). Joinville, Life of Saint Louis, on Louis IX burying the dead of Sidon (1253).

# Sidon, mid-13th century - crusader mass graves (bone evidence)

*Bones of at least 25 men killed when Mamluk (1253) or Mongol (1260) forces sacked crusader Sidon, later gathered into the town ditch. Killed only.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Each weapon column is that weapon's own recorded wounds from the source.

| Location | Zone | Sword | Axe | Spiked club | Spear | Dagger | Bow | Crossbow |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-03 | 01-03 | 01-10 | 01-19 | 01-05 | 01-05 | 01-05 |
| Skull (right) | head | 04-06 | 04-06 | 11-20 | 20-38 | 06-10 | 06-10 | 06-10 |
| Face | head | 07-09 | 07-09 | 21-32 | 39-60 | 11-16 | 11-16 | 11-16 |
| Neck | head | 10-48 | 10-48 | - | - | 17-40 | 17-40 | 17-40 |
| Chest (left) | torso | 49-51 | 49-51 | 33-36 | - | 41-43 | 41-43 | 41-43 |
| Chest (right) | torso | 52-54 | 52-54 | 37-40 | - | 44-46 | 44-46 | 44-46 |
| Groin / pelvis | torso | - | - | 41-43 | - | 47 | 47 | 47 |
| Upper back | torso | 55-56 | 55-56 | 44-45 | - | 48-49 | 48-49 | 48-49 |
| Lower back | torso | 57-58 | 57-58 | 46-47 | - | 50-51 | 50-51 | 50-51 |
| Shoulder (left) | arms | 59-63 | 59-63 | 48-51 | - | 52-56 | 52-56 | 52-56 |
| Shoulder (right) | arms | 64-68 | 64-68 | 52-55 | - | 57-61 | 57-61 | 57-61 |
| Upper arm (left) | arms | 69-70 | 69-70 | 56-57 | - | 62-63 | 62-63 | 62-63 |
| Upper arm (right) | arms | 71-72 | 71-72 | 58-59 | - | 64-65 | 64-65 | 64-65 |
| Forearm (left) | arms | 73-74 | 73-74 | 60-61 | - | 66-67 | 66-67 | 66-67 |
| Forearm (right) | arms | 75-76 | 75-76 | 62-63 | - | 68-69 | 68-69 | 68-69 |
| Hand (left) | arms | 77-84 | 77-84 | 64 | - | 70-75 | 70-75 | 70-75 |
| Hand (right) | arms | 85-92 | 85-92 | 65 | - | 76-81 | 76-81 | 76-81 |
| Thigh (left) | legs | 93-96 | 93-96 | 66-74 | 61-70 | 82-87 | 82-87 | 82-87 |
| Thigh (right) | legs | 97-00 | 97-00 | 75-83 | 71-80 | 88-93 | 88-93 | 88-93 |
| Knee (left) | legs | - | - | 84-85 | 81-83 | 94 | 94 | 94 |
| Knee (right) | legs | - | - | 86-87 | 84-86 | 95 | 95 | 95 |
| Lower leg (left) | legs | - | - | 88-92 | 87-93 | 96-97 | 96-97 | 96-97 |
| Lower leg (right) | legs | - | - | 93-97 | 94-00 | 98-99 | 98-99 | 98-99 |
| Foot (left) | legs | - | - | 98-99 | - | 00 | 00 | 00 |
| Foot (right) | legs | - | - | 00 | - | - | - | - |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head (cranium and mandible) (17) = 17.0%; Neck (cervical vertebrae) (24) = 24.0%; Shoulders (clavicle, scapula) (9) = 9.0%; Upper limbs (arm bones) (9) = 9.0%; Hands (11) = 11.0%; Torso (ribs, spine) (10) = 10.0%; Hips (1) = 1.0%; Thighs (12) = 12.0%; Legs (below the thigh) (6) = 6.0%; Feet (1) = 1.0% (n = 100). From the study's table of bones with definite battle injuries (100 bones; 62 sharp, 35 blunt and 5 penetrating injuries) by body region. Sides are not recorded, and each region is shared among its locations by body exposure. Swords and axes roll on the sharp-force bones, the mace (spiked club) on the blunt-force ones and the spear on the penetrating ones (spiked mace, arrow, lance or javelin, the authors suggest); other weapons use all bones together.

**Reading it.** Armour is already in these numbers. This is bone evidence: chest and belly wounds that missed bone left no mark. 24 of the 100 bones are blade cuts across the back of the neck, found on nearly every man; the authors think some may be executions of captives, so the neck is much higher here than in battle. For play, use the gameplay table, which leaves them out.

**Status.** Historical, from about 25 men killed in a sack, their remains left exposed for weeks before burial and partly disarticulated, so wounds cannot be matched to individuals or sides. A picture of a massacre after a city fell, not of a pitched battle.

**Sources.** Mikulski R.N.R., Schutkowski H., Smith M.J., Doumet-Serhal C. and Mitchell P.D., 'Weapon injuries in the crusader mass graves from a 13th century attack on the port city of Sidon (Lebanon)', PLoS ONE 16(8): e0256517 (2021), open access: S1 Table (elements with definite peri-mortem trauma by region). | Same study: more wounds on the back of the body than the front (men attacked as they fled); blows concentrated on the head, neck and shoulders, compatible with mounted attackers striking down at men on foot; one man had at least 12 injuries on 16 bones. | Tally: data/sources/sidon-1253-mass-graves.csv.

# Sidon, mid-13th century - gameplay-adjusted (before armour)

*Where blows land on a man fighting in the sack of a crusader town, before armour, re-weighted from the Sidon bone evidence without the neck cuts.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**.

| Location | Zone | Sword | Axe | Spiked club | Spear | Dagger | Bow | Crossbow |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-08 | 01-09 | 01-10 | 01-06 | 01-07 | 01-08 | 01-09 |
| Skull (right) | head | 09-16 | 10-18 | 11-20 | 07-12 | 08-14 | 09-16 | 10-18 |
| Face | head | 17-22 | 19-24 | 21-28 | 13-19 | 15-22 | 17-22 | 19-24 |
| Neck | head | 23-25 | 25-27 | 29-32 | 20-21 | 23-28 | 23-25 | 25-27 |
| Chest (left) | torso | 26-32 | 28-33 | 33-39 | 22-30 | 29-37 | 26-32 | 28-34 |
| Chest (right) | torso | 33-39 | 34-39 | 40-45 | 31-39 | 38-46 | 33-39 | 35-40 |
| Abdomen | torso | 40-46 | 40-44 | 46-51 | 40-48 | 47-55 | 40-46 | 41-46 |
| Groin / pelvis | torso | 47-49 | 45-46 | 52-54 | 49-52 | 56-59 | 47-49 | 47-49 |
| Upper back | torso | 50-52 | 47-48 | 55-57 | 53-56 | 60-63 | 50-52 | 50-52 |
| Lower back | torso | 53-55 | 49-50 | 58-60 | 57-60 | 64-67 | 53-55 | 53-55 |
| Shoulder (left) | arms | 56-58 | 51-53 | 61-63 | 61-63 | 68-70 | 56-58 | 56-58 |
| Shoulder (right) | arms | 59-61 | 54-56 | 64-66 | 64-66 | 71-73 | 59-61 | 59-61 |
| Upper arm (left) | arms | 62-63 | 57-58 | 67-68 | 67-68 | 74-75 | 62-63 | 62-63 |
| Upper arm (right) | arms | 64-65 | 59-60 | 69-70 | 69-70 | 76-77 | 64-65 | 64-65 |
| Forearm (left) | arms | 66-68 | 61-63 | 71-73 | 71-73 | 78-80 | 66-68 | 66-68 |
| Forearm (right) | arms | 69-71 | 64-66 | 74-76 | 74-76 | 81-83 | 69-71 | 69-71 |
| Hand (left) | arms | 72-75 | 67-70 | 77-80 | 77-79 | 84-87 | 72-75 | 72-75 |
| Hand (right) | arms | 76-79 | 71-74 | 81-84 | 80-82 | 88-91 | 76-79 | 76-79 |
| Thigh (left) | legs | 80-84 | 75-80 | 85-88 | 83-86 | 92-93 | 80-84 | 80-84 |
| Thigh (right) | legs | 85-89 | 81-86 | 89-92 | 87-90 | 94-95 | 85-89 | 85-89 |
| Knee (left) | legs | 90 | 87 | 93 | 91 | 96 | 90 | 90 |
| Knee (right) | legs | 91 | 88 | 94 | 92 | 97 | 91 | 91 |
| Lower leg (left) | legs | 92-94 | 89-92 | 95-96 | 93-95 | 98 | 92-94 | 92-94 |
| Lower leg (right) | legs | 95-97 | 93-96 | 97-98 | 96-97 | 99 | 95-97 | 95-97 |
| Foot (left) | legs | 98-99 | 97-99 | 99 | 98-99 | 00 | 98-99 | 98-99 |
| Foot (right) | legs | 00 | 00 | 00 | 00 | - | 00 | 00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** These weights are where blows LAND. Resolve armour separately with the target's kit (a mail hauberk for knights and many sergeants, padded coats for others).

**Status.** Corrections to the bone evidence: (1) The 24 blade cuts across the back of the neck are left out as probable executions of captives; the neck keeps a small share for battle blows. (2) Trunk raised to 30%: chest and belly wounds mostly miss bone (the evidence has 10 torso bones in 100). (3) Head kept near its share of the remaining bones (22%). (4) Arms lowered from 38% to 24% and legs and hips from 26% to 24% to make room for the trunk. Sides are not recorded, so left and right are near even.

**Sources.** Derived from sidon-1253-mass-graves (Mikulski et al. 2021, S1 Table).

# Visby 1361



*27 July 1361*

King Valdemar IV Atterdag of Denmark invaded Gotland, then a rich trading island, and met its rural defenders outside the walls of Visby. The Gotlanders were a peasant levy facing professional troops, and the fight became a slaughter. The dead were buried quickly in mass graves, many still in their armour, which is why Visby gives the best picture of medieval battle wounds: 1,185 skeletons were recovered from three graves.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Danish army** | Professional soldiers of Valdemar IV. | Mail, coats of plates and plate helmets; the best-equipped wore gauntlets and leg harness. Few of their dead are in the graves. |
| **Gotlandic levy** | Farmers and freeholders of rural Gotland, called out to defend the island. | Uneven. Many wore mail coifs and padded coats; some had coats of plates and gauntlets; many had little or nothing on the legs, which took most of the blade wounds. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Gotland levy** (picture) | Gotlandic levy | Gotland levy: mail coif, padded coat | Spear / lance | Coif and padded coat; legs bare. |
|  | **Gotland freeholder** (picture) | Gotlandic levy | Gotland, better equipped: coif, coat of plates | Axe (hand or long-hafted) | Wealthier farmer in a coat of plates. |
|  | **Danish man-at-arms** (picture) | Danish army | Danish man-at-arms (1361) | Sword (arming sword) | Fully harnessed professional. |
|  | **Danish sergeant** | Danish army | Coat of plates and mail (c.1350) | Spiked club / morning star (knout) | Coat of plates and mail; the spiked club matches square holes in Visby skulls. |
|  | **Danish crossbowman** (picture) | Danish army | Coat of plates and mail (c.1350) | Crossbow (bolts) | 126 arrow and bolt wounds were found among the Gotland dead. |

**Sources.** Swedish History Museum, 'Skull from the Battle of Visby'. Medievalists.net, 'Medieval battle injuries' (2024), summarising Ingelmark in Thordeman (1939).

# Visby 1361 - bone evidence

*Melee, Gotland peasant levy (mail, shields, few helmets) vs Danish professional troops.*

Confidence: **Fitted - estimated to match published summary figures**. Armour: **already reflected, do not apply**.

| Location | Zone | Sword | Axe | Spiked club | Spear | Dagger | Bow | Crossbow |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-13 | 01-12 | 01-18 | 01-11 | 01-16 | 01-13 | 01-14 |
| Skull (right) | head | 14-23 | 13-21 | 19-32 | 12-20 | 17-29 | 14-23 | 15-25 |
| Face | head | 24-27 | 22-25 | 33-38 | 21-25 | 30-37 | 24-27 | 26-29 |
| Neck | head | 28 | 26 | 39 | 26 | 38-39 | 28 | 30 |
| Chest (left) | torso | 29 | 27 | 40 | 27-28 | 40-41 | 29 | 31 |
| Chest (right) | torso | 30 | 28 | 41 | 29-30 | 42-43 | 30 | 32 |
| Upper back | torso | 31 | 29 | 42 | 31 | 44-45 | 31 | 33 |
| Shoulder (left) | arms | 32-33 | 30-31 | 43-44 | 32-33 | 46-48 | 32-33 | 34-35 |
| Shoulder (right) | arms | 34 | 32 | 45 | 34 | 49 | 34 | 36 |
| Upper arm (left) | arms | 35-36 | 33-34 | 46-47 | 35-36 | 50-52 | 35-36 | 37-38 |
| Upper arm (right) | arms | 37-38 | 35-36 | 48-49 | 37-38 | 53-55 | 37-38 | 39-40 |
| Forearm (left) | arms | 39-41 | 37-38 | 50-52 | 39-41 | 56-59 | 39-41 | 41-43 |
| Forearm (right) | arms | 42-44 | 39-40 | 53-55 | 42-44 | 60-63 | 42-44 | 44-46 |
| Hand (left) | arms | 45 | 41 | 56 | 45 | 64 | 45 | 47 |
| Hand (right) | arms | 46-47 | 42-43 | 57-58 | 46-47 | 65-67 | 46-47 | 48-49 |
| Thigh (left) | legs | 48-53 | 44-50 | 59-63 | 48-53 | 68-71 | 48-53 | 50-55 |
| Thigh (right) | legs | 54-57 | 51-54 | 64-66 | 54-57 | 72-73 | 54-57 | 56-59 |
| Knee (left) | legs | 58-62 | 55-59 | 67-70 | 58-62 | 74-76 | 58-62 | 60-64 |
| Knee (right) | legs | 63-65 | 60-62 | 71-72 | 63-65 | 77-78 | 63-65 | 65-67 |
| Lower leg (left) | legs | 66-87 | 63-86 | 73-89 | 66-87 | 79-92 | 66-87 | 68-88 |
| Lower leg (right) | legs | 88-97 | 87-97 | 90-97 | 88-97 | 93-98 | 88-97 | 89-97 |
| Foot (left) | legs | 98-99 | 98-99 | 98-99 | 98-99 | 99 | 98-99 | 98-99 |
| Foot (right) | legs | 00 | 00 | 00 | 00 | 00 | 00 | 00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** These weights are where blows left marks on bone, so the defenders' armour is already baked in (mail and shields kept blades off the torso). When using this table, do not also subtract torso armour - or use the adjusted table instead.

**Status.** PLACEHOLDER. Weights fitted to published summaries: lower legs (left heaviest) dominate, then skull, then arms; almost no blade marks on ribs. Replace with Ingelmark's per-bone counts when the chapter is available.

**Sources.** Thordeman, B. (ed.), Armour from the Battle of Wisby 1361, vol. I, 1939 - ch. IV, Ingelmark, B. E., 'The Skeletons', p. 149ff. (NOT YET READ - counts pending) | Summary figures: 1,185 individuals; 456 cut wounds; head wounds in 42.3% / 52.3% / 5.4% of graves 1-3 (Ingelmark, via Medievalists.net 2024). | Swedish History Museum: visible wounds concentrate on arms, legs and skulls; torso protected by mail and shields.

# Visby 1361 - gameplay-adjusted

*Same fight as the evidence table, re-weighted to where blows land before armour.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**.

| Location | Zone | Sword | Axe | Spiked club | Spear | Dagger | Bow | Crossbow | Sling |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-08 | 01-08 | 01-11 | 01-06 | 01-08 | 01-08 | 01-09 | 01-11 |
| Skull (right) | head | 09-15 | 09-15 | 12-21 | 07-11 | 09-15 | 09-15 | 10-17 | 12-21 |
| Face | head | 16-19 | 16-19 | 22-26 | 12-16 | 16-21 | 16-19 | 18-21 | 22-28 |
| Neck | head | 20-21 | 20-21 | 27-29 | 17 | 22-25 | 20-21 | 22-23 | 29-31 |
| Chest (left) | torso | 22-28 | 22-26 | 30-36 | 18-26 | 26-35 | 22-28 | 24-30 | 32-38 |
| Chest (right) | torso | 29-34 | 27-30 | 37-42 | 27-34 | 36-43 | 29-34 | 31-36 | 39-44 |
| Abdomen | torso | 35-41 | 31-35 | 43-49 | 35-43 | 44-53 | 35-41 | 37-43 | 45-51 |
| Groin / pelvis | torso | 42-43 | 36-37 | 50-51 | 44-46 | 54-56 | 42-43 | 44-45 | 52-53 |
| Upper back | torso | 44-46 | 38-39 | 52-54 | 47-50 | 57-60 | 44-46 | 46-48 | 54-56 |
| Lower back | torso | 47-48 | 40-41 | 55-56 | 51-53 | 61-63 | 47-48 | 49-50 | 57-58 |
| Shoulder (left) | arms | 49-51 | 42-44 | 57-59 | 54-56 | 64-66 | 49-51 | 51-53 | 59-61 |
| Shoulder (right) | arms | 52-54 | 45-47 | 60-62 | 57-59 | 67-69 | 52-54 | 54-56 | 62-64 |
| Upper arm (left) | arms | 55-56 | 48-49 | 63-64 | 60-61 | 70-71 | 55-56 | 57-58 | 65-66 |
| Upper arm (right) | arms | 57-58 | 50-51 | 65-66 | 62-63 | 72-73 | 57-58 | 59-60 | 67-68 |
| Forearm (left) | arms | 59-61 | 52-54 | 67-69 | 64-66 | 74-76 | 59-61 | 61-63 | 69-71 |
| Forearm (right) | arms | 62-64 | 55-57 | 70-72 | 67-69 | 77-79 | 62-64 | 64-66 | 72-73 |
| Hand (left) | arms | 65 | 58 | 73 | 70 | 80 | 65 | 67 | 74 |
| Hand (right) | arms | 66 | 59 | 74 | 71 | 81 | 66 | 68 | 75 |
| Thigh (left) | legs | 67-73 | 60-68 | 75-79 | 72-77 | 82-85 | 67-73 | 69-74 | 76-80 |
| Thigh (right) | legs | 74-78 | 69-74 | 80-83 | 78-81 | 86-88 | 74-78 | 75-79 | 81-84 |
| Knee (left) | legs | 79-81 | 75-78 | 84-85 | 82-83 | 89-90 | 79-81 | 80-82 | 85-86 |
| Knee (right) | legs | 82-83 | 79-80 | 86-87 | 84-85 | 91 | 82-83 | 83-84 | 87 |
| Lower leg (left) | legs | 84-92 | 81-91 | 88-94 | 86-93 | 92-96 | 84-92 | 85-92 | 88-94 |
| Lower leg (right) | legs | 93-97 | 92-97 | 95-98 | 94-97 | 97-99 | 93-97 | 93-97 | 95-98 |
| Foot (left) | legs | 98-99 | 98-99 | 99 | 98-99 | 00 | 98-99 | 98-99 | 99 |
| Foot (right) | legs | 00 | 00 | 00 | 00 | - | 00 | 00 | 00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** These weights are where blows LAND. Resolve armour separately using the location's armour slot (body.yaml). Use this table when your system tracks armour coverage.

**Status.** Corrections applied to the bone evidence: (1) Torso raised to ~27%: blows that mail stopped, and soft-tissue belly and chest wounds, leave no bone marks. (2) Thighs raised: deep muscle wounds often miss the femur. (3) Lower-leg dominance kept but reduced; the shield-side (left) bias kept. (4) Head lowered slightly: some skull wounds are blows struck after a man fell.

**Sources.** Derived from visby-1361-evidence.

# Wars of the Roses



*Towton, 29 March 1461*

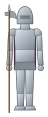
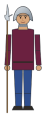
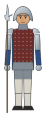
The Wars of the Roses (1455-1485) were fought between the houses of York and Lancaster for the English crown. At Towton in Yorkshire on Palm Sunday 1461, the Yorkist army of Edward IV defeated Henry VI's Lancastrians in a snowstorm, after an archery exchange and hours of fighting on foot; the Lancastrian army broke and was cut down in the rout. Contemporary claims of 28,000 dead are thought exaggerated, but it is often called the bloodiest battle fought in England. In 1996 building work at Towton Hall uncovered a mass grave of about 38 men, which gives the most detailed picture of medieval battle wounds yet recorded.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Yorkists** | Edward IV's army: his retinue and those of Warwick, Fauconberg and (late in the day) Norfolk. | Nobles and men-at-arms in full plate, fighting on foot; retainers in brigandines and sallets; archers and billmen in padded jacks and sallets. |
| **Lancastrians** | Henry VI's army under the Duke of Somerset, raised largely from the north. | The same mix as the Yorkists. The head wounds in the Towton grave suggest many of the dead had lost or taken off their helmets in the rout. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Yorkist man-at-arms** (picture) | Yorkists | Full plate harness (c.1450) | Poleaxe (axe, hammer and spike) | Full harness, fighting on foot with the poleaxe. |
|  | **Yorkist archer** (picture) | Yorkists | Jack and sallet (archer or billman, c.1460) | Longbow (English war bow, bodkin and broadhead arrows) | By the traditional account, shot with the wind and snow at his back. |
|  | **Yorkist billman** | Yorkists | Jack and sallet (archer or billman, c.1460) | Bill (English bill, hooked polearm) | Jack, sallet and bill: the common infantryman. |
|  | **Lancastrian man-at-arms** (picture) | Lancastrians | Full plate harness (c.1450) | Sword (arming sword) | Full harness; sword and dagger in the press. |
|  | **Lancastrian retainer** (picture) | Lancastrians | Brigandine, sallet and arm splints (c.1460) | Bill (English bill, hooked polearm) | A lord's better-equipped retainer. |
|  | **Lancastrian archer** (picture) | Lancastrians | Jack and sallet (archer or billman, c.1460) | Longbow (English war bow, bodkin and broadhead arrows) | By the traditional account, shot into the wind and fell short. |

**Sources.** Fiorato V., Boylston A. and Knüsel C. (eds), Blood Red Roses: the archaeology of a mass grave from the Battle of Towton AD 1461 (Oxbow, 2000; 2nd ed. 2007). Holst M. and Sutherland T., 'Towton Revisited', in Eickhoff and Schopper (eds), Schlachtfeld und Massengrab (2014), pp. 97-129.

# Towton 1461 - mass grave (bone evidence)

*Death wounds on the men buried at Towton Hall after the battle of 29 March 1461, most likely Lancastrians caught in the rout. Killed only.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Each weapon column is that weapon's own recorded wounds from the source.

| Location | Zone | Sword | Bill | Poleaxe | Dagger | Longbow | Bow |
|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-28 | 01-28 | 01-42 | 01-54 | 01-32 | 01-32 |
| Skull (right) | head | 29-49 | 29-49 | 43-58 | 55-00 | 33-54 | 33-54 |
| Face | head | 50-60 | 50-60 | 59-88 | - | 55-68 | 55-68 |
| Neck | head | 61-66 | 61-66 | 89-91 | - | 69-73 | 69-73 |
| Groin / pelvis | torso | 67 | 67 | - | - | 74 | 74 |
| Shoulder (left) | arms | 68-69 | 68-69 | - | - | 75 | 75 |
| Shoulder (right) | arms | 70 | 70 | - | - | 76 | 76 |
| Upper arm (left) | arms | 71 | 71 | - | - | 77 | 77 |
| Upper arm (right) | arms | 72 | 72 | - | - | 78 | 78 |
| Forearm (left) | arms | 73-75 | 73-75 | - | - | 79-80 | 79-80 |
| Forearm (right) | arms | 76-78 | 76-78 | 92-94 | - | 81-83 | 81-83 |
| Hand (left) | arms | 79-84 | 79-84 | - | - | 84-88 | 84-88 |
| Hand (right) | arms | 85-94 | 85-94 | 95-97 | - | 89-95 | 89-95 |
| Thigh (left) | legs | 95 | 95 | - | - | 96 | 96 |
| Thigh (right) | legs | 96 | 96 | - | - | 97 | 97 |
| Knee (right) | legs | 97 | 97 | - | - | - | - |
| Lower leg (left) | legs | 98 | 98 | 98-00 | - | 98 | 98 |
| Foot (left) | legs | 99-00 | 99-00 | - | - | 99-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Face and jaw (26) = 14.4%; Skull, left (48) = 26.7%; Skull, midline or both sides (21) = 11.7%; Skull, right (29) = 16.1%; Foot, left (3) = 1.7%; Forearm, left (4) = 2.2%; Forearm, right (5) = 2.8%; Pelvis (1) = 0.6%; Hand, left (8) = 4.4%; Hand, side not recorded (1) = 0.6%; Hand, right (13) = 7.2%; Knee, right (1) = 0.6%; Lower leg, left (2) = 1.1%; Neck (cervical vertebrae) (9) = 5.0%; Shoulder, left (2) = 1.1%; Shoulder, right (1) = 0.6%; Thigh, left (1) = 0.6%; Thigh, right (2) = 1.1%; Upper arm, left (2) = 1.1%; Upper arm, right (1) = 0.6% (n = 180). Counted wound by wound from the study's tables: 124 head wounds (79 blade, 32 blunt, 13 penetrating) on 31 skulls and 56 injured bones below the head (52 blade, 4 blunt). Head wounds are counted per wound and body injuries per bone, as published, so the body is slightly under-counted. Blade weapons roll on the blade wounds, the poleaxe on the blunt ones and the dagger on the penetrating ones (arrow, bolt, dagger and poleaxe-spike wounds to the skull). Bows use all the wounds together. Unsided skull wounds (frontal, occipital, both parietals) are shared between left and right.

**Reading it.** Armour is already in these numbers: the head wounds suggest many men had lost or taken off their helmets in the rout, and the wounds fall where they were unprotected. Do not also subtract armour on this table; use the all-hits table for play. This is bone evidence, so wounds to the chest and belly that touched no bone left no mark, and arrows (which mostly wounded soft tissue) are under-counted.

**Status.** Historical, from one grave of about 38 men killed in the rout after the battle. Their wounds are extreme: one man (skeleton 32) had 14 head wounds, many blows fell on the back of the skull and the neck, and the forearms and hands show defence wounds. It is a picture of men cut down while fleeing or after they fell, not of the fighting in the line.

**Sources.** Holst M. and Sutherland T., 'Towton Revisited - Analysis of the Human Remains from the Battle of Towton 1461', in Eickhoff S. and Schopper F. (eds), Schlachtfeld und Massengrab: Spektren interdisziplinärer Auswertung von Orten der Gewalt (Zossen 2014), pp. 97-129: Tab. 9 (post-cranial injuries) and Tab. 12 (peri-mortem cranial weapon trauma). | Same chapter: 36% of the skeletons had peri-mortem injuries below the head (56 in all), mostly to the hands and forearms; the back of the skull, left parietal and left frontal took the most blade wounds. Its text gives 33 blunt (25%) and 13 penetrating (10%) head wounds; its table lists 124 head wounds, which this table follows. | Fiorato V., Boylston A. and Knüsel C. (eds), Blood Red Roses: the archaeology of a mass grave from the Battle of Towton AD 1461 (Oxbow, 2000; 2nd ed. 2007): the original excavation and analysis (Novak, 'Battle-related trauma'). | Wound-by-wound tally: data/sources/towton-1461-mass-grave.csv.

# Towton 1461 - gameplay-adjusted (before armour)

*Where blows and arrows land on a man in the press at Towton, before armour, re-weighted from the bone evidence. Use for the Wars of the Roses.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**.

| Location | Zone | Sword | Bill | Poleaxe | Axe | Dagger | Spear | Longbow | Bow | Crossbow |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-12 | 01-13 | 01-14 | 01-13 | 01-11 | 01-09 | 01-12 | 01-12 | 01-13 |
| Skull (right) | head | 13-21 | 14-23 | 15-25 | 14-23 | 12-19 | 10-16 | 13-21 | 13-21 | 14-23 |
| Face | head | 22-27 | 24-30 | 26-32 | 24-29 | 20-27 | 17-23 | 22-27 | 22-27 | 24-29 |
| Neck | head | 28-31 | 31-34 | 33-37 | 30-33 | 28-34 | 24-26 | 28-31 | 28-31 | 30-33 |
| Chest (left) | torso | 32-38 | 35-40 | 38-43 | 34-39 | 35-43 | 27-35 | 32-38 | 32-38 | 34-39 |
| Chest (right) | torso | 39-44 | 41-45 | 44-48 | 40-44 | 44-50 | 36-43 | 39-44 | 39-44 | 40-45 |
| Abdomen | torso | 45-51 | 46-51 | 49-54 | 45-49 | 51-59 | 44-52 | 45-51 | 45-51 | 46-51 |
| Groin / pelvis | torso | 52-54 | 52-54 | 55-57 | 50-51 | 60-63 | 53-56 | 52-54 | 52-54 | 52-54 |
| Upper back | torso | 55-58 | 55-58 | 58-61 | 52-54 | 64-68 | 57-61 | 55-58 | 55-58 | 55-58 |
| Lower back | torso | 59-61 | 59-61 | 62-64 | 55-56 | 69-72 | 62-65 | 59-61 | 59-61 | 59-61 |
| Shoulder (left) | arms | 62-63 | 62-63 | 65-66 | 57-58 | 73-74 | 66-67 | 62-63 | 62-63 | 62-63 |
| Shoulder (right) | arms | 64 | 64 | 67 | 59 | 75 | 68 | 64 | 64 | 64 |
| Upper arm (left) | arms | 65-66 | 65-66 | 68-69 | 60-61 | 76-77 | 69-70 | 65-66 | 65-66 | 65-66 |
| Upper arm (right) | arms | 67-68 | 67-68 | 70-71 | 62-63 | 78-79 | 71-72 | 67-68 | 67-68 | 67-68 |
| Forearm (left) | arms | 69-71 | 69-71 | 72-74 | 64-66 | 80-82 | 73-75 | 69-71 | 69-71 | 69-71 |
| Forearm (right) | arms | 72-74 | 72-74 | 75-77 | 67-69 | 83-85 | 76-78 | 72-74 | 72-74 | 72-74 |
| Hand (left) | arms | 75-78 | 75-78 | 78-81 | 70-73 | 86-89 | 79-81 | 75-78 | 75-78 | 75-78 |
| Hand (right) | arms | 79-82 | 79-82 | 82-84 | 74-77 | 90-93 | 82-84 | 79-82 | 79-82 | 79-82 |
| Thigh (left) | legs | 83-86 | 83-86 | 85-87 | 78-82 | 94-95 | 85-87 | 83-86 | 83-86 | 83-86 |
| Thigh (right) | legs | 87-90 | 87-90 | 88-90 | 83-87 | 96-97 | 88-90 | 87-90 | 87-90 | 87-90 |
| Knee (left) | legs | 91 | 91 | 91 | 88 | - | 91 | 91 | 91 | 91 |
| Knee (right) | legs | 92 | 92 | 92 | 89 | - | 92 | 92 | 92 | 92 |
| Lower leg (left) | legs | 93-95 | 93-95 | 93-95 | 90-93 | 98 | 93-95 | 93-95 | 93-95 | 93-95 |
| Lower leg (right) | legs | 96-97 | 96-97 | 96-97 | 94-96 | 99 | 96-97 | 96-97 | 96-97 | 96-97 |
| Foot (left) | legs | 98-99 | 98-99 | 98-99 | 97-99 | 00 | 98-99 | 98-99 | 98-99 | 98-99 |
| Foot (right) | legs | 00 | 00 | 00 | 00 | - | 00 | 00 | 00 | 00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spiked club: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** These weights are where blows LAND. Resolve armour separately with the target's kit: a man-at-arms in full harness, a retainer in brigandine and sallet, or an archer or billman in a jack.

**Status.** The bone evidence is 74% head and neck, 21% arms, 5% legs and almost no trunk. Corrections: (1) Trunk raised to 30%: blade, spike and arrow wounds to the chest and belly mostly miss bone, and the evidence shows almost none. (2) Head and neck lowered to 31%: many Towton skull wounds were struck after a man fell or lost his helmet in the rout (14 on one skull). (3) Arms kept at about a fifth, hands and forearms highest (defence wounds). (4) Legs raised to 18%: deep thigh wounds often miss the femur. The left-over-right skull bias (48 against 29 wounds) is kept.

**Sources.** Derived from towton-1461-mass-grave (Holst and Sutherland 2014, Tab. 9 and 12).

# Thirty Years' War



*1618-1648*

A war of religion and power across the Holy Roman Empire: the Catholic Emperor and the Catholic League against Protestant princes, joined by Denmark, then Sweden under Gustavus Adolphus, and finally France. It was the age of pike and shot: squares of pikemen with musketeers on their flanks, and cavalry that fought with pistols and carbines as much as swords. At Lützen, near Leipzig, on 16 November 1632 the Swedes beat Wallenstein's imperial army but Gustavus Adolphus was killed. A mass grave of 47 men found there in 2011 is most likely part of the Swedish 'Blue Brigade', surprised in the flank by imperial cavalry; it is the source of this group's table.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Sweden and Protestant allies** | Swedish and Finnish troops and many German and Scottish mercenaries in Swedish pay, with Saxony and other Protestant states. | Pikemen wore helmets and corslets when they kept them; musketeers wore none; cavalry rode in buff coats with helmets and breastplates. |
| **Imperial army and Catholic League** | The Emperor's army under Wallenstein and Pappenheim, Bavaria and the Catholic League, and Croat light horse. | Cuirassiers in three-quarter plate, harquebusiers in buff coat and breastplate, pikemen in corslets; Croats and musketeers unarmoured. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Blue Brigade musketeer** (picture) | Sweden and Protestant allies | Unarmoured | Matchlock musket | Matchlock and bandolier; clubs the musket in the press. |
|  | **Swedish pikeman** (picture) | Sweden and Protestant allies | Pikeman: morion and corslet | Spear / lance | Pike about 5 m long. |
|  | **Finnish hakkapeliitta** | Sweden and Protestant allies | Harquebusier: buff coat, breastplate, pot helmet | Sabre or broadsword | Light cavalry: pistols, then swords at the gallop. |
|  | **Imperial cuirassier** (picture) | Imperial army and Catholic League | Cuirassier, three-quarter armour (1630s) | Wheellock pistol or carbine (cavalry) | Pappenheim's heavy horse; pistols at close range, aimed at the head. |
|  | **Imperial harquebusier** (picture) | Imperial army and Catholic League | Harquebusier: buff coat, breastplate, pot helmet | Wheellock pistol or carbine (cavalry) | Carbine and pistols; most of the balls in the Lützen grave are carbine shot. |
|  | **Croat light horseman** | Imperial army and Catholic League | Unarmoured | Sabre or broadsword | Raiders and skirmishers. |

**Sources.** Nicklisch et al., PLoS ONE 12(5): e0178252 (2017): the Lützen mass grave, the Blue Brigade, and contemporary advice to aim for the head and left chest. Sides, armour and dress: general historical knowledge, not from the grave study.

# Lützen 1632 - mass grave (bone evidence)

*Battle injuries on 47 men buried in one grave at Lützen, most likely infantry of the Swedish Blue Brigade cut down by imperial cavalry. Killed only.*

Confidence: **Historical - taken directly from a source**. Armour: **allowed**. Each weapon column is that weapon's own recorded wounds from the source.

| Location | Zone | Pistol/carbine | Matchlock | Sabre | Musket butt |
|---|---|---|---|---|---|
| Skull (left) | head | 01-33 | 01-33 | 01-10 | 01-14 |
| Skull (right) | head | 34-66 | 34-66 | 11-19 | 15-19 |
| Face | head | 67-69 | 67-69 | 20-32 | 20-62 |
| Chest (left) | torso | 70-71 | 70-71 | - | 63-64 |
| Chest (right) | torso | 72-73 | 72-73 | - | 65-66 |
| Groin / pelvis | torso | 74-79 | 74-79 | 33-39 | - |
| Upper back | torso | 80-84 | 80-84 | 40-49 | 67 |
| Lower back | torso | 85 | 85 | 50-57 | - |
| Upper arm (left) | arms | - | - | 58-63 | - |
| Forearm (left) | arms | - | - | 64-69 | 68-76 |
| Forearm (right) | arms | - | - | 70-75 | 77-81 |
| Hand (left) | arms | - | - | - | 82-86 |
| Thigh (left) | legs | - | - | - | 87-95 |
| Thigh (right) | legs | 86-88 | 86-88 | 76-94 | - |
| Lower leg (left) | legs | 89-94 | 89-94 | - | - |
| Lower leg (right) | legs | 95-00 | 95-00 | 95-00 | - |
| Foot (right) | legs | - | - | - | 96-00 |

**Close combat.** Dagger: roll location on *Knife assault - stab wounds (forensic)*. Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** frontal R; parietal R; occipital R (11) = 15.9%; frontal L; parietal L; occipital L (13) = 18.8%; facial L; facial centre; facial R (12) = 17.4%; cranial base; parietal centre; occipital centre (4) = 5.8%; lower limb R (4) = 5.8%; lower limb R (3) = 4.3%; lower limb L (2) = 2.9%; torso (1) = 1.4%; torso (1) = 1.4%; torso (1) = 1.4%; torso (2) = 2.9%; upper limb R (2) = 2.9%; upper limb L (3) = 4.3%; upper limb L (1) = 1.4%; lower limb R (1) = 1.4%; lower limb L (2) = 2.9%; torso (1) = 1.4%; upper limb L (1) = 1.4%; torso (4) = 5.8% (n = 69). Counted from the study's tables of reliable battle injuries (69: 32 projectile, 21 blunt, 16 sharp), each weapon rolling on its own mechanism's injuries. Sides are as recorded; injuries the authors only call probable are left out. Where a bone could belong to more than one location (vertebra, ribcage, cranial base), it is shared by body exposure.

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-47 | Lead ball (pistol, carbine, musket) |
| 48-77 | Blunt force (musket butt, hilt, horse) |
| 78-00 | Sharp force (sword, rapier, halberd) |

**Reading it.** The men wore little or no armour ('insufficient head protection', the authors note), so armour can be applied. This is bone evidence: wounds to soft tissue, above all the belly and chest, left no mark, so the trunk is under-counted. Some blunt facial fractures may come from the burial rather than the battle. For play, use the all-hits table.

**Status.** Historical, from one small grave (47 men, 69 reliable injuries) and one moment of the battle: a cavalry charge with pistols and carbines aimed at the head, as contemporary instructions advised. Not a picture of the whole war.

**Sources.** Nicklisch N., Ramsthaler F., Meller H., Friederich S., Alt K.W., 'The face of war: Trauma analysis of a mass grave from the Battle of Lützen (1632)', PLoS ONE 12(5): e0178252 (2017), open access (PMC5439951): Tables 1, 3 and 4. | Same study: lead balls in the grave were mostly from carbines (12-14), then muskets (5) and pistols (1-3); 21 men had gunshot wounds to the head; the attack came mainly from the front and side. | Injury tally: data/sources/lutzen-1632-mass-grave.csv.

# Thirty Years' War - all hits (gameplay)

*Where shot and blows land on a man in the open, before death and burial sort the evidence. For pike-and-shot battles of the 1600s.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Matchlock | Pistol/carbine |
|---|---|---|---|
| Skull (left) | head | 01-09 | 01-09 |
| Skull (right) | head | 10-18 | 10-18 |
| Face | head | 19-24 | 19-24 |
| Neck | head | 25 | 25 |
| Chest (left) | torso | 26-27 | 26-27 |
| Chest (right) | torso | 28-29 | 28-29 |
| Abdomen | torso | 30-31 | 30-31 |
| Groin / pelvis | torso | 32-34 | 32-34 |
| Upper back | torso | 35 | 35 |
| Lower back | torso | 36 | 36 |
| Shoulder (left) | arms | 37-39 | 37-39 |
| Shoulder (right) | arms | 40-42 | 40-42 |
| Upper arm (left) | arms | 43-49 | 43-49 |
| Upper arm (right) | arms | 50-55 | 50-55 |
| Forearm (left) | arms | 56-57 | 56-57 |
| Forearm (right) | arms | 58-59 | 58-59 |
| Hand (left) | arms | 60-62 | 60-62 |
| Hand (right) | arms | 63-64 | 63-64 |
| Thigh (left) | legs | 65-71 | 65-71 |
| Thigh (right) | legs | 72-79 | 72-79 |
| Knee (left) | legs | 80-81 | 80-81 |
| Knee (right) | legs | 82 | 82 |
| Lower leg (left) | legs | 83-89 | 83-89 |
| Lower leg (right) | legs | 90-96 | 90-96 |
| Foot (left) | legs | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 |

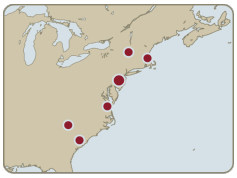
**Close combat.** Sabre, Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Musket butt, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Artillery: roll location on *Napoleonic era - all hits (gameplay)*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** Use this table for gunfire in play; armour applies (cuirassier, harquebusier and pikeman kits). Sabres and clubbed muskets roll on their own tables (the Peninsular records and the Visby melee table).

**Status.** Estimated. The killed part is real 1632 evidence but bone-only, so it under-counts trunk wounds; the wounded part is borrowed from a later war. Edit the share in tools/blend_tables.py if you prefer another.

**Sources.** Derived (tools/blend_tables.py): 74.5% peninsular-1808-14-officers (wounded) + 25.5% lutzen-1632-mass-grave (killed), location by location. | No wounded records survive for the Thirty Years' War; the Peninsular War officers' wounds (same smoothbore lead balls and swords, 180 years later) stand in. Killed share as for the Napoleonic blend (about 1 killed to 3 wounded).

# American Revolution



*1775-1783*

The thirteen colonies, joined by France from 1778 and later Spain and the Dutch Republic, fought Britain for independence. Battles were fought with smoothbore muskets and bayonets at close range, with artillery firing round shot, grape and canister; riflemen picked off officers from cover; dragoons fought with the sabre. Bayonet charges decided many fights, and some ended in killings of men trying to surrender, such as Paoli (1777) and Waxhaws (1780). Peckham counted 7,174 Americans killed and 8,241 wounded in action. The tables here are tallied from the men disabled for life who claimed federal invalid pensions in 1792-95.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Continental Army and militia** | Continental regulars, state militia, riflemen, light dragoons, and the Continental Navy and privateers. | None. Officers wore a small gorget as a mark of rank; light dragoons wore leather helmets. |
| **British Army and Loyalists** | British regulars and marines, Loyalist provincial regiments and rangers, and the Royal Navy. | None. Grenadiers' bearskin caps and light infantry caps gave no protection; light dragoons wore crested leather helmets. |
| **German auxiliaries** | Regiments hired from Hesse-Kassel, Brunswick and other German states ('Hessians'), including Jäger riflemen. | None; grenadiers wore brass-fronted mitre caps, which were not armour. |
| **French forces** | Rochambeau's expeditionary army and the French navy, from 1778; decisive at Yorktown in 1781. | None. |
| **Native nations** | Warriors on both sides: most of the Mohawk, Seneca and Cherokee fought with Britain; the Oneida and Tuscarora with the Americans. | None. Musket, tomahawk and knife; raids and ambushes on the frontier. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Continental infantryman** (picture) | Continental Army and militia | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Musket and bayonet. |
|  | **Morgan's rifleman** (picture) | Continental Army and militia | Unarmoured | Flintlock rifle (Pennsylvania, Baker, Jäger) | Long rifle, slow to load, deadly at range; no bayonet. |
|  | **Continental light dragoon** | Continental Army and militia | Dragoon or light dragoon helmet | Sabre or broadsword | Leather helmet and sabre. |
|  | **British grenadier** (picture) | British Army and Loyalists | Unarmoured | Bayonet (musket or rifle) | The bayonet charge was the British infantry's favoured tactic. |
|  | **Loyalist ranger** | British Army and Loyalists | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Frontier raids alongside Native allies. |

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **British light dragoon** (picture) | British Army and Loyalists | Dragoon or light dragoon helmet | Sabre or broadsword | Tarleton's British Legion fought with sabres at Waxhaws. |
|  | **Hessian Jäger** (picture) | German auxiliaries | Unarmoured | Flintlock rifle (Pennsylvania, Baker, Jäger) | German hunters' rifle, short and accurate. |
|  | **Hessian musketeer** (picture) | German auxiliaries | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Line infantry; musket and bayonet. |
|  | **French fusilier** | French forces | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Charleville musket; Yorktown 1781. |
|  | **Mohawk warrior** (picture) | Native nations | Unarmoured | Axe (hand or long-hafted) | Tomahawk and musket. |
|  | **Oneida scout** | Native nations | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Allied with the Americans. |

**Sources.** Peckham, The Toll of Independence (1974), via Journal of the American Revolution, 'The 25 Deadliest Battles of the Revolutionary War' (2014): 7,174 killed and 8,241 wounded. Federal Revolutionary War Invalid Pension Claims, 1792-1795, Magazine of Early American Datasets (Penn, 2021). Sides, alliances and dress: general historical knowledge, not from the pension data.

# American Revolution 1775-83 - disabled veterans' wounds

*Where Continental soldiers, militiamen and seamen were wounded, from the federal invalid pension lists of 1792-95. Men disabled for life who survived.*

Confidence: **Historical - taken directly from a source**. Armour: **allowed**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Musket | Rifle | Pistol | Artillery |
|---|---|---|---|---|---|
| Skull (left) | head | 01-03 | 01-03 | 01-03 | 01-03 |
| Skull (right) | head | 04-06 | 04-06 | 04-06 | 04-06 |
| Face | head | 07-09 | 07-09 | 07-09 | 07-09 |
| Neck | head | 10-11 | 10-11 | 10-11 | 10-11 |
| Chest (left) | torso | 12-17 | 12-17 | 12-17 | 12-17 |
| Chest (right) | torso | 18-23 | 18-23 | 18-23 | 18-23 |
| Abdomen | torso | 24-26 | 24-26 | 24-26 | 24-26 |
| Groin / pelvis | torso | 27-31 | 27-31 | 27-31 | 27-31 |
| Upper back | torso | 32-33 | 32-33 | 32-33 | 32-33 |
| Lower back | torso | 34-36 | 34-36 | 34-36 | 34-36 |
| Shoulder (left) | arms | 37-40 | 37-40 | 37-40 | 37-40 |
| Shoulder (right) | arms | 41-44 | 41-44 | 41-44 | 41-44 |
| Upper arm (left) | arms | 45-49 | 45-49 | 45-49 | 45-49 |
| Upper arm (right) | arms | 50-54 | 50-54 | 50-54 | 50-54 |
| Forearm (left) | arms | 55-59 | 55-59 | 55-59 | 55-59 |
| Forearm (right) | arms | 60-64 | 60-64 | 60-64 | 60-64 |
| Hand (left) | arms | 65-68 | 65-68 | 65-68 | 65-68 |
| Hand (right) | arms | 69-72 | 69-72 | 69-72 | 69-72 |
| Thigh (left) | legs | 73-77 | 73-77 | 73-77 | 73-77 |
| Thigh (right) | legs | 78-82 | 78-82 | 78-82 | 78-82 |
| Knee (left) | legs | 83 | 83 | 83 | 83 |
| Knee (right) | legs | 84 | 84 | 84 | 84 |
| Lower leg (left) | legs | 85-91 | 85-91 | 85-91 | 85-91 |
| Lower leg (right) | legs | 92-98 | 92-98 | 92-98 | 92-98 |
| Foot (left) | legs | 99 | 99 | 99 | 99 |
| Foot (right) | legs | 00 | 00 | 00 | 00 |

**Close combat.** Bayonet, Sabre: roll location on *Peninsular War 1808-14 - French officers' wounds*. Musket butt, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and scalp (24) = 6.3%; Face, eye, jaw (11) = 2.9%; Neck and throat (9) = 2.4%; Chest, breast and side (36) = 9.5%; Through the body (level not stated) (12) = 3.2%; Belly (8) = 2.1%; Hip, groin, testicles (18) = 4.7%; Back, shoulder blade (2) = 0.5%; Small of the back, loins (4) = 1.1%; Back, level not stated (9) = 2.4%; Shoulder and armpit (32) = 8.4%; Arm (not said which part) (49) = 12.9%; Upper arm and elbow (11) = 2.9%; Wrist and forearm (11) = 2.9%; Hand, fingers, thumb (32) = 8.4%; Thigh (40) = 10.6%; Knee (5) = 1.3%; Leg below the knee (55) = 14.5%; Foot, ankle, heel (11) = 2.9% (n = 379). Counted wound by wound from 344 men's entries (379 wounds whose site is named; 32 more not located). Sides are recorded for about half and were close to even (85 left, 99 right), so left and right are pooled and split by body exposure (design estimate). 'Arm' without more detail is shared between upper arm and forearm.

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-82 | Musket or rifle ball, buck shot |
| 83-90 | Cannon ball, grape, shell |
| 91-95 | Bayonet |
| 96-98 | Sword, broadsword, horseman's sword |
| 99-00 | Musket butt or muzzle, blows |

**Reading it.** Nobody wore armour, so these are hits before armour. But they are survivors with a lasting disability: wounds that killed (most head, chest and belly wounds) are missing, and limb wounds that left a man lame are over-represented. Use the all-hits table for play.

**Status.** Historical, from a pension roll rather than a battlefield count. Entries for sickness, ruptures, frostbite, falls, work accidents and accidental discharges were left out. The pattern (head and neck 12%, trunk 23%, arms 36%, legs 29%) is close to the Civil War wounded (11/18/36/35), which suggests the same survivor effect.

**Sources.** Federal Revolutionary War Invalid Pension Claims, 1792-1795 (573 veterans), Magazine of Early American Datasets, University of Pennsylvania, 2021; from the Secretary of War's lists printed in American State Papers, Claims, vol. 1. | Wound-by-wound tally: data/sources/revolution-invalid-pensions-1792-95.csv. The weapon mix counts the 251 wounds whose cause is stated.

# American Revolution 1775-83 - all hits (gameplay)

*Where musket balls and shot land on a man in the open, before survival sorts the records.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Musket | Rifle | Pistol | Artillery |
|---|---|---|---|---|---|
| Skull (left) | head | 01-07 | 01-07 | 01-07 | 01-07 |
| Skull (right) | head | 08-14 | 08-14 | 08-14 | 08-14 |
| Face | head | 15-21 | 15-21 | 15-21 | 15-21 |
| Neck | head | 22-26 | 22-26 | 22-26 | 22-26 |
| Chest (left) | torso | 27-34 | 27-34 | 27-34 | 27-34 |
| Chest (right) | torso | 35-42 | 35-42 | 35-42 | 35-42 |
| Abdomen | torso | 43-48 | 43-48 | 43-48 | 43-48 |
| Groin / pelvis | torso | 49-53 | 49-53 | 49-53 | 49-53 |
| Upper back | torso | 54-58 | 54-58 | 54-58 | 54-58 |
| Lower back | torso | 59-62 | 59-62 | 59-62 | 59-62 |
| Shoulder (left) | arms | 63-64 | 63-64 | 63-64 | 63-64 |
| Shoulder (right) | arms | 65-66 | 65-66 | 65-66 | 65-66 |
| Upper arm (left) | arms | 67-69 | 67-69 | 67-69 | 67-69 |
| Upper arm (right) | arms | 70-72 | 70-72 | 70-72 | 70-72 |
| Forearm (left) | arms | 73-75 | 73-75 | 73-75 | 73-75 |
| Forearm (right) | arms | 76-78 | 76-78 | 76-78 | 76-78 |
| Hand (left) | arms | 79-80 | 79-80 | 79-80 | 79-80 |
| Hand (right) | arms | 81-82 | 81-82 | 81-82 | 81-82 |
| Thigh (left) | legs | 83-85 | 83-85 | 83-85 | 83-85 |
| Thigh (right) | legs | 86-88 | 86-88 | 86-88 | 86-88 |
| Knee (left) | legs | 89 | 89 | 89 | 89 |
| Knee (right) | legs | 90 | 90 | 90 | 90 |
| Lower leg (left) | legs | 91-94 | 91-94 | 91-94 | 91-94 |
| Lower leg (right) | legs | 95-98 | 95-98 | 95-98 | 95-98 |
| Foot (left) | legs | 99 | 99 | 99 | 99 |
| Foot (right) | legs | 00 | 00 | 00 | 00 |

**Close combat.** Bayonet, Sabre: roll location on *Peninsular War 1808-14 - French officers' wounds*. Musket butt, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** Use this table for play. Nobody wore armour; a kit from another era can be applied.

**Status.** Estimated. No count of where the Revolution's dead were hit survives, so the Civil War killed-in-action table (also soft lead balls, 1,173 men) stands in for them. Peckham's killed count includes men killed after surrendering or in massacres, which raises the killed share; edit the share in tools/blend_tables.py if you prefer another.

**Sources.** Derived (tools/blend_tables.py): 53.5% revolution-1775-83-pensioners + 46.5% acw-1861-killed, location by location. | Killed share: Peckham, The Toll of Independence (1974), 7,174 Americans killed and 8,241 wounded in 1,331 land and 218 naval engagements (via Journal of the American Revolution, 'The 25 Deadliest Battles of the Revolutionary War', 2014).

# Napoleonic Wars



*1803-1815*

France under Napoleon against shifting coalitions of Britain, Austria, Prussia, Russia, Spain and Portugal. In the Peninsular War (1808-1814) a British and Portuguese army under Wellington, with Spanish armies and guerrillas, drove the French out of Spain; the wound tables come from French officers' files of that war. The War of 1812 in North America, fought with the same weapons, has its own entry, which borrows these records.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **French Empire and allies** | French line and light infantry, cavalry and artillery, with Polish, German and Italian allies. | Cuirassiers and carabiniers wore steel breastplates and backplates and helmets; dragoons wore brass helmets; infantry wore none. |
| **Britain and Peninsular allies** | Wellington's British and Portuguese army (including the 95th Rifles), Spanish regular armies and guerrillas. | None for infantry; heavy and light dragoons wore helmets. British Household Cavalry wore cuirasses only after 1815. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **French line infantryman** (picture) | French Empire and allies | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Charleville musket and bayonet. |
|  | **French cuirassier** (picture) | French Empire and allies | Cuirassier: steel cuirass and helmet | Sabre or broadsword | Heavy cavalry: straight sword, breastplate and helmet. |
|  | **Polish lancer** (picture) | French Empire and allies | Unarmoured | Cavalry lance | Vistula Legion lancers, feared at Albuera (1811). |
|  | **British redcoat** (picture) | Britain and Peninsular allies | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Brown Bess and bayonet, firing in two-deep line. |
|  | **95th rifleman** (picture) | Britain and Peninsular allies | Unarmoured | Flintlock rifle (Pennsylvania, Baker, Jäger) | Baker rifle; skirmisher and sniper. |
|  | **British heavy dragoon** (picture) | Britain and Peninsular allies | Dragoon or light dragoon helmet | Sabre or broadsword | 1796 heavy cavalry sword. |
|  | **Spanish guerrilla** (picture) | Britain and Peninsular allies | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Ambushes of convoys and couriers. |

**Sources.** Planas Campos and Grajal de Blas, British Journal for Military History 7(3), 2021 (Peninsular War wound database). Sides, armour and dress: general historical knowledge, not from the wound data.

# Peninsular War 1808-14 - French officers' wounds

*Where French officers were hit in Spain and Portugal, 1808-1814, by musket, sabre, bayonet, lance and cannon. Mostly men who survived.*

Confidence: **Historical - taken directly from a source**. Armour: **allowed**. Each weapon column is that weapon's own recorded wounds from the source.

| Location | Zone | Musket | Rifle | Pistol | Artillery | Sabre | Bayonet | Lance |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-04 | 01-04 | 01-04 | 01-06 | 01-17 | 01-05 | 01-07 |
| Skull (right) | head | 05-08 | 05-08 | 05-08 | 07-12 | 18-34 | 06-10 | 08-14 |
| Face | head | 09-10 | 09-10 | 09-10 | 13-15 | 35-37 | 11-12 | 15-19 |
| Neck | head | 11 | 11 | 11 | 16 | 38 | 13 | 20-21 |
| Chest (left) | torso | 12-13 | 12-13 | 12-13 | 17 | 39 | 14-17 | 22-28 |
| Chest (right) | torso | 14-15 | 14-15 | 14-15 | 18 | 40 | 18-21 | 29-35 |
| Abdomen | torso | 16-18 | 16-18 | 16-18 | 19-21 | 41 | 22-25 | - |
| Groin / pelvis | torso | 19-21 | 19-21 | 19-21 | 22-25 | - | 26-30 | 36-40 |
| Upper back | torso | - | - | - | - | - | 31 | 41 |
| Lower back | torso | 22 | 22 | 22 | 26 | - | 32 | 42-44 |
| Shoulder (left) | arms | 23-26 | 23-26 | 23-26 | 27-31 | 42-45 | 33-35 | 45-48 |
| Shoulder (right) | arms | 27-30 | 27-30 | 27-30 | 32-36 | 46-49 | 36-38 | 49-52 |
| Upper arm (left) | arms | 31-38 | 31-38 | 31-38 | 37-42 | 50-59 | 39-45 | 53-61 |
| Upper arm (right) | arms | 39-46 | 39-46 | 39-46 | 43-48 | 60-68 | 46-52 | 62-70 |
| Forearm (left) | arms | 47 | 47 | 47 | - | 69-71 | 53 | 71-72 |
| Forearm (right) | arms | 48 | 48 | 48 | - | 72-74 | 54 | 73-74 |
| Hand (left) | arms | 49-51 | 49-51 | 49-51 | 49-51 | 75-82 | 55-59 | 75-79 |
| Hand (right) | arms | 52-54 | 52-54 | 52-54 | 52-54 | 83-90 | 60-64 | 80-84 |
| Thigh (left) | legs | 55-64 | 55-64 | 55-64 | 55-62 | 91-92 | 65-76 | 85-86 |
| Thigh (right) | legs | 65-74 | 65-74 | 65-74 | 63-70 | 93-94 | 77-88 | 87-88 |
| Knee (left) | legs | 75-76 | 75-76 | 75-76 | 71-72 | 95 | 89-91 | 89-92 |
| Knee (right) | legs | 77-78 | 77-78 | 77-78 | 73-74 | 96 | 92-94 | 93-96 |
| Lower leg (left) | legs | 79-86 | 79-86 | 79-86 | 75-83 | 97 | 95-97 | 97 |
| Lower leg (right) | legs | 87-94 | 87-94 | 87-94 | 84-92 | 98 | 98-00 | 98 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 93-96 | 99 | - | 99 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 97-00 | 00 | - | 00 |

**Close combat.** Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Arm (bras) (678) = 17.0%; Hand (245) = 6.1%; Wrist (35) = 0.9%; Forearm (63) = 1.6%; Leg below the knee (621) = 15.5%; Thigh (687) = 17.2%; Hip (99) = 2.5%; Foot (201) = 5.0%; Knee (155) = 3.9%; Chest (169) = 4.2%; Head (390) = 9.8%; Eye (3) = 0.1%; Forehead (18) = 0.5%; Neck (46) = 1.2%; Cheek (42) = 1.1%; Jaw (41) = 1.0%; Abdomen (102) = 2.6%; Shoulder (341) = 8.5%; Buttock (16) = 0.4%; Back (10) = 0.3%; Loins (kidneys) (33) = 0.8% (n = 3,995). Location counts are from the source (3,995 wound events). Each weapon rolls on its own recorded distribution (musket 2,848, sword 284, bayonet 97, lance 43, ordnance 209 wounds). Left and right are not recorded and are split evenly; 'arm' (bras) is put on the upper arm because wrist and forearm are listed apart. Rifles and pistols use the musket data.

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-77 | Musket, rifle and pistol |
| 78-85 | Artillery (shell, round shot, canister, grape) |
| 86-92 | Sabre and sword |
| 93-95 | Bayonet |
| 96 | Lance |
| 97-00 | Contusion (spent ball, blow, fall) |

**Reading it.** Almost no one wore body armour (cuirassiers and carabiniers had steel cuirasses, and dragoons and heavy cavalry helmets), so armour can be applied. The records are of men who survived far more often than of men killed (146 killed and 84 died of wounds against 4,129 wounded), so head and trunk hits from gunfire are under-counted: use the all-hits table for gunfire in play. Edged weapons rarely killed (8 deaths among 424 sword, bayonet and lance casualties), so their columns are close to every hit they made.

**Status.** Historical. French officers, so mounted officers and men in the front rank are over-represented; the rank-and-file sample was hit in the head more often (24%). Tables 7 and 8 were read from the article's table images.

**Sources.** Planas Campos J. and Grajal de Blas A., 'Wounds & Weapons in the Napoleonic War: a database of the Peninsular War', British Journal for Military History 7(3), Nov 2021, pp. 117-127: Table 1 (weapons, 4,359 entries), Table 2 (locations, 3,995 events), Tables 6-8 (location by firearm, edged weapon and ordnance). Source registers: French Army officer files, SHD 2Yb. | Same article, Table 11: 491 rank and file of the 100e de ligne - arms 22%, legs 40%, chest 6%, head 24%, abdomen 1%, shoulder 7% (officers: 26/44/4/14/3/9). | Comparison quoted there: Austrian officers wounded at Austerlitz (Guiga 2021) - upper limbs 26%, lower limbs 45%, head 6%, upper body 9%, back 4%.

# Napoleonic era - all hits (gameplay)

*Where musket balls and shot land on a man in the open, before survival sorts the records. Use for the Peninsula and Waterloo; the War of 1812 has its own table, built from the same records.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Musket | Rifle | Pistol | Artillery |
|---|---|---|---|---|---|
| Skull (left) | head | 01-06 | 01-06 | 01-06 | 01-06 |
| Skull (right) | head | 07-12 | 07-12 | 07-12 | 07-12 |
| Face | head | 13-17 | 13-17 | 13-17 | 13-17 |
| Neck | head | 18-20 | 18-20 | 18-20 | 18-20 |
| Chest (left) | torso | 21-25 | 21-25 | 21-25 | 21-25 |
| Chest (right) | torso | 26-30 | 26-30 | 26-30 | 26-30 |
| Abdomen | torso | 31-35 | 31-35 | 31-35 | 31-35 |
| Groin / pelvis | torso | 36-38 | 36-38 | 36-38 | 36-38 |
| Upper back | torso | 39-40 | 39-40 | 39-40 | 39-40 |
| Lower back | torso | 41-42 | 41-42 | 41-42 | 41-42 |
| Shoulder (left) | arms | 43-45 | 43-45 | 43-45 | 43-45 |
| Shoulder (right) | arms | 46-48 | 46-48 | 46-48 | 46-48 |
| Upper arm (left) | arms | 49-54 | 49-54 | 49-54 | 49-54 |
| Upper arm (right) | arms | 55-60 | 55-60 | 55-60 | 55-60 |
| Forearm (left) | arms | 61 | 61 | 61 | 61 |
| Forearm (right) | arms | 62 | 62 | 62 | 62 |
| Hand (left) | arms | 63-64 | 63-64 | 63-64 | 63-64 |
| Hand (right) | arms | 65-66 | 65-66 | 65-66 | 65-66 |
| Thigh (left) | legs | 67-73 | 67-73 | 67-73 | 67-73 |
| Thigh (right) | legs | 74-80 | 74-80 | 74-80 | 74-80 |
| Knee (left) | legs | 81-82 | 81-82 | 81-82 | 81-82 |
| Knee (right) | legs | 83-84 | 83-84 | 83-84 | 83-84 |
| Lower leg (left) | legs | 85-90 | 85-90 | 85-90 | 85-90 |
| Lower leg (right) | legs | 91-96 | 91-96 | 91-96 | 91-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 | 99-00 |

**Close combat.** Sabre, Bayonet, Lance: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** Use this table for play. Armour applies (cuirassiers' breastplates and cavalry helmets). Sabres, bayonets and lances roll on the Peninsular records, where edged weapons rarely killed, so those records already include nearly every edged hit.

**Status.** Estimated. The Peninsular records hold only 146 killed outright (head and chest most often), too few to use, so the Civil War killed-in-action table (also soft lead balls) stands in for the dead. Edit the share in tools/blend_tables.py if you prefer another.

**Sources.** Derived (tools/blend_tables.py): 74.5% peninsular-1808-14-officers + 25.5% acw-1861-killed, location by location. | Killed share: about one killed for every three wounded in the Anglo-allied army at Waterloo (about 3,500 killed and 10,200 wounded; approximate, commonly cited totals).

# War of 1812



*1812-1815*

The United States fought Britain and its Canadian colonies, and Tecumseh's confederacy of Native nations, on the Great Lakes frontier, in the Chesapeake and at New Orleans. The weapons were those of the Napoleonic Wars: smoothbore muskets and bayonets, rifles, sabres and smoothbore artillery. Much of the fighting was in woods and from cover, riflemen were common on both sides, no one wore armour, and many British regulars of 1814-15 were Peninsular veterans. US forces lost about 2,260 killed in action and 4,505 wounded. No count of wounds by body region has been found for this war, so its table borrows the Peninsular War records; every place on the map is a hollow dot for that reason.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **United States (1812-15)** | US regulars, state militia (including Kentucky riflemen), US Navy and Marines. | None; some light dragoons wore leather helmets. |
| **British North America (1812-15)** | British regulars, Canadian fencibles and militia, the Royal Navy, and the Provincial Marine. | None. |
| **Tecumseh's confederacy** | Shawnee, Potawatomi, Kickapoo and other nations under Tecumseh, allied with Britain until his death at the Thames (1813). | None. Musket, rifle, tomahawk and war club. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **US regular, 1814** (picture) | United States (1812-15) | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Chippawa and Lundy's Lane; Springfield musket. |
|  | **Kentucky militia rifleman** (picture) | United States (1812-15) | Unarmoured | Flintlock rifle (Pennsylvania, Baker, Jäger) | River Raisin, the Thames, New Orleans. |
|  | **British regular, 1814** (picture) | British North America (1812-15) | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Often a Peninsular veteran. |
|  | **Canadian militiaman** | British North America (1812-15) | Unarmoured | Smoothbore musket (Brown Bess, Charleville) | Sedentary or embodied militia. |
|  | **Shawnee warrior** (picture) | Tecumseh's confederacy | Unarmoured | Axe (hand or long-hafted) | Tomahawk and musket. |

**Sources.** No wound-by-location record found for the War of 1812; the table borrows the Peninsular War data (Planas Campos and Grajal de Blas, British Journal for Military History 7(3), 2021). US losses of about 2,260 killed in action and 4,505 wounded: commonly cited totals (e.g. New World Encyclopedia, 'War of 1812'); British and Canadian losses are usually given only combined (about 4,400 killed or wounded). Sides, armour and dress: general historical knowledge, not from the wound data.

# War of 1812 - all hits (gameplay, borrowed records)

*Where musket balls, rifle balls and shot land on a man in the open or in the woods, before survival sorts the records. No wound count survives for this war; the table is built from the Peninsular War records.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Musket | Rifle | Pistol | Artillery |
|---|---|---|---|---|---|
| Skull (left) | head | 01-07 | 01-07 | 01-07 | 01-07 |
| Skull (right) | head | 08-14 | 08-14 | 08-14 | 08-14 |
| Face | head | 15-20 | 15-20 | 15-20 | 15-20 |
| Neck | head | 21-24 | 21-24 | 21-24 | 21-24 |
| Chest (left) | torso | 25-29 | 25-29 | 25-29 | 25-29 |
| Chest (right) | torso | 30-34 | 30-34 | 30-34 | 30-34 |
| Abdomen | torso | 35-39 | 35-39 | 35-39 | 35-39 |
| Groin / pelvis | torso | 40-43 | 40-43 | 40-43 | 40-43 |
| Upper back | torso | 44-45 | 44-45 | 44-45 | 44-45 |
| Lower back | torso | 46-48 | 46-48 | 46-48 | 46-48 |
| Shoulder (left) | arms | 49-51 | 49-51 | 49-51 | 49-51 |
| Shoulder (right) | arms | 52-54 | 52-54 | 52-54 | 52-54 |
| Upper arm (left) | arms | 55-60 | 55-60 | 55-60 | 55-60 |
| Upper arm (right) | arms | 61-66 | 61-66 | 61-66 | 61-66 |
| Forearm (left) | arms | 67 | 67 | 67 | 67 |
| Forearm (right) | arms | 68 | 68 | 68 | 68 |
| Hand (left) | arms | 69-70 | 69-70 | 69-70 | 69-70 |
| Hand (right) | arms | 71-72 | 71-72 | 71-72 | 71-72 |
| Thigh (left) | legs | 73-78 | 73-78 | 73-78 | 73-78 |
| Thigh (right) | legs | 79-84 | 79-84 | 79-84 | 79-84 |
| Knee (left) | legs | 85 | 85 | 85 | 85 |
| Knee (right) | legs | 86 | 86 | 86 | 86 |
| Lower leg (left) | legs | 87-91 | 87-91 | 87-91 | 87-91 |
| Lower leg (right) | legs | 92-96 | 92-96 | 92-96 | 92-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 | 99-00 |

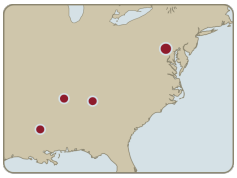
**Close combat.** Sabre, Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Reading it.** Use this table for play. No armour was worn in North America (some light dragoons had leather helmets), so leave it off. For a tomahawk or war club use the axe or spiked club, which roll on the Visby close-combat data. Much woodland and skirmish fighting makes 'behind cover' common.

**Status.** Estimated and borrowed. No count of wounds by body region has been found for the War of 1812. The weapons and drill were those of the Napoleonic Wars and many British regulars were Peninsular veterans, so the French officers' records stand in for the wounded and the Civil War killed-in-action table (also soft lead balls) for the dead. Edit the share in tools/blend_tables.py if you prefer another.

**Sources.** Derived (tools/blend_tables.py): 66.6% peninsular-1808-14-officers + 33.4% acw-1861-killed, location by location. | Killed share: US forces lost about 2,260 killed in action and 4,505 wounded (commonly cited totals; British and Canadian losses are usually given only combined). | Peninsular War records: Planas Campos and Grajal de Blas, British Journal for Military History 7(3), 2021.

# American Civil War



*1861-1865*

The United States (Union) against the seceded Confederate States. Most wounds came from the rifle-musket firing soft lead conoidal balls, which shattered bone and led to the war's many amputations. Artillery, especially canister at close range, did the rest; sabre and bayonet wounds were rare.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Union** | US Regular Army and state volunteer regiments. | None issued. In 1862 private 'bullet-proof vests' (Atwater, G&D; Cook) sold by the thousand; at about 8 lb most were thrown away on the march, and wearing one was thought unmanly. |
| **Confederacy** | Confederate States Army and state troops. | None issued; private vests were rarer still. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Union infantryman** (picture) | Union | Unarmoured | Rifle-musket (Minie ball) | Springfield rifle-musket, no armour. |
|  | **Union volunteer with steel vest, 1862** (picture) | Union | Private steel vest (Civil War, 1862) | Rifle-musket (Minie ball) | One of the few who kept his vest. |
|  | **Confederate cavalryman** (picture) | Confederacy | Unarmoured | Revolver | Revolver and sabre. |
|  | **Confederate infantryman** (picture) | Confederacy | Unarmoured | Rifle-musket (Minie ball) | Enfield or Springfield rifle-musket, no armour. |
|  | **Union cavalryman** (picture) | Union | Unarmoured | Sabre or broadsword | Sabre charge; carbine and revolver as well. |

**Sources.** Antietam on the Web, 'Saved by his steel vest' (2018): Atwater and G&D; Cook vests, sales and use.

# Civil War 1861-65 - wounded (surgeon's records)

*Union soldiers treated for gunshot and shell wounds. Men who reached a surgeon alive.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Rifle-musket | Artillery | Revolver |
|---|---|---|---|---|
| Skull (left) | head | 01-03 | 01-03 | 01-03 |
| Skull (right) | head | 04-06 | 04-06 | 04-06 |
| Face | head | 07-09 | 07-09 | 07-09 |
| Neck | head | 10-11 | 10-11 | 10-11 |
| Chest (left) | torso | 12-15 | 12-15 | 12-15 |
| Chest (right) | torso | 16-19 | 16-19 | 16-19 |
| Abdomen | torso | 20-23 | 20-23 | 20-23 |
| Groin / pelvis | torso | 24-25 | 24-25 | 24-25 |
| Upper back | torso | 26-28 | 26-28 | 26-28 |
| Lower back | torso | 29-30 | 29-30 | 29-30 |
| Shoulder (left) | arms | 31-34 | 31-34 | 31-34 |
| Shoulder (right) | arms | 35-38 | 35-38 | 35-38 |
| Upper arm (left) | arms | 39-43 | 39-43 | 39-43 |
| Upper arm (right) | arms | 44-48 | 44-48 | 44-48 |
| Forearm (left) | arms | 49-53 | 49-53 | 49-53 |
| Forearm (right) | arms | 54-58 | 54-58 | 54-58 |
| Hand (left) | arms | 59-62 | 59-62 | 59-62 |
| Hand (right) | arms | 63-66 | 63-66 | 63-66 |
| Thigh (left) | legs | 67-73 | 67-73 | 67-73 |
| Thigh (right) | legs | 74-80 | 74-80 | 74-80 |
| Knee (left) | legs | 81-82 | 81-82 | 81-82 |
| Knee (right) | legs | 83-84 | 83-84 | 83-84 |
| Lower leg (left) | legs | 85-90 | 85-90 | 85-90 |
| Lower leg (right) | legs | 91-96 | 91-96 | 91-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 |

**Close combat.** Sabre, Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, face, neck 11%; Trunk (chest, abdomen, back, pelvis) 18%; Upper extremities 35.6%; Lower extremities 35.2%. Region totals are from the source. The split inside each region uses body exposure (design estimate).

**Reading it.** Records of the wounded, so instant kills are missing: head and trunk hits are heavily under-counted. Pair with the killed table, or use the all-hits table.

**Status.** Historical at region level. One teaching source (Mutter Museum) gives different figures (head 15%, trunk 23%, upper extremities 31%) without a citation; the Medical and Surgical History figures are used here.

**Sources.** Medical and Surgical History of the War of the Rebellion (1870-88), vol. 3, p. 392: head and neck 11%, chest and abdomen 18%, arms and hands 36%, legs and feet 35% of wounds (via Encyclopedia.com, 'Battlefield Wounds'). | Same work: 35.6% upper extremities, 35.2% lower extremities (via Civil War Monitor, 'Wounded Warriors: Civil War Amputation').

# Civil War 1861-65 - killed in action

*Seat of the fatal wound in Union soldiers killed in action (n = 1,173).*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Rifle-musket | Artillery | Revolver |
|---|---|---|---|---|
| Skull (left) | head | 01-11 | 01-11 | 01-11 |
| Skull (right) | head | 12-21 | 12-21 | 12-21 |
| Face | head | 22-34 | 22-34 | 22-34 |
| Neck | head | 35-42 | 35-42 | 35-42 |
| Chest (left) | torso | 43-53 | 43-53 | 43-53 |
| Chest (right) | torso | 54-64 | 54-64 | 54-64 |
| Abdomen | torso | 65-74 | 65-74 | 65-74 |
| Groin / pelvis | torso | 75-79 | 75-79 | 75-79 |
| Upper back | torso | 80-86 | 80-86 | 80-86 |
| Lower back | torso | 87-92 | 87-92 | 87-92 |
| Upper arm (left) | arms | 93 | 93 | 93 |
| Upper arm (right) | arms | 94 | 94 | 94 |
| Forearm (left) | arms | 95 | 95 | 95 |
| Forearm (right) | arms | 96 | 96 | 96 |
| Thigh (left) | legs | 97 | 97 | 97 |
| Thigh (right) | legs | 98 | 98 | 98 |
| Lower leg (left) | legs | 99 | 99 | 99 |
| Lower leg (right) | legs | 00 | 00 | 00 |

**Close combat.** Sabre, Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 41.6%; Trunk 51.4%; Upper extremities 2.6%; Lower extremities 4.5%. Region totals are from the source. The split inside each region uses body exposure (design estimate).

**Reading it.** Only the killing wound is recorded. Use it for 'where did the fatal shot land', or for gunfire at men already exposed to aimed fire. Small sample.

**Status.** Historical at region level. The source notes that 'a thorough study of these cases was not made': raw tallies only.

**Sources.** Medical and Surgical History of the War of the Rebellion (1883): seat of injury in 1,173 killed - head and neck 487 (41.6%), trunk 603 (51.4%), upper extremities 30 (2.6%), lower extremities 53 (4.5%) (via US Army, Wound Ballistics in WWII, ch. 6).

# Civil War 1861-65 - all hits (gameplay)

*Where shots land on a man in the open, before survival sorts the records.*

Confidence: **Extrapolated - design estimate from analogous evidence**. Armour: **allowed**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Rifle-musket | Artillery | Revolver |
|---|---|---|---|---|
| Skull (left) | head | 01-04 | 01-04 | 01-04 |
| Skull (right) | head | 05-08 | 05-08 | 05-08 |
| Face | head | 09-13 | 09-13 | 09-13 |
| Neck | head | 14-16 | 14-16 | 14-16 |
| Chest (left) | torso | 17-22 | 17-22 | 17-22 |
| Chest (right) | torso | 23-27 | 23-27 | 23-27 |
| Abdomen | torso | 28-32 | 28-32 | 28-32 |
| Groin / pelvis | torso | 33-35 | 33-35 | 33-35 |
| Upper back | torso | 36-39 | 36-39 | 36-39 |
| Lower back | torso | 40-42 | 40-42 | 40-42 |
| Shoulder (left) | arms | 43-45 | 43-45 | 43-45 |
| Shoulder (right) | arms | 46-48 | 46-48 | 46-48 |
| Upper arm (left) | arms | 49-52 | 49-52 | 49-52 |
| Upper arm (right) | arms | 53-56 | 53-56 | 53-56 |
| Forearm (left) | arms | 57-60 | 57-60 | 57-60 |
| Forearm (right) | arms | 61-64 | 61-64 | 61-64 |
| Hand (left) | arms | 65-67 | 65-67 | 65-67 |
| Hand (right) | arms | 68-70 | 68-70 | 68-70 |
| Thigh (left) | legs | 71-76 | 71-76 | 71-76 |
| Thigh (right) | legs | 77-82 | 77-82 | 77-82 |
| Knee (left) | legs | 83-84 | 83-84 | 83-84 |
| Knee (right) | legs | 85-86 | 85-86 | 85-86 |
| Lower leg (left) | legs | 87-91 | 87-91 | 87-91 |
| Lower leg (right) | legs | 92-96 | 92-96 | 92-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 |

**Close combat.** Sabre, Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

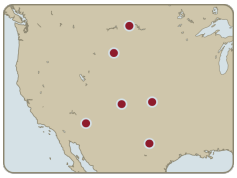
**Source totals.** Head and neck 17.1%; Trunk 24.7%; Upper extremities 29%; Lower extremities 29.1%. Blend of the two historical tables.

**Reading it.** Use this table for play. Armour is mostly irrelevant against a rifle ball; only plate gives a step.

**Status.** Blend of acw-1861-wounded and acw-1861-killed. The 4:1 ratio is approximate; edit the region totals here if you prefer another ratio.

**Sources.** Derived: 80% wounded table + 20% killed table, roughly one man killed outright for every four wounded in Union totals (approximate).

# American Indian Wars



*1862-1871 (arrow reports)*

After the Civil War the US Army garrisoned the Plains and the Southwest and fought a long series of campaigns against the nations whose land the railways and settlers were entering: on the Plains the Lakota (Sioux), Cheyenne, Arapaho, Kiowa and Comanche; in Arizona, New Mexico and Texas the Apache, Navajo and Lipan. Most fights were raids, ambushes of herders, mail escorts and wagon trains, and skirmishes between small parties. Warriors used firearms when they could get them; younger and poorer men still fought with the bow, which in skilled hands loosed several arrows in the time a soldier reloaded. The Army's surgeons wrote up 83 arrow cases from 1865-71; the table here is tallied from them.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **US Army** | Regular cavalry and infantry posted to frontier forts, with Indian scouts and civilian herders, teamsters and settlers. | None. Wool uniform; the cavalry fought mounted or dismounted with carbine, revolver and sabre. |
| **Plains and Southwest nations** | Lakota, Cheyenne, Arapaho, Kiowa, Comanche, Apache, Navajo and Lipan warriors, in raiding and war parties. | None worn. Many warriors carried a round shield of thick rawhide (buffalo hide on the Plains), which could turn an arrow or a glancing ball. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Plains warrior with bow** (picture) | Plains and Southwest nations | Rawhide shield, no armour (Plains warrior) | Bow (war arrows) | Iron trade-hoop or flint arrowheads; several arrows in quick succession. |
|  | **Comanche lancer** (picture) | Plains and Southwest nations | Rawhide shield, no armour (Plains warrior) | Spear / lance | Mounted with lance and shield. |
|  | **Apache raider** | Plains and Southwest nations | Unarmoured | Bow (war arrows) | Ambush from cover; hoop-iron arrowheads. |
|  | **US cavalry trooper** (picture) | US Army | Unarmoured | Revolver | No arrow-era gunshot table exists; his revolver rolls on a Civil War table. |
|  | **US infantryman** | US Army | Unarmoured | Rifle-musket (Minie ball) | Springfield rifle (rolls on a Civil War table). |
|  | **Indian scout** (picture) | US Army | Unarmoured | Bow (war arrows) | Several scouts in the reports were themselves hit by arrows. |

**Sources.** US Surgeon General's Office, Circular No. 3 (1871), 'Arrow-Wounds', pp. 144-161 (archive.org: reportofsurgical00unituoft). Places, tribes, units and the note on firearms and bows are from the case reports and Dr. Muller's note there. Rawhide shields: general ethnographic description, not from the report.

# Indian Wars 1865-71 - arrow wounds (US Army surgeons)

*Soldiers, scouts and civilians hit by arrows on the Plains and in the Southwest, mostly 1866-70. Killed and wounded together.*

Confidence: **Historical - taken directly from a source**. Armour: **allowed**.

| Location | Zone | Bow |
|---|---|---|
| Skull (left) | head | 01-03 |
| Skull (right) | head | 04-06 |
| Face | head | 07-08 |
| Neck | head | 09-11 |
| Chest (left) | torso | 12-21 |
| Chest (right) | torso | 22-31 |
| Abdomen | torso | 32-37 |
| Groin / pelvis | torso | 38-44 |
| Upper back | torso | 45-56 |
| Lower back | torso | 57-62 |
| Shoulder (left) | arms | 63-65 |
| Shoulder (right) | arms | 66-68 |
| Upper arm (left) | arms | 69-75 |
| Upper arm (right) | arms | 76-82 |
| Forearm (left) | arms | 83-85 |
| Forearm (right) | arms | 86-88 |
| Hand (left) | arms | 89-90 |
| Hand (right) | arms | 91-92 |
| Thigh (left) | legs | 93-94 |
| Thigh (right) | legs | 95-96 |
| Knee (left) | legs | 97 |
| Knee (right) | legs | 98 |
| Lower leg (left) | legs | 99 |
| Lower leg (right) | legs | 00 |

**Close combat.** Dagger: roll location on *Knife assault - stab wounds (forensic)*. Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Skull and scalp (8) = 6.6%; Face (3) = 2.5%; Neck (3) = 2.5%; Chest, front and sides (24) = 19.7%; Upper back (13) = 10.7%; Lower back, lumbar (6) = 4.9%; Back, level not stated (2) = 1.6%; Abdomen (7) = 5.7%; Groin, hip, buttock, pelvis (9) = 7.4%; Shoulders and armpit (7) = 5.7%; Upper arms and elbows (16) = 13.1%; Forearms (8) = 6.6%; Hands (5) = 4.1%; Thighs (6) = 4.9%; Knees (2) = 1.6%; Lower legs and feet (3) = 2.5% (n = 122). Counts are tallied wound by wound from the report's 83 cases (122 arrow wounds whose site is given; 12 more are not located). Left and right are pooled because the sample is small, then split by body exposure (design estimate).

**Reading it.** The men hit wore no armour, so these are hits before armour: armour can be applied. A mounted man's legs are partly covered by the horse, and many of these men were cavalry; that is one reason leg hits are rare here (9%) while Visby's arrow wounds were mostly in the legs.

**Status.** Historical, but a collection of case reports, not a census: striking trunk and skull cases were more likely to be written up. The 40 cases taken from routine casualty lists (none fatal) are less biased and still show 40% of wounds in the trunk. Several of the men had been hit by 3 to 9 arrows each.

**Sources.** US Surgeon General's Office, Circular No. 3: A Report of Surgical Cases Treated in the Army of the United States from 1865 to 1871 (Washington, 1871), 'Arrow-Wounds', pp. 144-161, cases 451-487. Full text: archive.org item reportofsurgical00unituoft. | Same report, p. 161: of 83 cases, 26 'proved fatal', including nearly all that entered the skull, chest or abdomen; 7 of 9 penetrating abdominal wounds were fatal. | Wound-by-wound tally: data/sources/indian-wars-arrows-1865-71.csv.

# WWI



*1914-1918*

The Allied powers (France, the British Empire, Russia, Italy, and from 1917 the United States) against the Central Powers (Germany, Austria-Hungary, the Ottoman Empire and Bulgaria). Trench warfare made artillery the main killer and the head the most exposed target, which drove the return of the steel helmet.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Allies** | French, British Empire, Russian, Italian and American armies. | Cloth caps and kepis in 1914. Steel helmets from 1915-16: the French Adrian, then the British Brodie. A little private-purchase body armour; Italian assault engineers used heavy plate. |
| **Central Powers** | German, Austro-Hungarian, Ottoman and Bulgarian armies. | German Stahlhelm from 1916. Heavy Sappenpanzer breastplates for sentries and machine gunners in fixed positions. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **French poilu, 1914** (picture) | Allies | Unarmoured | Service rifle (bolt or semi-auto, 1914-53) | Kepi, no helmet yet. |
|  | **British infantryman, 1916** (picture) | Allies | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | Brodie helmet, Lee-Enfield. |
|  | **German machine gunner** (picture) | Central Powers | Steel helmet and Sappenpanzer (WWI German sentry) | Machine gun | Stahlhelm and Sappenpanzer in a fixed position. |
|  | **Trench raider, 1917** (picture) | Allies | Steel helmet only (1915-1950s rifleman) | Sharpened entrenching tool (spade) | Night raid: sharpened spade, revolver and grenades; no rifle in the trench. |
|  | **Russian infantryman, 1915** (picture) | Allies | Unarmoured | Bayonet (musket or rifle) | Most Russian soldiers had no steel helmet; the army set great store by the bayonet. |
|  | **Italian Ardito, 1918** (picture) | Allies | Steel helmet only (1915-1950s rifleman) | Hand grenade | Assault trooper: Adrian helmet, grenades and dagger. |
|  | **American doughboy, 1918** (picture) | Allies | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | M1917 (Brodie-pattern) helmet, Springfield rifle. |

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **German stormtrooper, 1918** (picture) | Central Powers | Steel helmet only (1915-1950s rifleman) | Hand grenade | Stahlhelm; grenades, carbine and sharpened spade. |
|  | **Austro-Hungarian infantryman** (picture) | Central Powers | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | Steel helmet from 1916-17; Mannlicher rifle. |
|  | **Ottoman infantryman, Gallipoli** (picture) | Central Powers | Unarmoured | Service rifle (bolt or semi-auto, 1914-53) | Cloth headgear, no steel helmet; Mauser rifle. |
|  | **Bulgarian infantryman** (picture) | Central Powers | Unarmoured | Service rifle (bolt or semi-auto, 1914-53) | Mostly without steel helmets; Mannlicher rifle. |

**Sources.** Military Wiki, 'Stahlhelm' (introduction of German and Allied steel helmets).

# WWI 1914-18 - penetrating wounds

*Primary penetrating wounds in WWI casualties.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar |
|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-05 | 01-05 | 01-05 | 01-05 |
| Skull (right) | head | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 |
| Face | head | 11-15 | 11-15 | 11-15 | 11-15 | 11-15 |
| Neck | head | 16-19 | 16-19 | 16-19 | 16-19 | 16-19 |
| Chest (left) | torso | 20-21 | 20-21 | 20-21 | 20-21 | 20-21 |
| Chest (right) | torso | 22-23 | 22-23 | 22-23 | 22-23 | 22-23 |
| Abdomen | torso | 24 | 24 | 24 | 24 | 24 |
| Upper back | torso | 25 | 25 | 25 | 25 | 25 |
| Lower back | torso | 26 | 26 | 26 | 26 | 26 |
| Shoulder (left) | arms | 27-29 | 27-29 | 27-29 | 27-29 | 27-29 |
| Shoulder (right) | arms | 30-32 | 30-32 | 30-32 | 30-32 | 30-32 |
| Upper arm (left) | arms | 33-36 | 33-36 | 33-36 | 33-36 | 33-36 |
| Upper arm (right) | arms | 37-40 | 37-40 | 37-40 | 37-40 | 37-40 |
| Forearm (left) | arms | 41-44 | 41-44 | 41-44 | 41-44 | 41-44 |
| Forearm (right) | arms | 45-48 | 45-48 | 45-48 | 45-48 | 45-48 |
| Hand (left) | arms | 49-51 | 49-51 | 49-51 | 49-51 | 49-51 |
| Hand (right) | arms | 52-54 | 52-54 | 52-54 | 52-54 | 52-54 |
| Thigh (left) | legs | 55-63 | 55-63 | 55-63 | 55-63 | 55-63 |
| Thigh (right) | legs | 64-72 | 64-72 | 64-72 | 64-72 | 64-72 |
| Knee (left) | legs | 73-75 | 73-75 | 73-75 | 73-75 | 73-75 |
| Knee (right) | legs | 76-78 | 76-78 | 76-78 | 76-78 | 76-78 |
| Lower leg (left) | legs | 79-86 | 79-86 | 79-86 | 79-86 | 79-86 |
| Lower leg (right) | legs | 87-94 | 87-94 | 87-94 | 87-94 | 87-94 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, neck, face 17%; Thorax 4%; Abdomen 2%; Upper extremities (40% of extremity) 28%; Lower extremities (60% of extremity) 42%. Region totals are from the source. The split inside each region uses body exposure (design estimate). Extremities are split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate).

**Reading it.** Primary penetrating wound of casualties; mostly survivors who reached care, so instant kills are under-counted.

**Status.** Historical at region level (primary penetrating wounds). The source lists 7% 'other' (not located); it is left out and the rest normalised.

**Sources.** Borden Institute, Weapons Effects and War Wounds, ch. 1, Table 1-2: anatomical distribution of primary penetrating wounds (%): head/neck/face 17, thorax 4, abdomen 2, extremity 70, other 7.

# WWII



*1939-1945 (Bougainville 1943-45; Italy 1943-45)*

The Allies against the Axis. This group's tables come from two theatres: the jungle campaign on Bougainville in the Solomon Islands (US Marines and Army against the Imperial Japanese Army) and the Italian campaign (US, British and Commonwealth forces against the German army). Mortars and artillery caused half of the Bougainville wounds.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Allies** | US Army and Marine Corps; British and Commonwealth armies in Italy; the Australian army, which took over Bougainville from late 1944. | Steel helmets (M1, Brodie) for all infantry, no body armour. Bomber crews wore steel-plate flak vests from 1943. |
| **Axis** | Imperial Japanese Army; German Wehrmacht. | Steel helmets (Japanese Type 90, German Stahlhelm); no infantry body armour. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **US Marine, Bougainville** (picture) | Allies | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | M1 helmet, M1 Garand. |
|  | **Japanese infantryman** (picture) | Axis | Steel helmet only (1915-1950s rifleman) | Hand grenade | Type 90 helmet; grenades and knee mortars at close range. |
|  | **German MG gunner, Italy** (picture) | Axis | Steel helmet only (1915-1950s rifleman) | Machine gun | Stahlhelm, MG 42. |
|  | **US bomber waist gunner** (picture) | Allies | Aircrew flak vest and helmet (WWII bomber crew) | Machine gun | Flak vest and helmet; not an infantry kit. |
|  | **British infantryman, Italy** (picture) | Allies | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | Brodie helmet, Lee-Enfield; Eighth Army. |
|  | **Australian infantryman, Bougainville 1945** (picture) | Allies | Unarmoured | Pistol or submachine gun | Often a slouch hat rather than a helmet in the jungle; Owen submachine gun. |
|  | **German paratrooper, Cassino** (picture) | Axis | Steel helmet only (1915-1950s rifleman) | Pistol or submachine gun | Rimless paratrooper helmet, MP 40; fought as infantry. |

**Sources.** Borden Institute, Weapons Effects and War Wounds, Table 1-1 (Bougainville causes of injury).

# WWII Italy 1944 - killed in action

*Wounds on 983 American soldiers killed by missiles (bullets and shell fragments), examined in Italian military cemeteries, 1944. 7,006 wounds.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-03 | 01-03 | 01-03 | 01-03 | 01-03 | 01-03 | 01-03 |
| Skull (right) | head | 04-06 | 04-06 | 04-06 | 04-06 | 04-06 | 04-06 | 04-06 |
| Face | head | 07-09 | 07-09 | 07-09 | 07-09 | 07-09 | 07-09 | 07-09 |
| Neck | head | 10-13 | 10-13 | 10-13 | 10-13 | 10-13 | 10-13 | 10-13 |
| Chest (left) | torso | 14-23 | 14-23 | 14-23 | 14-23 | 14-23 | 14-23 | 14-23 |
| Chest (right) | torso | 24-33 | 24-33 | 24-33 | 24-33 | 24-33 | 24-33 | 24-33 |
| Abdomen | torso | 34-37 | 34-37 | 34-37 | 34-37 | 34-37 | 34-37 | 34-37 |
| Groin / pelvis | torso | 38-39 | 38-39 | 38-39 | 38-39 | 38-39 | 38-39 | 38-39 |
| Upper back | torso | 40-45 | 40-45 | 40-45 | 40-45 | 40-45 | 40-45 | 40-45 |
| Lower back | torso | 46-50 | 46-50 | 46-50 | 46-50 | 46-50 | 46-50 | 46-50 |
| Shoulder (left) | arms | 51-52 | 51-52 | 51-52 | 51-52 | 51-52 | 51-52 | 51-52 |
| Shoulder (right) | arms | 53-54 | 53-54 | 53-54 | 53-54 | 53-54 | 53-54 | 53-54 |
| Upper arm (left) | arms | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 |
| Upper arm (right) | arms | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 |
| Forearm (left) | arms | 61-63 | 61-63 | 61-63 | 61-63 | 61-63 | 61-63 | 61-63 |
| Forearm (right) | arms | 64-66 | 64-66 | 64-66 | 64-66 | 64-66 | 64-66 | 64-66 |
| Hand (left) | arms | 67-68 | 67-68 | 67-68 | 67-68 | 67-68 | 67-68 | 67-68 |
| Hand (right) | arms | 69-70 | 69-70 | 69-70 | 69-70 | 69-70 | 69-70 | 69-70 |
| Thigh (left) | legs | 71-76 | 71-76 | 71-76 | 71-76 | 71-76 | 71-76 | 71-76 |
| Thigh (right) | legs | 77-82 | 77-82 | 77-82 | 77-82 | 77-82 | 77-82 | 77-82 |
| Knee (left) | legs | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 |
| Knee (right) | legs | 85-86 | 85-86 | 85-86 | 85-86 | 85-86 | 85-86 | 85-86 |
| Lower leg (left) | legs | 87-91 | 87-91 | 87-91 | 87-91 | 87-91 | 87-91 | 87-91 |
| Lower leg (right) | legs | 92-96 | 92-96 | 92-96 | 92-96 | 92-96 | 92-96 | 92-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head 8.8%; Neck 3.7%; Thorax 25.7%; Abdomen 6.4%; Buttocks 4.9%; Genitalia 0.3%; Upper extremities 20.8%; Lower extremities 29.7%. Region totals are from the source. The split inside each region uses body exposure (design estimate).

**Reading it.** Counts every wound on the dead (several per man), so it shows where missiles struck men who were killed, not only the fatal wound. Steel helmets were worn.

**Status.** Historical at region level. Buttocks are split between groin/pelvis and lower back.

**Sources.** US Army Medical Department, Wound Ballistics in World War II, ch. 6, Table 118: lower extremities 29.7%, thorax 25.7%, upper extremities 20.8%, head 8.8%, abdomen 6.4%, buttocks 4.9%, neck 3.7%, genitalia 0.3%.

# Bougainville 1944 - wounded

*US soldiers wounded by Japanese rifle, machine-gun, mortar, grenade and artillery fire, Bougainville, 1944 (1,162 single-region wounds).*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-04 | 01-05 | 01-04 | 01-06 | 01-02 | 01-05 |
| Skull (right) | head | 06-10 | 05-08 | 06-10 | 05-08 | 07-12 | 03-04 | 06-10 |
| Face | head | 11-16 | 09-13 | 11-16 | 09-13 | 13-19 | 05-06 | 11-16 |
| Neck | head | 17-20 | 14-17 | 17-20 | 14-16 | 20-23 | 07-08 | 17-20 |
| Chest (left) | torso | 21-25 | 18-22 | 21-25 | 17-21 | 24-28 | 09-10 | 21-25 |
| Chest (right) | torso | 26-30 | 23-27 | 26-30 | 22-26 | 29-33 | 11-12 | 26-30 |
| Abdomen | torso | 31-33 | 28-30 | 31-33 | 27-29 | 34-36 | 13 | 31-33 |
| Groin / pelvis | torso | 34 | 31 | 34 | 30 | 37 | 14 | 34 |
| Upper back | torso | 35-37 | 32-34 | 35-37 | 31-33 | 38-40 | 15-16 | 35-37 |
| Lower back | torso | 38-39 | 35-36 | 38-39 | 34-35 | 41-42 | 17 | 38-39 |
| Shoulder (left) | arms | 40-42 | 37-39 | 40-42 | 36-38 | 43-45 | 18 | 40-42 |
| Shoulder (right) | arms | 43-45 | 40-42 | 43-45 | 39-40 | 46-48 | 19 | 43-45 |
| Upper arm (left) | arms | 46-49 | 43-46 | 46-49 | 41-44 | 49-52 | 20-21 | 46-49 |
| Upper arm (right) | arms | 50-53 | 47-50 | 50-53 | 45-48 | 53-56 | 22-23 | 50-53 |
| Forearm (left) | arms | 54-57 | 51-54 | 54-57 | 49-52 | 57-60 | 24-25 | 54-57 |
| Forearm (right) | arms | 58-61 | 55-58 | 58-61 | 53-56 | 61-64 | 26-27 | 58-61 |
| Hand (left) | arms | 62-64 | 59-61 | 62-64 | 57-58 | 65-67 | 28 | 62-64 |
| Hand (right) | arms | 65-67 | 62-64 | 65-67 | 59-60 | 68-70 | 29 | 65-67 |
| Thigh (left) | legs | 68-74 | 65-71 | 68-74 | 61-68 | 71-76 | 30-42 | 68-74 |
| Thigh (right) | legs | 75-80 | 72-78 | 75-80 | 69-76 | 77-82 | 43-54 | 75-80 |
| Knee (left) | legs | 81-82 | 79-80 | 81-82 | 77-78 | 83-84 | 55-58 | 81-82 |
| Knee (right) | legs | 83-84 | 81-82 | 83-84 | 79-80 | 85-86 | 59-62 | 83-84 |
| Lower leg (left) | legs | 85-90 | 83-88 | 85-90 | 81-87 | 87-91 | 63-73 | 85-90 |
| Lower leg (right) | legs | 91-96 | 89-94 | 91-96 | 88-94 | 92-96 | 74-84 | 91-96 |
| Foot (left) | legs | 97-98 | 95-97 | 97-98 | 95-97 | 97-98 | 85-92 | 97-98 |
| Foot (right) | legs | 99-00 | 98-00 | 99-00 | 98-00 | 99-00 | 93-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, face, neck (240) = 20.7%; Chest (144) = 12.4%; Abdomen (66) = 5.7%; Upper extremities (319) = 27.5%; Lower extremities (393) = 33.8% (n = 1,162). Region counts are from the source. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-33 | Service rifle (bolt or semi-auto, 1914-53) |
| 34-83 | Mortar, rocket or artillery |
| 84-95 | Hand grenade |
| 96-97 | Mine, booby trap or IED |
| 98-00 | Other |

Borden Institute, Weapons Effects and War Wounds, Table 1-1: causes of battle injury, Bougainville

**Reading it.** Survivors only: head and trunk are under-counted because those hits killed. Pair with the killed table, or use the all-hits table. Steel helmets were worn.

**Status.** Historical at region level (counts quoted from the study via a secondary post; the NLM scan is the primary). Abdomen is taken to include the lower back and pelvis.

**Sources.** Oughterson et al., Wound Ballistics study of the Bougainville campaign (US Army Medical Department, 1944; NLM scan nlmuid-46921110R): living wounded by region - head, face, neck 240; chest 144; abdomen 66; upper extremities 319; lower extremities 393. Excludes 332 multi-region wounds and 547 minor wounds returned to duty.

# Bougainville 1944 - killed

*US soldiers killed by single-region missile wounds, Bougainville, 1944 (294 men).*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-12 | 01-11 | 01-12 | 01-11 | 01-13 | 01-09 | 01-12 |
| Skull (right) | head | 13-24 | 12-22 | 13-24 | 12-22 | 14-26 | 10-18 | 13-24 |
| Face | head | 25-38 | 23-36 | 25-38 | 23-35 | 27-41 | 19-28 | 25-38 |
| Neck | head | 39-48 | 37-45 | 39-48 | 36-44 | 42-51 | 29-35 | 39-48 |
| Chest (left) | torso | 49-60 | 46-57 | 49-60 | 45-56 | 52-62 | 36-46 | 49-60 |
| Chest (right) | torso | 61-72 | 58-69 | 61-72 | 57-68 | 63-73 | 47-57 | 61-72 |
| Abdomen | torso | 73-80 | 70-77 | 73-80 | 69-77 | 74-80 | 58-65 | 73-80 |
| Groin / pelvis | torso | 81-84 | 78-81 | 81-84 | 78-81 | 81-84 | 66-71 | 81-84 |
| Upper back | torso | 85-91 | 82-89 | 85-91 | 82-89 | 85-91 | 72-78 | 85-91 |
| Lower back | torso | 92-96 | 90-94 | 92-96 | 90-94 | 92-96 | 79-82 | 92-96 |
| Thigh (left) | legs | 97 | 95 | 97 | 95 | 97 | 83-85 | 97 |
| Thigh (right) | legs | 98 | 96 | 98 | 96 | 98 | 86-88 | 98 |
| Knee (left) | legs | - | - | - | - | - | 89 | - |
| Knee (right) | legs | - | - | - | - | - | 90 | - |
| Lower leg (left) | legs | 99 | 97 | 99 | 97 | 99 | 91-93 | 99 |
| Lower leg (right) | legs | 00 | 98 | 00 | 98 | 00 | 94-96 | 00 |
| Foot (left) | legs | - | 99 | - | 99 | - | 97-98 | - |
| Foot (right) | legs | - | 00 | - | 00 | - | 99-00 | - |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, face, neck (144) = 49.0%; Chest (87) = 29.6%; Abdomen (48) = 16.3%; Upper extremities (1) = 0.3%; Lower extremities (14) = 4.8% (n = 294). Region counts are from the source. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-33 | Service rifle (bolt or semi-auto, 1914-53) |
| 34-83 | Mortar, rocket or artillery |
| 84-95 | Hand grenade |
| 96-97 | Mine, booby trap or IED |
| 98-00 | Other |

Borden Institute, Weapons Effects and War Wounds, Table 1-1: causes of battle injury, Bougainville

**Reading it.** Where the killing wound was. Use for fatal hits only.

**Status.** Historical at region level. Small sample (294).

**Sources.** Oughterson et al., Bougainville wound ballistics study (1944): killed by region - head, face, neck 144; chest 87; abdomen 48; upper extremities 1; lower extremities 14.

# Bougainville 1944 - all hits

*Every single-region missile hit in the study, wounded and killed together (1,456 hits).*

Confidence: **Historical - taken directly from a source**. Armour: **allowed**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-07 | 01-06 | 01-07 | 01-06 | 01-07 | 01-03 | 01-07 |
| Skull (right) | head | 08-14 | 07-12 | 08-14 | 07-11 | 08-14 | 04-06 | 08-14 |
| Face | head | 15-22 | 13-19 | 15-22 | 12-18 | 15-23 | 07-09 | 15-22 |
| Neck | head | 23-27 | 20-24 | 23-27 | 19-22 | 24-29 | 10-11 | 23-27 |
| Chest (left) | torso | 28-33 | 25-30 | 28-33 | 23-28 | 30-35 | 12-14 | 28-33 |
| Chest (right) | torso | 34-39 | 31-36 | 34-39 | 29-34 | 36-41 | 15-17 | 34-39 |
| Abdomen | torso | 40-43 | 37-40 | 40-43 | 35-38 | 42-45 | 18-19 | 40-43 |
| Groin / pelvis | torso | 44-45 | 41-42 | 44-45 | 39-40 | 46-47 | 20-21 | 44-45 |
| Upper back | torso | 46-49 | 43-46 | 46-49 | 41-44 | 48-51 | 22-23 | 46-49 |
| Lower back | torso | 50-51 | 47-49 | 50-51 | 45-46 | 52-53 | 24 | 50-51 |
| Shoulder (left) | arms | 52-53 | 50-51 | 52-53 | 47-48 | 54-55 | 25 | 52-53 |
| Shoulder (right) | arms | 54-55 | 52-53 | 54-55 | 49-50 | 56-57 | 26 | 54-55 |
| Upper arm (left) | arms | 56-58 | 54-56 | 56-58 | 51-53 | 58-61 | 27-28 | 56-58 |
| Upper arm (right) | arms | 59-61 | 57-59 | 59-61 | 54-56 | 62-64 | 29-30 | 59-61 |
| Forearm (left) | arms | 62-64 | 60-62 | 62-64 | 57-59 | 65-67 | 31-32 | 62-64 |
| Forearm (right) | arms | 65-67 | 63-65 | 65-67 | 60-62 | 68-70 | 33-34 | 65-67 |
| Hand (left) | arms | 68-69 | 66-67 | 68-69 | 63-64 | 71-72 | 35 | 68-69 |
| Hand (right) | arms | 70-71 | 68-69 | 70-71 | 65-66 | 73-74 | 36 | 70-71 |
| Thigh (left) | legs | 72-77 | 70-75 | 72-77 | 67-73 | 75-79 | 37-47 | 72-77 |
| Thigh (right) | legs | 78-82 | 76-81 | 78-82 | 74-80 | 80-84 | 48-58 | 78-82 |
| Knee (left) | legs | 83-84 | 82-83 | 83-84 | 81-82 | 85-86 | 59-61 | 83-84 |
| Knee (right) | legs | 85-86 | 84-85 | 85-86 | 83-84 | 87-88 | 62-64 | 85-86 |
| Lower leg (left) | legs | 87-91 | 86-91 | 87-91 | 85-90 | 89-92 | 65-74 | 87-91 |
| Lower leg (right) | legs | 92-96 | 92-96 | 92-96 | 91-96 | 93-96 | 75-84 | 92-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 | 85-92 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 | 93-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, face, neck (384) = 26.4%; Chest (231) = 15.9%; Abdomen (114) = 7.8%; Upper extremities (320) = 22.0%; Lower extremities (407) = 28.0% (n = 1,456). Region counts are from the source (wounded + killed). The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-33 | Service rifle (bolt or semi-auto, 1914-53) |
| 34-83 | Mortar, rocket or artillery |
| 84-95 | Hand grenade |
| 96-97 | Mine, booby trap or IED |
| 98-00 | Other |

Borden Institute, Weapons Effects and War Wounds, Table 1-1: causes of battle injury, Bougainville

**Reading it.** The closest thing in the data set to 'where bullets and fragments land'. Unlike the Civil War blend, this needs no assumed ratio: the study counted both groups. Share of hits that killed: head/face/neck 37.5%, chest 37.7%, abdomen 42.1%, arms 0.3%, legs 3.4%.

**Status.** Historical at region level. Still excludes multi-region and minor wounds.

**Sources.** Oughterson et al., Bougainville wound ballistics study (1944): totals by region - head, face, neck 384; chest 231; abdomen 114; upper extremities 320; lower extremities 407 (20.2% killed overall).

# WWII 1941-45 - hospitalized (Owens comparison set)

*US casualties hospitalized in WWII.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-05 | 01-05 | 01-04 | 01-06 | 01-02 | 01-05 |
| Skull (right) | head | 06-10 | 06-10 | 06-10 | 05-08 | 07-12 | 03-04 | 06-10 |
| Face | head | 11-16 | 11-15 | 11-16 | 09-13 | 13-19 | 05-06 | 11-16 |
| Neck | head | 17-20 | 16-19 | 17-20 | 14-16 | 20-23 | 07 | 17-20 |
| Chest (left) | torso | 21-25 | 20-24 | 21-25 | 17-21 | 24-28 | 08-10 | 21-25 |
| Chest (right) | torso | 26-30 | 25-29 | 26-30 | 22-26 | 29-33 | 11-12 | 26-30 |
| Abdomen | torso | 31-34 | 30-33 | 31-34 | 27-30 | 34-37 | 13-14 | 31-34 |
| Groin / pelvis | torso | 35-36 | 34-35 | 35-36 | 31-32 | 38-39 | 15 | 35-36 |
| Upper back | torso | 37-39 | 36-38 | 37-39 | 33-35 | 40-42 | 16-17 | 37-39 |
| Lower back | torso | 40-41 | 39-40 | 40-41 | 36-37 | 43-44 | 18 | 40-41 |
| Shoulder (left) | arms | 42-43 | 41-42 | 42-43 | 38-39 | 45-46 | 19 | 42-43 |
| Shoulder (right) | arms | 44-45 | 43-44 | 44-45 | 40-41 | 47-48 | 20 | 44-45 |
| Upper arm (left) | arms | 46-49 | 45-47 | 46-49 | 42-44 | 49-52 | 21 | 46-49 |
| Upper arm (right) | arms | 50-53 | 48-50 | 50-53 | 45-47 | 53-56 | 22 | 50-53 |
| Forearm (left) | arms | 54-57 | 51-53 | 54-57 | 48-50 | 57-60 | 23 | 54-57 |
| Forearm (right) | arms | 58-61 | 54-56 | 58-61 | 51-53 | 61-64 | 24 | 58-61 |
| Hand (left) | arms | 62-63 | 57-58 | 62-63 | 54-55 | 65-66 | 25 | 62-63 |
| Hand (right) | arms | 64-65 | 59-60 | 64-65 | 56-57 | 67-68 | 26 | 64-65 |
| Thigh (left) | legs | 66-72 | 61-68 | 66-72 | 58-66 | 69-74 | 27-39 | 66-72 |
| Thigh (right) | legs | 73-79 | 69-76 | 73-79 | 67-74 | 75-80 | 40-52 | 73-79 |
| Knee (left) | legs | 80-81 | 77-78 | 80-81 | 75-77 | 81-82 | 53-56 | 80-81 |
| Knee (right) | legs | 82-83 | 79-80 | 82-83 | 78-80 | 83-84 | 57-60 | 82-83 |
| Lower leg (left) | legs | 84-89 | 81-87 | 84-89 | 81-87 | 85-90 | 61-71 | 84-89 |
| Lower leg (right) | legs | 90-95 | 88-94 | 90-95 | 88-94 | 91-96 | 72-82 | 90-95 |
| Foot (left) | legs | 96-98 | 95-97 | 96-98 | 95-97 | 97-98 | 83-91 | 96-98 |
| Foot (right) | legs | 99-00 | 98-00 | 99-00 | 98-00 | 99-00 | 92-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 21%; Thorax 14%; Abdomen 8%; Upper extremities (40% of 58) 23.2%; Lower extremities (60% of 58) 34.8%. Region totals are from the source. Extremities split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate), the same rule for every table in this set. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-27 | Service rifle (bolt or semi-auto, 1914-53) |
| 28-00 | Fragments, unspecified (shell, grenade, mine) |

Owens et al. 2008 (National Academies 2014, Table 3-4): share of injuries from gunshot vs explosion

**Reading it.** Hospitalized casualties including those who later died of wounds; men killed outright are not counted, so head and trunk are still under-counted.

**Status.** Historical at region level. One of four tables from the same source with the same counting rule (see the comparison on the Reference page) (hospitalizations, including men who later died of wounds), same extremity split.

**Sources.** Owens BD et al., 'Combat wounds in Operation Iraqi Freedom and Operation Enduring Freedom', J Trauma 2008, comparison table (reproduced in National Academies, Review of DoD Test Protocols for Combat Helmets, 2014, Table 3-2): head and neck 21, thorax 14, abdomen 8, extremities 58.

# Korea



*1950-1953*

The United Nations Command, led by the United States with South Korea, the British Commonwealth and others, against North Korea and, from late 1950, the Chinese People's Volunteer Army. Fighting swung from mobile war to static trench lines, where body armour first came into general use.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **UN Command** | US, South Korean, British Commonwealth and other UN forces. | Steel helmets. The US Marines fielded the M1951 vest with Doron plates; the Army's nylon M-1952 reached front-line troops in late 1952, in small numbers. |
| **North Korea and China** | Korean People's Army; Chinese People's Volunteer Army. | No body armour; many Chinese soldiers had no steel helmet. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **US Army rifleman, 1950** (picture) | UN Command | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | M1 helmet, no vest. |
|  | **US Marine, 1952** (picture) | UN Command | M1 helmet, Doron vest (USMC M1951/M1955, 1951-75) | Service rifle (bolt or semi-auto, 1914-53) | M1951 Doron vest. |
|  | **South Korean infantryman** (picture) | UN Command | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | ROK Army: M1 helmet, M1 Garand. |
|  | **British infantryman, Imjin 1951** (picture) | UN Command | Steel helmet only (1915-1950s rifleman) | Service rifle (bolt or semi-auto, 1914-53) | Commonwealth brigade: steel helmet, Lee-Enfield. |
|  | **North Korean infantryman, 1950** | North Korea and China | Unarmoured | Service rifle (bolt or semi-auto, 1914-53) | Korean People's Army: Soviet-pattern kit, often no steel helmet; Mosin rifle or PPSh. |
|  | **Chinese infantryman** (picture) | North Korea and China | Unarmoured | Pistol or submachine gun | Quilted uniform, PPSh submachine gun, grenades. |

**Sources.** Wikipedia, 'M-1952 Flak Jacket'.

# Korea 1950-53 - hospitalized wounded

*Location of wounds in US Army casualties admitted to hospital, Korean War.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-04 | 01-05 | 01-03 | 01-05 | 01-02 | 01-05 |
| Skull (right) | head | 06-09 | 05-08 | 06-09 | 04-06 | 06-10 | 03-04 | 06-09 |
| Face | head | 10-14 | 09-13 | 10-14 | 07-10 | 11-16 | 05-06 | 10-14 |
| Neck | head | 15-18 | 14-16 | 15-18 | 11-13 | 17-20 | 07 | 15-18 |
| Chest (left) | torso | 19-21 | 17-19 | 19-21 | 14-16 | 21-23 | 08 | 19-21 |
| Chest (right) | torso | 22-24 | 20-22 | 22-24 | 17-19 | 24-25 | 09 | 22-24 |
| Abdomen | torso | 25-28 | 23-25 | 25-28 | 20-22 | 26-28 | 10-11 | 25-28 |
| Groin / pelvis | torso | 29-30 | 26-27 | 29-30 | 23-24 | 29-30 | 12 | 29-30 |
| Upper back | torso | 31-32 | 28-29 | 31-32 | 25-26 | 31-32 | 13 | 31-32 |
| Lower back | torso | 33-34 | 30-31 | 33-34 | 27-28 | 33-34 | 14 | 33-34 |
| Shoulder (left) | arms | 35-37 | 32-34 | 35-37 | 29-31 | 35-37 | 15 | 35-37 |
| Shoulder (right) | arms | 38-40 | 35-37 | 38-40 | 32-34 | 38-40 | 16 | 38-40 |
| Upper arm (left) | arms | 41-45 | 38-42 | 41-45 | 35-38 | 41-45 | 17-18 | 41-45 |
| Upper arm (right) | arms | 46-50 | 43-46 | 46-50 | 39-42 | 46-50 | 19-20 | 46-50 |
| Forearm (left) | arms | 51-55 | 47-50 | 51-55 | 43-46 | 51-55 | 21-22 | 51-55 |
| Forearm (right) | arms | 56-60 | 51-54 | 56-60 | 47-50 | 56-60 | 23-24 | 56-60 |
| Hand (left) | arms | 61-63 | 55-57 | 61-63 | 51-53 | 61-63 | 25 | 61-63 |
| Hand (right) | arms | 64-66 | 58-60 | 64-66 | 54-56 | 64-66 | 26 | 64-66 |
| Thigh (left) | legs | 67-73 | 61-68 | 67-73 | 57-65 | 67-73 | 27-39 | 67-73 |
| Thigh (right) | legs | 74-80 | 69-76 | 74-80 | 66-74 | 74-80 | 40-52 | 74-80 |
| Knee (left) | legs | 81-82 | 77-78 | 81-82 | 75-77 | 81-82 | 53-56 | 81-82 |
| Knee (right) | legs | 83-84 | 79-80 | 83-84 | 78-80 | 83-84 | 57-60 | 83-84 |
| Lower leg (left) | legs | 85-90 | 81-87 | 85-90 | 81-87 | 85-90 | 61-71 | 85-90 |
| Lower leg (right) | legs | 91-96 | 88-94 | 91-96 | 88-94 | 91-96 | 72-82 | 91-96 |
| Foot (left) | legs | 97-98 | 95-97 | 97-98 | 95-97 | 97-98 | 83-91 | 97-98 |
| Foot (right) | legs | 99-00 | 98-00 | 99-00 | 98-00 | 99-00 | 92-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 17%; Thorax 7%; Abdomen 7%; Upper extremities 30%; Lower extremities 37%. Region totals are from the source. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-27 | Service rifle (bolt or semi-auto, 1914-53) |
| 28-88 | Fragments, unspecified (shell, grenade, mine) |
| 89-92 | Mine, booby trap or IED |
| 93-00 | Other |

Neel (1973) Table 6: wounds by agent, Korea

**Reading it.** Hospitalized casualties: men killed outright are missing, so head and trunk are under-counted.

**Status.** Historical at region level. 2% other/multiple left out and the rest normalised.

**Sources.** Neel, S., Medical Support of the U.S. Army in Vietnam 1965-1970 (Dept of the Army, 1973), ch. 3, Table 7: location of wounds in hospitalized casualties (%): Korea - head and neck 17, thorax 7, abdomen 7, upper extremities 30, lower extremities 37, other/multiple 2. | Neel (1973) Table 6, Korea: deaths 33% small arms, 59% fragments, 4% mines; wounds 27% small arms, 61% fragments, 4% mines.

# Korea 1950-53 - hospitalized (Owens comparison set)

*US casualties hospitalized in the Korean War.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Rifle | MG | Pistol/SMG | Grenade | Mortar | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-05 | 01-05 | 01-04 | 01-06 | 01-02 | 01-05 |
| Skull (right) | head | 06-10 | 06-10 | 06-10 | 05-08 | 07-12 | 03-04 | 06-10 |
| Face | head | 11-16 | 11-16 | 11-16 | 09-13 | 13-19 | 05-06 | 11-16 |
| Neck | head | 17-20 | 17-20 | 17-20 | 14-16 | 20-24 | 07-08 | 17-20 |
| Chest (left) | torso | 21-24 | 21-24 | 21-24 | 17-20 | 25-28 | 09-10 | 21-24 |
| Chest (right) | torso | 25-28 | 25-28 | 25-28 | 21-24 | 29-32 | 11-12 | 25-28 |
| Abdomen | torso | 29-32 | 29-32 | 29-32 | 25-28 | 33-36 | 13-14 | 29-32 |
| Groin / pelvis | torso | 33-34 | 33-34 | 33-34 | 29-30 | 37-38 | 15-16 | 33-34 |
| Upper back | torso | 35-37 | 35-36 | 35-37 | 31-32 | 39-40 | 17 | 35-37 |
| Lower back | torso | 38-40 | 37-39 | 38-40 | 33-34 | 41-42 | 18 | 38-40 |
| Shoulder (left) | arms | 41-42 | 40-41 | 41-42 | 35-36 | 43-44 | 19 | 41-42 |
| Shoulder (right) | arms | 43-44 | 42-43 | 43-44 | 37-38 | 45-46 | 20 | 43-44 |
| Upper arm (left) | arms | 45-48 | 44-47 | 45-48 | 39-41 | 47-50 | 21 | 45-48 |
| Upper arm (right) | arms | 49-52 | 48-50 | 49-52 | 42-44 | 51-54 | 22 | 49-52 |
| Forearm (left) | arms | 53-56 | 51-53 | 53-56 | 45-47 | 55-58 | 23 | 53-56 |
| Forearm (right) | arms | 57-60 | 54-56 | 57-60 | 48-50 | 59-62 | 24 | 57-60 |
| Hand (left) | arms | 61-62 | 57-58 | 61-62 | 51-52 | 63-64 | 25 | 61-62 |
| Hand (right) | arms | 63-64 | 59-60 | 63-64 | 53-54 | 65-66 | 26 | 63-64 |
| Thigh (left) | legs | 65-71 | 61-68 | 65-71 | 55-63 | 67-73 | 27-39 | 65-71 |
| Thigh (right) | legs | 72-78 | 69-76 | 72-78 | 64-72 | 74-80 | 40-52 | 72-78 |
| Knee (left) | legs | 79-80 | 77-78 | 79-80 | 73-75 | 81-82 | 53-56 | 79-80 |
| Knee (right) | legs | 81-82 | 79-80 | 81-82 | 76-78 | 83-84 | 57-60 | 81-82 |
| Lower leg (left) | legs | 83-88 | 81-87 | 83-88 | 79-86 | 85-90 | 61-71 | 83-88 |
| Lower leg (right) | legs | 89-94 | 88-94 | 89-94 | 87-94 | 91-96 | 72-82 | 89-94 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 97-98 | 83-91 | 95-97 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 99-00 | 92-00 | 98-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 21%; Thorax 10%; Abdomen 9%; Upper extremities (40% of 60) 24%; Lower extremities (60% of 60) 36%. Region totals are from the source. Extremities split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate), the same rule for every table in this set. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-31 | Service rifle (bolt or semi-auto, 1914-53) |
| 32-00 | Fragments, unspecified (shell, grenade, mine) |

Owens et al. 2008 (National Academies 2014, Table 3-4): share of injuries from gunshot vs explosion

**Reading it.** Hospitalized casualties including those who later died of wounds; men killed outright are not counted, so head and trunk are still under-counted.

**Status.** Historical at region level. One of four tables from the same source with the same counting rule (see the comparison on the Reference page) (hospitalizations, including men who later died of wounds), same extremity split.

**Sources.** Owens BD et al., 'Combat wounds in Operation Iraqi Freedom and Operation Enduring Freedom', J Trauma 2008, comparison table (reproduced in National Academies, Review of DoD Test Protocols for Combat Helmets, 2014, Table 3-2): head and neck 21, thorax 10, abdomen 9, extremities 60.

# Vietnam



*1965-1973 (major US involvement)*

South Vietnam, the United States and allies (Australia, New Zealand, South Korea, Thailand) against North Vietnam's army (PAVN) and the southern Viet Cong. Most wounds came from fragments; mines and booby traps caused 15% of wounds and punji stakes 2%, while bullets killed far more often than they wounded.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **US and allies** | South Vietnamese army (ARVN), US Army and Marine Corps, and Australian, New Zealand, South Korean and Thai contingents. | M1 steel helmet; Army M69 nylon vest, Marine M1955 vest with Doron plates. Both stopped fragments, not AK rounds, and in the heat they were often left behind on patrol. |
| **PAVN and Viet Cong** | People's Army of Vietnam regulars; Viet Cong guerrillas. | None. The PAVN pith helmet gave no ballistic protection. They relied on concealment, mines and booby traps. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **US Army infantryman, 1968** (picture) | US and allies | M1 helmet, nylon flak vest (US Army M1952/M69, 1952-75) | Assault rifle (M16, AK) | M1 helmet, M69 vest, M16. |
|  | **US Marine, Hue 1968** (picture) | US and allies | M1 helmet, Doron vest (USMC M1951/M1955, 1951-75) | Assault rifle (M16, AK) | M1955 vest, urban fighting. |
|  | **ARVN infantryman** (picture) | US and allies | Steel helmet only (1915-1950s rifleman) | Assault rifle (M16, AK) | South Vietnamese army: M1 helmet, usually no vest; M16 from 1968, carbines before. |
|  | **Australian infantryman, 1968** (picture) | US and allies | Unarmoured | Assault rifle (M16, AK) | Bush hat, no body armour; L1A1 SLR. |
|  | **South Korean marine** (picture) | US and allies | Steel helmet only (1915-1950s rifleman) | Assault rifle (M16, AK) | ROK Marine Corps: M1 helmet. |
|  | **PAVN regular** | PAVN and Viet Cong | Unarmoured | Assault rifle (M16, AK) | Pith helmet, AK-47. |
|  | **Viet Cong guerrilla** | PAVN and Viet Cong | Unarmoured | Mine, booby trap or IED | Booby traps, mines and punji pits. |

**Sources.** Neel, Medical Support of the U.S. Army in Vietnam (1973), ch. 3, Table 6. Wikipedia, 'Flak jacket' (M69 and M1955 protection).

# Vietnam 1965-70 - hospitalized wounded

*Location of wounds in US Army casualties admitted to hospital in Vietnam (24-month sample).*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED | Punji | Frags |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-04 | 01-05 | 01-04 | 01-05 | 01-05 | 01-02 | - | 01-05 |
| Skull (right) | head | 06-10 | 05-08 | 06-10 | 05-08 | 06-10 | 06-10 | 03-04 | - | 06-10 |
| Face | head | 11-16 | 09-13 | 11-16 | 09-13 | 11-16 | 11-16 | 05-06 | - | 11-16 |
| Neck | head | 17-20 | 14-17 | 17-20 | 14-16 | 17-20 | 17-20 | 07 | - | 17-20 |
| Chest (left) | torso | 21-24 | 18-21 | 21-24 | 17-19 | 21-24 | 21-24 | 08-09 | - | 21-24 |
| Chest (right) | torso | 25-28 | 22-25 | 25-28 | 20-22 | 25-27 | 25-28 | 10-11 | - | 25-28 |
| Abdomen | torso | 29-31 | 26-28 | 29-31 | 23-25 | 28-30 | 29-31 | 12-13 | - | 29-31 |
| Groin / pelvis | torso | 32-33 | 29-30 | 32-33 | 26 | 31-32 | 32-33 | 14 | - | 32-33 |
| Upper back | torso | 34-35 | 31-32 | 34-35 | 27-28 | 33-34 | 34-35 | 15 | - | 34-35 |
| Lower back | torso | 36-37 | 33-34 | 36-37 | 29-30 | 35-36 | 36-37 | 16 | - | 36-37 |
| Shoulder (left) | arms | 38-40 | 35-36 | 38-40 | 31-32 | 37-39 | 38-40 | 17 | - | 38-40 |
| Shoulder (right) | arms | 41-42 | 37-38 | 41-42 | 33-34 | 40-42 | 41-42 | 18 | - | 41-42 |
| Upper arm (left) | arms | 43-46 | 39-42 | 43-46 | 35-38 | 43-46 | 43-46 | 19-20 | - | 43-46 |
| Upper arm (right) | arms | 47-50 | 43-46 | 47-50 | 39-42 | 47-50 | 47-50 | 21-22 | - | 47-50 |
| Forearm (left) | arms | 51-54 | 47-50 | 51-54 | 43-45 | 51-54 | 51-54 | 23-24 | - | 51-54 |
| Forearm (right) | arms | 55-58 | 51-54 | 55-58 | 46-48 | 55-58 | 55-58 | 25-26 | - | 55-58 |
| Hand (left) | arms | 59-60 | 55-56 | 59-60 | 49-50 | 59-61 | 59-60 | 27 | - | 59-60 |
| Hand (right) | arms | 61-62 | 57-58 | 61-62 | 51-52 | 62-64 | 61-62 | 28 | - | 61-62 |
| Thigh (left) | legs | 63-70 | 59-67 | 63-70 | 53-62 | 65-72 | 63-70 | 29-42 | 01-13 | 63-70 |
| Thigh (right) | legs | 71-78 | 68-76 | 71-78 | 63-72 | 73-80 | 71-78 | 43-56 | 14-26 | 71-78 |
| Knee (left) | legs | 79-80 | 77-79 | 79-80 | 73-75 | 81-82 | 79-80 | 57-60 | 27-30 | 79-80 |
| Knee (right) | legs | 81-82 | 80-82 | 81-82 | 76-78 | 83-84 | 81-82 | 61-64 | 31-34 | 81-82 |
| Lower leg (left) | legs | 83-89 | 83-89 | 83-89 | 79-86 | 85-90 | 83-89 | 65-75 | 35-56 | 83-89 |
| Lower leg (right) | legs | 90-96 | 90-96 | 90-96 | 87-94 | 91-96 | 90-96 | 76-86 | 57-78 | 90-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 | 95-97 | 97-98 | 97-98 | 87-93 | 79-89 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 | 98-00 | 99-00 | 99-00 | 94-00 | 90-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 14%; Thorax 7%; Abdomen 5%; Upper extremities 18%; Lower extremities 36%. Region totals are from the source. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-16 | Assault rifle (M16, AK) |
| 17-81 | Fragments, unspecified (shell, grenade, mine) |
| 82-96 | Mine, booby trap or IED |
| 97-98 | Punji stake |
| 99-00 | Other |

Neel (1973) Table 6: wounds by agent, Vietnam (the hospitalized population this table counts)

**Reading it.** Hospitalized casualties: men killed outright are missing. Bullets killed (51% of deaths) far more than they wounded (16%), so for fatal shots lean on the head and thorax.

**Status.** Historical at region level. The source puts 22% of casualties under 'other/multiple sites' (several wounds); those are left out and the single-region rest normalised. The source's column sums to 102% (rounding).

**Sources.** Neel, S., Medical Support of the U.S. Army in Vietnam 1965-1970 (Dept of the Army, 1973), ch. 3, Table 7: location of wounds in hospitalized casualties (%): Vietnam - head and neck 14, thorax 7, abdomen 5, upper extremities 18, lower extremities 36, other/multiple 22. | Neel (1973) Table 6, Vietnam: deaths 51% small arms, 36% fragments, 11% booby traps/mines; wounds 16% small arms, 65% fragments, 15% booby traps/mines, 2% punji stakes. | Borden Institute, Weapons Effects and War Wounds, Table 1-1 (WDMET): causes of battle injury - bullet 30, mortar/rocket/artillery 22, grenade 11, RPG 12, booby trap 17, other 8.

# Vietnam 1965-73 - hospitalized (Owens comparison set)

*US casualties hospitalized in the Vietnam War.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED | Punji | Frags |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-04 | 01-03 | 01-04 | 01-03 | 01-04 | 01-04 | 01 | - | 01-04 |
| Skull (right) | head | 05-08 | 04-06 | 05-08 | 04-06 | 05-08 | 05-08 | 02 | - | 05-08 |
| Face | head | 09-13 | 07-10 | 09-13 | 07-10 | 09-13 | 09-13 | 03-04 | - | 09-13 |
| Neck | head | 14-16 | 11-13 | 14-16 | 11-13 | 14-16 | 14-16 | 05 | - | 14-16 |
| Chest (left) | torso | 17-21 | 14-18 | 17-21 | 14-18 | 17-21 | 17-21 | 06-07 | - | 17-21 |
| Chest (right) | torso | 22-26 | 19-23 | 22-26 | 19-23 | 22-26 | 22-26 | 08-09 | - | 22-26 |
| Abdomen | torso | 27-31 | 24-28 | 27-31 | 24-27 | 27-30 | 27-31 | 10-11 | - | 27-31 |
| Groin / pelvis | torso | 32-33 | 29-30 | 32-33 | 28-29 | 31-32 | 32-33 | 12-13 | - | 32-33 |
| Upper back | torso | 34-36 | 31-33 | 34-36 | 30-32 | 33-35 | 34-36 | 14 | - | 34-36 |
| Lower back | torso | 37-39 | 34-36 | 37-39 | 33-35 | 36-38 | 37-39 | 15 | - | 37-39 |
| Shoulder (left) | arms | 40-42 | 37-38 | 40-42 | 36-37 | 39-41 | 40-42 | 16 | - | 40-42 |
| Shoulder (right) | arms | 43-44 | 39-40 | 43-44 | 38-39 | 42-44 | 43-44 | 17 | - | 43-44 |
| Upper arm (left) | arms | 45-48 | 41-44 | 45-48 | 40-42 | 45-48 | 45-48 | 18-19 | - | 45-48 |
| Upper arm (right) | arms | 49-52 | 45-48 | 49-52 | 43-45 | 49-52 | 49-52 | 20-21 | - | 49-52 |
| Forearm (left) | arms | 53-56 | 49-52 | 53-56 | 46-48 | 53-56 | 53-56 | 22-23 | - | 53-56 |
| Forearm (right) | arms | 57-60 | 53-56 | 57-60 | 49-51 | 57-60 | 57-60 | 24 | - | 57-60 |
| Hand (left) | arms | 61-62 | 57-58 | 61-62 | 52-53 | 61-63 | 61-62 | 25 | - | 61-62 |
| Hand (right) | arms | 63-64 | 59-60 | 63-64 | 54-55 | 64-66 | 63-64 | 26 | - | 63-64 |
| Thigh (left) | legs | 65-71 | 61-68 | 65-71 | 56-64 | 67-73 | 65-71 | 27-39 | 01-12 | 65-71 |
| Thigh (right) | legs | 72-78 | 69-76 | 72-78 | 65-73 | 74-80 | 72-78 | 40-52 | 13-24 | 72-78 |
| Knee (left) | legs | 79-80 | 77-78 | 79-80 | 74-76 | 81-82 | 79-80 | 53-56 | 25-28 | 79-80 |
| Knee (right) | legs | 81-82 | 79-80 | 81-82 | 77-79 | 83-84 | 81-82 | 57-60 | 29-32 | 81-82 |
| Lower leg (left) | legs | 83-88 | 81-87 | 83-88 | 80-87 | 85-90 | 83-88 | 61-71 | 33-53 | 83-88 |
| Lower leg (right) | legs | 89-94 | 88-94 | 89-94 | 88-94 | 91-96 | 89-94 | 72-82 | 54-74 | 89-94 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 97-98 | 95-97 | 83-91 | 75-87 | 95-97 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 99-00 | 98-00 | 92-00 | 88-00 | 98-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 16%; Thorax 13%; Abdomen 10%; Upper extremities (40% of 61) 24.4%; Lower extremities (60% of 61) 36.6%. Region totals are from the source. Extremities split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate), the same rule for every table in this set. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-35 | Assault rifle (M16, AK) |
| 36-00 | Fragments, unspecified (shell, grenade, mine) |

Owens et al. 2008 (National Academies 2014, Table 3-4): share of injuries from gunshot vs explosion

**Reading it.** Hospitalized casualties including those who later died of wounds; men killed outright are not counted, so head and trunk are still under-counted.

**Status.** Historical at region level. One of four tables from the same source with the same counting rule (see the comparison on the Reference page) (hospitalizations, including men who later died of wounds), same extremity split.

**Sources.** Owens BD et al., 'Combat wounds in Operation Iraqi Freedom and Operation Enduring Freedom', J Trauma 2008, comparison table (reproduced in National Academies, Review of DoD Test Protocols for Combat Helmets, 2014, Table 3-2): head and neck 16, thorax 13, abdomen 10, extremities 61.

# Late 20th century



*1969-1996*

Six smaller wars with very different fighting. Northern Ireland (1969-98): the British Army and police against the Provisional IRA, mostly sniping and bombs. The Falklands (1982): Britain against Argentina, infantry fighting over open hills. The Gulf War (1991): a US-led coalition against Iraq. Somalia (1993): US Rangers and special forces against Somali militia in Mogadishu. Chechnya (1994-96 and later): Russia against Chechen separatists, with urban battles in Grozny.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Northern Ireland** | British Army and RUC; Provisional IRA and other paramilitaries. | British patrols wore helmets and soft body armour; the IRA wore none and used rifles, snipers and bombs. |
| **Falklands 1982** | British paratroopers, Royal Marines and infantry; Argentine conscripts and marines. | Steel helmets on both sides; little body armour. |
| **Gulf War 1991** | US-led coalition; Iraqi army. | US troops in PASGT Kevlar helmets and vests (fragments and pistol rounds); Iraqi troops in steel helmets. |
| **Somalia 1993** | US Task Force Ranger; Somali National Alliance militia. | Rangers wore IIIA soft vests with a single ceramic front plate and no back plate; militia were unarmoured, with AKs and RPGs. |
| **Chechnya** | Russian army and interior troops; Chechen separatist fighters. | Russian helmets and body armour, unevenly issued; Chechen fighters largely unarmoured. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **British soldier, Belfast** | Northern Ireland | PASGT vest and helmet (1980s-90s) | Assault rifle (M16, AK) | Soft body armour and helmet (PASGT used as the nearest kit). |
|  | **IRA sniper** | Northern Ireland | Unarmoured | Assault rifle (M16, AK) | Armalite, no armour. |
|  | **British Guardsman, Tumbledown** | Falklands 1982 | Steel helmet only (1915-1950s rifleman) | Bayonet (musket or rifle) | Steel helmet; the final assaults on the mountains ended with the bayonet. |
|  | **Argentine conscript** (picture) | Falklands 1982 | Steel helmet only (1915-1950s rifleman) | Assault rifle (M16, AK) | Steel helmet, FN FAL. |
|  | **US soldier, Desert Storm** (picture) | Gulf War 1991 | PASGT vest and helmet (1980s-90s) | Assault rifle (M16, AK) | PASGT helmet and vest, M16. |
|  | **British soldier, Desert Storm** (picture) | Gulf War 1991 | PASGT vest and helmet (1980s-90s) | Assault rifle (M16, AK) | Soft body armour and helmet (PASGT used as the nearest kit). |

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Iraqi conscript** (picture) | Gulf War 1991 | Steel helmet only (1915-1950s rifleman) | Assault rifle (M16, AK) | Steel helmet, AK-47. |
|  | **US Ranger, Mogadishu** (picture) | Somalia 1993 | Ranger Body Armor, front plate only (Mogadishu 1993) | Assault rifle (M16, AK) | Front plate only. |
|  | **Somali militiaman** | Somalia 1993 | Unarmoured | RPG (rocket-propelled grenade) | No armour; AK-47 and RPG-7. |
|  | **Russian conscript, Grozny 1995** | Chechnya | Steel helmet only (1915-1950s rifleman) | Assault rifle (M16, AK) | Steel helmet; body armour often missing. |
|  | **Chechen fighter, Grozny** (picture) | Chechnya | Unarmoured | RPG (rocket-propelled grenade) | Urban ambush teams. |

**Sources.** Wikipedia, 'Ranger Body Armor' (front plate only in 1993). Borden Institute, Weapons Effects and War Wounds, Table 1-2 (wound distributions).

# Northern Ireland - penetrating wounds

*Primary penetrating wounds, British forces and casualties in Northern Ireland (gunshot and bombs).*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-05 | 01-05 | 01-05 | 01-05 | 01-05 | 01-05 |
| Skull (right) | head | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 |
| Face | head | 11-16 | 11-16 | 11-16 | 11-16 | 11-16 | 11-16 | 11-16 |
| Neck | head | 17-20 | 17-20 | 17-20 | 17-20 | 17-20 | 17-20 | 17-20 |
| Chest (left) | torso | 21-26 | 21-26 | 21-26 | 21-26 | 21-26 | 21-26 | 21-26 |
| Chest (right) | torso | 27-32 | 27-32 | 27-32 | 27-32 | 27-32 | 27-32 | 27-32 |
| Abdomen | torso | 33-39 | 33-39 | 33-39 | 33-39 | 33-39 | 33-39 | 33-39 |
| Groin / pelvis | torso | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 |
| Upper back | torso | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 |
| Lower back | torso | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 |
| Shoulder (left) | arms | 51-52 | 51-52 | 51-52 | 51-52 | 51-52 | 51-52 | 51-52 |
| Shoulder (right) | arms | 53-54 | 53-54 | 53-54 | 53-54 | 53-54 | 53-54 | 53-54 |
| Upper arm (left) | arms | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 |
| Upper arm (right) | arms | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 |
| Forearm (left) | arms | 61-63 | 61-63 | 61-63 | 61-63 | 61-63 | 61-63 | 61-63 |
| Forearm (right) | arms | 64-66 | 64-66 | 64-66 | 64-66 | 64-66 | 64-66 | 64-66 |
| Hand (left) | arms | 67-68 | 67-68 | 67-68 | 67-68 | 67-68 | 67-68 | 67-68 |
| Hand (right) | arms | 69-70 | 69-70 | 69-70 | 69-70 | 69-70 | 69-70 | 69-70 |
| Thigh (left) | legs | 71-76 | 71-76 | 71-76 | 71-76 | 71-76 | 71-76 | 71-76 |
| Thigh (right) | legs | 77-82 | 77-82 | 77-82 | 77-82 | 77-82 | 77-82 | 77-82 |
| Knee (left) | legs | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 |
| Knee (right) | legs | 85-86 | 85-86 | 85-86 | 85-86 | 85-86 | 85-86 | 85-86 |
| Lower leg (left) | legs | 87-91 | 87-91 | 87-91 | 87-91 | 87-91 | 87-91 | 87-91 |
| Lower leg (right) | legs | 92-96 | 92-96 | 92-96 | 92-96 | 92-96 | 92-96 | 92-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, neck, face 20%; Thorax 15%; Abdomen 15%; Upper extremities (40% of extremity) 20%; Lower extremities (60% of extremity) 30%. Region totals are from the source. The split inside each region uses body exposure (design estimate). Extremities are split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate).

**Reading it.** Primary penetrating wound of casualties; mostly survivors who reached care, so instant kills are under-counted.

**Status.** Historical at region level (primary penetrating wounds).

**Sources.** Borden Institute, Weapons Effects and War Wounds, ch. 1, Table 1-2: anatomical distribution of primary penetrating wounds (%): head/neck/face 20, thorax 15, abdomen 15, extremity 50.

# Falklands 1982 - penetrating wounds

*Primary penetrating wounds, Falklands War casualties.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-04 | 01-04 | 01-04 | 01-04 | 01-04 | 01-04 | 01-04 |
| Skull (right) | head | 05-08 | 05-08 | 05-08 | 05-08 | 05-08 | 05-08 | 05-08 |
| Face | head | 09-13 | 09-13 | 09-13 | 09-13 | 09-13 | 09-13 | 09-13 |
| Neck | head | 14-16 | 14-16 | 14-16 | 14-16 | 14-16 | 14-16 | 14-16 |
| Chest (left) | torso | 17-22 | 17-22 | 17-22 | 17-22 | 17-22 | 17-22 | 17-22 |
| Chest (right) | torso | 23-28 | 23-28 | 23-28 | 23-28 | 23-28 | 23-28 | 23-28 |
| Abdomen | torso | 29-33 | 29-33 | 29-33 | 29-33 | 29-33 | 29-33 | 29-33 |
| Groin / pelvis | torso | 34-35 | 34-35 | 34-35 | 34-35 | 34-35 | 34-35 | 34-35 |
| Upper back | torso | 36-39 | 36-39 | 36-39 | 36-39 | 36-39 | 36-39 | 36-39 |
| Lower back | torso | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 |
| Shoulder (left) | arms | 43-44 | 43-44 | 43-44 | 43-44 | 43-44 | 43-44 | 43-44 |
| Shoulder (right) | arms | 45-46 | 45-46 | 45-46 | 45-46 | 45-46 | 45-46 | 45-46 |
| Upper arm (left) | arms | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 |
| Upper arm (right) | arms | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 |
| Forearm (left) | arms | 55-58 | 55-58 | 55-58 | 55-58 | 55-58 | 55-58 | 55-58 |
| Forearm (right) | arms | 59-62 | 59-62 | 59-62 | 59-62 | 59-62 | 59-62 | 59-62 |
| Hand (left) | arms | 63-64 | 63-64 | 63-64 | 63-64 | 63-64 | 63-64 | 63-64 |
| Hand (right) | arms | 65-66 | 65-66 | 65-66 | 65-66 | 65-66 | 65-66 | 65-66 |
| Thigh (left) | legs | 67-73 | 67-73 | 67-73 | 67-73 | 67-73 | 67-73 | 67-73 |
| Thigh (right) | legs | 74-80 | 74-80 | 74-80 | 74-80 | 74-80 | 74-80 | 74-80 |
| Knee (left) | legs | 81-82 | 81-82 | 81-82 | 81-82 | 81-82 | 81-82 | 81-82 |
| Knee (right) | legs | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 | 83-84 |
| Lower leg (left) | legs | 85-90 | 85-90 | 85-90 | 85-90 | 85-90 | 85-90 | 85-90 |
| Lower leg (right) | legs | 91-96 | 91-96 | 91-96 | 91-96 | 91-96 | 91-96 | 91-96 |
| Foot (left) | legs | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, neck, face 16%; Thorax 15%; Abdomen 10%; Upper extremities (40% of extremity) 23.6%; Lower extremities (60% of extremity) 35.4%. Region totals are from the source. The split inside each region uses body exposure (design estimate). Extremities are split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate).

**Reading it.** Primary penetrating wound of casualties; mostly survivors who reached care, so instant kills are under-counted.

**Status.** Historical at region level (primary penetrating wounds).

**Sources.** Borden Institute, Weapons Effects and War Wounds, ch. 1, Table 1-2: anatomical distribution of primary penetrating wounds (%): head/neck/face 16, thorax 15, abdomen 10, extremity 59.

# Gulf War 1991 (US) - penetrating wounds

*Primary penetrating wounds, US casualties, 1991 Gulf War.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-03 | 01-03 | 01-03 | 01-03 | 01-03 | 01-03 | 01-03 |
| Skull (right) | head | 04-06 | 04-06 | 04-06 | 04-06 | 04-06 | 04-06 | 04-06 |
| Face | head | 07-10 | 07-10 | 07-10 | 07-10 | 07-10 | 07-10 | 07-10 |
| Neck | head | 11-13 | 11-13 | 11-13 | 11-13 | 11-13 | 11-13 | 11-13 |
| Chest (left) | torso | 14-17 | 14-17 | 14-17 | 14-17 | 14-17 | 14-17 | 14-17 |
| Chest (right) | torso | 18-21 | 18-21 | 18-21 | 18-21 | 18-21 | 18-21 | 18-21 |
| Abdomen | torso | 22-25 | 22-25 | 22-25 | 22-25 | 22-25 | 22-25 | 22-25 |
| Groin / pelvis | torso | 26-27 | 26-27 | 26-27 | 26-27 | 26-27 | 26-27 | 26-27 |
| Upper back | torso | 28-29 | 28-29 | 28-29 | 28-29 | 28-29 | 28-29 | 28-29 |
| Lower back | torso | 30-31 | 30-31 | 30-31 | 30-31 | 30-31 | 30-31 | 30-31 |
| Shoulder (left) | arms | 32-34 | 32-34 | 32-34 | 32-34 | 32-34 | 32-34 | 32-34 |
| Shoulder (right) | arms | 35-37 | 35-37 | 35-37 | 35-37 | 35-37 | 35-37 | 35-37 |
| Upper arm (left) | arms | 38-41 | 38-41 | 38-41 | 38-41 | 38-41 | 38-41 | 38-41 |
| Upper arm (right) | arms | 42-45 | 42-45 | 42-45 | 42-45 | 42-45 | 42-45 | 42-45 |
| Forearm (left) | arms | 46-49 | 46-49 | 46-49 | 46-49 | 46-49 | 46-49 | 46-49 |
| Forearm (right) | arms | 50-53 | 50-53 | 50-53 | 50-53 | 50-53 | 50-53 | 50-53 |
| Hand (left) | arms | 54-56 | 54-56 | 54-56 | 54-56 | 54-56 | 54-56 | 54-56 |
| Hand (right) | arms | 57-59 | 57-59 | 57-59 | 57-59 | 57-59 | 57-59 | 57-59 |
| Thigh (left) | legs | 60-67 | 60-67 | 60-67 | 60-67 | 60-67 | 60-67 | 60-67 |
| Thigh (right) | legs | 68-75 | 68-75 | 68-75 | 68-75 | 68-75 | 68-75 | 68-75 |
| Knee (left) | legs | 76-78 | 76-78 | 76-78 | 76-78 | 76-78 | 76-78 | 76-78 |
| Knee (right) | legs | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 |
| Lower leg (left) | legs | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 |
| Lower leg (right) | legs | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, neck, face 11%; Thorax 8%; Abdomen 7%; Upper extremities (40% of extremity) 22.4%; Lower extremities (60% of extremity) 33.6%. Region totals are from the source. The split inside each region uses body exposure (design estimate). Extremities are split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate).

**Reading it.** Primary penetrating wound of casualties; mostly survivors who reached care, so instant kills are under-counted.

**Status.** Historical at region level (primary penetrating wounds). The source lists 18% 'other' (not located); it is left out and the rest normalised.

**Sources.** Borden Institute, Weapons Effects and War Wounds, ch. 1, Table 1-2: anatomical distribution of primary penetrating wounds (%): head/neck/face 11, thorax 8, abdomen 7, extremity 56, other 18.

# Gulf War 1991 (UK) - penetrating wounds

*Primary penetrating wounds, UK casualties, 1991 Gulf War.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01 | 01 | 01 | 01 | 01 | 01 | 01 |
| Skull (right) | head | 02 | 02 | 02 | 02 | 02 | 02 | 02 |
| Face | head | 03-04 | 03-04 | 03-04 | 03-04 | 03-04 | 03-04 | 03-04 |
| Neck | head | 05 | 05 | 05 | 05 | 05 | 05 | 05 |
| Chest (left) | torso | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 |
| Chest (right) | torso | 11-15 | 11-15 | 11-15 | 11-15 | 11-15 | 11-15 | 11-15 |
| Abdomen | torso | 16-20 | 16-20 | 16-20 | 16-20 | 16-20 | 16-20 | 16-20 |
| Groin / pelvis | torso | 21-23 | 21-23 | 21-23 | 21-23 | 21-23 | 21-23 | 21-23 |
| Upper back | torso | 24-26 | 24-26 | 24-26 | 24-26 | 24-26 | 24-26 | 24-26 |
| Lower back | torso | 27-29 | 27-29 | 27-29 | 27-29 | 27-29 | 27-29 | 27-29 |
| Shoulder (left) | arms | 30-32 | 30-32 | 30-32 | 30-32 | 30-32 | 30-32 | 30-32 |
| Shoulder (right) | arms | 33-35 | 33-35 | 33-35 | 33-35 | 33-35 | 33-35 | 33-35 |
| Upper arm (left) | arms | 36-39 | 36-39 | 36-39 | 36-39 | 36-39 | 36-39 | 36-39 |
| Upper arm (right) | arms | 40-43 | 40-43 | 40-43 | 40-43 | 40-43 | 40-43 | 40-43 |
| Forearm (left) | arms | 44-47 | 44-47 | 44-47 | 44-47 | 44-47 | 44-47 | 44-47 |
| Forearm (right) | arms | 48-51 | 48-51 | 48-51 | 48-51 | 48-51 | 48-51 | 48-51 |
| Hand (left) | arms | 52-54 | 52-54 | 52-54 | 52-54 | 52-54 | 52-54 | 52-54 |
| Hand (right) | arms | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 |
| Thigh (left) | legs | 58-66 | 58-66 | 58-66 | 58-66 | 58-66 | 58-66 | 58-66 |
| Thigh (right) | legs | 67-74 | 67-74 | 67-74 | 67-74 | 67-74 | 67-74 | 67-74 |
| Knee (left) | legs | 75-77 | 75-77 | 75-77 | 75-77 | 75-77 | 75-77 | 75-77 |
| Knee (right) | legs | 78-80 | 78-80 | 78-80 | 78-80 | 78-80 | 78-80 | 78-80 |
| Lower leg (left) | legs | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 |
| Lower leg (right) | legs | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, neck, face 6%; Thorax 12%; Abdomen 11%; Upper extremities (40% of extremity) 28.4%; Lower extremities (60% of extremity) 42.6%. Region totals are from the source. The split inside each region uses body exposure (design estimate). Extremities are split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate).

**Reading it.** Primary penetrating wound of casualties; mostly survivors who reached care, so instant kills are under-counted.

**Status.** Historical at region level (primary penetrating wounds).

**Sources.** Borden Institute, Weapons Effects and War Wounds, ch. 1, Table 1-2: anatomical distribution of primary penetrating wounds (%): head/neck/face 6, thorax 12, abdomen 11, extremity 71.

# Somalia 1993 - penetrating wounds

*Primary penetrating wounds, US casualties in Somalia (urban fighting, Mogadishu).*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-05 | 01-05 | 01-05 | 01-05 | 01-05 | 01-05 | 01-05 |
| Skull (right) | head | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 | 06-10 |
| Face | head | 11-16 | 11-16 | 11-16 | 11-16 | 11-16 | 11-16 | 11-16 |
| Neck | head | 17-20 | 17-20 | 17-20 | 17-20 | 17-20 | 17-20 | 17-20 |
| Chest (left) | torso | 21-23 | 21-23 | 21-23 | 21-23 | 21-23 | 21-23 | 21-23 |
| Chest (right) | torso | 24-26 | 24-26 | 24-26 | 24-26 | 24-26 | 24-26 | 24-26 |
| Abdomen | torso | 27-28 | 27-28 | 27-28 | 27-28 | 27-28 | 27-28 | 27-28 |
| Groin / pelvis | torso | 29 | 29 | 29 | 29 | 29 | 29 | 29 |
| Upper back | torso | 30-31 | 30-31 | 30-31 | 30-31 | 30-31 | 30-31 | 30-31 |
| Lower back | torso | 32 | 32 | 32 | 32 | 32 | 32 | 32 |
| Shoulder (left) | arms | 33-35 | 33-35 | 33-35 | 33-35 | 33-35 | 33-35 | 33-35 |
| Shoulder (right) | arms | 36-38 | 36-38 | 36-38 | 36-38 | 36-38 | 36-38 | 36-38 |
| Upper arm (left) | arms | 39-42 | 39-42 | 39-42 | 39-42 | 39-42 | 39-42 | 39-42 |
| Upper arm (right) | arms | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 |
| Forearm (left) | arms | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 |
| Forearm (right) | arms | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 |
| Hand (left) | arms | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 | 55-57 |
| Hand (right) | arms | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 | 58-60 |
| Thigh (left) | legs | 61-68 | 61-68 | 61-68 | 61-68 | 61-68 | 61-68 | 61-68 |
| Thigh (right) | legs | 69-76 | 69-76 | 69-76 | 69-76 | 69-76 | 69-76 | 69-76 |
| Knee (left) | legs | 77-78 | 77-78 | 77-78 | 77-78 | 77-78 | 77-78 | 77-78 |
| Knee (right) | legs | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 |
| Lower leg (left) | legs | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 | 81-87 |
| Lower leg (right) | legs | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head, neck, face 20%; Thorax 8%; Abdomen 5%; Upper extremities (40% of extremity) 26%; Lower extremities (60% of extremity) 39%. Region totals are from the source. The split inside each region uses body exposure (design estimate). Extremities are split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate).

**Reading it.** Primary penetrating wound of casualties; mostly survivors who reached care, so instant kills are under-counted.

**Status.** Historical at region level (primary penetrating wounds). The source lists 2% 'other' (not located); it is left out and the rest normalised.

**Sources.** Borden Institute, Weapons Effects and War Wounds, ch. 1, Table 1-2: anatomical distribution of primary penetrating wounds (%): head/neck/face 20, thorax 8, abdomen 5, extremity 65, other 2.

# Chechnya - penetrating wounds

*Primary penetrating wounds, Russian casualties in Chechnya (urban and mountain fighting).*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Assault rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-06 | 01-06 | 01-06 | 01-06 | 01-06 | 01-06 | 01-06 |
| Skull (right) | head | 07-12 | 07-12 | 07-12 | 07-12 | 07-12 | 07-12 | 07-12 |
| Face | head | 13-19 | 13-19 | 13-19 | 13-19 | 13-19 | 13-19 | 13-19 |
| Neck | head | 20-24 | 20-24 | 20-24 | 20-24 | 20-24 | 20-24 | 20-24 |
| Chest (left) | torso | 25-27 | 25-27 | 25-27 | 25-27 | 25-27 | 25-27 | 25-27 |
| Chest (right) | torso | 28-30 | 28-30 | 28-30 | 28-30 | 28-30 | 28-30 | 28-30 |
| Abdomen | torso | 31-32 | 31-32 | 31-32 | 31-32 | 31-32 | 31-32 | 31-32 |
| Groin / pelvis | torso | 33 | 33 | 33 | 33 | 33 | 33 | 33 |
| Upper back | torso | 34-35 | 34-35 | 34-35 | 34-35 | 34-35 | 34-35 | 34-35 |
| Lower back | torso | 36 | 36 | 36 | 36 | 36 | 36 | 36 |
| Shoulder (left) | arms | 37-39 | 37-39 | 37-39 | 37-39 | 37-39 | 37-39 | 37-39 |
| Shoulder (right) | arms | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 | 40-42 |
| Upper arm (left) | arms | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 | 43-46 |
| Upper arm (right) | arms | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 | 47-50 |
| Forearm (left) | arms | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 | 51-54 |
| Forearm (right) | arms | 55-58 | 55-58 | 55-58 | 55-58 | 55-58 | 55-58 | 55-58 |
| Hand (left) | arms | 59-60 | 59-60 | 59-60 | 59-60 | 59-60 | 59-60 | 59-60 |
| Hand (right) | arms | 61-62 | 61-62 | 61-62 | 61-62 | 61-62 | 61-62 | 61-62 |
| Thigh (left) | legs | 63-70 | 63-70 | 63-70 | 63-70 | 63-70 | 63-70 | 63-70 |
| Thigh (right) | legs | 71-78 | 71-78 | 71-78 | 71-78 | 71-78 | 71-78 | 71-78 |
| Knee (left) | legs | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 | 79-80 |
| Knee (right) | legs | 81-82 | 81-82 | 81-82 | 81-82 | 81-82 | 81-82 | 81-82 |
| Lower leg (left) | legs | 83-88 | 83-88 | 83-88 | 83-88 | 83-88 | 83-88 | 83-88 |
| Lower leg (right) | legs | 89-94 | 89-94 | 89-94 | 89-94 | 89-94 | 89-94 | 89-94 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Spade, Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted* with Target has no shield applied. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

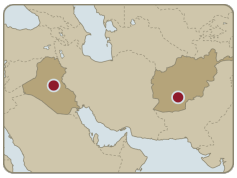
**Source totals.** Head, neck, face 24%; Thorax 9%; Abdomen 4%; Upper extremities (40% of extremity) 25.2%; Lower extremities (60% of extremity) 37.8%. Region totals are from the source. The split inside each region uses body exposure (design estimate). Extremities are split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate).

**Reading it.** Primary penetrating wound of casualties; mostly survivors who reached care, so instant kills are under-counted.

**Status.** Historical at region level (primary penetrating wounds).

**Sources.** Borden Institute, Weapons Effects and War Wounds, ch. 1, Table 1-2: anatomical distribution of primary penetrating wounds (%): head/neck/face 24, thorax 9, abdomen 4, extremity 63.

# Iraq & Afghanistan



*2001-2021*

US-led coalitions in Afghanistan from 2001 (against the Taliban and al-Qaeda) and in Iraq from 2003 (against the Iraqi army, then insurgents and militias). Roadside bombs became the main weapon against coalition troops, and modern body armour moved wounds from the torso to the head, face and limbs.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Coalition** | US, British and other allied forces; the Afghan and new Iraqi security forces they trained. | Interceptor vests with SAPI rifle plates and Kevlar helmets from the start of the Iraq war; ESAPI plates against armour-piercing rounds by the mid-2000s; lighter plate carriers later, especially in Afghanistan. |
| **Iraqi army, 2003** | Iraqi regular army and Republican Guard during the invasion. | Steel helmets for some; little body armour. |
| **Insurgents and Taliban** | Iraqi insurgents and militias; Taliban fighters. | Largely unarmoured. IEDs, RPGs, rifles and mortars. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **US soldier, Iraq 2004** (picture) | Coalition | Interceptor vest, SAPI plates, ACH (2000s) | Assault rifle (M16, AK) | Interceptor vest, SAPI plates, ACH. |
|  | **US soldier, Afghanistan 2011** (picture) | Coalition | Plate carrier, ESAPI plates, helmet (2010s-) | Assault rifle (M16, AK) | Plate carrier with ESAPI. |
|  | **British soldier, Helmand 2009** (picture) | Coalition | Interceptor vest, SAPI plates, ACH (2000s) | Assault rifle (M16, AK) | Osprey vest with plates (Interceptor used as the nearest kit). |
|  | **Afghan National Army soldier** (picture) | Coalition | PASGT vest and helmet (1980s-90s) | Assault rifle (M16, AK) | Helmet and vest when issued (PASGT used as the nearest kit). |
|  | **Iraqi soldier, 2003** | Iraqi army, 2003 | Steel helmet only (1915-1950s rifleman) | Assault rifle (M16, AK) | Steel helmet, AK-47. |
|  | **Insurgent, Iraq** | Insurgents and Taliban | Unarmoured | Mine, booby trap or IED | Roadside bombs. |
|  | **Taliban fighter** | Insurgents and Taliban | Unarmoured | Assault rifle (M16, AK) | AK-47, no armour. |

**Sources.** Owens et al., J Trauma 2008; Borden Institute, Table 1-1 (DoD Trauma Registry 2007-17).

# Iraq & Afghanistan 2001-05 - combat wounds

*6,609 combat wounds on 1,566 US soldiers, Operations Iraqi Freedom and Enduring Freedom, Oct 2001 - Jan 2005.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Assault rifle | AP rifle | MG | Pistol/S MG | Grenade | Mortar | RPG | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-07 | 01-07 | 01-07 | 01-07 | 01-06 | 01-08 | 01-07 | 01-05 | 01-07 |
| Skull (right) | head | 08-14 | 08-14 | 08-14 | 08-14 | 07-12 | 09-16 | 08-14 | 06-09 | 08-14 |
| Face | head | 15-35 | 15-35 | 15-33 | 15-35 | 13-30 | 17-38 | 15-35 | 10-22 | 15-35 |
| Neck | head | 36-39 | 36-39 | 34-37 | 36-39 | 31-33 | 39-42 | 36-39 | 23-24 | 36-39 |
| Chest (left) | torso | 40-42 | 40-42 | 38-40 | 40-42 | 34-36 | 43-44 | 40-42 | 25-26 | 40-42 |
| Chest (right) | torso | 43-45 | 43-45 | 41-43 | 43-45 | 37-39 | 45-46 | 43-45 | 27-28 | 43-45 |
| Abdomen | torso | 46-51 | 46-51 | 44-49 | 46-51 | 40-45 | 47-51 | 46-51 | 29-33 | 46-51 |
| Groin / pelvis | torso | 52-53 | 52-53 | 50-51 | 52-53 | 46-47 | 52-53 | 52-53 | 34-36 | 52-53 |
| Upper back | torso | 54-55 | 54-55 | 52-53 | 54-55 | 48-49 | 54-55 | 54-55 | 37 | 54-55 |
| Lower back | torso | 56-58 | 56-58 | 54-56 | 56-58 | 50-53 | 56-58 | 56-58 | 38-40 | 56-58 |
| Shoulder (left) | arms | 59-61 | 59-61 | 57-59 | 59-61 | 54-56 | 59-61 | 59-61 | 41-42 | 59-61 |
| Shoulder (right) | arms | 62-64 | 62-64 | 60-62 | 62-64 | 57-59 | 62-64 | 62-64 | 43-44 | 62-64 |
| Upper arm (left) | arms | 65-69 | 65-69 | 63-67 | 65-69 | 60-64 | 65-69 | 65-69 | 45-48 | 65-69 |
| Upper arm (right) | arms | 70-74 | 70-74 | 68-72 | 70-74 | 65-69 | 70-74 | 70-74 | 49-52 | 70-74 |
| Forearm (left) | arms | 75-79 | 75-79 | 73-77 | 75-79 | 70-74 | 75-79 | 75-79 | 53-56 | 75-79 |
| Forearm (right) | arms | 80-84 | 80-84 | 78-82 | 80-84 | 75-79 | 80-84 | 80-84 | 57-60 | 80-84 |
| Hand (left) | arms | 85-87 | 85-87 | 83-85 | 85-87 | 80-82 | 85-87 | 85-87 | 61-62 | 85-87 |
| Hand (right) | arms | 88-90 | 88-90 | 86-88 | 88-90 | 83-85 | 88-90 | 88-90 | 63-64 | 88-90 |
| Thigh (left) | legs | 91-92 | 91-92 | 89-91 | 91-92 | 86-88 | 91-92 | 91-92 | 65-71 | 91-92 |
| Thigh (right) | legs | 93-94 | 93-94 | 92-94 | 93-94 | 89-91 | 93-94 | 93-94 | 72-78 | 93-94 |
| Knee (left) | legs | 95 | 95 | 95 | 95 | 92 | 95 | 95 | 79-80 | 95 |
| Knee (right) | legs | 96 | 96 | 96 | 96 | 93 | 96 | 96 | 81-82 | 96 |
| Lower leg (left) | legs | 97-98 | 97-98 | 97-98 | 97-98 | 94-96 | 97-98 | 97-98 | 83-88 | 97-98 |
| Lower leg (right) | legs | 99-00 | 99-00 | 99-00 | 99-00 | 97-98 | 99-00 | 99-00 | 89-94 | 99-00 |
| Foot (left) | legs | - | - | - | - | 99 | - | - | 95-97 | - |
| Foot (right) | legs | - | - | - | - | 00 | - | - | 98-00 | - |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and ears (8 + 3) 11%; Face and eyes (10 + 6) 16%; Neck 3%; Thorax 6%; Abdomen 11%; Upper extremity (half of 54) 27%; Lower extremity (half of 54) 27%. Region totals are from the source. Extremities are split evenly between arms and legs (the companion study found fractures 50/50); the split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-18 | Assault rifle (M16, AK) |
| 19-79 | Mine, booby trap or IED |
| 80-87 | Mortar, rocket or artillery |
| 88-92 | Hand grenade |
| 93-96 | RPG (rocket-propelled grenade) |
| 97-00 | Other |

Owens et al. 2008: gunshot 18%, explosion 78%; explosions split by the DoD Trauma Registry 2007-17 shares (Borden Table 1-1: IED 60.4, mortar/rocket/artillery 7.8, grenade 4.6, RPG 4.5)

**Reading it.** Soldiers wore helmets and plate body armour, so the torso is already protected in these numbers (thorax only 6%). Don't apply armour again. Counts all wounds, wounded and killed, several per soldier.

**Status.** Historical at region level. Eyes are folded into face and ears into the skull. Abdomen is taken to include the lower back and pelvis.

**Sources.** Owens BD et al., 'Combat wounds in Operation Iraqi Freedom and Operation Enduring Freedom', J Trauma 2008: head 8%, eyes 6%, ears 3%, face 10%, neck 3%, thorax 6%, abdomen 11%, extremity 54%; explosions 78%, gunshot 18%. | Owens BD et al., 'Characterization of extremity wounds in OIF and OEF', J Orthop Trauma 2007: 915 extremity fractures, 461 upper (50%) and 454 lower (50%).

# Iraq & Afghanistan 2001-05 - hospitalized (Owens comparison set)

*US casualties hospitalized from Operations Iraqi Freedom and Enduring Freedom.*

Confidence: **Historical - taken directly from a source**. Armour: **already reflected, do not apply**. Weapon columns differ, but across the weapon mix below they average back to the source totals.

| Location | Zone | Assault rifle | AP rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED | Frags |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-08 | 01-08 | 01-07 | 01-08 | 01-06 | 01-08 | 01-08 | 01-03 | 01-08 |
| Skull (right) | head | 09-16 | 09-16 | 08-14 | 09-16 | 07-12 | 09-16 | 09-16 | 04-06 | 09-16 |
| Face | head | 17-25 | 17-25 | 15-22 | 17-25 | 13-19 | 17-26 | 17-25 | 07-09 | 17-25 |
| Neck | head | 26-31 | 26-31 | 23-27 | 26-31 | 20-24 | 27-33 | 26-31 | 10-11 | 26-31 |
| Chest (left) | torso | 32-33 | 32-33 | 28-29 | 32-33 | 25-26 | 34-35 | 32-33 | 12 | 32-33 |
| Chest (right) | torso | 34-35 | 34-35 | 30-31 | 34-35 | 27-28 | 36-37 | 34-35 | 13 | 34-35 |
| Abdomen | torso | 36-39 | 36-39 | 32-35 | 36-39 | 29-32 | 38-41 | 36-39 | 14-15 | 36-39 |
| Groin / pelvis | torso | 40-41 | 40-41 | 36-37 | 40-41 | 33-34 | 42-43 | 40-41 | 16-17 | 40-41 |
| Upper back | torso | 42-43 | 42-43 | 38-39 | 42-43 | 35 | 44 | 42-43 | 18 | 42-43 |
| Lower back | torso | 44-46 | 44-46 | 40-42 | 44-46 | 36-38 | 45-47 | 44-46 | 19 | 44-46 |
| Shoulder (left) | arms | 47-48 | 47-48 | 43-44 | 47-48 | 39-40 | 48-49 | 47-48 | 20 | 47-48 |
| Shoulder (right) | arms | 49-50 | 49-50 | 45-46 | 49-50 | 41-42 | 50-51 | 49-50 | 21 | 49-50 |
| Upper arm (left) | arms | 51-53 | 51-53 | 47-49 | 51-53 | 43-45 | 52-55 | 51-53 | 22-23 | 51-53 |
| Upper arm (right) | arms | 54-56 | 54-56 | 50-52 | 54-56 | 46-48 | 56-59 | 54-56 | 24 | 54-56 |
| Forearm (left) | arms | 57-59 | 57-59 | 53-55 | 57-59 | 49-51 | 60-63 | 57-59 | 25 | 57-59 |
| Forearm (right) | arms | 60-62 | 60-62 | 56-58 | 60-62 | 52-54 | 64-66 | 60-62 | 26 | 60-62 |
| Hand (left) | arms | 63-64 | 63-64 | 59-60 | 63-64 | 55-56 | 67-68 | 63-64 | 27 | 63-64 |
| Hand (right) | arms | 65-66 | 65-66 | 61-62 | 65-66 | 57-58 | 69-70 | 65-66 | 28 | 65-66 |
| Thigh (left) | legs | 67-73 | 67-73 | 63-70 | 67-73 | 59-66 | 71-76 | 67-73 | 29-40 | 67-73 |
| Thigh (right) | legs | 74-80 | 74-80 | 71-78 | 74-80 | 67-74 | 77-82 | 74-80 | 41-52 | 74-80 |
| Knee (left) | legs | 81-82 | 81-82 | 79-80 | 81-82 | 75-77 | 83-84 | 81-82 | 53-56 | 81-82 |
| Knee (right) | legs | 83-84 | 83-84 | 81-82 | 83-84 | 78-80 | 85-86 | 83-84 | 57-60 | 83-84 |
| Lower leg (left) | legs | 85-90 | 85-90 | 83-88 | 85-90 | 81-87 | 87-91 | 85-90 | 61-71 | 85-90 |
| Lower leg (right) | legs | 91-96 | 91-96 | 89-94 | 91-96 | 88-94 | 92-96 | 91-96 | 72-82 | 91-96 |
| Foot (left) | legs | 97-98 | 97-98 | 95-97 | 97-98 | 95-97 | 97-98 | 97-98 | 83-91 | 97-98 |
| Foot (right) | legs | 99-00 | 99-00 | 98-00 | 99-00 | 98-00 | 99-00 | 99-00 | 92-00 | 99-00 |

**Close combat.** Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Dagger: roll location on *Knife assault - stab wounds (forensic)*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 30%; Thorax 6%; Abdomen 9%; Upper extremities (40% of 55) 22%; Lower extremities (60% of 55) 33%. Region totals are from the source. Extremities split 40% arms / 60% legs, the average of the WWII, Korea and Vietnam hospital figures in Neel (1973) Table 7 (design estimate), the same rule for every table in this set. The split inside each region uses body exposure (design estimate).

## What hit them? (d100)

| d100 | Cause |
|---|---|
| 01-19 | Assault rifle (M16, AK) |
| 20-00 | Fragments, unspecified (shell, grenade, mine) |

Owens et al. 2008 (National Academies 2014, Table 3-4): share of injuries from gunshot vs explosion

**Reading it.** Hospitalized casualties including those who later died of wounds; men killed outright are not counted, so head and trunk are still under-counted. Helmets and plate body armour were worn: don't apply armour again.

**Status.** Historical at region level. One of four tables from the same source with the same counting rule (see the comparison on the Reference page) (hospitalizations, including men who later died of wounds), same extremity split.

**Sources.** Owens BD et al., 'Combat wounds in Operation Iraqi Freedom and Operation Enduring Freedom', J Trauma 2008, comparison table (reproduced in National Academies, Review of DoD Test Protocols for Combat Helmets, 2014, Table 3-2): head and neck 30, thorax 6, abdomen 9, extremities 55.

# Reference

Not a war. The random-hit table shows where hits would land if every square inch of the body were equally likely: use it as a neutral table with any armour, or to see how far a real war's pattern moves away from chance. Below it, the four-war comparison shows how American wound patterns changed from WWII to Iraq and Afghanistan, counted the same way each time; each of those four tables is listed under its own war as the Owens comparison set.

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Modern soldier** | - | Plate carrier, ESAPI plates, helmet (2010s-) | Assault rifle (M16, AK) | Plate carrier and helmet. |
|  | **Police officer** | - | Concealable soft vest (police IIIA) | Pistol or submachine gun | Concealable IIIA vest. |
|  | **Civilian** | - | Unarmoured | Pistol or submachine gun | No armour. |

## US wars compared: where hospitalized wounded were hit

| | Head & neck | Thorax | Abdomen | Extremities | Gunshot / explosion |
|---|---|---|---|---|---|
| **Body surface area (chance)** | 12% | 16% | 11% | 61% | - |
| **WWII** | 21% | 14% | 8% | 58% | 27 / 73 |
| **Korea** | 21% | 10% | 9% | 60% | 31 / 69 |
| **Vietnam** | 16% | 13% | 10% | 61% | 35 / 65 |
| **Iraq & Afghanistan** | 30% | 6% | 9% | 55% | 19 / 81 |

Same source and counting rule for every row: hospitalized casualties, including men who later died of wounds. With modern body armour the thorax share fell to 6% and the head and neck rose to 30%, because plates cover the chest and nothing covers the face and neck. Gunshot vs explosion shows how the weapon mix shifted.

**Source.** Owens et al., 'Combat wounds in Operation Iraqi Freedom and Operation Enduring Freedom', J Trauma 2008 (National Academies 2014, Tables 3-2 and 3-4).

# Baseline - random hit (body surface area)

*Each region's share of body surface area: where hits would land if every square inch were equally likely.*

Confidence: **Fitted - estimated to match published summary figures**. Armour: **allowed**.

| Location | Zone | Rifle | Assault rifle | AP rifle | MG | Pistol/SMG | Grenade | Mortar | RPG | Mine/IED | Rifle-musket | Revolver | Bow | Crossbow | Sling |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | head | 01-03 | 01-03 | 01-03 | 01-02 | 01-03 | 01-02 | 01-03 | 01-03 | 01 | 01-03 | 01-03 | 01-03 | 01-03 | 01-04 |
| Skull (right) | head | 04-06 | 04-06 | 04-06 | 03-04 | 04-06 | 03-04 | 04-06 | 04-06 | 02 | 04-06 | 04-06 | 04-06 | 04-06 | 05-08 |
| Face | head | 07-10 | 07-10 | 07-10 | 05-07 | 07-10 | 05-07 | 07-10 | 07-10 | 03 | 07-10 | 07-10 | 07-10 | 07-10 | 09-15 |
| Neck | head | 11-12 | 11-12 | 11-12 | 08-09 | 11-12 | 08-09 | 11-13 | 11-12 | 04 | 11-12 | 11-12 | 11-12 | 11-12 | 16-19 |
| Chest (left) | torso | 13-18 | 13-18 | 13-18 | 10-15 | 13-18 | 10-14 | 14-19 | 13-18 | 05-07 | 13-18 | 13-18 | 13-18 | 13-18 | 20-25 |
| Chest (right) | torso | 19-24 | 19-24 | 19-24 | 16-21 | 19-24 | 15-19 | 20-25 | 19-24 | 08-10 | 19-24 | 19-24 | 19-24 | 19-24 | 26-31 |
| Abdomen | torso | 25-29 | 25-29 | 25-29 | 22-26 | 25-29 | 20-24 | 26-30 | 25-29 | 11-12 | 25-29 | 25-29 | 25-29 | 25-29 | 32-36 |
| Groin / pelvis | torso | 30-32 | 30-32 | 30-32 | 27-28 | 30-32 | 25-26 | 31-33 | 30-32 | 13-14 | 30-32 | 30-32 | 30-32 | 30-32 | 37-39 |
| Upper back | torso | 33-36 | 33-36 | 33-36 | 29-32 | 33-36 | 27-29 | 34-37 | 33-36 | 15-16 | 33-36 | 33-36 | 33-36 | 33-36 | 40-43 |
| Lower back | torso | 37-39 | 37-39 | 37-39 | 33-35 | 37-39 | 30-32 | 38-40 | 37-39 | 17-18 | 37-39 | 37-39 | 37-39 | 37-39 | 44-46 |
| Shoulder (left) | arms | 40-41 | 40-41 | 40-41 | 36-37 | 40-41 | 33-34 | 41-42 | 40-41 | 19 | 40-41 | 40-41 | 40-41 | 40-41 | 47-48 |
| Shoulder (right) | arms | 42-43 | 42-43 | 42-43 | 38-39 | 42-43 | 35-36 | 43-44 | 42-43 | 20 | 42-43 | 42-43 | 42-43 | 42-43 | 49-50 |
| Upper arm (left) | arms | 44-46 | 44-46 | 44-46 | 40-42 | 44-46 | 37-39 | 45-47 | 44-46 | 21 | 44-46 | 44-46 | 44-46 | 44-46 | 51-53 |
| Upper arm (right) | arms | 47-49 | 47-49 | 47-49 | 43-45 | 47-49 | 40-42 | 48-50 | 47-49 | 22 | 47-49 | 47-49 | 47-49 | 47-49 | 54-56 |
| Forearm (left) | arms | 50-52 | 50-52 | 50-52 | 46-48 | 50-52 | 43-45 | 51-53 | 50-52 | 23 | 50-52 | 50-52 | 50-52 | 50-52 | 57-59 |
| Forearm (right) | arms | 53-55 | 53-55 | 53-55 | 49-51 | 53-55 | 46-48 | 54-56 | 53-55 | 24 | 53-55 | 53-55 | 53-55 | 53-55 | 60-62 |
| Hand (left) | arms | 56-57 | 56-57 | 56-57 | 52-53 | 56-57 | 49-50 | 57-58 | 56-57 | 25 | 56-57 | 56-57 | 56-57 | 56-57 | 63-64 |
| Hand (right) | arms | 58-59 | 58-59 | 58-59 | 54-55 | 58-59 | 51-52 | 59-60 | 58-59 | 26 | 58-59 | 58-59 | 58-59 | 58-59 | 65-66 |
| Thigh (left) | legs | 60-67 | 60-67 | 60-67 | 56-64 | 60-67 | 53-62 | 61-68 | 60-67 | 27-39 | 60-67 | 60-67 | 60-67 | 60-67 | 67-73 |
| Thigh (right) | legs | 68-75 | 68-75 | 68-75 | 65-73 | 68-75 | 63-72 | 69-76 | 68-75 | 40-52 | 68-75 | 68-75 | 68-75 | 68-75 | 74-80 |
| Knee (left) | legs | 76-78 | 76-78 | 76-78 | 74-76 | 76-78 | 73-75 | 77-78 | 76-78 | 53-56 | 76-78 | 76-78 | 76-78 | 76-78 | 81-82 |
| Knee (right) | legs | 79-80 | 79-80 | 79-80 | 77-79 | 79-80 | 76-78 | 79-80 | 79-80 | 57-60 | 79-80 | 79-80 | 79-80 | 79-80 | 83-84 |
| Lower leg (left) | legs | 81-87 | 81-87 | 81-87 | 80-87 | 81-87 | 79-86 | 81-87 | 81-87 | 61-71 | 81-87 | 81-87 | 81-87 | 81-87 | 85-90 |
| Lower leg (right) | legs | 88-94 | 88-94 | 88-94 | 88-94 | 88-94 | 87-94 | 88-94 | 88-94 | 72-82 | 88-94 | 88-94 | 88-94 | 88-94 | 91-96 |
| Foot (left) | legs | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 95-97 | 83-91 | 95-97 | 95-97 | 95-97 | 95-97 | 97-98 |
| Foot (right) | legs | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 98-00 | 92-00 | 98-00 | 98-00 | 98-00 | 98-00 | 99-00 |

**Close combat.** Dagger: roll location on *Knife assault - stab wounds (forensic)*. Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head and neck 12%; Thorax 16%; Abdomen 11%; Upper extremities (1/3 of 61) 20.33%; Lower extremities (2/3 of 61) 40.67%. Region totals are from the source. Extremities split 1/3 arms, 2/3 legs by surface (rule of nines: arms 18%, legs 36%). The split inside each region uses body exposure (design estimate).

**Reading it.** Use to compare a war's table against pure chance (a head share above 12% means heads were exposed or aimed at), or as a neutral table when nothing else fits.

**Status.** Body surface area from the same source table. Not wound data: a neutral reference.

**Sources.** Owens BD et al., 'Combat wounds in Operation Iraqi Freedom and Operation Enduring Freedom', J Trauma 2008, comparison table (reproduced in National Academies, Review of DoD Test Protocols for Combat Helmets, 2014, Table 3-2): body surface area - head and neck 12, thorax 16, abdomen 11, extremities 61.

# Close combat



*Stockholm, 1983-1993*

Knife killings recorded by forensic pathologists: usually one attacker against an unarmoured, often surprised victim, with several wounds each. The table stands in for knives and bayonets on every non-medieval table.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Attacker** | Single assailant with a knife. | None. |
| **Victim** | Usually unarmoured civilians; 41% had defensive wounds on the hands and arms. | None. Soft ballistic vests give little protection against a stab; plates stop one. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Knife attacker** | Attacker | Unarmoured | Dagger / knife | Single assailant. |
|  | **Police officer** | Victim | Concealable soft vest (police IIIA) | Dagger / knife | IIIA vest: stops bullets, not knives. |
|  | **Soldier in plates** | Victim | Plate carrier, ESAPI plates, helmet (2010s-) | Bayonet (musket or rifle) | Plates stop most stabs to the chest. |
|  | **Civilian** | Victim | Unarmoured | Dagger / knife | No armour. |

**Sources.** Karlsson 1998, via forensicmed.co.uk.

# Knife assault - stab wounds (forensic)

*Where stab and cut wounds were found on homicide victims, Stockholm 1983-1993. Used for knives and bayonets on any non-medieval table.*

Confidence: **Historical - taken directly from a source**. Armour: **allowed**. Weapons share the source's pooled odds (no weapon mix to separate them).

| Location | Zone | Dagger | Bayonet |
|---|---|---|---|
| Skull (left) | head | 01-04 | 01-04 |
| Skull (right) | head | 05-08 | 05-08 |
| Face | head | 09-13 | 09-13 |
| Neck | head | 14-28 | 14-28 |
| Chest (left) | torso | 29-36 | 29-36 |
| Chest (right) | torso | 37-44 | 37-44 |
| Abdomen | torso | 45-52 | 45-52 |
| Groin / pelvis | torso | 53-56 | 53-56 |
| Upper back | torso | 57-64 | 57-64 |
| Lower back | torso | 65-71 | 65-71 |
| Shoulder (left) | arms | 72 | 72 |
| Shoulder (right) | arms | 73 | 73 |
| Upper arm (left) | arms | 74-76 | 74-76 |
| Upper arm (right) | arms | 77-79 | 77-79 |
| Forearm (left) | arms | 80-83 | 80-83 |
| Forearm (right) | arms | 84-87 | 84-87 |
| Hand (left) | arms | 88-90 | 88-90 |
| Hand (right) | arms | 91-93 | 91-93 |
| Thigh (left) | legs | 94-95 | 94-95 |
| Thigh (right) | legs | 96-97 | 96-97 |
| Lower leg (left) | legs | 98 | 98 |
| Lower leg (right) | legs | 99 | 99 |
| Foot (left) | legs | 00 | 00 |

**Close combat.** Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*. Punch, Kick, Elbow/knee: roll location on *Unarmed - landed strikes (MMA)*.

**Source totals.** Head (27% of victims) = 12.3%; Neck (33%) = 15.1%; Upper extremity (46%) = 21.0%; Chest (34%) = 15.5%; Back (34%) = 15.5%; Abdomen (26%) = 11.9%; Lower extremity (19%) = 8.7% (n = 219). Region figures are from the source (share of victims wounded in each region; victims usually had several wounds, so they sum to more than 100 and are normalised). Arm wounds favour forearms and hands (defensive wounds); other splits use body exposure (design estimate).

**Reading it.** Victims were mostly unarmoured civilians, so these are hits before armour: armour can be applied. 41% of victims had defensive wounds to the hands and forearms, already included here, so 'warding' doubles up. Soft ballistic vests give little protection against a stab (pierce 0); plates stop it.

**Status.** Historical at region level. Homicide, not battle: single attacker, unarmoured victim, often surprised.

**Sources.** Karlsson T., 'Sharp force homicides in the Stockholm area, 1983-1992', Forensic Sci Int 1998, via forensicmed.co.uk 'Patterns of sharp force trauma': share of homicide victims with wounds by region - head 27%, neck 33%, upper extremity 46%, chest 34%, back 34%, abdomen 26%, lower extremity 19%; defensive injuries in 41% of victims.

# Unarmed

*UFC, 2005-2026*

Professional mixed martial arts under the Unified Rules: four-ounce gloves, no strikes to the groin, spine or back of the head. The table also stands in for punches, kicks, elbows and knees on every other table.

## Sides and armour

| Side | Who fought | How they used armour |
|---|---|---|
| **Fighters** | Professional MMA fighters. | Gloves, mouthguard and groin cup only. |

## Example combatants

The kit is what they wear when hit (use it on a table that allows armour); the weapon is what they fight with. 'Picture' links to a period image of such a combatant on Wikimedia Commons.

| | Example | Side | Wears (armour kit) | Fights with | Notes |
|---|---|---|---|---|---|
|  | **Striker** | Fighters | Unarmoured | Punch | Boxer-style; head hunter. |
|  | **Kickboxer** | Fighters | Unarmoured | Kick | Low kicks to the lead leg. |

**Sources.** Fightshow, UFC Strike Map.

# Unarmed - landed strikes (MMA)

*Where landed significant strikes go in professional MMA: all UFC divisions, 2005-2026, 648,783 strikes.*

Confidence: **Historical - taken directly from a source**. Armour: **allowed**.

| Location | Zone | Punch | Kick | Elbow/knee |
|---|---|---|---|---|
| Skull (left) | head | 01-17 | 01-05 | 01-14 |
| Skull (right) | head | 18-30 | 06-08 | 15-25 |
| Face | head | 31-81 | 09-22 | 26-68 |
| Neck | head | 82-83 | 23 | 69-70 |
| Chest (left) | torso | 84-87 | 24-31 | 71-77 |
| Chest (right) | torso | 88-92 | 32-41 | 78-85 |
| Abdomen | torso | 93-99 | 42-53 | 86-95 |
| Upper back | torso | 00 | 54 | 96 |
| Thigh (left) | legs | - | 55-70 | 97 |
| Thigh (right) | legs | - | 71-83 | 98 |
| Knee (left) | legs | - | 84-85 | - |
| Knee (right) | legs | - | 86-87 | - |
| Lower leg (left) | legs | - | 88-95 | 99 |
| Lower leg (right) | legs | - | 96-00 | 00 |

**Close combat.** Dagger: roll location on *Knife assault - stab wounds (forensic)*. Bayonet: roll location on *Peninsular War 1808-14 - French officers' wounds*. Sword, Axe, Spiked club, Spear: roll location on *Visby 1361 - gameplay-adjusted*.

**Source totals.** Head 64%; Body 21%; Legs 15%. Head / body / legs totals are from the source. The split inside each region is a design estimate (orthodox stance, legal targets only).

**Reading it.** Sport rules: no strikes to the groin, spine or back of the head, and blocked strikes on the arms are not counted. A street fight would add groin, back and arm hits. Totals pool all strike types; the weapon columns are design estimates.

**Status.** Historical at head / body / legs level. Within-region split favours the face and the target's lead (left) side and liver (right) side.

**Sources.** Fightshow, UFC Strike Map (UFC round statistics 2005-2026, data through 3 Oct 2026): head 64%, body 21%, legs 15% of 648,783 landed significant strikes.

# Situations

Each situation reshapes where blows land. The roller and CLI stack any number of them (one per group: facing, height, posture, cover). On paper, each column shows one situation alone. Each table offers only the situations that make sense for it.

| Situation | Column | Effect on weights | Notes |
|---|---|---|---|
| **Attacker on target's shield side** *group: facing* | Shield side | left side x1.8, right side x0.5 | Attacker works round to the target's left (shield) side. |
| **Attacker on target's weapon side** *group: facing* | Weapon side | left side x0.5, right side x1.8 | Attacker is off the target's right (weapon-arm) side. |
| **Attacker higher (mounted, uphill, on a wall)** *group: height* | Higher | head x1.6, arms x1.2, legs x0.4, shoulder (left) x1.3, shoulder (right) x1.3 | Blows come down onto the head and shoulders; legs out of reach. |
| **Attacker lower (in a ditch, kneeling, vs. rider)** *group: height* | Lower | head x0.5, legs x1.6, groin / pelvis x1.8, abdomen x1.3 | Blows come up into the legs, groin and belly. |
| **Target down (fallen, kneeling, helpless)** *group: posture* | Down | head x2, torso x0.6, legs x0.6, upper back x3, lower back x2, forearm (left) x1.6, forearm (right) x1.6, hand (left) x1.6, hand (right) x1.6, face x0.7 | Towton pattern: repeated blows to the head and back; arms raised to ward. Shaped on Towton (1461): 96% of crania with weapon wounds, far fewer below the skull. |
| **Target fleeing (back turned)** *group: posture* | Fleeing | head x1.2, arms x0.6, legs x1.3, upper back x4, lower back x3, face x0.1, chest (left) x0.15, chest (right) x0.15, abdomen x0.15, groin / pelvis x0.2 | Rout: back, back of the head and legs exposed; front almost unreachable. |
| **Target flanked (two or more attackers)** | Flanked | legs x1.3, upper back x2.5, lower back x2.5, skull (left) x1.2, skull (right) x1.2 | Visby team-fighting: one engages the front while others strike from behind and low. From the Visby reading of attackers working in teams (Randolph-Quinney et al., 2026). |
| **Target has no shield** | No shield | head x1.1, torso x1.4, forearm (left) x0.7, hand (left) x0.7 | Tables assume a shield. Without one the body and head are more open. |
| **Target warding with arms (unarmed or disarmed)** | Warding | arms x1.5, forearm (left) x1.6, forearm (right) x1.6, hand (left) x1.6, hand (right) x1.6 | Defensive wounds: forearms and hands raised to block. Defensive forearm and hand wounds are recorded at Towton and in modern sharp-force forensics. |
| **Target behind cover (breastwork, wall, trench)** *group: cover* | Cover | head x1.5, torso x0.3, legs x0.05, shoulder (left) x1.5, shoulder (right) x1.5, abdomen x0.3, groin / pelvis x0.2, lower back x0.3 | Only head, shoulders and arms show over the parapet. For entrenched or firing-line fighting. Gunfire tables only by default. |

**Called shot.** Roll the location twice; keep whichever result lands in the called zone. If both or neither do, keep the first. Your system sets any to-hit penalty.

# Sidon, mid-13th century - crusader mass graves (bone evidence): Sword / Axe

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-03 | 01 | 01-03 | 01 | 01-03 | 01-03 | 01-03 | 01-02 |
| Skull (right) | 04-06 | 02-05 | 04-06 | 02 | 04-06 | 04-06 | 04-06 | 03-04 |
| Face | 07-09 | 06-08 | 07-10 | 03-04 | 07-09 | - | 07-09 | 05-06 |
| Neck | 10-48 | 09-44 | 11-57 | 05-28 | 10-59 | 07-53 | 10-45 | 07-35 |
| Chest (left) | 49-51 | 45 | 58-59 | 29-32 | 60 | - | 46-48 | 36-37 |
| Chest (right) | 52-54 | 46-50 | 60-61 | 33-36 | 61 | - | 49-51 | 38-39 |
| Upper back | 55-56 | 51-52 | 62-63 | 37-38 | 62-63 | 54-61 | 52-56 | 40-41 |
| Lower back | 57-58 | 53-54 | 64 | 39-40 | 64 | 62-66 | 57-60 | 42 |
| Shoulder (left) | 59-63 | 55-56 | 65-70 | 41-46 | 65-67 | 67-69 | 61-64 | 43-47 |
| Shoulder (right) | 64-68 | 57-64 | 71-76 | 47-52 | 68-70 | 70-72 | 65-68 | 48-52 |
| Upper arm (left) | 69-70 | 65 | 77-78 | 53-55 | 71-72 | 73-74 | 69-70 | 53-55 |
| Upper arm (right) | 71-72 | 66-69 | 79-80 | 56-58 | 73-74 | 75-76 | 71-72 | 56-58 |
| Forearm (left) | 73-74 | 70 | 81-82 | 59-61 | 75-77 | 77-78 | 73-74 | 59-62 |
| Forearm (right) | 75-76 | 71-74 | 83-84 | 62-64 | 78-80 | 79-80 | 75-76 | 63-66 |
| Hand (left) | 77-84 | 75-78 | 85-91 | 65-74 | 81-88 | 81-85 | 77-83 | 67-80 |
| Hand (right) | 85-92 | 79-91 | 92-98 | 75-84 | 89-96 | 86-90 | 84-90 | 81-94 |
| Thigh (left) | 93-96 | 92-93 | 99 | 85-92 | 97-98 | 91-95 | 91-95 | 95-97 |
| Thigh (right) | 97-00 | 94-00 | 00 | 93-00 | 99-00 | 96-00 | 96-00 | 98-00 |

# Sidon, mid-13th century - crusader mass graves (bone evidence): Spiked club

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-10 | 01-04 | 01-15 | 01-05 | 01-17 | 01-12 | 01-10 | 01-08 |
| Skull (right) | 11-20 | 05-20 | 16-30 | 06-10 | 18-34 | 13-24 | 11-20 | 09-16 |
| Face | 21-32 | 21-30 | 31-48 | 11-15 | 35-49 | 25 | 21-30 | 17-26 |
| Chest (left) | 33-36 | 31-32 | 49-51 | 16-18 | 50-51 | 26 | 31-33 | 27-29 |
| Chest (right) | 37-40 | 33-38 | 52-54 | 19-21 | 52-53 | 27 | 34-36 | 30-32 |
| Groin / pelvis | 41-43 | 39-41 | 55-57 | 22-26 | 54 | 28 | 37-38 | 33-34 |
| Upper back | 44-45 | 42-43 | 58-59 | 27-28 | 55-58 | 29-37 | 39-43 | 35-36 |
| Lower back | 46-47 | 44-45 | 60-61 | 29-30 | 59-60 | 38-43 | 44-47 | 37-38 |
| Shoulder (left) | 48-51 | 46-47 | 62-67 | 31-34 | 61-64 | 44-46 | 48-51 | 39-44 |
| Shoulder (right) | 52-55 | 48-54 | 68-73 | 35-38 | 65-68 | 47-49 | 52-54 | 45-50 |
| Upper arm (left) | 56-57 | 55 | 74-76 | 39-40 | 69-70 | 50 | 55-56 | 51-53 |
| Upper arm (right) | 58-59 | 56-58 | 77-78 | 41-42 | 71-72 | 51 | 57-58 | 54-56 |
| Forearm (left) | 60-61 | 59 | 79-80 | 43-44 | 73-75 | 52 | 59-60 | 57-60 |
| Forearm (right) | 62-63 | 60-62 | 81-82 | 45-46 | 76-78 | 53 | 61-62 | 61-64 |
| Hand (left) | 64 | 63 | 83-84 | 47 | 79-80 | 54 | 63 | 65-67 |
| Hand (right) | 65 | 64-65 | 85-86 | 48 | 81-82 | 55 | 64 | 68-70 |
| Thigh (left) | 66-74 | 66-69 | 87-89 | 49-61 | 83-86 | 56-66 | 65-73 | 71-77 |
| Thigh (right) | 75-83 | 70-83 | 90-92 | 62-74 | 87-90 | 67-77 | 74-82 | 78-84 |
| Knee (left) | 84-85 | 84 | 93 | 75-77 | 91 | 78-80 | 83-84 | 85-86 |
| Knee (right) | 86-87 | 85-87 | 94 | 78-80 | 92 | 81-82 | 85-86 | 87-88 |
| Lower leg (left) | 88-92 | 88-89 | 95-96 | 81-88 | 93-95 | 83-89 | 87-92 | 89-93 |
| Lower leg (right) | 93-97 | 90-97 | 97-98 | 89-96 | 96-98 | 90-96 | 93-98 | 94-98 |
| Foot (left) | 98-99 | 98 | 99 | 97-98 | 99 | 97-98 | 99 | 99 |
| Foot (right) | 00 | 99-00 | 00 | 99-00 | 00 | 99-00 | 00 | 00 |

# Sidon, mid-13th century - crusader mass graves (bone evidence): Spear

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-19 | 01-09 | 01-27 | 01-10 | 01-29 | 01-23 | 01-19 | 01-19 |
| Skull (right) | 20-38 | 10-39 | 28-54 | 11-20 | 30-58 | 24-45 | 20-38 | 20-38 |
| Face | 39-60 | 40-59 | 55-86 | 21-32 | 59-82 | 46-48 | 39-56 | 39-60 |
| Thigh (left) | 61-70 | 60-64 | 87-89 | 33-49 | 83-87 | 49-61 | 57-67 | 61-70 |
| Thigh (right) | 71-80 | 65-80 | 90-92 | 50-66 | 88-92 | 62-74 | 68-78 | 71-80 |
| Knee (left) | 81-83 | 81 | 93 | 67-70 | 93 | 75-77 | 79-81 | 81-83 |
| Knee (right) | 84-86 | 82-85 | 94 | 71-74 | 94 | 78-80 | 82-84 | 84-86 |
| Lower leg (left) | 87-93 | 86-88 | 95-97 | 75-87 | 95-97 | 81-90 | 85-92 | 87-93 |
| Lower leg (right) | 94-00 | 89-00 | 98-00 | 88-00 | 98-00 | 91-00 | 93-00 | 94-00 |

# Sidon, mid-13th century - crusader mass graves (bone evidence): Dagger / Bow / Crossbow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-02 | 01-07 | 01-03 | 01-08 | 01-07 | 01-06 | 01-04 |
| Skull (right) | 06-10 | 03-11 | 08-14 | 04-06 | 09-16 | 08-13 | 07-12 | 05-08 |
| Face | 11-16 | 12-17 | 15-22 | 07-10 | 17-22 | 14 | 13-18 | 09-13 |
| Neck | 17-40 | 18-39 | 23-54 | 11-23 | 23-57 | 15-43 | 19-39 | 14-32 |
| Chest (left) | 41-43 | 40 | 55-57 | 24-26 | 58 | 44 | 40-42 | 33-34 |
| Chest (right) | 44-46 | 41-45 | 58-60 | 27-29 | 59 | 45 | 43-45 | 35-36 |
| Groin / pelvis | 47 | 46 | 61 | 30-31 | - | - | 46 | 37 |
| Upper back | 48-49 | 47-48 | 62-63 | 32-33 | 60-62 | 46-53 | 47-50 | 38-39 |
| Lower back | 50-51 | 49-50 | 64 | 34-35 | 63-64 | 54-58 | 51-54 | 40 |
| Shoulder (left) | 52-56 | 51-52 | 65-70 | 36-40 | 65-67 | 59-61 | 55-58 | 41-45 |
| Shoulder (right) | 57-61 | 53-59 | 71-76 | 41-45 | 68-70 | 62-64 | 59-62 | 46-50 |
| Upper arm (left) | 62-63 | 60 | 77-78 | 46-48 | 71-72 | 65 | 63-64 | 51-53 |
| Upper arm (right) | 64-65 | 61-64 | 79-80 | 49-50 | 73-74 | 66 | 65-66 | 54-56 |
| Forearm (left) | 66-67 | 65 | 81-82 | 51-52 | 75-77 | 67 | 67-68 | 57-60 |
| Forearm (right) | 68-69 | 66-69 | 83-84 | 53-54 | 78-80 | 68 | 69-70 | 61-64 |
| Hand (left) | 70-75 | 70-71 | 85-89 | 55-60 | 81-86 | 69-71 | 71-75 | 65-74 |
| Hand (right) | 76-81 | 72-80 | 90-94 | 61-66 | 87-92 | 72-74 | 76-80 | 75-84 |
| Thigh (left) | 82-87 | 81-83 | 95-96 | 67-77 | 93-95 | 75-82 | 81-87 | 85-89 |
| Thigh (right) | 88-93 | 84-93 | 97-98 | 78-88 | 96-98 | 83-90 | 88-94 | 90-94 |
| Knee (left) | 94 | - | - | 89 | - | 91 | 95 | 95 |
| Knee (right) | 95 | 94 | - | 90 | - | 92 | 96 | 96 |
| Lower leg (left) | 96-97 | 95 | 99 | 91-94 | 99 | 93-95 | 97-98 | 97-98 |
| Lower leg (right) | 98-99 | 96-99 | 00 | 95-98 | 00 | 96-98 | 99-00 | 99-00 |
| Foot (left) | 00 | - | - | 99 | - | 99 | - | - |
| Foot (right) | - | 00 | - | 00 | - | 00 | - | - |

# Sidon, mid-13th century - gameplay-adjusted (before armour): Sword / Bow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-08 | 01-13 | 01-04 | 01-12 | 01-04 | 01-14 | 01-11 | 01-08 | 01-08 | 01-06 | 01-17 |
| Skull (right) | 09-16 | 14-17 | 05-17 | 13-24 | 05-08 | 15-28 | 12-22 | 09-16 | 09-16 | 07-12 | 18-34 |
| Face | 17-22 | 18-22 | 18-22 | 25-33 | 09-11 | 29-35 | 23 | 17-21 | 17-22 | 13-17 | 35-47 |
| Neck | 23-25 | 23-25 | 23-25 | 34-37 | 12 | 36-40 | 24-27 | 22-24 | 23-25 | 18-19 | 48-53 |
| Chest (left) | 26-32 | 26-36 | 26-28 | 38-44 | 13-19 | 41-44 | 28 | 25-30 | 26-34 | 20-25 | 54-56 |
| Chest (right) | 33-39 | 37-39 | 29-39 | 45-51 | 20-26 | 45-48 | 29 | 31-36 | 35-43 | 26-31 | 57-59 |
| Abdomen | 40-46 | 40-45 | 40-45 | 52-58 | 27-35 | 49-52 | 30 | 37-42 | 44-52 | 32-37 | 60 |
| Groin / pelvis | 47-49 | 46-48 | 46-48 | 59-61 | 36-40 | 53 | 31 | 43-45 | 53-56 | 38-39 | - |
| Upper back | 50-52 | 49-51 | 49-51 | 62-64 | 41-43 | 54-58 | 32-44 | 46-51 | 57-60 | 40-41 | 61 |
| Lower back | 53-55 | 52-54 | 52-54 | 65-67 | 44-46 | 59-61 | 45-54 | 52-57 | 61-64 | 42-43 | 62 |
| Shoulder (left) | 56-58 | 55-59 | 55 | 68-71 | 47-49 | 62-64 | 55-56 | 58-60 | 65-67 | 44-47 | 63-68 |
| Shoulder (right) | 59-61 | 60 | 56-60 | 72-75 | 50-52 | 65-67 | 57-58 | 61-63 | 68-70 | 48-51 | 69-74 |
| Upper arm (left) | 62-63 | 61-63 | 61 | 76-77 | 53-54 | 68-69 | 59 | 64-65 | 71-72 | 52-53 | 75-77 |
| Upper arm (right) | 64-65 | 64 | 62-64 | 78-79 | 55-56 | 70-71 | 60 | 66-67 | 73-74 | 54-55 | 78-80 |
| Forearm (left) | 66-68 | 65-69 | 65 | 80-82 | 57-59 | 72-75 | 61-62 | 68-70 | 75-76 | 56-61 | 81-84 |
| Forearm (right) | 69-71 | 70 | 66-70 | 83-85 | 60-62 | 76-79 | 63-64 | 71-73 | 77-78 | 62-67 | 85-88 |
| Hand (left) | 72-75 | 71-76 | 71-72 | 86-89 | 63-66 | 80-85 | 65-67 | 74-76 | 79-80 | 68-75 | 89-94 |
| Hand (right) | 76-79 | 77-78 | 73-79 | 90-93 | 67-70 | 86-91 | 68-70 | 77-79 | 81-83 | 76-83 | 95-00 |
| Thigh (left) | 80-84 | 79-86 | 80-81 | 94-95 | 71-78 | 92-94 | 71-77 | 80-84 | 84-87 | 84-87 | - |
| Thigh (right) | 85-89 | 87-88 | 82-89 | 96-97 | 79-86 | 95-97 | 78-84 | 85-89 | 88-91 | 88-91 | - |
| Knee (left) | 90 | 89-90 | - | - | 87 | - | 85-86 | 90 | 92 | 92 | - |
| Knee (right) | 91 | 91 | 90-91 | - | 88 | - | 87-88 | 91 | 93 | 93 | - |
| Lower leg (left) | 92-94 | 92-96 | 92 | 98 | 89-92 | 98 | 89-92 | 92-94 | 94-95 | 94-95 | - |
| Lower leg (right) | 95-97 | 97 | 93-97 | 99 | 93-96 | 99 | 93-96 | 95-97 | 96-97 | 96-97 | - |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 97-99 | 00 | 97-99 | 98-99 | 98-99 | 98-99 | - |
| Foot (right) | 00 | - | 99-00 | - | 00 | - | 00 | 00 | 00 | 00 | - |

# Sidon, mid-13th century - gameplay-adjusted (before armour): Axe

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-09 | 01-14 | 01-04 | 01-13 | 01-04 | 01-15 | 01-11 | 01-09 | 01-09 | 01-07 | 01-18 |
| Skull (right) | 10-18 | 15-18 | 05-18 | 14-26 | 05-08 | 16-30 | 12-22 | 10-18 | 10-18 | 08-14 | 19-36 |
| Face | 19-24 | 19-24 | 19-24 | 27-36 | 09-11 | 31-38 | 23 | 19-23 | 19-24 | 15-19 | 37-49 |
| Neck | 25-27 | 25-27 | 25-27 | 37-41 | 12 | 39-44 | 24-27 | 24-26 | 25-27 | 20-22 | 50-56 |
| Chest (left) | 28-33 | 28-36 | 28-29 | 42-46 | 13-17 | 45-47 | 28 | 27-31 | 28-34 | 23-26 | 57-58 |
| Chest (right) | 34-39 | 37-38 | 30-38 | 47-51 | 18-22 | 48-50 | 29 | 32-36 | 35-41 | 27-30 | 59-60 |
| Abdomen | 40-44 | 39-43 | 39-43 | 52-56 | 23-29 | 51-53 | 30 | 37-41 | 42-48 | 31-34 | 61 |
| Groin / pelvis | 45-46 | 44-45 | 44-45 | 57-58 | 30-33 | 54 | - | 42-43 | 49-51 | 35-36 | - |
| Upper back | 47-48 | 46-47 | 46-47 | 59-60 | 34-35 | 55-58 | 31-40 | 44-48 | 52-54 | 37-38 | 62 |
| Lower back | 49-50 | 48-49 | 48-49 | 61-62 | 36-37 | 59-60 | 41-48 | 49-53 | 55-57 | 39-40 | - |
| Shoulder (left) | 51-53 | 50-54 | 50 | 63-66 | 38-40 | 61-62 | 49-50 | 54-56 | 58-60 | 41-44 | 63-68 |
| Shoulder (right) | 54-56 | 55 | 51-55 | 67-70 | 41-43 | 63-64 | 51-52 | 57-58 | 61-63 | 45-47 | 69-74 |
| Upper arm (left) | 57-58 | 56-58 | 56 | 71-72 | 44-45 | 65-66 | 53 | 59-60 | 64-65 | 48-49 | 75-77 |
| Upper arm (right) | 59-60 | 59 | 57-59 | 73-74 | 46-47 | 67-68 | 54 | 61-62 | 66-67 | 50-51 | 78-80 |
| Forearm (left) | 61-63 | 60-64 | 60 | 75-77 | 48-50 | 69-72 | 55-56 | 63-64 | 68-69 | 52-57 | 81-84 |
| Forearm (right) | 64-66 | 65 | 61-65 | 78-80 | 51-53 | 73-76 | 57-58 | 65-66 | 70-72 | 58-63 | 85-88 |
| Hand (left) | 67-70 | 66-71 | 66-67 | 81-84 | 54-57 | 77-81 | 59-60 | 67-69 | 73-74 | 64-71 | 89-93 |
| Hand (right) | 71-74 | 72-73 | 68-73 | 85-88 | 58-60 | 82-86 | 61-62 | 70-72 | 75-77 | 72-79 | 94-98 |
| Thigh (left) | 75-80 | 74-83 | 74-76 | 89-90 | 61-69 | 87-89 | 63-71 | 73-79 | 78-83 | 80-84 | 99 |
| Thigh (right) | 81-86 | 84-86 | 77-86 | 91-92 | 70-78 | 90-92 | 72-80 | 80-86 | 84-89 | 85-89 | 00 |
| Knee (left) | 87 | 87-88 | 87 | 93 | 79-80 | 93 | 81-82 | 87 | 90 | 90 | - |
| Knee (right) | 88 | - | 88-89 | 94 | 81-82 | 94 | 83-84 | 88 | 91 | 91 | - |
| Lower leg (left) | 89-92 | 89-94 | 90-91 | 95-96 | 83-88 | 95-96 | 85-89 | 89-92 | 92-94 | 92-94 | - |
| Lower leg (right) | 93-96 | 95-96 | 92-97 | 97-98 | 89-94 | 97-98 | 90-94 | 93-96 | 95-97 | 95-97 | - |
| Foot (left) | 97-99 | 97-00 | 98 | 99 | 95-98 | 99 | 95-98 | 97-99 | 98-99 | 98-99 | - |
| Foot (right) | 00 | - | 99-00 | 00 | 99-00 | 00 | 99-00 | 00 | 00 | 00 | - |

# Sidon, mid-13th century - gameplay-adjusted (before armour): Spiked club

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-10 | 01-17 | 01-05 | 01-15 | 01-05 | 01-17 | 01-14 | 01-11 | 01-10 | 01-09 | 01-19 |
| Skull (right) | 11-20 | 18-22 | 06-22 | 16-29 | 06-10 | 18-34 | 15-28 | 12-22 | 11-20 | 10-18 | 20-38 |
| Face | 21-28 | 23-29 | 23-29 | 30-40 | 11-14 | 35-43 | 29 | 23-29 | 21-28 | 19-24 | 39-53 |
| Neck | 29-32 | 30-33 | 30-33 | 41-45 | 15-16 | 44-50 | 30-34 | 30-32 | 29-32 | 25-27 | 54-60 |
| Chest (left) | 33-39 | 34-44 | 34-36 | 46-51 | 17-23 | 51-53 | 35 | 33-38 | 33-40 | 28-32 | 61-63 |
| Chest (right) | 40-45 | 45-47 | 37-47 | 52-57 | 24-30 | 54-56 | 36 | 39-44 | 41-48 | 33-37 | 64-66 |
| Abdomen | 46-51 | 48-53 | 48-53 | 58-63 | 31-39 | 57-59 | 37 | 45-50 | 49-56 | 38-42 | 67 |
| Groin / pelvis | 52-54 | 54-56 | 54-56 | 64-65 | 40-44 | 60 | 38 | 51-52 | 57-60 | 43-44 | - |
| Upper back | 55-57 | 57-58 | 57-58 | 66-67 | 45-47 | 61-64 | 39-51 | 53-58 | 61-64 | 45-46 | 68 |
| Lower back | 58-60 | 59-60 | 59-60 | 68-69 | 48-50 | 65-67 | 52-61 | 59-64 | 65-68 | 47-48 | - |
| Shoulder (left) | 61-63 | 61-65 | 61 | 70-73 | 51-53 | 68-69 | 62-63 | 65-66 | 69-71 | 49-52 | 69-73 |
| Shoulder (right) | 64-66 | 66 | 62-66 | 74-77 | 54-56 | 70-71 | 64-65 | 67-68 | 72-73 | 53-56 | 74-78 |
| Upper arm (left) | 67-68 | 67-69 | 67 | 78-79 | 57-58 | 72-73 | 66 | 69-70 | 74-75 | 57-58 | 79-80 |
| Upper arm (right) | 69-70 | 70 | 68-70 | 80-81 | 59-60 | 74-75 | 67 | 71-72 | 76-77 | 59-60 | 81-82 |
| Forearm (left) | 71-73 | 71-75 | 71 | 82-84 | 61-63 | 76-79 | 68-69 | 73-74 | 78-79 | 61-66 | 83-86 |
| Forearm (right) | 74-76 | 76 | 72-76 | 85-87 | 64-66 | 80-83 | 70-71 | 75-76 | 80-81 | 67-72 | 87-90 |
| Hand (left) | 77-80 | 77-82 | 77-78 | 88-91 | 67-70 | 84-88 | 72-74 | 77-79 | 82-83 | 73-79 | 91-95 |
| Hand (right) | 81-84 | 83-84 | 79-84 | 92-95 | 71-74 | 89-93 | 75-77 | 80-82 | 84-86 | 80-86 | 96-00 |
| Thigh (left) | 85-88 | 85-90 | 85-86 | 96 | 75-80 | 94-95 | 78-83 | 83-86 | 87-89 | 87-89 | - |
| Thigh (right) | 89-92 | 91-92 | 87-92 | 97 | 81-86 | 96-97 | 84-89 | 87-90 | 90-92 | 90-92 | - |
| Knee (left) | 93 | 93 | - | - | 87 | - | 90 | 91 | 93 | 93 | - |
| Knee (right) | 94 | - | 93 | - | 88 | - | 91 | 92 | 94 | 94 | - |
| Lower leg (left) | 95-96 | 94-97 | 94 | 98 | 89-92 | 98 | 92-94 | 93-95 | 95-96 | 95-96 | - |
| Lower leg (right) | 97-98 | 98 | 95-98 | 99 | 93-96 | 99 | 95-97 | 96-97 | 97-98 | 97-98 | - |
| Foot (left) | 99 | 99-00 | 99 | 00 | 97-99 | 00 | 98-99 | 98-99 | 99 | 99 | - |
| Foot (right) | 00 | - | 00 | - | 00 | - | 00 | 00 | 00 | 00 | - |

# Sidon, mid-13th century - gameplay-adjusted (before armour): Spear

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-06 | 01-10 | 01-03 | 01-09 | 01-03 | 01-11 | 01-09 | 01-06 | 01-06 | 01-05 | 01-14 |
| Skull (right) | 07-12 | 11-13 | 04-13 | 10-18 | 04-06 | 12-22 | 10-17 | 07-12 | 07-12 | 06-10 | 15-28 |
| Face | 13-19 | 14-19 | 14-19 | 19-28 | 07-09 | 23-31 | 18 | 13-18 | 13-19 | 11-16 | 29-44 |
| Neck | 20-21 | 20-21 | 20-21 | 29-31 | 10 | 32-35 | 19-21 | 19-20 | 20-21 | 17-18 | 45-49 |
| Chest (left) | 22-30 | 22-36 | 22-25 | 32-40 | 11-19 | 36-40 | 22-23 | 21-28 | 22-32 | 19-26 | 50-54 |
| Chest (right) | 31-39 | 37-40 | 26-40 | 41-49 | 20-28 | 41-45 | 24-25 | 29-36 | 33-43 | 27-34 | 55-59 |
| Abdomen | 40-48 | 41-48 | 41-49 | 50-58 | 29-39 | 46-50 | 26-27 | 37-44 | 44-54 | 35-42 | 60 |
| Groin / pelvis | 49-52 | 49-52 | 50-53 | 59-62 | 40-46 | 51-52 | 28 | 45-47 | 55-59 | 43-45 | 61 |
| Upper back | 53-56 | 53-56 | 54-57 | 63-66 | 47-50 | 53-59 | 29-46 | 48-55 | 60-64 | 46-48 | 62-63 |
| Lower back | 57-60 | 57-60 | 58-61 | 67-70 | 51-54 | 60-64 | 47-60 | 56-63 | 65-69 | 49-51 | 64 |
| Shoulder (left) | 61-63 | 61-64 | 62 | 71-74 | 55-56 | 65-66 | 61-62 | 64-65 | 70-71 | 52-54 | 65-70 |
| Shoulder (right) | 64-66 | 65 | 63-66 | 75-78 | 57-58 | 67-68 | 63-64 | 66-67 | 72-73 | 55-57 | 71-76 |
| Upper arm (left) | 67-68 | 66-68 | 67 | 79-80 | 59-60 | 69-70 | 65 | 68 | 74 | 58-59 | 77-79 |
| Upper arm (right) | 69-70 | 69 | 68-70 | 81-82 | 61-62 | 71-72 | 66 | 69 | 75 | 60-61 | 80-82 |
| Forearm (left) | 71-73 | 70-73 | 71 | 83-85 | 63-64 | 73-76 | 67-68 | 70-71 | 76-77 | 62-66 | 83-86 |
| Forearm (right) | 74-76 | 74 | 72-75 | 86-88 | 65-66 | 77-80 | 69-70 | 72-73 | 78-79 | 67-71 | 87-90 |
| Hand (left) | 77-79 | 75-80 | 76-77 | 89-92 | 67-69 | 81-85 | 71-72 | 74-76 | 80-81 | 72-78 | 91-95 |
| Hand (right) | 80-82 | 81-82 | 78-83 | 93-96 | 70-72 | 86-90 | 73-74 | 77-79 | 82-84 | 79-85 | 96-00 |
| Thigh (left) | 83-86 | 83-89 | 84-85 | 97 | 73-79 | 91-92 | 75-80 | 80-84 | 85-88 | 86-89 | - |
| Thigh (right) | 87-90 | 90-91 | 86-92 | 98 | 80-86 | 93-94 | 81-86 | 85-89 | 89-92 | 90-92 | - |
| Knee (left) | 91 | 92 | - | - | 87 | 95 | 87 | 90 | 93 | 93 | - |
| Knee (right) | 92 | - | 93 | - | 88 | 96 | 88 | 91 | 94 | 94 | - |
| Lower leg (left) | 93-95 | 93-96 | 94 | 99 | 89-92 | 97 | 89-92 | 92-94 | 95-96 | 95-96 | - |
| Lower leg (right) | 96-97 | 97 | 95-98 | 00 | 93-96 | 98 | 93-96 | 95-97 | 97-98 | 97-98 | - |
| Foot (left) | 98-99 | 98-00 | 99 | - | 97-99 | 99 | 97-99 | 98-99 | 99 | 99 | - |
| Foot (right) | 00 | - | 00 | - | 00 | 00 | 00 | 00 | 00 | 00 | - |

# Sidon, mid-13th century - gameplay-adjusted (before armour): Dagger

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-07 | 01-12 | 01-03 | 01-10 | 01-04 | 01-12 | 01-10 | 01-07 | 01-07 | 01-06 | 01-14 |
| Skull (right) | 08-14 | 13-15 | 04-15 | 11-20 | 05-08 | 13-24 | 11-20 | 08-14 | 08-14 | 07-12 | 15-28 |
| Face | 15-22 | 16-22 | 16-22 | 21-31 | 09-12 | 25-34 | 21 | 15-21 | 15-22 | 13-19 | 29-44 |
| Neck | 23-28 | 23-27 | 23-27 | 32-38 | 13-15 | 35-43 | 22-29 | 22-26 | 23-27 | 20-23 | 45-55 |
| Chest (left) | 29-37 | 28-42 | 28-31 | 39-46 | 16-24 | 44-48 | 30-31 | 27-34 | 28-38 | 24-30 | 56-58 |
| Chest (right) | 38-46 | 43-46 | 32-46 | 47-54 | 25-33 | 49-52 | 32-33 | 35-42 | 39-49 | 31-37 | 59-61 |
| Abdomen | 47-55 | 47-54 | 47-54 | 55-62 | 34-45 | 53-56 | 34-35 | 43-50 | 50-60 | 38-44 | 62 |
| Groin / pelvis | 56-59 | 55-58 | 55-58 | 63-65 | 46-52 | 57-58 | 36 | 51-53 | 61-65 | 45-47 | - |
| Upper back | 60-63 | 59-61 | 59-62 | 66-68 | 53-56 | 59-64 | 37-54 | 54-61 | 66-70 | 48-50 | 63-64 |
| Lower back | 64-67 | 62-64 | 63-65 | 69-71 | 57-60 | 65-68 | 55-68 | 62-69 | 71-75 | 51-53 | - |
| Shoulder (left) | 68-70 | 65-69 | 66 | 72-75 | 61-63 | 69-71 | 69-70 | 70-72 | 76-78 | 54-57 | 65-70 |
| Shoulder (right) | 71-73 | 70 | 67-71 | 76-79 | 64-66 | 72-74 | 71-72 | 73-75 | 79-80 | 58-61 | 71-76 |
| Upper arm (left) | 74-75 | 71-73 | 72 | 80-81 | 67-68 | 75-76 | 73 | 76-77 | 81-82 | 62-63 | 77-79 |
| Upper arm (right) | 76-77 | 74 | 73-75 | 82-83 | 69-70 | 77-78 | 74 | 78-79 | 83-84 | 64-65 | 80-82 |
| Forearm (left) | 78-80 | 75-79 | 76 | 84-86 | 71-73 | 79-82 | 75-76 | 80-82 | 85-86 | 66-71 | 83-86 |
| Forearm (right) | 81-83 | 80 | 77-81 | 87-89 | 74-76 | 83-86 | 77-78 | 83-85 | 87-88 | 72-77 | 87-90 |
| Hand (left) | 84-87 | 81-87 | 82-83 | 90-93 | 77-80 | 87-91 | 79-81 | 86-88 | 89-90 | 78-85 | 91-95 |
| Hand (right) | 88-91 | 88-89 | 84-90 | 94-97 | 81-84 | 92-96 | 82-84 | 89-91 | 91-93 | 86-93 | 96-00 |
| Thigh (left) | 92-93 | 90-93 | 91 | 98 | 85-88 | 97 | 85-88 | 92-94 | 94-95 | 94-95 | - |
| Thigh (right) | 94-95 | 94 | 92-95 | 99 | 89-92 | 98 | 89-92 | 95-97 | 96-97 | 96-97 | - |
| Knee (left) | 96 | 95 | - | - | 93 | - | 93 | - | - | - | - |
| Knee (right) | 97 | - | 96 | - | 94 | - | 94 | - | - | - | - |
| Lower leg (left) | 98 | 96-97 | 97 | 00 | 95-96 | 99 | 95-96 | 98 | 98 | 98 | - |
| Lower leg (right) | 99 | 98 | 98-99 | - | 97-98 | 00 | 97-98 | 99 | 99 | 99 | - |
| Foot (left) | 00 | 99-00 | - | - | 99 | - | 99 | 00 | 00 | 00 | - |
| Foot (right) | - | - | 00 | - | 00 | - | 00 | - | - | - | - |

# Sidon, mid-13th century - gameplay-adjusted (before armour): Crossbow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-09 | 01-14 | 01-04 | 01-13 | 01-04 | 01-15 | 01-12 | 01-09 | 01-09 | 01-07 | 01-18 |
| Skull (right) | 10-18 | 15-18 | 05-18 | 14-26 | 05-08 | 16-30 | 13-24 | 10-18 | 10-18 | 08-14 | 19-36 |
| Face | 19-24 | 19-24 | 19-24 | 27-36 | 09-11 | 31-38 | 25 | 19-24 | 19-24 | 15-19 | 37-50 |
| Neck | 25-27 | 25-27 | 25-27 | 37-40 | 12 | 39-43 | 26-29 | 25-27 | 25-27 | 20-21 | 51-56 |
| Chest (left) | 28-34 | 28-38 | 28-30 | 41-46 | 13-19 | 44-46 | 30 | 28-33 | 28-35 | 22-27 | 57-59 |
| Chest (right) | 35-40 | 39-41 | 31-41 | 47-52 | 20-26 | 47-49 | 31 | 34-39 | 36-43 | 28-32 | 60-62 |
| Abdomen | 41-46 | 42-47 | 42-47 | 53-58 | 27-35 | 50-52 | 32 | 40-45 | 44-51 | 33-37 | 63 |
| Groin / pelvis | 47-49 | 48-50 | 48-50 | 59-61 | 36-40 | 53-54 | 33 | 46-48 | 52-55 | 38-39 | - |
| Upper back | 50-52 | 51-53 | 51-53 | 62-64 | 41-43 | 55-58 | 34-46 | 49-54 | 56-59 | 40-41 | 64 |
| Lower back | 53-55 | 54-56 | 54-56 | 65-67 | 44-46 | 59-61 | 47-56 | 55-60 | 60-63 | 42-43 | - |
| Shoulder (left) | 56-58 | 57-61 | 57 | 68-71 | 47-49 | 62-64 | 57-58 | 61-63 | 64-66 | 44-47 | 65-70 |
| Shoulder (right) | 59-61 | 62 | 58-62 | 72-75 | 50-52 | 65-66 | 59-60 | 64-65 | 67-69 | 48-51 | 71-76 |
| Upper arm (left) | 62-63 | 63-65 | 63 | 76-77 | 53-54 | 67-68 | 61 | 66-67 | 70-71 | 52-53 | 77-79 |
| Upper arm (right) | 64-65 | 66 | 64-66 | 78-79 | 55-56 | 69-70 | 62 | 68-69 | 72-73 | 54-55 | 80-82 |
| Forearm (left) | 66-68 | 67-71 | 67 | 80-82 | 57-59 | 71-74 | 63-64 | 70-71 | 74-75 | 56-61 | 83-86 |
| Forearm (right) | 69-71 | 72 | 68-72 | 83-85 | 60-62 | 75-78 | 65-66 | 72-73 | 76-78 | 62-67 | 87-90 |
| Hand (left) | 72-75 | 73-78 | 73-74 | 86-89 | 63-66 | 79-83 | 67-69 | 74-76 | 79-80 | 68-75 | 91-95 |
| Hand (right) | 76-79 | 79-80 | 75-80 | 90-93 | 67-70 | 84-88 | 70-72 | 77-79 | 81-83 | 76-83 | 96-00 |
| Thigh (left) | 80-84 | 81-88 | 81-82 | 94-95 | 71-78 | 89-90 | 73-79 | 80-84 | 84-87 | 84-87 | - |
| Thigh (right) | 85-89 | 89-90 | 83-90 | 96-97 | 79-86 | 91-92 | 80-86 | 85-89 | 88-91 | 88-91 | - |
| Knee (left) | 90 | 91 | - | - | 87 | 93 | 87 | 90 | 92 | 92 | - |
| Knee (right) | 91 | - | 91-92 | - | 88 | 94 | 88 | 91 | 93 | 93 | - |
| Lower leg (left) | 92-94 | 92-96 | 93 | 98 | 89-92 | 95-96 | 89-92 | 92-94 | 94-95 | 94-95 | - |
| Lower leg (right) | 95-97 | 97 | 94-98 | 99 | 93-96 | 97-98 | 93-96 | 95-97 | 96-97 | 96-97 | - |
| Foot (left) | 98-99 | 98-00 | 99 | 00 | 97-99 | 99 | 97-99 | 98-99 | 98-99 | 98-99 | - |
| Foot (right) | 00 | - | 00 | - | 00 | 00 | 00 | 00 | 00 | 00 | - |

# Visby 1361 - bone evidence: Sword / Bow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-13 | 01-18 | 01-06 | 01-23 | 01-05 | 01-24 | 01-14 | 01-13 | 01-14 | 01-11 |
| Skull (right) | 14-23 | 19-22 | 07-24 | 24-41 | 06-09 | 25-42 | 15-25 | 14-23 | 15-25 | 12-20 |
| Face | 24-27 | 23-25 | 25-28 | 42-48 | 10-11 | 43-47 | - | 24-26 | 26-29 | 21-23 |
| Neck | 28 | 26 | 29 | 49-50 | - | 48-49 | 26 | 27 | 30 | 24 |
| Chest (left) | 29 | 27-28 | 30 | 51 | 12 | 50 | - | 28 | 31 | 25 |
| Chest (right) | 30 | - | 31-32 | 52 | 13 | 51 | - | 29 | 32 | 26 |
| Upper back | 31 | 29 | 33 | 53 | 14 | 52-53 | 27-30 | 30-31 | 33 | 27 |
| Shoulder (left) | 32-33 | 30-32 | 34 | 54-57 | 15-16 | 54-55 | 31 | 32-33 | 34-35 | 28-30 |
| Shoulder (right) | 34 | - | 35-36 | 58-59 | 17 | 56 | 32 | 34 | 36 | 31 |
| Upper arm (left) | 35-36 | 33-35 | 37 | 60-62 | 18-19 | 57-58 | 33 | 35-36 | 37-38 | 32-34 |
| Upper arm (right) | 37-38 | 36 | 38-41 | 63-65 | 20-21 | 59-60 | 34 | 37-38 | 39-40 | 35-37 |
| Forearm (left) | 39-41 | 37-40 | 42 | 66-69 | 22-23 | 61-64 | 35-36 | 39-40 | 41-42 | 38-43 |
| Forearm (right) | 42-44 | 41 | 43-47 | 70-73 | 24-25 | 65-68 | 37-38 | 41-42 | 43-45 | 44-49 |
| Hand (left) | 45 | 42 | - | 74 | 26 | 69 | - | 43 | 46 | 50-51 |
| Hand (right) | 46-47 | 43 | 48-51 | 75-77 | 27-28 | 70-72 | 39 | 44-45 | 47-48 | 52-55 |
| Thigh (left) | 48-53 | 44-52 | 52-54 | 78-80 | 29-36 | 73-75 | 40-46 | 46-51 | 49-54 | 56-60 |
| Thigh (right) | 54-57 | 53-54 | 55-61 | 81-82 | 37-41 | 76-77 | 47-51 | 52-55 | 55-58 | 61-63 |
| Knee (left) | 58-62 | 55-61 | 62-63 | 83-84 | 42-48 | 78-80 | 52-57 | 56-60 | 59-63 | 64-67 |
| Knee (right) | 63-65 | 62 | 64-68 | 85 | 49-52 | 81-82 | 58-60 | 61-63 | 64-66 | 68-69 |
| Lower leg (left) | 66-87 | 63-93 | 69-79 | 86-95 | 53-82 | 83-94 | 61-85 | 64-86 | 67-87 | 70-88 |
| Lower leg (right) | 88-97 | 94-97 | 80-97 | 96-99 | 83-96 | 95-99 | 86-97 | 87-97 | 88-97 | 89-97 |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 97-99 | 00 | 98-99 | 98-99 | 98-99 | 98-99 |
| Foot (right) | 00 | - | 99-00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Visby 1361 - bone evidence: Axe

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-12 | 01-17 | 01-06 | 01-23 | 01-05 | 01-23 | 01-13 | 01-12 | 01-13 | 01-11 |
| Skull (right) | 13-21 | 18-21 | 07-23 | 24-41 | 06-09 | 24-41 | 14-23 | 13-21 | 14-23 | 12-19 |
| Face | 22-25 | 22-24 | 24-27 | 42-48 | 10-11 | 42-46 | - | 22-24 | 24-27 | 20-22 |
| Neck | 26 | 25 | 28 | 49-50 | - | 47-48 | 24 | 25 | 28 | 23 |
| Chest (left) | 27 | 26 | - | 51 | 12 | - | - | 26 | 29 | 24 |
| Chest (right) | 28 | - | 29 | 52 | 13 | - | - | 27 | 30 | 25 |
| Upper back | 29 | 27 | 30 | 53 | 14 | 49 | 25-26 | 28 | 31 | 26 |
| Shoulder (left) | 30-31 | 28-29 | 31 | 54-56 | 15 | 50-51 | 27 | 29 | 32-33 | 27-28 |
| Shoulder (right) | 32 | - | 32 | 57-58 | 16 | 52 | - | 30 | 34 | 29 |
| Upper arm (left) | 33-34 | 30-31 | 33 | 59-60 | 17 | 53-54 | 28 | 31 | 35-36 | 30-31 |
| Upper arm (right) | 35-36 | 32 | 34-36 | 61-62 | 18 | 55-56 | 29 | 32 | 37-38 | 32-33 |
| Forearm (left) | 37-38 | 33-36 | 37 | 63-66 | 19-20 | 57-60 | 30 | 33-34 | 39-40 | 34-38 |
| Forearm (right) | 39-40 | 37 | 38-42 | 67-70 | 21-22 | 61-64 | 31 | 35-36 | 41-42 | 39-43 |
| Hand (left) | 41 | 38 | - | 71 | 23 | 65 | - | 37 | - | 44-45 |
| Hand (right) | 42-43 | 39 | 43-45 | 72-73 | 24 | 66-67 | 32 | 38 | 43-44 | 46-48 |
| Thigh (left) | 44-50 | 40-48 | 46-48 | 74-76 | 25-33 | 68-71 | 33-40 | 39-45 | 45-50 | 49-54 |
| Thigh (right) | 51-54 | 49-50 | 49-56 | 77-78 | 34-39 | 72-73 | 41-45 | 46-50 | 51-54 | 55-58 |
| Knee (left) | 55-59 | 51-58 | 57-59 | 79-81 | 40-46 | 74-76 | 46-51 | 51-56 | 55-59 | 59-63 |
| Knee (right) | 60-62 | 59 | 60-65 | 82-83 | 47-50 | 77-78 | 52-55 | 57-59 | 60-62 | 64-66 |
| Lower leg (left) | 63-86 | 60-93 | 66-77 | 84-94 | 51-82 | 79-92 | 56-83 | 60-85 | 63-86 | 67-87 |
| Lower leg (right) | 87-97 | 94-97 | 78-97 | 95-99 | 83-96 | 93-98 | 84-96 | 86-97 | 87-97 | 88-97 |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 97-99 | 99 | 97-99 | 98-99 | 98-99 | 98-99 |
| Foot (right) | 00 | - | 99-00 | - | 00 | 00 | 00 | 00 | 00 | 00 |

# Visby 1361 - bone evidence: Spiked club

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-18 | 01-26 | 01-09 | 01-28 | 01-08 | 01-29 | 01-20 | 01-18 | 01-19 | 01-16 |
| Skull (right) | 19-32 | 27-32 | 10-33 | 29-50 | 09-14 | 30-51 | 21-35 | 19-32 | 20-34 | 17-28 |
| Face | 33-38 | 33-37 | 34-38 | 51-59 | 15-17 | 52-57 | 36 | 33-37 | 35-40 | 29-33 |
| Neck | 39 | 38 | 39 | 60-61 | 18 | 58-59 | 37 | 38 | 41 | 34 |
| Chest (left) | 40 | 39-40 | 40 | 62 | 19 | 60 | - | 39 | 42 | 35 |
| Chest (right) | 41 | - | 41-42 | 63 | 20 | - | - | 40 | 43 | 36 |
| Upper back | 42 | 41 | 43 | 64 | 21 | 61 | 38-41 | 41-42 | 44 | 37 |
| Shoulder (left) | 43-44 | 42-44 | 44 | 65-67 | 22-23 | 62-63 | 42 | 43-44 | 45-46 | 38-40 |
| Shoulder (right) | 45 | - | 45-46 | 68-69 | 24 | 64 | - | 45 | 47 | 41 |
| Upper arm (left) | 46-47 | 45-47 | 47 | 70-71 | 25-26 | 65-66 | 43 | 46-47 | 48-49 | 42-44 |
| Upper arm (right) | 48-49 | 48 | 48-51 | 72-73 | 27-28 | 67-68 | 44 | 48-49 | 50-51 | 45-46 |
| Forearm (left) | 50-52 | 49-52 | 52 | 74-77 | 29-31 | 69-72 | 45-46 | 50-51 | 52-53 | 47-52 |
| Forearm (right) | 53-55 | 53 | 53-57 | 78-80 | 32-34 | 73-76 | 47-48 | 52-53 | 54-56 | 53-58 |
| Hand (left) | 56 | 54 | - | 81 | 35 | 77 | - | 54 | 57 | 59-60 |
| Hand (right) | 57-58 | 55 | 58-61 | 82-83 | 36-37 | 78-80 | 49 | 55-56 | 58-59 | 61-64 |
| Thigh (left) | 59-63 | 56-62 | 62-63 | 84-85 | 38-44 | 81-82 | 50-55 | 57-61 | 60-64 | 65-68 |
| Thigh (right) | 64-66 | 63 | 64-69 | 86 | 45-49 | 83-84 | 56-59 | 62-64 | 65-67 | 69-71 |
| Knee (left) | 67-70 | 64-69 | 70-71 | 87-88 | 50-55 | 85-86 | 60-64 | 65-68 | 68-71 | 72-74 |
| Knee (right) | 71-72 | 70 | 72-75 | 89 | 56-59 | 87 | 65-67 | 69-70 | 72-73 | 75-76 |
| Lower leg (left) | 73-89 | 71-95 | 76-84 | 90-96 | 60-85 | 88-95 | 68-88 | 71-89 | 74-90 | 77-91 |
| Lower leg (right) | 90-97 | 96-98 | 85-98 | 97-99 | 86-97 | 96-99 | 89-97 | 90-97 | 91-98 | 92-98 |
| Foot (left) | 98-99 | 99-00 | 99 | 00 | 98-99 | 00 | 98-99 | 98-99 | 99 | 99 |
| Foot (right) | 00 | - | 00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Visby 1361 - bone evidence: Spear

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-11 | 01-16 | 01-06 | 01-21 | 01-05 | 01-21 | 01-12 | 01-11 | 01-12 | 01-10 |
| Skull (right) | 12-20 | 17-20 | 07-22 | 22-37 | 06-09 | 22-37 | 13-22 | 12-20 | 13-21 | 11-18 |
| Face | 21-25 | 21-24 | 23-27 | 38-47 | 10-11 | 38-44 | 23 | 21-24 | 22-27 | 19-23 |
| Neck | 26 | 25 | 28 | 48 | - | 45-46 | 24 | 25 | 28 | 24 |
| Chest (left) | 27-28 | 26-27 | 29 | 49-50 | 12 | 47 | - | 26 | 29-30 | 25 |
| Chest (right) | 29-30 | 28 | 30-32 | 51-52 | 13 | 48 | - | 27 | 31-32 | 26 |
| Upper back | 31 | 29 | 33 | 53-54 | 14 | 49-51 | 25-30 | 28-30 | 33-34 | 27 |
| Shoulder (left) | 32-33 | 30-32 | 34 | 55-57 | 15-16 | 52-53 | 31 | 31-32 | 35-36 | 28-30 |
| Shoulder (right) | 34 | - | 35-36 | 58-59 | 17 | 54 | - | 33 | 37 | 31 |
| Upper arm (left) | 35-36 | 33-35 | 37 | 60-62 | 18-19 | 55-56 | 32 | 34-35 | 38-39 | 32-34 |
| Upper arm (right) | 37-38 | 36 | 38-41 | 63-65 | 20-21 | 57-58 | 33 | 36-37 | 40-41 | 35-37 |
| Forearm (left) | 39-41 | 37-40 | 42 | 66-69 | 22-24 | 59-62 | 34-35 | 38-40 | 42-43 | 38-43 |
| Forearm (right) | 42-44 | 41 | 43-47 | 70-73 | 25-26 | 63-66 | 36-37 | 41-42 | 44-46 | 44-49 |
| Hand (left) | 45 | 42 | - | 74 | 27 | 67 | - | 43 | 47 | 50-51 |
| Hand (right) | 46-47 | 43 | 48-51 | 75-77 | 28-29 | 68-70 | 38 | 44-45 | 48-49 | 52-55 |
| Thigh (left) | 48-53 | 44-52 | 52-54 | 78-80 | 30-37 | 71-73 | 39-45 | 46-51 | 50-55 | 56-60 |
| Thigh (right) | 54-57 | 53-54 | 55-61 | 81-82 | 38-42 | 74-75 | 46-50 | 52-55 | 56-59 | 61-63 |
| Knee (left) | 58-62 | 55-61 | 62-63 | 83-84 | 43-49 | 76-78 | 51-56 | 56-60 | 60-64 | 64-67 |
| Knee (right) | 63-65 | 62 | 64-68 | 85 | 50-53 | 79-80 | 57-59 | 61-63 | 65-67 | 68-69 |
| Lower leg (left) | 66-87 | 63-93 | 69-79 | 86-95 | 54-83 | 81-92 | 60-85 | 64-86 | 68-88 | 70-88 |
| Lower leg (right) | 88-97 | 94-97 | 80-97 | 96-99 | 84-96 | 93-98 | 86-97 | 87-97 | 89-97 | 89-97 |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 97-99 | 99 | 98-99 | 98-99 | 98-99 | 98-99 |
| Foot (right) | 00 | - | 99-00 | - | 00 | 00 | 00 | 00 | 00 | 00 |

# Visby 1361 - bone evidence: Dagger

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-16 | 01-24 | 01-08 | 01-24 | 01-08 | 01-26 | 01-19 | 01-17 | 01-17 | 01-13 |
| Skull (right) | 17-29 | 25-29 | 09-30 | 25-42 | 09-14 | 27-46 | 20-34 | 18-30 | 18-30 | 14-23 |
| Face | 30-37 | 30-35 | 31-37 | 43-53 | 15-18 | 47-54 | 35 | 31-36 | 31-38 | 24-29 |
| Neck | 38-39 | 36-37 | 38-39 | 54-57 | 19 | 55-58 | 36-38 | 37-38 | 39-41 | 30-31 |
| Chest (left) | 40-41 | 38-40 | 40 | 58-59 | 20-21 | 59 | - | 39-40 | 42-43 | 32-33 |
| Chest (right) | 42-43 | 41 | 41-43 | 60-61 | 22-23 | 60 | - | 41-42 | 44-45 | 34-35 |
| Upper back | 44-45 | 42 | 44-45 | 62-63 | 24-25 | 61-62 | 39-45 | 43-46 | 46-47 | 36-37 |
| Shoulder (left) | 46-48 | 43-46 | 46 | 64-67 | 26-28 | 63-64 | 46-47 | 47-48 | 48-50 | 38-40 |
| Shoulder (right) | 49 | 47 | 47-48 | 68-69 | 29 | 65 | 48 | 49 | 51 | 41-42 |
| Upper arm (left) | 50-52 | 48-51 | 49 | 70-72 | 30-32 | 66-67 | 49-50 | 50-51 | 52-54 | 43-45 |
| Upper arm (right) | 53-55 | 52 | 50-54 | 73-75 | 33-35 | 68-69 | 51-52 | 52-53 | 55-57 | 46-48 |
| Forearm (left) | 56-59 | 53-58 | 55-56 | 76-80 | 36-39 | 70-74 | 53-54 | 54-57 | 58-60 | 49-56 |
| Forearm (right) | 60-63 | 59-60 | 57-63 | 81-85 | 40-43 | 75-79 | 55-56 | 58-61 | 61-64 | 57-64 |
| Hand (left) | 64 | 61-62 | 64 | 86 | 44 | 80-81 | 57 | 62 | 65 | 65-67 |
| Hand (right) | 65-67 | 63 | 65-69 | 87-89 | 45-47 | 82-84 | 58-59 | 63-64 | 66-68 | 68-72 |
| Thigh (left) | 68-71 | 64-68 | 70-71 | 90 | 48-53 | 85-86 | 60-64 | 65-68 | 69-72 | 73-75 |
| Thigh (right) | 72-73 | 69 | 72-75 | 91 | 54-57 | 87 | 65-67 | 69-71 | 73-74 | 76-77 |
| Knee (left) | 74-76 | 70-74 | 76-77 | 92 | 58-62 | 88-89 | 68-71 | 72-74 | 75-77 | 78-80 |
| Knee (right) | 77-78 | 75 | 78-80 | 93 | 63-65 | 90 | 72-73 | 75-76 | 78-79 | 81-82 |
| Lower leg (left) | 79-92 | 76-95 | 81-87 | 94-98 | 66-87 | 91-96 | 74-90 | 77-91 | 80-92 | 83-93 |
| Lower leg (right) | 93-98 | 96-98 | 88-98 | 99-00 | 88-97 | 97-99 | 91-98 | 92-98 | 93-98 | 94-98 |
| Foot (left) | 99 | 99-00 | 99 | - | 98-99 | 00 | 99 | 99 | 99 | 99 |
| Foot (right) | 00 | - | 00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Visby 1361 - bone evidence: Crossbow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-14 | 01-20 | 01-07 | 01-25 | 01-06 | 01-25 | 01-15 | 01-14 | 01-15 | 01-12 |
| Skull (right) | 15-25 | 21-24 | 08-27 | 26-44 | 07-11 | 26-45 | 16-27 | 15-25 | 16-27 | 13-22 |
| Face | 26-29 | 25-28 | 28-31 | 45-52 | 12-13 | 46-50 | - | 26-29 | 28-32 | 23-26 |
| Neck | 30 | 29 | 32 | 53-54 | - | 51-52 | 28 | 30 | 33 | 27 |
| Chest (left) | 31 | 30 | 33 | 55 | 14 | - | - | 31 | 34 | 28 |
| Chest (right) | 32 | - | 34-35 | 56 | 15 | - | - | 32 | 35 | 29 |
| Upper back | 33 | 31 | 36 | 57 | 16 | 53-54 | 29-31 | 33-34 | 36 | 30 |
| Shoulder (left) | 34-35 | 32-34 | 37 | 58-60 | 17-18 | 55-56 | 32 | 35-36 | 37-38 | 31-33 |
| Shoulder (right) | 36 | - | 38-39 | 61-62 | 19 | 57 | 33 | 37 | 39 | 34 |
| Upper arm (left) | 37-38 | 35-37 | 40 | 63-65 | 20-21 | 58-59 | 34 | 38-39 | 40-41 | 35-37 |
| Upper arm (right) | 39-40 | 38 | 41-43 | 66-67 | 22-23 | 60-61 | 35 | 40-41 | 42-43 | 38-39 |
| Forearm (left) | 41-43 | 39-42 | 44 | 68-71 | 24-25 | 62-65 | 36-37 | 42-43 | 44-45 | 40-45 |
| Forearm (right) | 44-46 | 43 | 45-49 | 72-75 | 26-27 | 66-69 | 38-39 | 44-45 | 46-48 | 46-51 |
| Hand (left) | 47 | 44 | 50 | 76 | 28 | 70 | 40 | 46 | 49 | 52-53 |
| Hand (right) | 48-49 | 45 | 51-53 | 77-78 | 29-30 | 71-73 | 41 | 47 | 50-51 | 54-57 |
| Thigh (left) | 50-55 | 46-53 | 54-56 | 79-81 | 31-38 | 74-76 | 42-48 | 48-53 | 52-56 | 58-62 |
| Thigh (right) | 56-59 | 54-55 | 57-63 | 82-83 | 39-43 | 77-78 | 49-52 | 54-57 | 57-60 | 63-65 |
| Knee (left) | 60-64 | 56-62 | 64-65 | 84-85 | 44-50 | 79-81 | 53-58 | 58-62 | 61-65 | 66-69 |
| Knee (right) | 65-67 | 63 | 66-70 | 86 | 51-54 | 82-83 | 59-61 | 63-65 | 66-68 | 70-71 |
| Lower leg (left) | 68-88 | 64-93 | 71-80 | 87-95 | 55-83 | 84-94 | 62-86 | 66-87 | 69-88 | 72-89 |
| Lower leg (right) | 89-97 | 94-97 | 81-97 | 96-99 | 84-96 | 95-99 | 87-97 | 88-97 | 89-97 | 90-97 |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 97-99 | 00 | 98-99 | 98-99 | 98-99 | 98-99 |
| Foot (right) | 00 | - | 99-00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Visby 1361 - gameplay-adjusted: Sword / Bow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-08 | 01-12 | 01-04 | 01-13 | 01-03 | 01-15 | 01-10 | 01-08 | 01-08 | 01-07 |
| Skull (right) | 09-15 | 13-15 | 05-16 | 14-24 | 04-06 | 16-28 | 11-19 | 09-15 | 09-15 | 08-13 |
| Face | 16-19 | 16-18 | 17-20 | 25-31 | 07-08 | 29-33 | - | 16-18 | 16-19 | 14-16 |
| Neck | 20-21 | 19-20 | 21-22 | 32-34 | 09 | 34-37 | 20-21 | 19-20 | 20-21 | 17-18 |
| Chest (left) | 22-28 | 21-31 | 23-25 | 35-41 | 10-15 | 38-41 | 22 | 21-26 | 22-30 | 19-24 |
| Chest (right) | 29-34 | 32-33 | 26-35 | 42-47 | 16-20 | 42-44 | 23 | 27-31 | 31-37 | 25-29 |
| Abdomen | 35-41 | 34-39 | 36-42 | 48-54 | 21-28 | 45-48 | 24 | 32-37 | 38-46 | 30-35 |
| Groin / pelvis | 42-43 | 40-41 | 43-44 | 55-56 | 29-31 | 49 | - | 38-39 | 47-48 | 36-37 |
| Upper back | 44-46 | 42-43 | 45-47 | 57-59 | 32-34 | 50-54 | 25-36 | 40-45 | 49-52 | 38-40 |
| Lower back | 47-48 | 44-45 | 48-49 | 60-61 | 35-36 | 55-56 | 37-42 | 46-49 | 53-54 | 41-42 |
| Shoulder (left) | 49-51 | 46-50 | 50 | 62-66 | 37-39 | 57-59 | 43-44 | 50-52 | 55-57 | 43-46 |
| Shoulder (right) | 52-54 | 51 | 51-55 | 67-71 | 40-42 | 60-62 | 45-46 | 53-54 | 58-60 | 47-50 |
| Upper arm (left) | 55-56 | 52-54 | 56 | 72-74 | 43-44 | 63-64 | 47 | 55-56 | 61-62 | 51-53 |
| Upper arm (right) | 57-58 | 55 | 57-59 | 75-76 | 45-46 | 65-66 | 48 | 57-58 | 63-64 | 54-55 |
| Forearm (left) | 59-61 | 56-60 | 60 | 77-80 | 47-48 | 67-71 | 49-50 | 59-60 | 65-66 | 56-61 |
| Forearm (right) | 62-64 | 61 | 61-65 | 81-84 | 49-50 | 72-76 | 51-52 | 61-62 | 67-69 | 62-67 |
| Hand (left) | 65 | 62 | 66 | 85 | 51 | 77-78 | 53 | 63 | - | 68-69 |
| Hand (right) | 66 | - | 67-68 | 86 | 52 | 79-80 | 54 | 64 | 70 | 70-71 |
| Thigh (left) | 67-73 | 63-73 | 69-71 | 87-89 | 53-62 | 81-84 | 55-63 | 65-72 | 71-76 | 72-77 |
| Thigh (right) | 74-78 | 74-75 | 72-80 | 90-91 | 63-69 | 85-87 | 64-70 | 73-77 | 77-80 | 78-81 |
| Knee (left) | 79-81 | 76-80 | 81 | 92 | 70-73 | 88-89 | 71-74 | 78-80 | 81-83 | 82-83 |
| Knee (right) | 82-83 | 81 | 82-84 | 93 | 74-76 | 90 | 75-77 | 81-82 | 84-85 | 84-85 |
| Lower leg (left) | 84-92 | 82-95 | 85-88 | 94-97 | 77-89 | 91-95 | 78-89 | 83-92 | 86-93 | 86-93 |
| Lower leg (right) | 93-97 | 96-97 | 89-97 | 98-99 | 90-96 | 96-98 | 90-96 | 93-97 | 94-97 | 94-97 |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 97-99 | 99 | 97-99 | 98-99 | 98-99 | 98-99 |
| Foot (right) | 00 | - | 99-00 | - | 00 | 00 | 00 | 00 | 00 | 00 |

# Visby 1361 - gameplay-adjusted: Axe

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-08 | 01-12 | 01-04 | 01-14 | 01-04 | 01-16 | 01-10 | 01-08 | 01-08 | 01-07 |
| Skull (right) | 09-15 | 13-15 | 05-16 | 15-26 | 05-07 | 17-30 | 11-18 | 09-15 | 09-15 | 08-13 |
| Face | 16-19 | 16-18 | 17-20 | 27-33 | 08-09 | 31-36 | - | 16-18 | 16-19 | 14-17 |
| Neck | 20-21 | 19-20 | 21-22 | 34-36 | 10 | 37-40 | 19-20 | 19-20 | 20-21 | 18-19 |
| Chest (left) | 22-26 | 21-28 | 23-25 | 37-42 | 11-15 | 41-43 | 21 | 21-24 | 22-28 | 20-24 |
| Chest (right) | 27-30 | 29-30 | 26-33 | 43-47 | 16-19 | 44-46 | 22 | 25-28 | 29-34 | 25-28 |
| Abdomen | 31-35 | 31-34 | 34-38 | 48-53 | 20-25 | 47-49 | 23 | 29-32 | 35-41 | 29-33 |
| Groin / pelvis | 36-37 | 35 | 39-40 | 54-55 | 26-27 | 50 | - | 33 | 42-43 | 34 |
| Upper back | 38-39 | 36-37 | 41-42 | 56-57 | 28-29 | 51-54 | 24-32 | 34-38 | 44-46 | 35-36 |
| Lower back | 40-41 | 38 | 43 | 58-59 | 30 | 55-56 | 33-36 | 39-41 | 47-48 | 37 |
| Shoulder (left) | 42-44 | 39-42 | 44 | 60-64 | 31-32 | 57-59 | 37-38 | 42-43 | 49-50 | 38-41 |
| Shoulder (right) | 45-47 | 43 | 45-49 | 65-69 | 33-34 | 60-62 | 39-40 | 44-45 | 51-52 | 42-45 |
| Upper arm (left) | 48-49 | 44-46 | 50 | 70-71 | 35-36 | 63-64 | 41 | 46-47 | 53-54 | 46-47 |
| Upper arm (right) | 50-51 | 47 | 51-53 | 72-73 | 37-38 | 65-66 | 42 | 48-49 | 55-56 | 48-49 |
| Forearm (left) | 52-54 | 48-51 | 54 | 74-77 | 39-40 | 67-70 | 43-44 | 50-51 | 57-58 | 50-55 |
| Forearm (right) | 55-57 | 52 | 55-59 | 78-80 | 41-42 | 71-74 | 45-46 | 52-53 | 59-60 | 56-61 |
| Hand (left) | 58 | 53 | 60 | 81 | 43 | 75 | - | 54 | 61 | 62-63 |
| Hand (right) | 59 | - | 61-62 | 82 | 44 | 76 | - | 55 | 62 | 64-65 |
| Thigh (left) | 60-68 | 54-66 | 63-66 | 83-86 | 45-56 | 77-81 | 47-57 | 56-64 | 63-70 | 66-72 |
| Thigh (right) | 69-74 | 67-69 | 67-76 | 87-89 | 57-64 | 82-85 | 58-65 | 65-71 | 71-76 | 73-77 |
| Knee (left) | 75-78 | 70-75 | 77-78 | 90 | 65-69 | 86-87 | 66-70 | 72-75 | 77-79 | 78-80 |
| Knee (right) | 79-80 | 76 | 79-82 | 91 | 70-72 | 88 | 71-73 | 76-78 | 80-81 | 81-82 |
| Lower leg (left) | 81-91 | 77-92 | 83-87 | 92-96 | 73-87 | 89-94 | 74-87 | 79-90 | 82-91 | 83-92 |
| Lower leg (right) | 92-97 | 93-95 | 88-97 | 97-99 | 88-95 | 95-98 | 88-95 | 91-96 | 92-97 | 93-97 |
| Foot (left) | 98-99 | 96-99 | 98 | 00 | 96-98 | 99 | 96-98 | 97-99 | 98-99 | 98-99 |
| Foot (right) | 00 | 00 | 99-00 | - | 99-00 | 00 | 99-00 | 00 | 00 | 00 |

# Visby 1361 - gameplay-adjusted: Spiked club

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-11 | 01-17 | 01-05 | 01-17 | 01-05 | 01-19 | 01-14 | 01-11 | 01-11 | 01-09 |
| Skull (right) | 12-21 | 18-21 | 06-21 | 18-31 | 06-10 | 20-36 | 15-26 | 12-21 | 12-20 | 10-17 |
| Face | 22-26 | 22-26 | 22-26 | 32-39 | 11-13 | 37-43 | 27 | 22-26 | 21-25 | 18-22 |
| Neck | 27-29 | 27-28 | 27-29 | 40-43 | 14 | 44-48 | 28-31 | 27-28 | 26-28 | 23-24 |
| Chest (left) | 30-36 | 29-39 | 30-32 | 44-49 | 15-20 | 49-52 | 32 | 29-34 | 29-37 | 25-30 |
| Chest (right) | 37-42 | 40-41 | 33-42 | 50-55 | 21-26 | 53-55 | 33 | 35-39 | 38-44 | 31-35 |
| Abdomen | 43-49 | 42-47 | 43-49 | 56-61 | 27-34 | 56-59 | 34 | 40-45 | 45-53 | 36-41 |
| Groin / pelvis | 50-51 | 48-49 | 50-51 | 62-63 | 35-37 | 60 | - | 46-47 | 54-55 | 42-43 |
| Upper back | 52-54 | 50-51 | 52-54 | 64-66 | 38-40 | 61-65 | 35-46 | 48-53 | 56-59 | 44-46 |
| Lower back | 55-56 | 52-53 | 55-56 | 67-68 | 41-42 | 66-67 | 47-52 | 54-57 | 60-61 | 47-48 |
| Shoulder (left) | 57-59 | 54-58 | 57 | 69-72 | 43-45 | 68-70 | 53-54 | 58-59 | 62-64 | 49-52 |
| Shoulder (right) | 60-62 | 59 | 58-62 | 73-76 | 46-48 | 71-72 | 55-56 | 60-61 | 65-67 | 53-56 |
| Upper arm (left) | 63-64 | 60-62 | 63 | 77-78 | 49-50 | 73-74 | 57 | 62-63 | 68-69 | 57-59 |
| Upper arm (right) | 65-66 | 63 | 64-66 | 79-80 | 51-52 | 75-76 | 58 | 64-65 | 70-71 | 60-62 |
| Forearm (left) | 67-69 | 64-68 | 67 | 81-83 | 53-55 | 77-80 | 59-60 | 66-67 | 72-73 | 63-68 |
| Forearm (right) | 70-72 | 69 | 68-72 | 84-86 | 56-58 | 81-84 | 61-62 | 68-69 | 74-76 | 69-74 |
| Hand (left) | 73 | 70 | - | 87 | 59 | 85 | 63 | 70 | 77 | 75-76 |
| Hand (right) | 74 | - | 73-74 | 88 | 60 | 86 | 64 | 71 | 78 | 77-78 |
| Thigh (left) | 75-79 | 71-78 | 75-77 | 89-90 | 61-68 | 87-89 | 65-72 | 72-77 | 79-83 | 79-83 |
| Thigh (right) | 80-83 | 79-80 | 78-84 | 91-92 | 69-74 | 90-91 | 73-77 | 78-81 | 84-86 | 84-86 |
| Knee (left) | 84-85 | 81-84 | 85 | 93 | 75-78 | 92 | 78-80 | 82-83 | 87-88 | 87-88 |
| Knee (right) | 86-87 | 85 | 86-88 | 94 | 79-80 | 93 | 81-82 | 84-85 | 89 | 89 |
| Lower leg (left) | 88-94 | 86-96 | 89-91 | 95-97 | 81-91 | 94-97 | 83-92 | 86-93 | 90-95 | 90-95 |
| Lower leg (right) | 95-98 | 97-98 | 92-98 | 98-99 | 92-97 | 98-99 | 93-97 | 94-97 | 96-98 | 96-98 |
| Foot (left) | 99 | 99-00 | 99 | 00 | 98-99 | 00 | 98-99 | 98-99 | 99 | 99 |
| Foot (right) | 00 | - | 00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Visby 1361 - gameplay-adjusted: Spear

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-06 | 01-10 | 01-03 | 01-10 | 01-03 | 01-12 | 01-08 | 01-06 | 01-06 | 01-05 |
| Skull (right) | 07-11 | 11-12 | 04-12 | 11-19 | 04-05 | 13-23 | 09-15 | 07-11 | 07-11 | 06-10 |
| Face | 12-16 | 13-16 | 13-16 | 20-27 | 06-07 | 24-29 | 16 | 12-15 | 12-15 | 11-14 |
| Neck | 17 | 17 | 17 | 28-30 | 08 | 30-32 | 17-18 | 16 | 16 | 15 |
| Chest (left) | 18-26 | 18-32 | 18-22 | 31-40 | 09-16 | 33-38 | 19 | 17-24 | 17-28 | 16-23 |
| Chest (right) | 27-34 | 33-36 | 23-36 | 41-48 | 17-23 | 39-43 | 20 | 25-31 | 29-38 | 24-30 |
| Abdomen | 35-43 | 37-44 | 37-45 | 49-58 | 24-34 | 44-49 | 21 | 32-39 | 39-50 | 31-38 |
| Groin / pelvis | 44-46 | 45-46 | 46-48 | 59-61 | 35-38 | 50-51 | 22 | 40-41 | 51-53 | 39-40 |
| Upper back | 47-50 | 47-50 | 49-52 | 62-65 | 39-42 | 52-58 | 23-39 | 42-49 | 54-58 | 41-44 |
| Lower back | 51-53 | 51-52 | 53-55 | 66-68 | 43-44 | 59-61 | 40-48 | 50-55 | 59-61 | 45-46 |
| Shoulder (left) | 54-56 | 53-56 | 56 | 69-72 | 45-46 | 62-64 | 49-50 | 56-57 | 62-63 | 47-50 |
| Shoulder (right) | 57-59 | 57 | 57-61 | 73-76 | 47-48 | 65-67 | 51-52 | 58-59 | 64-65 | 51-53 |
| Upper arm (left) | 60-61 | 58-60 | 62 | 77-78 | 49-50 | 68-69 | 53 | 60 | 66-67 | 54-55 |
| Upper arm (right) | 62-63 | 61 | 63-65 | 79-80 | 51-52 | 70-71 | 54 | 61 | 68-69 | 56-57 |
| Forearm (left) | 64-66 | 62-65 | 66 | 81-83 | 53-54 | 72-75 | 55-56 | 62-63 | 70-71 | 58-63 |
| Forearm (right) | 67-69 | 66 | 67-70 | 84-86 | 55-56 | 76-79 | 57-58 | 64-65 | 72-73 | 64-69 |
| Hand (left) | 70 | 67 | - | 87 | 57 | 80 | 59 | 66 | 74 | 70-71 |
| Hand (right) | 71 | - | 71-72 | 88 | 58 | 81 | 60 | 67 | 75 | 72-73 |
| Thigh (left) | 72-77 | 68-76 | 73-75 | 89-90 | 59-67 | 82-85 | 61-68 | 68-74 | 76-80 | 74-78 |
| Thigh (right) | 78-81 | 77-78 | 76-82 | 91-92 | 68-73 | 86-88 | 69-74 | 75-79 | 81-84 | 79-82 |
| Knee (left) | 82-83 | 79-82 | 83 | 93 | 74-77 | 89-90 | 75-78 | 80-82 | 85-86 | 83-84 |
| Knee (right) | 84-85 | 83 | 84-86 | 94 | 78-80 | 91 | 79-80 | 83-84 | 87 | 85-86 |
| Lower leg (left) | 86-93 | 84-95 | 87-90 | 95-97 | 81-91 | 92-96 | 81-91 | 85-92 | 88-94 | 87-93 |
| Lower leg (right) | 94-97 | 96-97 | 91-97 | 98-99 | 92-97 | 97-99 | 92-97 | 93-97 | 95-98 | 94-97 |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 98-99 | 00 | 98-99 | 98-99 | 99 | 98-99 |
| Foot (right) | 00 | - | 99-00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Visby 1361 - gameplay-adjusted: Dagger

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-08 | 01-13 | 01-04 | 01-12 | 01-04 | 01-14 | 01-11 | 01-08 | 01-07 | 01-07 |
| Skull (right) | 09-15 | 14-16 | 05-16 | 13-22 | 05-07 | 15-27 | 12-21 | 09-15 | 08-14 | 08-13 |
| Face | 16-21 | 17-21 | 17-22 | 23-31 | 08-10 | 28-35 | 22 | 16-20 | 15-20 | 14-18 |
| Neck | 22-25 | 22-25 | 23-26 | 32-37 | 11-12 | 36-42 | 23-27 | 21-23 | 21-24 | 19-22 |
| Chest (left) | 26-35 | 26-40 | 27-31 | 38-46 | 13-22 | 43-47 | 28-29 | 24-31 | 25-36 | 23-30 |
| Chest (right) | 36-43 | 41-44 | 32-45 | 47-54 | 23-30 | 48-51 | 30 | 32-38 | 37-46 | 31-37 |
| Abdomen | 44-53 | 45-53 | 46-54 | 55-63 | 31-42 | 52-56 | 31-32 | 39-46 | 47-58 | 38-45 |
| Groin / pelvis | 54-56 | 54-55 | 55-57 | 64-65 | 43-47 | 57 | 33 | 47-48 | 59-61 | 46-47 |
| Upper back | 57-60 | 56-59 | 58-61 | 66-69 | 48-51 | 58-64 | 34-52 | 49-57 | 62-66 | 48-51 |
| Lower back | 61-63 | 60-61 | 62-64 | 70-71 | 52-54 | 65-67 | 53-62 | 58-63 | 67-69 | 52-53 |
| Shoulder (left) | 64-66 | 62-66 | 65 | 72-76 | 55-57 | 68-70 | 63-64 | 64-66 | 70-72 | 54-57 |
| Shoulder (right) | 67-69 | 67 | 66-71 | 77-81 | 58-60 | 71-73 | 65-66 | 67-69 | 73-75 | 58-61 |
| Upper arm (left) | 70-71 | 68-71 | 72 | 82-83 | 61-62 | 74-75 | 67-68 | 70-71 | 76-77 | 62-64 |
| Upper arm (right) | 72-73 | 72 | 73-76 | 84-85 | 63-64 | 76-77 | 69 | 72-73 | 78-79 | 65-67 |
| Forearm (left) | 74-76 | 73-77 | 77 | 86-89 | 65-67 | 78-82 | 70-71 | 74-76 | 80-81 | 68-74 |
| Forearm (right) | 77-79 | 78 | 78-82 | 90-93 | 68-70 | 83-87 | 72-73 | 77-79 | 82-84 | 75-81 |
| Hand (left) | 80 | 79-80 | - | 94 | 71 | 88-89 | 74 | 80 | 85 | 82-83 |
| Hand (right) | 81 | 81 | 83-84 | 95 | 72 | 90-91 | 75 | 81 | 86 | 84-85 |
| Thigh (left) | 82-85 | 82-87 | 85-86 | 96 | 73-78 | 92-93 | 76-80 | 82-85 | 87-89 | 86-88 |
| Thigh (right) | 86-88 | 88 | 87-90 | 97 | 79-82 | 94 | 81-84 | 86-88 | 90-91 | 89-90 |
| Knee (left) | 89-90 | 89-90 | 91 | - | 83-84 | 95 | 85-86 | 89-90 | 92 | 91 |
| Knee (right) | 91 | - | 92-93 | - | 85-86 | 96 | 87 | 91 | 93 | 92 |
| Lower leg (left) | 92-96 | 91-97 | 94-95 | 98-99 | 87-93 | 97-98 | 88-94 | 92-96 | 94-97 | 93-96 |
| Lower leg (right) | 97-99 | 98 | 96-99 | 00 | 94-97 | 99 | 95-98 | 97-99 | 98-99 | 97-98 |
| Foot (left) | 00 | 99-00 | - | - | 98-99 | 00 | 99 | 00 | 00 | 99 |
| Foot (right) | - | - | 00 | - | 00 | - | 00 | - | - | 00 |

# Visby 1361 - gameplay-adjusted: Crossbow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-09 | 01-14 | 01-04 | 01-14 | 01-04 | 01-17 | 01-11 | 01-09 | 01-09 | 01-08 |
| Skull (right) | 10-17 | 15-17 | 05-17 | 15-27 | 05-07 | 18-32 | 12-21 | 10-17 | 10-17 | 09-15 |
| Face | 18-21 | 18-21 | 18-21 | 28-34 | 08-09 | 33-38 | - | 18-21 | 18-21 | 16-19 |
| Neck | 22-23 | 22-23 | 22-23 | 35-37 | 10 | 39-42 | 22-23 | 22-23 | 22-23 | 20-21 |
| Chest (left) | 24-30 | 24-33 | 24-26 | 38-44 | 11-16 | 43-46 | 24 | 24-29 | 24-31 | 22-27 |
| Chest (right) | 31-36 | 34-36 | 27-36 | 45-50 | 17-21 | 47-49 | 25 | 30-34 | 32-38 | 28-32 |
| Abdomen | 37-43 | 37-42 | 37-43 | 51-57 | 22-29 | 50-53 | 26 | 35-40 | 39-46 | 33-38 |
| Groin / pelvis | 44-45 | 43-44 | 44-45 | 58-59 | 30-32 | 54 | - | 41-42 | 47-48 | 39-40 |
| Upper back | 46-48 | 45-47 | 46-48 | 60-62 | 33-35 | 55-59 | 27-38 | 43-48 | 49-52 | 41-42 |
| Lower back | 49-50 | 48-49 | 49-50 | 63-64 | 36-37 | 60-61 | 39-44 | 49-52 | 53-54 | 43-44 |
| Shoulder (left) | 51-53 | 50-53 | 51-52 | 65-69 | 38-40 | 62-64 | 45-46 | 53-54 | 55-57 | 45-48 |
| Shoulder (right) | 54-56 | 54 | 53-57 | 70-74 | 41-43 | 65-67 | 47-48 | 55-56 | 58-60 | 49-52 |
| Upper arm (left) | 57-58 | 55-57 | 58 | 75-76 | 44-45 | 68-69 | 49 | 57-58 | 61-62 | 53-54 |
| Upper arm (right) | 59-60 | 58 | 59-61 | 77-78 | 46-47 | 70-71 | 50 | 59-60 | 63-64 | 55-56 |
| Forearm (left) | 61-63 | 59-62 | 62 | 79-81 | 48-49 | 72-75 | 51-52 | 61-62 | 65-66 | 57-62 |
| Forearm (right) | 64-66 | 63 | 63-67 | 82-84 | 50-51 | 76-79 | 53-54 | 63-64 | 67-69 | 63-68 |
| Hand (left) | 67 | 64-65 | 68 | 85 | 52 | 80 | 55 | 65 | - | 69-70 |
| Hand (right) | 68 | - | 69-70 | 86 | 53 | 81 | 56 | 66 | 70 | 71-72 |
| Thigh (left) | 69-74 | 66-75 | 71-73 | 87-89 | 54-63 | 82-85 | 57-65 | 67-73 | 71-76 | 73-78 |
| Thigh (right) | 75-79 | 76-77 | 74-81 | 90-91 | 64-70 | 86-88 | 66-71 | 74-78 | 77-80 | 79-82 |
| Knee (left) | 80-82 | 78-81 | 82 | 92 | 71-74 | 89-90 | 72-75 | 79-81 | 81-83 | 83-84 |
| Knee (right) | 83-84 | 82 | 83-85 | 93 | 75-77 | 91 | 76-78 | 82-83 | 84-85 | 85-86 |
| Lower leg (left) | 85-92 | 83-95 | 86-89 | 94-97 | 78-89 | 92-96 | 79-90 | 84-92 | 86-93 | 87-93 |
| Lower leg (right) | 93-97 | 96-97 | 90-97 | 98-99 | 90-96 | 97-99 | 91-96 | 93-97 | 94-97 | 94-97 |
| Foot (left) | 98-99 | 98-00 | 98 | 00 | 97-99 | 00 | 97-99 | 98-99 | 98-99 | 98-99 |
| Foot (right) | 00 | - | 99-00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Visby 1361 - gameplay-adjusted: Sling

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Shield side | Weapon side | Higher | Lower | Down | Fleeing | Flanked | No shield | Warding |
|---|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-11 | 01-18 | 01-05 | 01-17 | 01-06 | 01-20 | 01-15 | 01-12 | 01-11 | 01-10 |
| Skull (right) | 12-21 | 19-22 | 06-22 | 18-32 | 07-11 | 21-38 | 16-28 | 13-22 | 12-21 | 11-19 |
| Face | 22-28 | 23-28 | 23-29 | 33-43 | 12-15 | 39-47 | 29 | 23-28 | 22-28 | 20-26 |
| Neck | 29-31 | 29-31 | 30-32 | 44-47 | 16 | 48-52 | 30-33 | 29-30 | 29-31 | 27-29 |
| Chest (left) | 32-38 | 32-41 | 33-35 | 48-53 | 17-23 | 53-56 | 34 | 31-36 | 32-39 | 30-35 |
| Chest (right) | 39-44 | 42-43 | 36-45 | 54-58 | 24-29 | 57-59 | 35 | 37-41 | 40-46 | 36-40 |
| Abdomen | 45-51 | 44-49 | 46-51 | 59-64 | 30-37 | 60-62 | 36 | 42-47 | 47-54 | 41-46 |
| Groin / pelvis | 52-53 | 50-51 | 52-53 | 65-66 | 38-40 | 63 | - | 48-49 | 55-56 | 47-48 |
| Upper back | 54-56 | 52-53 | 54-56 | 67-69 | 41-43 | 64-68 | 37-48 | 50-55 | 57-60 | 49-51 |
| Lower back | 57-58 | 54-55 | 57-58 | 70-71 | 44-45 | 69-70 | 49-54 | 56-59 | 61-62 | 52-53 |
| Shoulder (left) | 59-61 | 56-59 | 59 | 72-75 | 46-48 | 71-72 | 55-56 | 60-61 | 63-64 | 54-56 |
| Shoulder (right) | 62-64 | 60 | 60-63 | 76-79 | 49-50 | 73-74 | 57-58 | 62-63 | 65-66 | 57-59 |
| Upper arm (left) | 65-66 | 61-63 | 64 | 80-81 | 51-52 | 75 | 59 | 64 | 67-68 | 60-61 |
| Upper arm (right) | 67-68 | 64 | 65-67 | 82-83 | 53-54 | 76 | 60 | 65 | 69-70 | 62-63 |
| Forearm (left) | 69-71 | 65-68 | 68 | 84-86 | 55-56 | 77-80 | 61-62 | 66-67 | 71-72 | 64-69 |
| Forearm (right) | 72-73 | 69 | 69-72 | 87-89 | 57-58 | 81-84 | 63-64 | 68-69 | 73-74 | 70-74 |
| Hand (left) | 74 | 70 | - | 90 | 59 | 85 | 65 | 70 | 75 | 75-76 |
| Hand (right) | 75 | - | 73-74 | 91 | 60 | 86 | 66 | 71 | 76 | 77-78 |
| Thigh (left) | 76-80 | 71-78 | 75-77 | 92-93 | 61-68 | 87-89 | 67-73 | 72-77 | 77-81 | 79-83 |
| Thigh (right) | 81-84 | 79-80 | 78-84 | 94 | 69-74 | 90-91 | 74-78 | 78-81 | 82-85 | 84-86 |
| Knee (left) | 85-86 | 81-84 | 85 | 95 | 75-78 | 92 | 79-81 | 82-83 | 86-87 | 87-88 |
| Knee (right) | 87 | 85 | 86-88 | 96 | 79-80 | 93 | 82-83 | 84-85 | 88 | 89 |
| Lower leg (left) | 88-94 | 86-96 | 89-91 | 97-98 | 81-91 | 94-97 | 84-92 | 86-93 | 89-94 | 90-95 |
| Lower leg (right) | 95-98 | 97-98 | 92-98 | 99 | 92-97 | 98-99 | 93-97 | 94-97 | 95-98 | 96-98 |
| Foot (left) | 99 | 99-00 | 99 | 00 | 98-99 | 00 | 98-99 | 98-99 | 99 | 99 |
| Foot (right) | 00 | - | 00 | - | 00 | - | 00 | 00 | 00 | 00 |

# Towton 1461 - mass grave (bone evidence): Sword / Bill

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-28 | 01-13 | 01-31 | 01-20 | 01-33 | 01-37 | 01-30 | 01-21 |
| Skull (right) | 29-49 | 14-48 | 32-55 | 21-35 | 34-58 | 38-65 | 31-52 | 22-37 |
| Face | 50-60 | 49-59 | 56-68 | 36-43 | 59-67 | 66-67 | 53-62 | 38-46 |
| Neck | 61-66 | 60-65 | 69-75 | 44-47 | 68-74 | 68-75 | 63-67 | 47-51 |
| Groin / pelvis | 67 | 66 | - | 48-49 | - | - | 68 | 52 |
| Shoulder (left) | 68-69 | 67 | 76-77 | 50-51 | 75 | 76 | 69 | 53-54 |
| Shoulder (right) | 70 | 68 | 78 | 52 | 76 | 77 | 70 | 55 |
| Upper arm (left) | 71 | 69 | 79 | 53-54 | 77 | 78 | 71 | 56-57 |
| Upper arm (right) | 72 | 70 | 80 | 55 | - | - | 72 | 58 |
| Forearm (left) | 73-75 | 71 | 81-83 | 56-59 | 78-80 | 79-80 | 73-75 | 59-63 |
| Forearm (right) | 76-78 | 72-76 | 84-86 | 60-63 | 81-83 | 81-82 | 76-78 | 64-68 |
| Hand (left) | 79-84 | 77-79 | 87-91 | 64-72 | 84-89 | 83-86 | 79-84 | 69-80 |
| Hand (right) | 85-94 | 80-95 | 92-99 | 73-85 | 90-98 | 87-92 | 85-92 | 81-97 |
| Thigh (left) | 95 | - | - | 86-87 | - | 93 | 93 | - |
| Thigh (right) | 96 | 96-98 | - | 88-91 | 99 | 94-95 | 94-95 | 98 |
| Knee (right) | 97 | 99 | - | 92-93 | - | 96 | 96 | - |
| Lower leg (left) | 98 | - | - | 94-95 | - | 97 | 97 | - |
| Foot (left) | 99-00 | 00 | 00 | 96-00 | 00 | 98-00 | 98-00 | 99-00 |

# Towton 1461 - mass grave (bone evidence): Poleaxe

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-42 | 01-22 | 01-43 | 01-37 | 01-47 | 01-60 | 01-44 | 01-39 |
| Skull (right) | 43-58 | 23-53 | 44-60 | 38-52 | 48-66 | 61-84 | 45-62 | 40-54 |
| Face | 59-88 | 54-85 | 61-92 | 53-79 | 67-90 | 85-88 | 63-89 | 55-82 |
| Neck | 89-91 | 86-88 | 93-95 | 80-82 | 91-93 | 89-92 | 90-91 | 83-85 |
| Forearm (right) | 92-94 | 89-93 | 96-97 | 83-87 | 94-96 | 93-94 | 92-94 | 86-91 |
| Hand (right) | 95-97 | 94-98 | 98-99 | 88-92 | 97-99 | 95-96 | 95-97 | 92-97 |
| Lower leg (left) | 98-00 | 99-00 | 00 | 93-00 | 00 | 97-00 | 98-00 | 98-00 |

# Towton 1461 - mass grave (bone evidence): Dagger

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-54 | 01-24 | 01-54 | 01-54 | 01-54 | 01-54 | 01-54 | 01-54 |
| Skull (right) | 55-00 | 25-00 | 55-00 | 55-00 | 55-00 | 55-00 | 55-00 | 55-00 |

# Towton 1461 - mass grave (bone evidence): Longbow / Bow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-32 | 01-15 | 01-36 | 01-24 | 01-38 | 01-42 | 01-35 | 01-26 |
| Skull (right) | 33-54 | 16-52 | 37-60 | 25-40 | 39-63 | 43-71 | 36-58 | 27-44 |
| Face | 55-68 | 53-66 | 61-76 | 41-51 | 64-75 | 72-73 | 59-71 | 45-56 |
| Neck | 69-73 | 67-71 | 77-82 | 52-55 | 76-81 | 74-79 | 72-75 | 57-60 |
| Groin / pelvis | 74 | 72 | - | 56-57 | - | - | 76 | 61 |
| Shoulder (left) | 75 | 73 | 83 | 58-59 | 82 | 80 | 77 | 62 |
| Shoulder (right) | 76 | 74 | 84 | 60 | - | - | 78 | 63 |
| Upper arm (left) | 77 | - | 85 | 61-62 | 83 | 81 | 79 | 64 |
| Upper arm (right) | 78 | 75 | 86 | 63 | - | - | - | 65 |
| Forearm (left) | 79-80 | 76 | 87-88 | 64-66 | 84-85 | 82 | 80-81 | 66-69 |
| Forearm (right) | 81-83 | 77-81 | 89-90 | 67-70 | 86-88 | 83-84 | 82-83 | 70-74 |
| Hand (left) | 84-88 | 82-83 | 91-94 | 71-77 | 89-92 | 85-87 | 84-87 | 75-83 |
| Hand (right) | 89-95 | 84-96 | 95-00 | 78-88 | 93-99 | 88-92 | 88-94 | 84-97 |
| Thigh (left) | 96 | - | - | 89 | - | 93 | 95 | - |
| Thigh (right) | 97 | 97-98 | - | 90-92 | - | 94-95 | 96 | 98 |
| Knee (right) | - | 99 | - | 93 | - | 96 | 97 | - |
| Lower leg (left) | 98 | - | - | 94-96 | - | 97-98 | 98 | 99 |
| Foot (left) | 99-00 | 00 | - | 97-00 | 00 | 99-00 | 99-00 | 00 |

# Towton 1461 - gameplay-adjusted (before armour): Sword / Longbow / Bow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-12 | 01-06 | 01-17 | 01-06 | 01-19 | 01-15 | 01-12 | 01-10 | 01-24 |
| Skull (right) | 13-21 | 07-21 | 18-30 | 07-11 | 20-34 | 16-26 | 13-21 | 11-17 | 25-42 |
| Face | 22-27 | 22-27 | 31-39 | 12-14 | 35-41 | 27 | 22-26 | 18-22 | 43-54 |
| Neck | 28-31 | 28-31 | 40-45 | 15-16 | 42-48 | 28-32 | 27-29 | 23-25 | 55-62 |
| Chest (left) | 32-38 | 32-34 | 46-51 | 17-23 | 49-51 | 33 | 30-35 | 26-31 | 63-65 |
| Chest (right) | 39-44 | 35-44 | 52-56 | 24-29 | 52-54 | 34 | 36-40 | 32-36 | 66-67 |
| Abdomen | 45-51 | 45-50 | 57-62 | 30-38 | 55-57 | 35 | 41-46 | 37-42 | 68 |
| Groin / pelvis | 52-54 | 51-53 | 63-65 | 39-43 | 58 | 36 | 47-49 | 43-45 | - |
| Upper back | 55-58 | 54-57 | 66-69 | 44-47 | 59-64 | 37-53 | 50-57 | 46-48 | 69-70 |
| Lower back | 59-61 | 58-60 | 70-72 | 48-50 | 65-67 | 54-62 | 58-63 | 49-50 | - |
| Shoulder (left) | 62-63 | 61 | 73-75 | 51-52 | 68-69 | 63 | 64-65 | 51-52 | 71-74 |
| Shoulder (right) | 64 | 62-63 | 76 | 53 | 70 | 64 | 66 | 53 | 75-76 |
| Upper arm (left) | 65-66 | 64 | 77-78 | 54-55 | 71-72 | 65 | 67-68 | 54-55 | 77-79 |
| Upper arm (right) | 67-68 | 65-67 | 79-80 | 56-57 | 73-74 | 66 | 69-70 | 56-57 | 80-82 |
| Forearm (left) | 69-71 | 68 | 81-83 | 58-60 | 75-78 | 67-68 | 71-73 | 58-63 | 83-86 |
| Forearm (right) | 72-74 | 69-73 | 84-86 | 61-63 | 79-82 | 69-70 | 74-76 | 64-69 | 87-90 |
| Hand (left) | 75-78 | 74-75 | 87-90 | 64-67 | 83-87 | 71-73 | 77-79 | 70-77 | 91-95 |
| Hand (right) | 79-82 | 76-82 | 91-94 | 68-71 | 88-92 | 74-76 | 80-82 | 78-85 | 96-00 |
| Thigh (left) | 83-86 | 83-84 | 95-96 | 72-77 | 93-94 | 77-82 | 83-86 | 86-88 | - |
| Thigh (right) | 87-90 | 85-91 | 97 | 78-83 | 95-96 | 83-87 | 87-90 | 89-91 | - |
| Knee (left) | 91 | - | - | 84-85 | 97 | 88 | 91 | 92 | - |
| Knee (right) | 92 | 92-93 | - | 86-87 | - | 89 | 92 | 93 | - |
| Lower leg (left) | 93-95 | 94 | 98 | 88-92 | 98 | 90-93 | 93-95 | 94-95 | - |
| Lower leg (right) | 96-97 | 95-97 | 99 | 93-95 | 99 | 94-96 | 96-97 | 96-97 | - |
| Foot (left) | 98-99 | 98 | 00 | 96-98 | 00 | 97-99 | 98-99 | 98-99 | - |
| Foot (right) | 00 | 99-00 | - | 99-00 | - | 00 | 00 | 00 | - |

# Towton 1461 - gameplay-adjusted (before armour): Bill

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-13 | 01-06 | 01-19 | 01-07 | 01-21 | 01-17 | 01-13 | 01-11 | 01-25 |
| Skull (right) | 14-23 | 07-23 | 20-33 | 08-12 | 22-37 | 18-29 | 14-23 | 12-19 | 26-44 |
| Face | 24-30 | 24-29 | 34-42 | 13-15 | 38-44 | 30 | 24-29 | 20-25 | 45-57 |
| Neck | 31-34 | 30-33 | 43-48 | 16-17 | 45-51 | 31-36 | 30-33 | 26-29 | 58-65 |
| Chest (left) | 35-40 | 34-36 | 49-54 | 18-23 | 52-54 | 37 | 34-38 | 30-34 | 66-68 |
| Chest (right) | 41-45 | 37-45 | 55-59 | 24-29 | 55-57 | 38 | 39-43 | 35-39 | 69-70 |
| Abdomen | 46-51 | 46-51 | 60-65 | 30-37 | 58-60 | 39 | 44-48 | 40-44 | 71 |
| Groin / pelvis | 52-54 | 52-54 | 66-68 | 38-42 | 61 | 40 | 49-50 | 45-46 | - |
| Upper back | 55-58 | 55-58 | 69-71 | 43-46 | 62-66 | 41-55 | 51-58 | 47-49 | 72 |
| Lower back | 59-61 | 59-61 | 72-73 | 47-49 | 67-69 | 56-64 | 59-64 | 50-51 | - |
| Shoulder (left) | 62-63 | 62 | 74-76 | 50-51 | 70 | 65 | 65-66 | 52-53 | 73-76 |
| Shoulder (right) | 64 | 63-64 | 77 | 52 | 71 | 66 | 67 | 54 | 77-78 |
| Upper arm (left) | 65-66 | 65 | 78-79 | 53-54 | 72 | 67 | 68-69 | 55-56 | 79-80 |
| Upper arm (right) | 67-68 | 66-68 | 80-81 | 55-56 | 73 | 68 | 70-71 | 57-58 | 81-82 |
| Forearm (left) | 69-71 | 69 | 82-84 | 57-59 | 74-77 | 69-70 | 72-73 | 59-64 | 83-86 |
| Forearm (right) | 72-74 | 70-74 | 85-87 | 60-62 | 78-81 | 71-72 | 74-75 | 65-70 | 87-90 |
| Hand (left) | 75-78 | 75-76 | 88-91 | 63-66 | 82-86 | 73-74 | 76-78 | 71-77 | 91-95 |
| Hand (right) | 79-82 | 77-82 | 92-95 | 67-70 | 87-91 | 75-76 | 79-81 | 78-84 | 96-00 |
| Thigh (left) | 83-86 | 83-84 | 96 | 71-77 | 92-93 | 77-82 | 82-86 | 85-87 | - |
| Thigh (right) | 87-90 | 85-91 | 97 | 78-83 | 94-95 | 83-87 | 87-90 | 88-90 | - |
| Knee (left) | 91 | - | - | 84-85 | 96 | 88 | 91 | 91 | - |
| Knee (right) | 92 | 92-93 | - | 86-87 | 97 | 89 | 92 | 92 | - |
| Lower leg (left) | 93-95 | 94 | 98 | 88-92 | 98 | 90-93 | 93-95 | 93-95 | - |
| Lower leg (right) | 96-97 | 95-97 | 99 | 93-95 | 99 | 94-96 | 96-97 | 96-97 | - |
| Foot (left) | 98-99 | 98 | 00 | 96-98 | 00 | 97-99 | 98-99 | 98-99 | - |
| Foot (right) | 00 | 99-00 | - | 99-00 | - | 00 | 00 | 00 | - |

# Towton 1461 - gameplay-adjusted (before armour): Poleaxe

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-14 | 01-07 | 01-20 | 01-07 | 01-22 | 01-18 | 01-14 | 01-12 | 01-26 |
| Skull (right) | 15-25 | 08-25 | 21-35 | 08-13 | 23-39 | 19-31 | 15-25 | 13-21 | 27-46 |
| Face | 26-32 | 26-32 | 36-45 | 14-17 | 40-47 | 32 | 26-31 | 22-27 | 47-59 |
| Neck | 33-37 | 33-36 | 46-52 | 18-19 | 48-54 | 33-38 | 32-35 | 28-31 | 60-68 |
| Chest (left) | 38-43 | 37-39 | 53-57 | 20-26 | 55-57 | 39 | 36-40 | 32-36 | 69-70 |
| Chest (right) | 44-48 | 40-48 | 58-62 | 27-32 | 58-60 | 40 | 41-45 | 37-41 | 71-72 |
| Abdomen | 49-54 | 49-54 | 63-67 | 33-41 | 61-63 | 41 | 46-50 | 42-46 | 73 |
| Groin / pelvis | 55-57 | 55-57 | 68-69 | 42-46 | 64 | 42 | 51-52 | 47-48 | - |
| Upper back | 58-61 | 58-60 | 70-72 | 47-50 | 65-69 | 43-57 | 53-60 | 49-51 | 74-75 |
| Lower back | 62-64 | 61-63 | 73-74 | 51-53 | 70-72 | 58-66 | 61-66 | 52-53 | - |
| Shoulder (left) | 65-66 | 64 | 75-76 | 54-55 | 73 | 67 | 67-68 | 54-55 | 76-78 |
| Shoulder (right) | 67 | 65-66 | 77 | 56 | 74 | 68 | 69 | 56 | 79-80 |
| Upper arm (left) | 68-69 | 67 | 78-79 | 57-58 | 75 | 69 | 70-71 | 57-58 | 81-82 |
| Upper arm (right) | 70-71 | 68-70 | 80-81 | 59-60 | 76 | 70 | 72 | 59-60 | 83-84 |
| Forearm (left) | 72-74 | 71 | 82-84 | 61-63 | 77-79 | 71-72 | 73-74 | 61-66 | 85-87 |
| Forearm (right) | 75-77 | 72-76 | 85-87 | 64-66 | 80-82 | 73-74 | 75-76 | 67-71 | 88-90 |
| Hand (left) | 78-81 | 77-78 | 88-91 | 67-70 | 83-87 | 75-76 | 77-79 | 72-78 | 91-95 |
| Hand (right) | 82-84 | 79-84 | 92-95 | 71-74 | 88-92 | 77-78 | 80-82 | 79-85 | 96-00 |
| Thigh (left) | 85-87 | 85-86 | 96 | 75-80 | 93-94 | 79-83 | 83-86 | 86-88 | - |
| Thigh (right) | 88-90 | 87-92 | 97 | 81-86 | 95-96 | 84-88 | 87-90 | 89-91 | - |
| Knee (left) | 91 | - | - | 87 | 97 | 89 | 91 | 92 | - |
| Knee (right) | 92 | 93-94 | - | 88 | - | 90 | 92 | 93 | - |
| Lower leg (left) | 93-95 | 95 | 98 | 89-93 | 98 | 91-94 | 93-95 | 94-95 | - |
| Lower leg (right) | 96-97 | 96-98 | 99 | 94-96 | 99 | 95-97 | 96-97 | 96-97 | - |
| Foot (left) | 98-99 | 99 | 00 | 97-99 | 00 | 98-99 | 98-99 | 98-99 | - |
| Foot (right) | 00 | 00 | - | 00 | - | 00 | 00 | 00 | - |

# Towton 1461 - gameplay-adjusted (before armour): Axe

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-13 | 01-06 | 01-19 | 01-06 | 01-21 | 01-16 | 01-13 | 01-11 | 01-25 |
| Skull (right) | 14-23 | 07-22 | 20-33 | 07-11 | 22-36 | 17-28 | 14-23 | 12-19 | 26-44 |
| Face | 24-29 | 23-28 | 34-42 | 12-14 | 37-43 | 29 | 24-28 | 20-24 | 45-56 |
| Neck | 30-33 | 29-32 | 43-48 | 15-16 | 44-50 | 30-34 | 29-32 | 25-28 | 57-64 |
| Chest (left) | 34-39 | 33-35 | 49-53 | 17-22 | 51-53 | 35 | 33-37 | 29-32 | 65-66 |
| Chest (right) | 40-44 | 36-43 | 54-57 | 23-27 | 54-55 | 36 | 38-41 | 33-36 | 67-68 |
| Abdomen | 45-49 | 44-48 | 58-62 | 28-34 | 56-58 | 37 | 42-46 | 37-40 | 69 |
| Groin / pelvis | 50-51 | 49-50 | 63-64 | 35-38 | 59 | - | 47-48 | 41-42 | - |
| Upper back | 52-54 | 51-53 | 65-67 | 39-41 | 60-63 | 38-50 | 49-55 | 43-45 | 70 |
| Lower back | 55-56 | 54-55 | 68-69 | 42-43 | 64-65 | 51-57 | 56-60 | 46-47 | - |
| Shoulder (left) | 57-58 | 56 | 70-72 | 44-45 | 66-67 | 58 | 61-62 | 48-49 | 71-74 |
| Shoulder (right) | 59 | 57-58 | 73 | 46 | 68 | 59 | 63 | 50 | 75-76 |
| Upper arm (left) | 60-61 | 59 | 74-75 | 47-48 | 69-70 | 60 | 64-65 | 51-52 | 77-79 |
| Upper arm (right) | 62-63 | 60-62 | 76-77 | 49-50 | 71 | 61 | 66-67 | 53-54 | 80-82 |
| Forearm (left) | 64-66 | 63 | 78-80 | 51-53 | 72-75 | 62-63 | 68-69 | 55-60 | 83-86 |
| Forearm (right) | 67-69 | 64-68 | 81-83 | 54-56 | 76-79 | 64-65 | 70-71 | 61-66 | 87-90 |
| Hand (left) | 70-73 | 69-70 | 84-87 | 57-60 | 80-84 | 66-67 | 72-74 | 67-74 | 91-95 |
| Hand (right) | 74-77 | 71-77 | 88-91 | 61-64 | 85-89 | 68-69 | 75-77 | 75-82 | 96-00 |
| Thigh (left) | 78-82 | 78-79 | 92-93 | 65-72 | 90-91 | 70-76 | 78-82 | 83-86 | - |
| Thigh (right) | 83-87 | 80-88 | 94-95 | 73-80 | 92-93 | 77-83 | 83-87 | 87-90 | - |
| Knee (left) | 88 | 89 | 96 | 81-82 | 94 | 84-85 | 88 | 91 | - |
| Knee (right) | 89 | 90-91 | 97 | 83-84 | 95 | 86-87 | 89 | 92 | - |
| Lower leg (left) | 90-93 | 92-93 | 98 | 85-90 | 96-97 | 88-92 | 90-93 | 93-95 | - |
| Lower leg (right) | 94-96 | 94-97 | 99 | 91-94 | 98 | 93-95 | 94-96 | 96-97 | - |
| Foot (left) | 97-99 | 98 | 00 | 95-98 | 99 | 96-98 | 97-99 | 98-99 | - |
| Foot (right) | 00 | 99-00 | - | 99-00 | 00 | 99-00 | 00 | 00 | - |

# Towton 1461 - gameplay-adjusted (before armour): Dagger

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-11 | 01-05 | 01-14 | 01-06 | 01-17 | 01-14 | 01-11 | 01-09 | 01-20 |
| Skull (right) | 12-19 | 06-19 | 15-25 | 07-10 | 18-30 | 15-25 | 12-19 | 10-16 | 21-35 |
| Face | 20-27 | 20-27 | 26-36 | 11-14 | 31-39 | 26 | 20-26 | 17-23 | 36-50 |
| Neck | 28-34 | 28-34 | 37-46 | 15-18 | 40-50 | 27-35 | 27-32 | 24-29 | 51-63 |
| Chest (left) | 35-43 | 35-38 | 47-53 | 19-28 | 51-54 | 36 | 33-39 | 30-36 | 64-66 |
| Chest (right) | 44-50 | 39-51 | 54-59 | 29-36 | 55-57 | 37 | 40-45 | 37-42 | 67-69 |
| Abdomen | 51-59 | 52-59 | 60-66 | 37-48 | 58-61 | 38 | 46-52 | 43-49 | 70 |
| Groin / pelvis | 60-63 | 60-62 | 67-69 | 49-55 | 62-63 | 39 | 53-55 | 50-52 | - |
| Upper back | 64-68 | 63-67 | 70-73 | 56-60 | 64-70 | 40-61 | 56-65 | 53-56 | 71-72 |
| Lower back | 69-72 | 68-70 | 74-76 | 61-64 | 71-73 | 62-73 | 66-73 | 57-59 | - |
| Shoulder (left) | 73-74 | 71 | 77-79 | 65-66 | 74-75 | 74 | 74-75 | 60-61 | 73-76 |
| Shoulder (right) | 75 | 72-73 | 80 | 67 | 76 | 75 | 76 | 62 | 77-78 |
| Upper arm (left) | 76-77 | 74 | 81-82 | 68-69 | 77-78 | 76 | 77-78 | 63-64 | 79-80 |
| Upper arm (right) | 78-79 | 75-77 | 83-84 | 70-71 | 79 | 77 | 79-80 | 65-66 | 81-82 |
| Forearm (left) | 80-82 | 78 | 85-87 | 72-74 | 80-83 | 78-79 | 81-82 | 67-72 | 83-86 |
| Forearm (right) | 83-85 | 79-83 | 88-90 | 75-77 | 84-87 | 80-81 | 83-84 | 73-78 | 87-90 |
| Hand (left) | 86-89 | 84-85 | 91-94 | 78-81 | 88-92 | 82-84 | 85-87 | 79-86 | 91-95 |
| Hand (right) | 90-93 | 86-92 | 95-98 | 82-85 | 93-97 | 85-87 | 88-90 | 87-94 | 96-00 |
| Thigh (left) | 94-95 | 93 | 99 | 86-88 | 98 | 88-90 | 91-92 | 95-96 | - |
| Thigh (right) | 96-97 | 94-96 | 00 | 89-91 | 99 | 91-93 | 93-94 | 97 | - |
| Knee (left) | - | - | - | 92 | - | 94 | 95 | - | - |
| Knee (right) | - | 97 | - | 93 | - | 95 | 96 | - | - |
| Lower leg (left) | 98 | 98 | - | 94-95 | 00 | 96-97 | 97 | 98 | - |
| Lower leg (right) | 99 | 99 | - | 96-97 | - | 98 | 98 | 99 | - |
| Foot (left) | 00 | - | - | 98-99 | - | 99 | 99 | 00 | - |
| Foot (right) | - | 00 | - | 00 | - | 00 | 00 | - | - |

# Towton 1461 - gameplay-adjusted (before armour): Spear

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-09 | 01-04 | 01-13 | 01-05 | 01-16 | 01-12 | 01-09 | 01-08 | 01-21 |
| Skull (right) | 10-16 | 05-16 | 14-23 | 06-08 | 17-28 | 13-21 | 10-16 | 09-14 | 22-36 |
| Face | 17-23 | 17-23 | 24-33 | 09-11 | 29-36 | 22 | 17-22 | 15-20 | 37-51 |
| Neck | 24-26 | 24-26 | 34-37 | 12-13 | 37-41 | 23-26 | 23-24 | 21-23 | 52-58 |
| Chest (left) | 27-35 | 27-30 | 38-46 | 14-22 | 42-46 | 27-28 | 25-32 | 24-31 | 59-62 |
| Chest (right) | 36-43 | 31-44 | 47-53 | 23-30 | 47-50 | 29 | 33-38 | 32-38 | 63-66 |
| Abdomen | 44-52 | 45-53 | 54-61 | 31-42 | 51-55 | 30 | 39-46 | 39-46 | 67 |
| Groin / pelvis | 53-56 | 54-57 | 62-65 | 43-49 | 56-57 | 31 | 47-49 | 47-49 | - |
| Upper back | 57-61 | 58-62 | 66-70 | 50-54 | 58-65 | 32-54 | 50-60 | 50-53 | 68-69 |
| Lower back | 62-65 | 63-66 | 71-74 | 55-58 | 66-69 | 55-67 | 61-68 | 54-56 | 70 |
| Shoulder (left) | 66-67 | 67 | 75-76 | 59-60 | 70-71 | 68 | 69 | 57-58 | 71-74 |
| Shoulder (right) | 68 | 68 | 77 | 61 | 72 | 69 | 70 | 59 | 75-76 |
| Upper arm (left) | 69-70 | 69 | 78-79 | 62-63 | 73-74 | 70 | 71 | 60-61 | 77-79 |
| Upper arm (right) | 71-72 | 70-72 | 80-81 | 64-65 | 75 | 71 | 72 | 62-63 | 80-82 |
| Forearm (left) | 73-75 | 73 | 82-84 | 66-68 | 76-79 | 72-73 | 73-74 | 64-68 | 83-86 |
| Forearm (right) | 76-78 | 74-77 | 85-87 | 69-71 | 80-83 | 74-75 | 75-76 | 69-73 | 87-90 |
| Hand (left) | 79-81 | 78-79 | 88-91 | 72-74 | 84-88 | 76-77 | 77-79 | 74-80 | 91-95 |
| Hand (right) | 82-84 | 80-85 | 92-95 | 75-77 | 89-93 | 78-79 | 80-82 | 81-87 | 96-00 |
| Thigh (left) | 85-87 | 86-87 | 96 | 78-82 | 94-95 | 80-84 | 83-86 | 88-90 | - |
| Thigh (right) | 88-90 | 88-93 | 97 | 83-87 | 96-97 | 85-89 | 87-90 | 91-93 | - |
| Knee (left) | 91 | - | - | 88 | - | 90 | 91 | 94 | - |
| Knee (right) | 92 | 94 | - | 89 | - | 91 | 92 | 95 | - |
| Lower leg (left) | 93-95 | 95 | 98 | 90-93 | 98 | 92-95 | 93-95 | 96-97 | - |
| Lower leg (right) | 96-97 | 96-98 | 99 | 94-96 | 99 | 96-97 | 96-97 | 98 | - |
| Foot (left) | 98-99 | 99 | 00 | 97-99 | 00 | 98-99 | 98-99 | 99 | - |
| Foot (right) | 00 | 00 | - | 00 | - | 00 | 00 | 00 | - |

# Towton 1461 - gameplay-adjusted (before armour): Crossbow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Weapon side | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-13 | 01-06 | 01-19 | 01-07 | 01-21 | 01-17 | 01-13 | 01-11 | 01-25 |
| Skull (right) | 14-23 | 07-23 | 20-33 | 08-12 | 22-37 | 18-30 | 14-23 | 12-19 | 26-44 |
| Face | 24-29 | 24-29 | 34-42 | 13-15 | 38-44 | 31 | 24-29 | 20-24 | 45-57 |
| Neck | 30-33 | 30-33 | 43-47 | 16-17 | 45-50 | 32-36 | 30-32 | 25-27 | 58-64 |
| Chest (left) | 34-39 | 34-36 | 48-53 | 18-24 | 51-53 | 37 | 33-38 | 28-33 | 65-67 |
| Chest (right) | 40-45 | 37-46 | 54-58 | 25-30 | 54-56 | 38 | 39-43 | 34-38 | 68-69 |
| Abdomen | 46-51 | 47-52 | 59-64 | 31-39 | 57-59 | 39 | 44-49 | 39-43 | 70 |
| Groin / pelvis | 52-54 | 53-55 | 65-67 | 40-44 | 60 | 40 | 50-51 | 44-45 | - |
| Upper back | 55-58 | 56-59 | 68-70 | 45-48 | 61-65 | 41-56 | 52-59 | 46-48 | 71-72 |
| Lower back | 59-61 | 60-62 | 71-73 | 49-51 | 66-68 | 57-65 | 60-65 | 49-50 | - |
| Shoulder (left) | 62-63 | 63 | 74-76 | 52-53 | 69-70 | 66 | 66-67 | 51-52 | 73-76 |
| Shoulder (right) | 64 | 64-65 | 77 | 54 | 71 | 67 | 68 | 53 | 77-78 |
| Upper arm (left) | 65-66 | 66 | 78-79 | 55-56 | 72-73 | 68 | 69-70 | 54-55 | 79-80 |
| Upper arm (right) | 67-68 | 67-69 | 80-81 | 57-58 | 74-75 | 69 | 71-72 | 56-57 | 81-82 |
| Forearm (left) | 69-71 | 70 | 82-84 | 59-61 | 76-79 | 70-71 | 73-74 | 58-63 | 83-86 |
| Forearm (right) | 72-74 | 71-75 | 85-87 | 62-64 | 80-83 | 72-73 | 75-76 | 64-69 | 87-90 |
| Hand (left) | 75-78 | 76-77 | 88-91 | 65-68 | 84-88 | 74-75 | 77-79 | 70-77 | 91-95 |
| Hand (right) | 79-82 | 78-83 | 92-95 | 69-72 | 89-93 | 76-77 | 80-82 | 78-85 | 96-00 |
| Thigh (left) | 83-86 | 84-85 | 96 | 73-78 | 94-95 | 78-82 | 83-86 | 86-88 | - |
| Thigh (right) | 87-90 | 86-91 | 97 | 79-84 | 96-97 | 83-87 | 87-90 | 89-91 | - |
| Knee (left) | 91 | - | - | 85-86 | - | 88 | 91 | 92 | - |
| Knee (right) | 92 | 92-93 | - | 87-88 | - | 89 | 92 | 93 | - |
| Lower leg (left) | 93-95 | 94 | 98 | 89-93 | 98 | 90-93 | 93-95 | 94-95 | - |
| Lower leg (right) | 96-97 | 95-97 | 99 | 94-96 | 99 | 94-96 | 96-97 | 96-97 | - |
| Foot (left) | 98-99 | 98 | 00 | 97-99 | 00 | 97-99 | 98-99 | 98-99 | - |
| Foot (right) | 00 | 99-00 | - | 00 | - | 00 | 00 | 00 | - |

# Lützen 1632 - mass grave (bone evidence): Pistol/carbine / Matchlock

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-33 | 01-40 | 01-21 | 01-41 | 01-32 | 01-31 | 01-33 | 01-46 |
| Skull (right) | 34-66 | 41-80 | 22-42 | 42-82 | 33-64 | 32-62 | 34-66 | 47-92 |
| Face | 67-69 | 81-84 | 43-44 | 83-85 | - | 63-65 | 67-69 | 93-97 |
| Chest (left) | 70-71 | 85 | 45-46 | 86 | - | 66 | 70-71 | 98 |
| Chest (right) | 72-73 | 86 | 47-48 | 87 | - | 67 | 72-73 | 99 |
| Groin / pelvis | 74-79 | 87-91 | 49-62 | 88-89 | 65 | 68-72 | 74-79 | - |
| Upper back | 80-84 | 92-94 | 63-68 | 90-94 | 66-80 | 73-82 | 80-84 | 00 |
| Lower back | 85 | 95 | 69-70 | 95 | 81-83 | 83-85 | 85 | - |
| Thigh (right) | 86-88 | 96 | 71-76 | 96 | 84-86 | 86-88 | 86-88 | - |
| Lower leg (left) | 89-94 | 97-98 | 77-88 | 97-98 | 87-93 | 89-94 | 89-94 | - |
| Lower leg (right) | 95-00 | 99-00 | 89-00 | 99-00 | 94-00 | 95-00 | 95-00 | - |

# Thirty Years' War - all hits (gameplay): Matchlock / Pistol/carbine

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Flanked | Cover |
|---|---|---|---|---|---|---|
| Skull (left) | 01-09 | 01-15 | 01-04 | 01-12 | 01-09 | 01-19 |
| Skull (right) | 10-18 | 16-28 | 05-08 | 13-23 | 10-18 | 20-37 |
| Face | 19-24 | 29-37 | 09-11 | 24 | 19-23 | 38-50 |
| Neck | 25 | 38 | - | 25 | 24 | 51-52 |
| Chest (left) | 26-27 | 39-40 | 12-13 | - | 25-26 | 53 |
| Chest (right) | 28-29 | 41-42 | 14-15 | - | 27-28 | 54 |
| Abdomen | 30-31 | 43-44 | 16-17 | - | 29-30 | - |
| Groin / pelvis | 32-34 | 45-47 | 18-22 | 26 | 31-33 | - |
| Upper back | 35 | 48 | 23 | 27-31 | 34-36 | 55 |
| Lower back | 36 | 49 | 24 | 32-35 | 37-39 | - |
| Shoulder (left) | 37-39 | 50-54 | 25-27 | 36-37 | 40-42 | 56-62 |
| Shoulder (right) | 40-42 | 55-59 | 28-30 | 38-39 | 43-45 | 63-69 |
| Upper arm (left) | 43-49 | 60-67 | 31-36 | 40-43 | 46-51 | 70-78 |
| Upper arm (right) | 50-55 | 68-74 | 37-42 | 44-47 | 52-56 | 79-87 |
| Forearm (left) | 56-57 | 75-76 | 43-44 | 48 | 57-58 | 88-90 |
| Forearm (right) | 58-59 | 77-78 | 45-46 | 49 | 59 | 91-92 |
| Hand (left) | 60-62 | 79-81 | 47-48 | 50-51 | 60-61 | 93-96 |
| Hand (right) | 63-64 | 82-84 | 49-50 | 52 | 62-63 | 97-99 |
| Thigh (left) | 65-71 | 85-87 | 51-60 | 53-62 | 64-71 | - |
| Thigh (right) | 72-79 | 88-90 | 61-71 | 63-73 | 72-80 | 00 |
| Knee (left) | 80-81 | 91 | 72-73 | 74-75 | 81 | - |
| Knee (right) | 82 | 92 | 74-75 | 76-77 | 82 | - |
| Lower leg (left) | 83-89 | 93-95 | 76-84 | 78-86 | 83-89 | - |
| Lower leg (right) | 90-96 | 96-98 | 85-94 | 87-95 | 90-96 | - |
| Foot (left) | 97-98 | 99 | 95-97 | 96-97 | 97-98 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | 99-00 | - |

# American Revolution 1775-83 - disabled veterans' wounds: Musket / Rifle / Pistol / Artillery

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Flanked | Cover |
|---|---|---|---|---|---|---|
| Skull (left) | 01-03 | 01-05 | 01 | 01-04 | 01-03 | 01-08 |
| Skull (right) | 04-06 | 06-10 | 02 | 05-08 | 04-06 | 09-16 |
| Face | 07-09 | 11-15 | 03 | - | 07-08 | 17-23 |
| Neck | 10-11 | 16-19 | 04 | 09-11 | 09-10 | 24-29 |
| Chest (left) | 12-17 | 20-24 | 05-09 | 12 | 11-15 | 30-32 |
| Chest (right) | 18-23 | 25-29 | 10-14 | 13 | 16-20 | 33-35 |
| Abdomen | 24-26 | 30-32 | 15-17 | 14 | 21-22 | - |
| Groin / pelvis | 27-31 | 33-37 | 18-24 | 15 | 23-26 | - |
| Upper back | 32-33 | 38-39 | 25-26 | 16-25 | 27-31 | 36 |
| Lower back | 34-36 | 40-42 | 27-28 | 26-34 | 32-36 | - |
| Shoulder (left) | 37-40 | 43-49 | 29-32 | 35-37 | 37-40 | 37-46 |
| Shoulder (right) | 41-44 | 50-56 | 33-36 | 38-40 | 41-44 | 47-56 |
| Upper arm (left) | 45-49 | 57-62 | 37-40 | 41-43 | 45-48 | 57-63 |
| Upper arm (right) | 50-54 | 63-68 | 41-44 | 44-46 | 49-52 | 64-70 |
| Forearm (left) | 55-59 | 69-74 | 45-48 | 47-49 | 53-56 | 71-77 |
| Forearm (right) | 60-64 | 75-80 | 49-52 | 50-52 | 57-60 | 78-84 |
| Hand (left) | 65-68 | 81-85 | 53-56 | 53-55 | 61-63 | 85-91 |
| Hand (right) | 69-72 | 86-90 | 57-60 | 56-58 | 64-66 | 92-98 |
| Thigh (left) | 73-77 | 91-92 | 61-67 | 59-66 | 67-72 | - |
| Thigh (right) | 78-82 | 93-94 | 68-74 | 67-74 | 73-78 | - |
| Knee (left) | 83 | - | 75 | 75 | 79 | - |
| Knee (right) | 84 | - | 76 | 76 | 80 | - |
| Lower leg (left) | 85-91 | 95-97 | 77-86 | 77-86 | 81-88 | 99 |
| Lower leg (right) | 92-98 | 98-00 | 87-96 | 87-96 | 89-96 | 00 |
| Foot (left) | 99 | - | 97-98 | 97-98 | 97-98 | - |
| Foot (right) | 00 | - | 99-00 | 99-00 | 99-00 | - |

# American Revolution 1775-83 - all hits (gameplay): Musket / Rifle / Pistol / Artillery

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Flanked | Cover |
|---|---|---|---|---|---|---|
| Skull (left) | 01-07 | 01-10 | 01-03 | 01-08 | 01-06 | 01-14 |
| Skull (right) | 08-14 | 11-19 | 04-06 | 09-16 | 07-12 | 15-28 |
| Face | 15-21 | 20-30 | 07-10 | 17 | 13-18 | 29-44 |
| Neck | 22-26 | 31-37 | 11-12 | 18-24 | 19-22 | 45-55 |
| Chest (left) | 27-34 | 38-44 | 13-20 | 25 | 23-29 | 56-59 |
| Chest (right) | 35-42 | 45-51 | 21-28 | 26 | 30-36 | 60-63 |
| Abdomen | 43-48 | 52-57 | 29-36 | 27 | 37-41 | 64 |
| Groin / pelvis | 49-53 | 58-62 | 37-45 | 28 | 42-45 | - |
| Upper back | 54-58 | 63-66 | 46-49 | 29-48 | 46-54 | 65-66 |
| Lower back | 59-62 | 67-70 | 50-53 | 49-62 | 55-63 | 67 |
| Shoulder (left) | 63-64 | 71-73 | 54-55 | 63-64 | 64-65 | 68-72 |
| Shoulder (right) | 65-66 | 74-76 | 56-57 | 65-66 | 66-67 | 73-77 |
| Upper arm (left) | 67-69 | 77-79 | 58-60 | 67-68 | 68-69 | 78-81 |
| Upper arm (right) | 70-72 | 80-82 | 61-63 | 69-70 | 70-71 | 82-85 |
| Forearm (left) | 73-75 | 83-85 | 64-66 | 71-72 | 72-73 | 86-89 |
| Forearm (right) | 76-78 | 86-88 | 67-69 | 73-74 | 74-75 | 90-93 |
| Hand (left) | 79-80 | 89-91 | 70-71 | 75 | 76-77 | 94-97 |
| Hand (right) | 81-82 | 92-94 | 72-73 | 76 | 78-79 | 98-00 |
| Thigh (left) | 83-85 | 95 | 74-78 | 77-80 | 80-83 | - |
| Thigh (right) | 86-88 | 96 | 79-83 | 81-84 | 84-86 | - |
| Knee (left) | 89 | - | 84 | 85 | 87 | - |
| Knee (right) | 90 | - | 85 | 86 | 88 | - |
| Lower leg (left) | 91-94 | 97-98 | 86-92 | 87-92 | 89-93 | - |
| Lower leg (right) | 95-98 | 99-00 | 93-98 | 93-98 | 94-98 | - |
| Foot (left) | 99 | - | 99 | 99 | 99 | - |
| Foot (right) | 00 | - | 00 | 00 | 00 | - |

# Peninsular War 1808-14 - French officers' wounds: Musket / Rifle / Pistol

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Down | Fleeing | Flanked | Warding | Cover |
|---|---|---|---|---|---|---|---|---|
| Skull (left) | 01-04 | 01-07 | 01-02 | 01-08 | 01-05 | 01-04 | 01-03 | 01-10 |
| Skull (right) | 05-08 | 08-14 | 03 | 09-16 | 06-10 | 05-08 | 04-06 | 11-20 |
| Face | 09-10 | 15-18 | 04 | 17-19 | - | 09-10 | 07-08 | 21-25 |
| Neck | 11 | 19-20 | - | 20-22 | 11-12 | 11 | 09 | 26-28 |
| Chest (left) | 12-13 | 21-22 | 05-06 | 23 | - | 12-13 | 10-11 | 29 |
| Chest (right) | 14-15 | 23-24 | 07-08 | 24 | - | 14-15 | 12-13 | 30 |
| Abdomen | 16-18 | 25-27 | 09-11 | 25-26 | - | 16-17 | 14-15 | - |
| Groin / pelvis | 19-21 | 28-30 | 12-15 | 27-28 | 13 | 18-20 | 16-17 | - |
| Upper back | - | - | - | - | 14 | - | - | - |
| Lower back | 22 | 31 | 16 | 29-30 | 15-18 | 21-22 | 18 | - |
| Shoulder (left) | 23-26 | 32-39 | 17-20 | 31-35 | 19-21 | 23-26 | 19-23 | 31-42 |
| Shoulder (right) | 27-30 | 40-47 | 21-24 | 36-40 | 22-24 | 27-30 | 24-28 | 43-54 |
| Upper arm (left) | 31-38 | 48-58 | 25-31 | 41-49 | 25-29 | 31-37 | 29-38 | 55-68 |
| Upper arm (right) | 39-46 | 59-69 | 32-38 | 50-58 | 30-34 | 38-44 | 39-48 | 69-82 |
| Forearm (left) | 47 | 70-71 | 39 | 59-60 | 35 | 45 | 49-50 | 83-84 |
| Forearm (right) | 48 | 72 | 40 | 61-62 | 36 | 46 | 51-52 | 85-86 |
| Hand (left) | 49-51 | 73-76 | 41-42 | 63-67 | 37-38 | 47-48 | 53-57 | 87-91 |
| Hand (right) | 52-54 | 77-80 | 43-44 | 68-72 | 39-40 | 49-50 | 58-62 | 92-96 |
| Thigh (left) | 55-64 | 81-84 | 45-56 | 73-78 | 41-53 | 51-61 | 63-70 | 97 |
| Thigh (right) | 65-74 | 85-88 | 57-68 | 79-84 | 54-66 | 62-72 | 71-78 | 98 |
| Knee (left) | 75-76 | 89 | 69-71 | 85 | 67-69 | 73-74 | 79-80 | - |
| Knee (right) | 77-78 | 90 | 72-74 | 86 | 70-72 | 75-76 | 81-82 | - |
| Lower leg (left) | 79-86 | 91-94 | 75-84 | 87-91 | 73-83 | 77-85 | 83-89 | 99 |
| Lower leg (right) | 87-94 | 95-98 | 85-94 | 92-96 | 84-94 | 86-94 | 90-96 | 00 |
| Foot (left) | 95-97 | 99 | 95-97 | 97-98 | 95-97 | 95-97 | 97-98 | - |
| Foot (right) | 98-00 | 00 | 98-00 | 99-00 | 98-00 | 98-00 | 99-00 | - |

# Napoleonic era - all hits (gameplay): Musket / Rifle / Pistol / Artillery

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Flanked | Cover |
|---|---|---|---|---|---|---|
| Skull (left) | 01-06 | 01-10 | 01-03 | 01-08 | 01-06 | 01-15 |
| Skull (right) | 07-12 | 11-20 | 04-06 | 09-16 | 07-12 | 16-30 |
| Face | 13-17 | 21-28 | 07-08 | 17 | 13-16 | 31-41 |
| Neck | 18-20 | 29-33 | 09 | 18-21 | 17-18 | 42-48 |
| Chest (left) | 21-25 | 34-37 | 10-13 | 22 | 19-22 | 49-50 |
| Chest (right) | 26-30 | 38-41 | 14-17 | 23 | 23-26 | 51-52 |
| Abdomen | 31-35 | 42-45 | 18-22 | 24 | 27-30 | 53 |
| Groin / pelvis | 36-38 | 46-48 | 23-28 | 25 | 31-33 | - |
| Upper back | 39-40 | 49-50 | 29-30 | 26-33 | 34-37 | 54 |
| Lower back | 41-42 | 51-52 | 31-32 | 34-40 | 38-42 | - |
| Shoulder (left) | 43-45 | 53-57 | 33-35 | 41-42 | 43-45 | 55-61 |
| Shoulder (right) | 46-48 | 58-62 | 36-38 | 43-44 | 46-48 | 62-68 |
| Upper arm (left) | 49-54 | 63-70 | 39-44 | 45-48 | 49-53 | 69-78 |
| Upper arm (right) | 55-60 | 71-78 | 45-50 | 49-52 | 54-58 | 79-88 |
| Forearm (left) | 61 | 79 | 51 | 53 | 59 | 89-90 |
| Forearm (right) | 62 | 80 | 52 | 54 | 60 | 91-92 |
| Hand (left) | 63-64 | 81-83 | 53-54 | 55 | 61-62 | 93-96 |
| Hand (right) | 65-66 | 84-86 | 55-56 | 56 | 63-64 | 97-00 |
| Thigh (left) | 67-73 | 87-89 | 57-65 | 57-65 | 65-71 | - |
| Thigh (right) | 74-80 | 90-92 | 66-74 | 66-74 | 72-78 | - |
| Knee (left) | 81-82 | 93 | 75-76 | 75-76 | 79-80 | - |
| Knee (right) | 83-84 | 94 | 77-78 | 77-78 | 81-82 | - |
| Lower leg (left) | 85-90 | 95-96 | 79-86 | 79-86 | 83-89 | - |
| Lower leg (right) | 91-96 | 97-98 | 87-94 | 87-94 | 90-96 | - |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | 97-98 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | 99-00 | - |

# War of 1812 - all hits (gameplay, borrowed records): Musket / Rifle / Pistol / Artillery

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Flanked | Cover |
|---|---|---|---|---|---|---|
| Skull (left) | 01-07 | 01-11 | 01-03 | 01-09 | 01-07 | 01-15 |
| Skull (right) | 08-14 | 12-22 | 04-06 | 10-18 | 08-14 | 16-30 |
| Face | 15-20 | 23-31 | 07-08 | 19 | 15-19 | 31-43 |
| Neck | 21-24 | 32-36 | 09-10 | 20-23 | 20-22 | 44-51 |
| Chest (left) | 25-29 | 37-41 | 11-15 | 24 | 23-26 | 52-53 |
| Chest (right) | 30-34 | 42-46 | 16-20 | 25 | 27-30 | 54-55 |
| Abdomen | 35-39 | 47-51 | 21-26 | 26 | 31-34 | 56 |
| Groin / pelvis | 40-43 | 52-55 | 27-32 | 27 | 35-37 | - |
| Upper back | 44-45 | 56-57 | 33-34 | 28-38 | 38-42 | 57 |
| Lower back | 46-48 | 58-60 | 35-36 | 39-46 | 43-48 | - |
| Shoulder (left) | 49-51 | 61-64 | 37-39 | 47-48 | 49-50 | 58-64 |
| Shoulder (right) | 52-54 | 65-68 | 40-42 | 49-50 | 51-52 | 65-71 |
| Upper arm (left) | 55-60 | 69-75 | 43-47 | 51-54 | 53-57 | 72-80 |
| Upper arm (right) | 61-66 | 76-82 | 48-52 | 55-58 | 58-62 | 81-89 |
| Forearm (left) | 67 | 83 | 53 | 59 | 63 | 90-91 |
| Forearm (right) | 68 | 84 | 54 | 60 | 64 | 92 |
| Hand (left) | 69-70 | 85-86 | 55-56 | 61 | 65-66 | 93-95 |
| Hand (right) | 71-72 | 87-88 | 57-58 | 62 | 67-68 | 96-98 |
| Thigh (left) | 73-78 | 89-90 | 59-67 | 63-70 | 69-75 | 99 |
| Thigh (right) | 79-84 | 91-92 | 68-76 | 71-78 | 76-82 | 00 |
| Knee (left) | 85 | 93 | 77-78 | 79-80 | 83 | - |
| Knee (right) | 86 | 94 | 79-80 | 81-82 | 84 | - |
| Lower leg (left) | 87-91 | 95-96 | 81-88 | 83-89 | 85-90 | - |
| Lower leg (right) | 92-96 | 97-98 | 89-96 | 90-96 | 91-96 | - |
| Foot (left) | 97-98 | 99 | 97-98 | 97-98 | 97-98 | - |
| Foot (right) | 99-00 | 00 | 99-00 | 99-00 | 99-00 | - |

# Civil War 1861-65 - wounded (surgeon's records): Rifle-musket / Artillery / Revolver

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-03 | 01-05 | 01 | 01-04 | 01-07 |
| Skull (right) | 04-06 | 06-10 | 02 | 05-08 | 08-14 |
| Face | 07-09 | 11-15 | 03 | - | 15-22 |
| Neck | 10-11 | 16-19 | 04 | 09-11 | 23-27 |
| Chest (left) | 12-15 | 20-23 | 05-07 | 12 | 28-29 |
| Chest (right) | 16-19 | 24-27 | 08-10 | 13 | 30-31 |
| Abdomen | 20-23 | 28-31 | 11-14 | 14 | - |
| Groin / pelvis | 24-25 | 32-33 | 15-17 | - | - |
| Upper back | 26-28 | 34-36 | 18-19 | 15-25 | 32 |
| Lower back | 29-30 | 37-38 | 20-21 | 26-32 | - |
| Shoulder (left) | 31-34 | 39-44 | 22-24 | 33-34 | 33-41 |
| Shoulder (right) | 35-38 | 45-50 | 25-27 | 35-36 | 42-50 |
| Upper arm (left) | 39-43 | 51-57 | 28-32 | 37-39 | 51-59 |
| Upper arm (right) | 44-48 | 58-64 | 33-37 | 40-42 | 60-68 |
| Forearm (left) | 49-53 | 65-71 | 38-42 | 43-45 | 69-77 |
| Forearm (right) | 54-58 | 72-78 | 43-46 | 46-48 | 78-86 |
| Hand (left) | 59-62 | 79-82 | 47-49 | 49-50 | 87-92 |
| Hand (right) | 63-66 | 83-86 | 50-52 | 51-52 | 93-98 |
| Thigh (left) | 67-73 | 87-89 | 53-62 | 53-62 | 99 |
| Thigh (right) | 74-80 | 90-92 | 63-72 | 63-72 | 00 |
| Knee (left) | 81-82 | 93 | 73-75 | 73-75 | - |
| Knee (right) | 83-84 | 94 | 76-78 | 76-78 | - |
| Lower leg (left) | 85-90 | 95-96 | 79-86 | 79-86 | - |
| Lower leg (right) | 91-96 | 97-98 | 87-94 | 87-94 | - |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Civil War 1861-65 - killed in action: Rifle-musket / Artillery / Revolver

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-11 | 01-14 | 01-06 | 01-13 | 01-20 |
| Skull (right) | 12-21 | 15-28 | 07-12 | 14-26 | 21-40 |
| Face | 22-34 | 29-44 | 13-19 | 27-28 | 41-65 |
| Neck | 35-42 | 45-55 | 20-24 | 29-38 | 66-81 |
| Chest (left) | 43-53 | 56-64 | 25-37 | 39-40 | 82-85 |
| Chest (right) | 54-64 | 65-73 | 38-50 | 41-42 | 86-89 |
| Abdomen | 65-74 | 74-81 | 51-65 | 43-44 | 90 |
| Groin / pelvis | 75-79 | 82-85 | 66-75 | 45 | - |
| Upper back | 80-86 | 86-91 | 76-83 | 46-75 | 91-93 |
| Lower back | 87-92 | 92-96 | 84-90 | 76-94 | 94 |
| Shoulder (left) | - | - | - | - | 95 |
| Shoulder (right) | - | - | - | - | 96 |
| Upper arm (left) | 93 | 97 | - | - | 97 |
| Upper arm (right) | 94 | 98 | - | - | 98 |
| Forearm (left) | 95 | 99 | - | - | 99 |
| Forearm (right) | 96 | 00 | - | - | 00 |
| Thigh (left) | 97 | - | 91-92 | 95 | - |
| Thigh (right) | 98 | - | 93-94 | 96 | - |
| Knee (left) | - | - | 95 | - | - |
| Knee (right) | - | - | 96 | - | - |
| Lower leg (left) | 99 | - | 97 | 97 | - |
| Lower leg (right) | 00 | - | 98 | 98 | - |
| Foot (left) | - | - | 99 | 99 | - |
| Foot (right) | - | - | 00 | 00 | - |

# Civil War 1861-65 - all hits (gameplay): Rifle-musket / Artillery / Revolver

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-04 | 01-07 | 01-02 | 01-05 | 01-10 |
| Skull (right) | 05-08 | 08-14 | 03-04 | 06-10 | 11-20 |
| Face | 09-13 | 15-22 | 05-06 | - | 21-32 |
| Neck | 14-16 | 23-27 | 07 | 11-14 | 33-40 |
| Chest (left) | 17-22 | 28-32 | 08-12 | 15 | 41-43 |
| Chest (right) | 23-27 | 33-37 | 13-17 | 16 | 44-46 |
| Abdomen | 28-32 | 38-42 | 18-23 | 17 | 47 |
| Groin / pelvis | 33-35 | 43-44 | 24-27 | - | - |
| Upper back | 36-39 | 45-47 | 28-30 | 18-31 | 48-49 |
| Lower back | 40-42 | 48-50 | 31-33 | 32-40 | - |
| Shoulder (left) | 43-45 | 51-55 | 34-36 | 41-42 | 50-56 |
| Shoulder (right) | 46-48 | 56-60 | 37-39 | 43-44 | 57-63 |
| Upper arm (left) | 49-52 | 61-65 | 40-43 | 45-47 | 64-70 |
| Upper arm (right) | 53-56 | 66-70 | 44-47 | 48-50 | 71-77 |
| Forearm (left) | 57-60 | 71-75 | 48-51 | 51-53 | 78-84 |
| Forearm (right) | 61-64 | 76-80 | 52-55 | 54-56 | 85-91 |
| Hand (left) | 65-67 | 81-84 | 56-58 | 57-58 | 92-96 |
| Hand (right) | 68-70 | 85-88 | 59-60 | 59-60 | 97-00 |
| Thigh (left) | 71-76 | 89-90 | 61-68 | 61-68 | - |
| Thigh (right) | 77-82 | 91-92 | 69-76 | 69-76 | - |
| Knee (left) | 83-84 | 93 | 77-78 | 77-78 | - |
| Knee (right) | 85-86 | 94 | 79-80 | 79-80 | - |
| Lower leg (left) | 87-91 | 95-96 | 81-87 | 81-87 | - |
| Lower leg (right) | 92-96 | 97-98 | 88-94 | 88-94 | - |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Indian Wars 1865-71 - arrow wounds (US Army surgeons): Bow

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Flanked | Cover |
|---|---|---|---|---|---|---|
| Skull (left) | 01-03 | 01-05 | 01-02 | 01-04 | 01-03 | 01-08 |
| Skull (right) | 04-06 | 06-10 | 03 | 05-08 | 04-06 | 09-16 |
| Face | 07-08 | 11-14 | 04 | - | 07-08 | 17-22 |
| Neck | 09-11 | 15-18 | 05 | 09-11 | 09-10 | 23-28 |
| Chest (left) | 12-21 | 19-27 | 06-14 | 12 | 11-18 | 29-33 |
| Chest (right) | 22-31 | 28-36 | 15-23 | 13 | 19-26 | 34-38 |
| Abdomen | 32-37 | 37-41 | 24-30 | 14 | 27-30 | 39 |
| Groin / pelvis | 38-44 | 42-48 | 31-42 | 15 | 31-36 | 40 |
| Upper back | 45-56 | 49-59 | 43-53 | 16-57 | 37-58 | 41-46 |
| Lower back | 57-62 | 60-64 | 54-58 | 58-73 | 59-69 | 47 |
| Shoulder (left) | 63-65 | 65-68 | 59-61 | 74-75 | 70-71 | 48-54 |
| Shoulder (right) | 66-68 | 69-72 | 62-64 | 76 | 72-73 | 55-61 |
| Upper arm (left) | 69-75 | 73-79 | 65-70 | 77-80 | 74-78 | 62-72 |
| Upper arm (right) | 76-82 | 80-86 | 71-76 | 81-84 | 79-83 | 73-83 |
| Forearm (left) | 83-85 | 87-90 | 77-79 | 85-86 | 84-86 | 84-89 |
| Forearm (right) | 86-88 | 91-94 | 80-82 | 87-88 | 87-88 | 90-94 |
| Hand (left) | 89-90 | 95-96 | 83-84 | 89 | 89-90 | 95-97 |
| Hand (right) | 91-92 | 97-98 | 85-86 | 90 | 91-92 | 98-00 |
| Thigh (left) | 93-94 | 99 | 87-90 | 91-93 | 93-94 | - |
| Thigh (right) | 95-96 | 00 | 91-94 | 94-96 | 95-96 | - |
| Knee (left) | 97 | - | 95 | 97 | 97 | - |
| Knee (right) | 98 | - | 96 | 98 | 98 | - |
| Lower leg (left) | 99 | - | 97 | 99 | 99 | - |
| Lower leg (right) | 00 | - | 98 | 00 | 00 | - |
| Foot (left) | - | - | 99 | - | - | - |
| Foot (right) | - | - | 00 | - | - | - |

# WWI 1914-18 - penetrating wounds: Rifle / MG / Pistol/SMG / Grenade / Mortar

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-08 | 01-02 | 01-05 | 01-11 |
| Skull (right) | 06-10 | 09-16 | 03-04 | 06-10 | 12-22 |
| Face | 11-15 | 17-26 | 05-06 | 11 | 23-35 |
| Neck | 16-19 | 27-32 | 07-08 | 12-15 | 36-43 |
| Chest (left) | 20-21 | 33-34 | 09 | - | 44 |
| Chest (right) | 22-23 | 35-36 | 10 | - | 45 |
| Abdomen | 24 | 37 | 11 | - | - |
| Groin / pelvis | - | 38 | 12 | - | - |
| Upper back | 25 | 39 | 13 | 16-19 | - |
| Lower back | 26 | 40 | - | 20-21 | - |
| Shoulder (left) | 27-29 | 41-45 | 14-16 | 22-23 | 46-52 |
| Shoulder (right) | 30-32 | 46-50 | 17-19 | 24-25 | 53-59 |
| Upper arm (left) | 33-36 | 51-56 | 20-23 | 26-28 | 60-66 |
| Upper arm (right) | 37-40 | 57-62 | 24-27 | 29-31 | 67-73 |
| Forearm (left) | 41-44 | 63-68 | 28-31 | 32-34 | 74-80 |
| Forearm (right) | 45-48 | 69-74 | 32-35 | 35-37 | 81-87 |
| Hand (left) | 49-51 | 75-78 | 36-38 | 38-39 | 88-92 |
| Hand (right) | 52-54 | 79-82 | 39-40 | 40-41 | 93-97 |
| Thigh (left) | 55-63 | 83-86 | 41-52 | 42-53 | 98 |
| Thigh (right) | 64-72 | 87-90 | 53-64 | 54-65 | 99 |
| Knee (left) | 73-75 | 91 | 65-68 | 66-69 | - |
| Knee (right) | 76-78 | 92 | 69-72 | 70-72 | - |
| Lower leg (left) | 79-86 | 93-95 | 73-82 | 73-82 | 00 |
| Lower leg (right) | 87-94 | 96-98 | 83-92 | 83-92 | - |
| Foot (left) | 95-97 | 99 | 93-96 | 93-96 | - |
| Foot (right) | 98-00 | 00 | 97-00 | 97-00 | - |

# WWII Italy 1944 - killed in action: Rifle / MG / Pistol/SMG / Grenade / Mortar / Mine/IED / Frags

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-03 | 01-05 | 01 | 01-03 | 01-08 |
| Skull (right) | 04-06 | 06-10 | 02 | 04-06 | 09-16 |
| Face | 07-09 | 11-16 | 03 | - | 17-25 |
| Neck | 10-13 | 17-22 | 04-05 | 07-10 | 26-36 |
| Chest (left) | 14-23 | 23-32 | 06-13 | 11-12 | 37-42 |
| Chest (right) | 24-33 | 33-42 | 14-21 | 13 | 43-48 |
| Abdomen | 34-37 | 43-46 | 22-26 | 14 | 49 |
| Groin / pelvis | 38-39 | 47-49 | 27-30 | 15 | - |
| Upper back | 40-45 | 50-55 | 31-35 | 16-38 | 50-53 |
| Lower back | 46-50 | 56-60 | 36-39 | 39-52 | 54 |
| Shoulder (left) | 51-52 | 61-63 | 40-41 | 53 | 55-60 |
| Shoulder (right) | 53-54 | 64-66 | 42-43 | 54 | 61-66 |
| Upper arm (left) | 55-57 | 67-70 | 44-46 | 55-56 | 67-72 |
| Upper arm (right) | 58-60 | 71-74 | 47-49 | 57-58 | 73-78 |
| Forearm (left) | 61-63 | 75-78 | 50-52 | 59-60 | 79-84 |
| Forearm (right) | 64-66 | 79-82 | 53-55 | 61-62 | 85-90 |
| Hand (left) | 67-68 | 83-85 | 56-57 | 63 | 91-94 |
| Hand (right) | 69-70 | 86-88 | 58-59 | 64 | 95-98 |
| Thigh (left) | 71-76 | 89-90 | 60-67 | 65-71 | 99 |
| Thigh (right) | 77-82 | 91-92 | 68-75 | 72-78 | 00 |
| Knee (left) | 83-84 | 93 | 76-78 | 79-80 | - |
| Knee (right) | 85-86 | 94 | 79-80 | 81-82 | - |
| Lower leg (left) | 87-91 | 95-96 | 81-87 | 83-88 | - |
| Lower leg (right) | 92-96 | 97-98 | 88-94 | 89-94 | - |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Bougainville 1944 - wounded: Rifle / Pistol/SMG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-08 | 01-02 | 01-06 | 01-12 |
| Skull (right) | 06-10 | 09-16 | 03-04 | 07-12 | 13-24 |
| Face | 11-16 | 17-26 | 05-07 | 13 | 25-38 |
| Neck | 17-20 | 27-33 | 08-09 | 14-18 | 39-47 |
| Chest (left) | 21-25 | 34-38 | 10-13 | 19 | 48-49 |
| Chest (right) | 26-30 | 39-43 | 14-17 | 20 | 50-51 |
| Abdomen | 31-33 | 44-46 | 18-20 | - | - |
| Groin / pelvis | 34 | 47 | 21-22 | - | - |
| Upper back | 35-37 | 48-50 | 23-25 | 21-33 | 52 |
| Lower back | 38-39 | 51-52 | 26-27 | 34-38 | - |
| Shoulder (left) | 40-42 | 53-56 | 28-30 | 39-40 | 53-58 |
| Shoulder (right) | 43-45 | 57-60 | 31-32 | 41-42 | 59-64 |
| Upper arm (left) | 46-49 | 61-65 | 33-36 | 43-44 | 65-70 |
| Upper arm (right) | 50-53 | 66-70 | 37-40 | 45-46 | 71-76 |
| Forearm (left) | 54-57 | 71-75 | 41-44 | 47-48 | 77-82 |
| Forearm (right) | 58-61 | 76-80 | 45-48 | 49-50 | 83-88 |
| Hand (left) | 62-64 | 81-83 | 49-50 | 51-52 | 89-92 |
| Hand (right) | 65-67 | 84-86 | 51-52 | 53-54 | 93-96 |
| Thigh (left) | 68-74 | 87-89 | 53-62 | 55-63 | 97 |
| Thigh (right) | 75-80 | 90-92 | 63-72 | 64-72 | 98 |
| Knee (left) | 81-82 | 93 | 73-75 | 73-75 | - |
| Knee (right) | 83-84 | 94 | 76-78 | 76-78 | - |
| Lower leg (left) | 85-90 | 95-96 | 79-86 | 79-86 | 99 |
| Lower leg (right) | 91-96 | 97-98 | 87-94 | 87-94 | 00 |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Bougainville 1944 - killed: Rifle / Pistol/SMG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-12 | 01-15 | 01-07 | 01-15 | 01-22 |
| Skull (right) | 13-24 | 16-30 | 08-14 | 16-30 | 23-44 |
| Face | 25-38 | 31-48 | 15-23 | 31-32 | 45-70 |
| Neck | 39-48 | 49-60 | 24-29 | 33-44 | 71-87 |
| Chest (left) | 49-60 | 61-69 | 30-43 | 45-46 | 88-91 |
| Chest (right) | 61-72 | 70-78 | 44-57 | 47-48 | 92-95 |
| Abdomen | 73-80 | 79-85 | 58-69 | 49 | 96 |
| Groin / pelvis | 81-84 | 86-88 | 70-77 | 50 | - |
| Upper back | 85-91 | 89-94 | 78-86 | 51-80 | 97-99 |
| Lower back | 92-96 | 95-98 | 87-92 | 81-95 | 00 |
| Thigh (left) | 97 | 99 | 93-94 | 96 | - |
| Thigh (right) | 98 | 00 | 95-96 | 97 | - |
| Lower leg (left) | 99 | - | 97 | 98 | - |
| Lower leg (right) | 00 | - | 98 | 99 | - |
| Foot (left) | - | - | 99 | 00 | - |
| Foot (right) | - | - | 00 | - | - |

# Bougainville 1944 - all hits: Rifle / Pistol/SMG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-07 | 01-10 | 01-03 | 01-08 | 01-14 |
| Skull (right) | 08-14 | 11-20 | 04-06 | 09-16 | 15-28 |
| Face | 15-22 | 21-32 | 07-10 | 17 | 29-45 |
| Neck | 23-27 | 33-40 | 11-12 | 18-24 | 46-56 |
| Chest (left) | 28-33 | 41-46 | 13-18 | 25 | 57-59 |
| Chest (right) | 34-39 | 47-52 | 19-24 | 26 | 60-62 |
| Abdomen | 40-43 | 53-56 | 25-29 | 27 | - |
| Groin / pelvis | 44-45 | 57-58 | 30-32 | 28 | - |
| Upper back | 46-49 | 59-62 | 33-36 | 29-45 | 63-64 |
| Lower back | 50-51 | 63-64 | 37-38 | 46-52 | - |
| Shoulder (left) | 52-53 | 65-67 | 39-40 | 53 | 65-69 |
| Shoulder (right) | 54-55 | 68-70 | 41-42 | 54 | 70-74 |
| Upper arm (left) | 56-58 | 71-74 | 43-45 | 55-56 | 75-79 |
| Upper arm (right) | 59-61 | 75-78 | 46-48 | 57-58 | 80-84 |
| Forearm (left) | 62-64 | 79-82 | 49-51 | 59-60 | 85-89 |
| Forearm (right) | 65-67 | 83-86 | 52-54 | 61-62 | 90-94 |
| Hand (left) | 68-69 | 87-88 | 55-56 | 63 | 95-97 |
| Hand (right) | 70-71 | 89-90 | 57-58 | 64 | 98-00 |
| Thigh (left) | 72-77 | 91-92 | 59-66 | 65-71 | - |
| Thigh (right) | 78-82 | 93-94 | 67-74 | 72-78 | - |
| Knee (left) | 83-84 | - | 75-77 | 79-80 | - |
| Knee (right) | 85-86 | - | 78-80 | 81-82 | - |
| Lower leg (left) | 87-91 | 95-96 | 81-87 | 83-88 | - |
| Lower leg (right) | 92-96 | 97-98 | 88-94 | 89-94 | - |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# WWII 1941-45 - hospitalized (Owens comparison set): Rifle / Pistol/SMG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-09 | 01-02 | 01-06 | 01-12 |
| Skull (right) | 06-10 | 10-18 | 03-04 | 07-12 | 13-24 |
| Face | 11-16 | 19-28 | 05-07 | 13 | 25-39 |
| Neck | 17-20 | 29-35 | 08-09 | 14-18 | 40-49 |
| Chest (left) | 21-25 | 36-40 | 10-14 | 19 | 50-52 |
| Chest (right) | 26-30 | 41-45 | 15-19 | 20 | 53-55 |
| Abdomen | 31-34 | 46-49 | 20-23 | 21 | 56 |
| Groin / pelvis | 35-36 | 50-51 | 24-26 | - | - |
| Upper back | 37-39 | 52-54 | 27-29 | 22-35 | 57-58 |
| Lower back | 40-41 | 55-56 | 30-31 | 36-42 | - |
| Shoulder (left) | 42-43 | 57-60 | 32-33 | 43 | 59-64 |
| Shoulder (right) | 44-45 | 61-64 | 34-35 | 44 | 65-70 |
| Upper arm (left) | 46-49 | 65-68 | 36-38 | 45-46 | 71-75 |
| Upper arm (right) | 50-53 | 69-72 | 39-41 | 47-48 | 76-80 |
| Forearm (left) | 54-57 | 73-76 | 42-44 | 49-50 | 81-85 |
| Forearm (right) | 58-61 | 77-80 | 45-47 | 51-52 | 86-90 |
| Hand (left) | 62-63 | 81-83 | 48-49 | 53 | 91-94 |
| Hand (right) | 64-65 | 84-86 | 50-51 | 54 | 95-98 |
| Thigh (left) | 66-72 | 87-89 | 52-61 | 55-63 | 99 |
| Thigh (right) | 73-79 | 90-92 | 62-71 | 64-72 | 00 |
| Knee (left) | 80-81 | 93 | 72-74 | 73-75 | - |
| Knee (right) | 82-83 | 94 | 75-77 | 76-78 | - |
| Lower leg (left) | 84-89 | 95-96 | 78-85 | 79-86 | - |
| Lower leg (right) | 90-95 | 97-98 | 86-93 | 87-94 | - |
| Foot (left) | 96-98 | 99 | 94-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Korea 1950-53 - hospitalized wounded: Rifle / Pistol/SMG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-07 | 01-02 | 01-06 | 01-10 |
| Skull (right) | 06-09 | 08-14 | 03-04 | 07-12 | 11-20 |
| Face | 10-14 | 15-23 | 05-06 | 13 | 21-32 |
| Neck | 15-18 | 24-29 | 07-08 | 14-17 | 33-40 |
| Chest (left) | 19-21 | 30-32 | 09-10 | - | 41 |
| Chest (right) | 22-24 | 33-35 | 11-12 | - | 42 |
| Abdomen | 25-28 | 36-38 | 13-16 | 18 | 43 |
| Groin / pelvis | 29-30 | 39-40 | 17-19 | - | - |
| Upper back | 31-32 | 41-42 | 20 | 19-25 | 44 |
| Lower back | 33-34 | 43-44 | 21-22 | 26-32 | - |
| Shoulder (left) | 35-37 | 45-49 | 23-25 | 33-34 | 45-51 |
| Shoulder (right) | 38-40 | 50-54 | 26-28 | 35-36 | 52-58 |
| Upper arm (left) | 41-45 | 55-60 | 29-32 | 37-39 | 59-65 |
| Upper arm (right) | 46-50 | 61-66 | 33-36 | 40-42 | 66-72 |
| Forearm (left) | 51-55 | 67-72 | 37-40 | 43-45 | 73-79 |
| Forearm (right) | 56-60 | 73-78 | 41-44 | 46-48 | 80-86 |
| Hand (left) | 61-63 | 79-82 | 45-47 | 49-50 | 87-91 |
| Hand (right) | 64-66 | 83-86 | 48-50 | 51-52 | 92-96 |
| Thigh (left) | 67-73 | 87-89 | 51-60 | 53-62 | 97 |
| Thigh (right) | 74-80 | 90-92 | 61-70 | 63-72 | 98 |
| Knee (left) | 81-82 | 93 | 71-73 | 73-75 | - |
| Knee (right) | 83-84 | 94 | 74-76 | 76-78 | - |
| Lower leg (left) | 85-90 | 95-96 | 77-85 | 79-86 | 99 |
| Lower leg (right) | 91-96 | 97-98 | 86-94 | 87-94 | 00 |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Korea 1950-53 - hospitalized (Owens comparison set): Rifle / Pistol/SMG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-09 | 01-02 | 01-06 | 01-12 |
| Skull (right) | 06-10 | 10-18 | 03-04 | 07-12 | 13-24 |
| Face | 11-16 | 19-28 | 05-07 | 13 | 25-39 |
| Neck | 17-20 | 29-35 | 08-09 | 14-18 | 40-49 |
| Chest (left) | 21-24 | 36-39 | 10-12 | 19 | 50-51 |
| Chest (right) | 25-28 | 40-43 | 13-15 | 20 | 52-53 |
| Abdomen | 29-32 | 44-47 | 16-20 | 21 | 54 |
| Groin / pelvis | 33-34 | 48-49 | 21-24 | - | - |
| Upper back | 35-37 | 50-51 | 25-26 | 22-31 | 55 |
| Lower back | 38-40 | 52-54 | 27-28 | 32-39 | - |
| Shoulder (left) | 41-42 | 55-58 | 29-30 | 40-41 | 56-61 |
| Shoulder (right) | 43-44 | 59-62 | 31-32 | 42-43 | 62-67 |
| Upper arm (left) | 45-48 | 63-66 | 33-35 | 44-45 | 68-73 |
| Upper arm (right) | 49-52 | 67-70 | 36-38 | 46-47 | 74-79 |
| Forearm (left) | 53-56 | 71-74 | 39-41 | 48-49 | 80-85 |
| Forearm (right) | 57-60 | 75-78 | 42-44 | 50-51 | 86-91 |
| Hand (left) | 61-62 | 79-81 | 45-46 | 52-53 | 92-95 |
| Hand (right) | 63-64 | 82-84 | 47-48 | 54 | 96-99 |
| Thigh (left) | 65-71 | 85-87 | 49-58 | 55-63 | 00 |
| Thigh (right) | 72-78 | 88-90 | 59-68 | 64-72 | - |
| Knee (left) | 79-80 | 91 | 69-71 | 73-75 | - |
| Knee (right) | 81-82 | 92 | 72-74 | 76-78 | - |
| Lower leg (left) | 83-88 | 93-95 | 75-83 | 79-86 | - |
| Lower leg (right) | 89-94 | 96-98 | 84-92 | 87-94 | - |
| Foot (left) | 95-97 | 99 | 93-96 | 95-97 | - |
| Foot (right) | 98-00 | 00 | 97-00 | 98-00 | - |

# Vietnam 1965-70 - hospitalized wounded: Assault rifle / Pistol/SMG / RPG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-08 | 01-02 | 01-06 | 01-12 |
| Skull (right) | 06-10 | 09-16 | 03-04 | 07-12 | 13-24 |
| Face | 11-16 | 17-26 | 05-07 | 13 | 25-38 |
| Neck | 17-20 | 27-33 | 08-09 | 14-18 | 39-47 |
| Chest (left) | 21-24 | 34-37 | 10-12 | 19 | 48-49 |
| Chest (right) | 25-28 | 38-41 | 13-15 | 20 | 50-51 |
| Abdomen | 29-31 | 42-44 | 16-19 | - | - |
| Groin / pelvis | 32-33 | 45-46 | 20-22 | - | - |
| Upper back | 34-35 | 47-48 | 23-24 | 21-29 | 52 |
| Lower back | 36-37 | 49-50 | 25-26 | 30-35 | - |
| Shoulder (left) | 38-40 | 51-54 | 27-28 | 36-37 | 53-58 |
| Shoulder (right) | 41-42 | 55-58 | 29-30 | 38-39 | 59-64 |
| Upper arm (left) | 43-46 | 59-63 | 31-33 | 40-41 | 65-70 |
| Upper arm (right) | 47-50 | 64-68 | 34-36 | 42-43 | 71-76 |
| Forearm (left) | 51-54 | 69-73 | 37-39 | 44-45 | 77-82 |
| Forearm (right) | 55-58 | 74-78 | 40-42 | 46-47 | 83-88 |
| Hand (left) | 59-60 | 79-81 | 43-44 | 48-49 | 89-92 |
| Hand (right) | 61-62 | 82-84 | 45-46 | 50 | 93-96 |
| Thigh (left) | 63-70 | 85-87 | 47-57 | 51-60 | 97 |
| Thigh (right) | 71-78 | 88-90 | 58-68 | 61-70 | 98 |
| Knee (left) | 79-80 | 91 | 69-72 | 71-73 | - |
| Knee (right) | 81-82 | 92 | 73-76 | 74-76 | - |
| Lower leg (left) | 83-89 | 93-95 | 77-85 | 77-85 | 99 |
| Lower leg (right) | 90-96 | 96-98 | 86-94 | 86-94 | 00 |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Vietnam 1965-73 - hospitalized (Owens comparison set): Assault rifle / Pistol/SMG / RPG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-04 | 01-07 | 01-02 | 01-05 | 01-11 |
| Skull (right) | 05-08 | 08-14 | 03-04 | 06-10 | 12-22 |
| Face | 09-13 | 15-22 | 05-06 | 11 | 23-35 |
| Neck | 14-16 | 23-27 | 07 | 12-15 | 36-43 |
| Chest (left) | 17-21 | 28-32 | 08-11 | 16 | 44-46 |
| Chest (right) | 22-26 | 33-37 | 12-15 | 17 | 47-49 |
| Abdomen | 27-31 | 38-42 | 16-20 | 18 | 50 |
| Groin / pelvis | 32-33 | 43-44 | 21-24 | - | - |
| Upper back | 34-36 | 45-47 | 25-27 | 19-31 | 51-52 |
| Lower back | 37-39 | 48-50 | 28-29 | 32-40 | - |
| Shoulder (left) | 40-42 | 51-54 | 30-31 | 41 | 53-58 |
| Shoulder (right) | 43-44 | 55-58 | 32-33 | 42 | 59-64 |
| Upper arm (left) | 45-48 | 59-63 | 34-36 | 43-44 | 65-70 |
| Upper arm (right) | 49-52 | 64-68 | 37-39 | 45-46 | 71-76 |
| Forearm (left) | 53-56 | 69-73 | 40-42 | 47-48 | 77-82 |
| Forearm (right) | 57-60 | 74-78 | 43-45 | 49-50 | 83-88 |
| Hand (left) | 61-62 | 79-81 | 46-47 | 51 | 89-92 |
| Hand (right) | 63-64 | 82-84 | 48-49 | 52 | 93-96 |
| Thigh (left) | 65-71 | 85-87 | 50-59 | 53-62 | 97 |
| Thigh (right) | 72-78 | 88-90 | 60-69 | 63-72 | 98 |
| Knee (left) | 79-80 | 91 | 70-72 | 73-75 | - |
| Knee (right) | 81-82 | 92 | 73-75 | 76-78 | - |
| Lower leg (left) | 83-88 | 93-95 | 76-84 | 79-86 | 99 |
| Lower leg (right) | 89-94 | 96-98 | 85-92 | 87-94 | 00 |
| Foot (left) | 95-97 | 99 | 93-96 | 95-97 | - |
| Foot (right) | 98-00 | 00 | 97-00 | 98-00 | - |

# Northern Ireland - penetrating wounds: Assault rifle / MG / Pistol/SMG / Grenade / Mortar / RPG / Mine/IED

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-08 | 01-02 | 01-06 | 01-13 |
| Skull (right) | 06-10 | 09-16 | 03-04 | 07-12 | 14-26 |
| Face | 11-16 | 17-26 | 05-07 | 13 | 27-41 |
| Neck | 17-20 | 27-32 | 08-09 | 14-18 | 42-51 |
| Chest (left) | 21-26 | 33-38 | 10-14 | 19 | 52-54 |
| Chest (right) | 27-32 | 39-44 | 15-19 | 20 | 55-57 |
| Abdomen | 33-39 | 45-51 | 20-27 | 21 | 58 |
| Groin / pelvis | 40-42 | 52-54 | 28-33 | 22 | - |
| Upper back | 43-46 | 55-58 | 34-36 | 23-36 | 59-60 |
| Lower back | 47-50 | 59-62 | 37-40 | 37-49 | 61 |
| Shoulder (left) | 51-52 | 63-65 | 41-42 | 50 | 62-66 |
| Shoulder (right) | 53-54 | 66-68 | 43-44 | 51 | 67-71 |
| Upper arm (left) | 55-57 | 69-72 | 45-47 | 52-53 | 72-76 |
| Upper arm (right) | 58-60 | 73-76 | 48-50 | 54-55 | 77-81 |
| Forearm (left) | 61-63 | 77-80 | 51-53 | 56-57 | 82-86 |
| Forearm (right) | 64-66 | 81-84 | 54-56 | 58-59 | 87-91 |
| Hand (left) | 67-68 | 85-86 | 57-58 | 60 | 92-94 |
| Hand (right) | 69-70 | 87-88 | 59-60 | 61 | 95-97 |
| Thigh (left) | 71-76 | 89-90 | 61-68 | 62-69 | 98 |
| Thigh (right) | 77-82 | 91-92 | 69-76 | 70-77 | 99 |
| Knee (left) | 83-84 | 93 | 77-78 | 78-79 | - |
| Knee (right) | 85-86 | 94 | 79-80 | 80-81 | - |
| Lower leg (left) | 87-91 | 95-96 | 81-87 | 82-88 | 00 |
| Lower leg (right) | 92-96 | 97-98 | 88-94 | 89-94 | - |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Falklands 1982 - penetrating wounds: Assault rifle / MG / Pistol/SMG / Grenade / Mortar / RPG / Mine/IED

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-04 | 01-07 | 01-02 | 01-05 | 01-11 |
| Skull (right) | 05-08 | 08-14 | 03-04 | 06-10 | 12-22 |
| Face | 09-13 | 15-22 | 05-06 | 11 | 23-35 |
| Neck | 14-16 | 23-27 | 07 | 12-15 | 36-43 |
| Chest (left) | 17-22 | 28-33 | 08-12 | 16 | 44-46 |
| Chest (right) | 23-28 | 34-39 | 13-17 | 17 | 47-49 |
| Abdomen | 29-33 | 40-44 | 18-22 | 18 | 50 |
| Groin / pelvis | 34-35 | 45-47 | 23-26 | 19 | - |
| Upper back | 36-39 | 48-51 | 27-29 | 20-33 | 51-52 |
| Lower back | 40-42 | 52-54 | 30-32 | 34-42 | - |
| Shoulder (left) | 43-44 | 55-58 | 33-34 | 43 | 53-58 |
| Shoulder (right) | 45-46 | 59-62 | 35-36 | 44 | 59-64 |
| Upper arm (left) | 47-50 | 63-66 | 37-39 | 45-46 | 65-70 |
| Upper arm (right) | 51-54 | 67-70 | 40-42 | 47-48 | 71-76 |
| Forearm (left) | 55-58 | 71-74 | 43-45 | 49-50 | 77-82 |
| Forearm (right) | 59-62 | 75-78 | 46-48 | 51-52 | 83-88 |
| Hand (left) | 63-64 | 79-81 | 49-50 | 53 | 89-92 |
| Hand (right) | 65-66 | 82-84 | 51-52 | 54 | 93-96 |
| Thigh (left) | 67-73 | 85-87 | 53-62 | 55-63 | 97 |
| Thigh (right) | 74-80 | 88-90 | 63-72 | 64-72 | 98 |
| Knee (left) | 81-82 | 91 | 73-75 | 73-75 | - |
| Knee (right) | 83-84 | 92 | 76-78 | 76-78 | - |
| Lower leg (left) | 85-90 | 93-95 | 79-86 | 79-86 | 99 |
| Lower leg (right) | 91-96 | 96-98 | 87-94 | 87-94 | 00 |
| Foot (left) | 97-98 | 99 | 95-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Gulf War 1991 (US) - penetrating wounds: Assault rifle / MG / Pistol/SMG / Grenade / Mortar / RPG / Mine/IED

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-03 | 01-06 | 01 | 01-04 | 01-09 |
| Skull (right) | 04-06 | 07-12 | 02 | 05-08 | 10-18 |
| Face | 07-10 | 13-19 | 03-04 | - | 19-29 |
| Neck | 11-13 | 20-24 | 05 | 09-11 | 30-36 |
| Chest (left) | 14-17 | 25-28 | 06-08 | 12 | 37-38 |
| Chest (right) | 18-21 | 29-32 | 09-11 | 13 | 39-40 |
| Abdomen | 22-25 | 33-36 | 12-15 | 14 | 41 |
| Groin / pelvis | 26-27 | 37-38 | 16-18 | - | - |
| Upper back | 28-29 | 39-41 | 19-20 | 15-23 | 42 |
| Lower back | 30-31 | 42-44 | 21-22 | 24-30 | 43 |
| Shoulder (left) | 32-34 | 45-49 | 23-24 | 31-32 | 44-51 |
| Shoulder (right) | 35-37 | 50-54 | 25-26 | 33-34 | 52-58 |
| Upper arm (left) | 38-41 | 55-59 | 27-30 | 35-36 | 59-65 |
| Upper arm (right) | 42-45 | 60-64 | 31-34 | 37-38 | 66-72 |
| Forearm (left) | 46-49 | 65-69 | 35-38 | 39-40 | 73-79 |
| Forearm (right) | 50-53 | 70-74 | 39-42 | 41-42 | 80-86 |
| Hand (left) | 54-56 | 75-78 | 43-44 | 43-44 | 87-91 |
| Hand (right) | 57-59 | 79-82 | 45-46 | 45-46 | 92-96 |
| Thigh (left) | 60-67 | 83-86 | 47-57 | 47-57 | 97 |
| Thigh (right) | 68-75 | 87-90 | 58-68 | 58-68 | 98 |
| Knee (left) | 76-78 | 91 | 69-71 | 69-71 | - |
| Knee (right) | 79-80 | 92 | 72-74 | 72-74 | - |
| Lower leg (left) | 81-87 | 93-95 | 75-83 | 75-83 | 99 |
| Lower leg (right) | 88-94 | 96-98 | 84-92 | 84-92 | 00 |
| Foot (left) | 95-97 | 99 | 93-96 | 93-96 | - |
| Foot (right) | 98-00 | 00 | 97-00 | 97-00 | - |

# Gulf War 1991 (UK) - penetrating wounds: Assault rifle / MG / Pistol/SMG / Grenade / Mortar / RPG / Mine/IED

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01 | 01-03 | 01 | 01-02 | 01-05 |
| Skull (right) | 02 | 04-06 | 02 | 03-04 | 06-10 |
| Face | 03-04 | 07-09 | 03 | - | 11-16 |
| Neck | 05 | 10-11 | 04 | 05 | 17-20 |
| Chest (left) | 06-10 | 12-16 | 05-08 | 06 | 21-23 |
| Chest (right) | 11-15 | 17-21 | 09-12 | 07 | 24-26 |
| Abdomen | 16-20 | 22-27 | 13-17 | 08 | 27 |
| Groin / pelvis | 21-23 | 28-30 | 18-21 | - | - |
| Upper back | 24-26 | 31-33 | 22-23 | 09-19 | 28-29 |
| Lower back | 27-29 | 34-37 | 24-26 | 20-28 | 30 |
| Shoulder (left) | 30-32 | 38-42 | 27-28 | 29-30 | 31-39 |
| Shoulder (right) | 33-35 | 43-47 | 29-30 | 31-32 | 40-48 |
| Upper arm (left) | 36-39 | 48-53 | 31-33 | 33-35 | 49-57 |
| Upper arm (right) | 40-43 | 54-59 | 34-36 | 36-38 | 58-66 |
| Forearm (left) | 44-47 | 60-65 | 37-39 | 39-40 | 67-75 |
| Forearm (right) | 48-51 | 66-71 | 40-42 | 41-42 | 76-84 |
| Hand (left) | 52-54 | 72-75 | 43-44 | 43-44 | 85-90 |
| Hand (right) | 55-57 | 76-79 | 45-46 | 45-46 | 91-96 |
| Thigh (left) | 58-66 | 80-83 | 47-57 | 47-57 | 97 |
| Thigh (right) | 67-74 | 84-87 | 58-68 | 58-68 | 98 |
| Knee (left) | 75-77 | 88 | 69-71 | 69-71 | - |
| Knee (right) | 78-80 | 89 | 72-74 | 72-74 | - |
| Lower leg (left) | 81-87 | 90-93 | 75-83 | 75-83 | 99 |
| Lower leg (right) | 88-94 | 94-96 | 84-92 | 84-92 | 00 |
| Foot (left) | 95-97 | 97-98 | 93-96 | 93-96 | - |
| Foot (right) | 98-00 | 99-00 | 97-00 | 97-00 | - |

# Somalia 1993 - penetrating wounds: Assault rifle / MG / Pistol/SMG / Grenade / Mortar / RPG / Mine/IED

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-05 | 01-09 | 01-02 | 01-06 | 01-12 |
| Skull (right) | 06-10 | 10-18 | 03-04 | 07-12 | 13-24 |
| Face | 11-16 | 19-28 | 05-07 | 13 | 25-38 |
| Neck | 17-20 | 29-35 | 08-09 | 14-18 | 39-48 |
| Chest (left) | 21-23 | 36-38 | 10-12 | 19 | 49-50 |
| Chest (right) | 24-26 | 39-41 | 13-15 | 20 | 51 |
| Abdomen | 27-28 | 42-44 | 16-18 | - | - |
| Groin / pelvis | 29 | 45 | 19-20 | - | - |
| Upper back | 30-31 | 46-47 | 21-22 | 21-28 | 52 |
| Lower back | 32 | 48-49 | 23 | 29-32 | - |
| Shoulder (left) | 33-35 | 50-53 | 24-25 | 33-34 | 53-58 |
| Shoulder (right) | 36-38 | 54-57 | 26-27 | 35-36 | 59-64 |
| Upper arm (left) | 39-42 | 58-62 | 28-31 | 37-38 | 65-70 |
| Upper arm (right) | 43-46 | 63-67 | 32-35 | 39-40 | 71-76 |
| Forearm (left) | 47-50 | 68-72 | 36-39 | 41-42 | 77-82 |
| Forearm (right) | 51-54 | 73-77 | 40-42 | 43-44 | 83-88 |
| Hand (left) | 55-57 | 78-81 | 43-44 | 45-46 | 89-92 |
| Hand (right) | 58-60 | 82-84 | 45-46 | 47-48 | 93-96 |
| Thigh (left) | 61-68 | 85-87 | 47-57 | 49-58 | 97 |
| Thigh (right) | 69-76 | 88-90 | 58-68 | 59-68 | 98 |
| Knee (left) | 77-78 | 91 | 69-71 | 69-71 | - |
| Knee (right) | 79-80 | 92 | 72-74 | 72-74 | - |
| Lower leg (left) | 81-87 | 93-95 | 75-83 | 75-83 | 99 |
| Lower leg (right) | 88-94 | 96-98 | 84-92 | 84-92 | 00 |
| Foot (left) | 95-97 | 99 | 93-96 | 93-96 | - |
| Foot (right) | 98-00 | 00 | 97-00 | 97-00 | - |

# Chechnya - penetrating wounds: Assault rifle / MG / Pistol/SMG / Grenade / Mortar / RPG / Mine/IED

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-06 | 01-10 | 01-03 | 01-07 | 01-13 |
| Skull (right) | 07-12 | 11-20 | 04-06 | 08-14 | 14-26 |
| Face | 13-19 | 21-32 | 07-09 | 15 | 27-42 |
| Neck | 20-24 | 33-40 | 10-11 | 16-21 | 43-52 |
| Chest (left) | 25-27 | 41-43 | 12-14 | 22 | 53 |
| Chest (right) | 28-30 | 44-46 | 15-17 | 23 | 54 |
| Abdomen | 31-32 | 47-48 | 18-19 | - | - |
| Groin / pelvis | 33 | 49 | 20-21 | - | - |
| Upper back | 34-35 | 50-51 | 22-23 | 24-32 | 55 |
| Lower back | 36 | 52 | 24 | 33-35 | - |
| Shoulder (left) | 37-39 | 53-56 | 25-26 | 36-37 | 56-61 |
| Shoulder (right) | 40-42 | 57-60 | 27-28 | 38-39 | 62-67 |
| Upper arm (left) | 43-46 | 61-65 | 29-32 | 40-41 | 68-73 |
| Upper arm (right) | 47-50 | 66-70 | 33-36 | 42-43 | 74-79 |
| Forearm (left) | 51-54 | 71-74 | 37-39 | 44-45 | 80-85 |
| Forearm (right) | 55-58 | 75-78 | 40-42 | 46-47 | 86-90 |
| Hand (left) | 59-60 | 79-81 | 43-44 | 48-49 | 91-94 |
| Hand (right) | 61-62 | 82-84 | 45-46 | 50-51 | 95-98 |
| Thigh (left) | 63-70 | 85-87 | 47-57 | 52-61 | 99 |
| Thigh (right) | 71-78 | 88-90 | 58-68 | 62-71 | 00 |
| Knee (left) | 79-80 | 91 | 69-71 | 72-74 | - |
| Knee (right) | 81-82 | 92 | 72-74 | 75-77 | - |
| Lower leg (left) | 83-88 | 93-95 | 75-83 | 78-85 | - |
| Lower leg (right) | 89-94 | 96-98 | 84-92 | 86-93 | - |
| Foot (left) | 95-97 | 99 | 93-96 | 94-97 | - |
| Foot (right) | 98-00 | 00 | 97-00 | 98-00 | - |

# Iraq & Afghanistan 2001-05 - combat wounds: Assault rifle / AP rifle / Pistol/SMG / RPG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Cover |
|---|---|---|---|---|
| Skull (left) | 01-07 | 01-09 | 01-04 | 01-11 |
| Skull (right) | 08-14 | 10-18 | 05-08 | 12-22 |
| Face | 15-35 | 19-44 | 09-20 | 23-54 |
| Neck | 36-39 | 45-49 | 21-22 | 55-60 |
| Chest (left) | 40-42 | 50-51 | 23-25 | 61 |
| Chest (right) | 43-45 | 52-53 | 26-28 | 62 |
| Abdomen | 46-51 | 54-58 | 29-36 | 63 |
| Groin / pelvis | 52-53 | 59-60 | 37-40 | - |
| Upper back | 54-55 | 61 | 41-42 | 64 |
| Lower back | 56-58 | 62-64 | 43-46 | - |
| Shoulder (left) | 59-61 | 65-68 | 47-50 | 65-69 |
| Shoulder (right) | 62-64 | 69-72 | 51-54 | 70-74 |
| Upper arm (left) | 65-69 | 73-77 | 55-59 | 75-79 |
| Upper arm (right) | 70-74 | 78-82 | 60-64 | 80-84 |
| Forearm (left) | 75-79 | 83-86 | 65-69 | 85-89 |
| Forearm (right) | 80-84 | 87-90 | 70-74 | 90-94 |
| Hand (left) | 85-87 | 91-93 | 75-78 | 95-97 |
| Hand (right) | 88-90 | 94-96 | 79-82 | 98-00 |
| Thigh (left) | 91-92 | 97 | 83-86 | - |
| Thigh (right) | 93-94 | 98 | 87-90 | - |
| Knee (left) | 95 | - | 91 | - |
| Knee (right) | 96 | - | 92 | - |
| Lower leg (left) | 97-98 | 99 | 93-95 | - |
| Lower leg (right) | 99-00 | 00 | 96-98 | - |
| Foot (left) | - | - | 99 | - |
| Foot (right) | - | - | 00 | - |

# Iraq & Afghanistan 2001-05 - hospitalized (Owens comparison set): Assault rifle / AP rifle / Pistol/SMG / RPG / Frags

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-08 | 01-11 | 01-04 | 01-09 | 01-15 |
| Skull (right) | 09-16 | 12-22 | 05-08 | 10-18 | 16-30 |
| Face | 17-25 | 23-36 | 09-12 | 19 | 31-48 |
| Neck | 26-31 | 37-45 | 13-15 | 20-26 | 49-60 |
| Chest (left) | 32-33 | 46-47 | 16-17 | - | 61 |
| Chest (right) | 34-35 | 48-49 | 18-19 | - | 62 |
| Abdomen | 36-39 | 50-53 | 20-24 | 27 | 63 |
| Groin / pelvis | 40-41 | 54-55 | 25-28 | 28 | - |
| Upper back | 42-43 | 56 | 29 | 29-34 | 64 |
| Lower back | 44-46 | 57-58 | 30-31 | 35-42 | - |
| Shoulder (left) | 47-48 | 59-61 | 32-33 | 43 | 65-69 |
| Shoulder (right) | 49-50 | 62-64 | 34-35 | 44 | 70-74 |
| Upper arm (left) | 51-53 | 65-68 | 36-38 | 45-46 | 75-79 |
| Upper arm (right) | 54-56 | 69-72 | 39-41 | 47-48 | 80-84 |
| Forearm (left) | 57-59 | 73-76 | 42-44 | 49-50 | 85-89 |
| Forearm (right) | 60-62 | 77-80 | 45-47 | 51-52 | 90-94 |
| Hand (left) | 63-64 | 81-83 | 48-49 | 53 | 95-97 |
| Hand (right) | 65-66 | 84-86 | 50-51 | 54 | 98-00 |
| Thigh (left) | 67-73 | 87-89 | 52-61 | 55-63 | - |
| Thigh (right) | 74-80 | 90-92 | 62-71 | 64-72 | - |
| Knee (left) | 81-82 | 93 | 72-74 | 73-75 | - |
| Knee (right) | 83-84 | 94 | 75-77 | 76-78 | - |
| Lower leg (left) | 85-90 | 95-96 | 78-85 | 79-86 | - |
| Lower leg (right) | 91-96 | 97-98 | 86-93 | 87-94 | - |
| Foot (left) | 97-98 | 99 | 94-97 | 95-97 | - |
| Foot (right) | 99-00 | 00 | 98-00 | 98-00 | - |

# Baseline - random hit (body surface area): Rifle / Assault rifle / AP rifle / Pistol/SMG / RPG / Rifle-musket / Revolver / Bow

By situation. Roll d100 in the column that fits; 00 = 100. Other weapons on this table: use the roller or CLI.

| Location | Normal | Higher | Lower | Fleeing | Cover |
|---|---|---|---|---|---|
| Skull (left) | 01-03 | 01-05 | 01 | 01-03 | 01-10 |
| Skull (right) | 04-06 | 06-10 | 02 | 04-06 | 11-20 |
| Face | 07-10 | 11-17 | 03 | - | 21-31 |
| Neck | 11-12 | 18-21 | 04 | 07-09 | 32-39 |
| Chest (left) | 13-18 | 22-28 | 05-09 | 10 | 40-43 |
| Chest (right) | 19-24 | 29-35 | 10-14 | 11 | 44-47 |
| Abdomen | 25-29 | 36-41 | 15-20 | 12 | 48 |
| Groin / pelvis | 30-32 | 42-44 | 21-24 | - | - |
| Upper back | 33-36 | 45-48 | 25-27 | 13-27 | 49-51 |
| Lower back | 37-39 | 49-52 | 28-30 | 28-36 | 52 |
| Shoulder (left) | 40-41 | 53-56 | 31-32 | 37 | 53-58 |
| Shoulder (right) | 42-43 | 57-60 | 33-34 | 38 | 59-64 |
| Upper arm (left) | 44-46 | 61-64 | 35-36 | 39-40 | 65-70 |
| Upper arm (right) | 47-49 | 65-68 | 37-38 | 41-42 | 71-76 |
| Forearm (left) | 50-52 | 69-72 | 39-40 | 43-44 | 77-82 |
| Forearm (right) | 53-55 | 73-76 | 41-42 | 45-46 | 83-88 |
| Hand (left) | 56-57 | 77-79 | 43-44 | 47 | 89-92 |
| Hand (right) | 58-59 | 80-82 | 45-46 | 48 | 93-96 |
| Thigh (left) | 60-67 | 83-86 | 47-57 | 49-58 | 97 |
| Thigh (right) | 68-75 | 87-90 | 58-68 | 59-68 | 98 |
| Knee (left) | 76-78 | 91 | 69-71 | 69-71 | - |
| Knee (right) | 79-80 | 92 | 72-74 | 72-74 | - |
| Lower leg (left) | 81-87 | 93-95 | 75-83 | 75-83 | 99 |
| Lower leg (right) | 88-94 | 96-98 | 84-92 | 84-92 | 00 |
| Foot (left) | 95-97 | 99 | 93-96 | 93-96 | - |
| Foot (right) | 98-00 | 00 | 97-00 | 97-00 | - |

# Knife assault - stab wounds (forensic): Dagger / Bayonet

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Higher | Lower | Down | Fleeing | Flanked | Warding |
|---|---|---|---|---|---|---|---|
| Skull (left) | 01-04 | 01-05 | 01-02 | 01-06 | 01-04 | 01-04 | 01-03 |
| Skull (right) | 05-08 | 06-10 | 03-04 | 07-12 | 05-08 | 05-08 | 04-06 |
| Face | 09-13 | 11-16 | 05-06 | 13-17 | 09 | 09-12 | 07-10 |
| Neck | 14-28 | 17-37 | 07-14 | 18-41 | 10-25 | 13-24 | 11-22 |
| Chest (left) | 29-36 | 38-44 | 15-22 | 42-45 | 26 | 25-30 | 23-28 |
| Chest (right) | 37-44 | 45-51 | 23-30 | 46-49 | 27 | 31-36 | 29-34 |
| Abdomen | 45-52 | 52-58 | 31-41 | 50-53 | 28 | 37-42 | 35-41 |
| Groin / pelvis | 53-56 | 59-61 | 42-48 | 54-55 | 29 | 43-45 | 42-44 |
| Upper back | 57-64 | 62-68 | 49-57 | 56-67 | 30-59 | 46-61 | 45-51 |
| Lower back | 65-71 | 69-74 | 58-64 | 68-74 | 60-78 | 62-75 | 52-57 |
| Shoulder (left) | 72 | 75-76 | 65 | 75 | 79 | 76 | 58-59 |
| Shoulder (right) | 73 | 77-78 | 66 | 76 | 80 | 77 | 60-61 |
| Upper arm (left) | 74-76 | 79-81 | 67-69 | 77-78 | 81 | 78-79 | 62-64 |
| Upper arm (right) | 77-79 | 82-84 | 70-72 | 79-80 | 82 | 80-81 | 65-67 |
| Forearm (left) | 80-83 | 85-88 | 73-76 | 81-85 | 83-84 | 82-84 | 68-75 |
| Forearm (right) | 84-87 | 89-92 | 77-80 | 86-90 | 85-86 | 85-87 | 76-83 |
| Hand (left) | 88-90 | 93-95 | 81-83 | 91-93 | 87 | 88-89 | 84-88 |
| Hand (right) | 91-93 | 96-98 | 84-86 | 94-96 | 88 | 90-91 | 89-93 |
| Thigh (left) | 94-95 | 99 | 87-89 | 97 | 89-90 | 92-93 | 94 |
| Thigh (right) | 96-97 | 00 | 90-92 | 98 | 91-92 | 94-95 | 95 |
| Knee (left) | - | - | 93 | - | 93 | 96 | 96 |
| Knee (right) | - | - | 94 | - | 94 | - | - |
| Lower leg (left) | 98 | - | 95-96 | 99 | 95-96 | 97 | 97 |
| Lower leg (right) | 99 | - | 97-98 | 00 | 97-98 | 98 | 98 |
| Foot (left) | 00 | - | 99 | - | 99 | 99 | 99 |
| Foot (right) | - | - | 00 | - | 00 | 00 | 00 |

# Unarmed - landed strikes (MMA): Punch

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Down |
|---|---|---|
| Skull (left) | 01-17 | 01-23 |
| Skull (right) | 18-30 | 24-40 |
| Face | 31-81 | 41-89 |
| Neck | 82-83 | 90-92 |
| Chest (left) | 84-87 | 93-94 |
| Chest (right) | 88-92 | 95-96 |
| Abdomen | 93-99 | 97-99 |
| Upper back | 00 | 00 |

# Unarmed - landed strikes (MMA): Kick

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Down |
|---|---|---|
| Skull (left) | 01-05 | 01-11 |
| Skull (right) | 06-08 | 12-19 |
| Face | 09-22 | 20-42 |
| Neck | 23 | 43-44 |
| Chest (left) | 24-31 | 45-50 |
| Chest (right) | 32-41 | 51-57 |
| Abdomen | 42-53 | 58-65 |
| Upper back | 54 | 66-67 |
| Thigh (left) | 55-70 | 68-78 |
| Thigh (right) | 71-83 | 79-88 |
| Knee (left) | 84-85 | 89 |
| Knee (right) | 86-87 | 90 |
| Lower leg (left) | 88-95 | 91-96 |
| Lower leg (right) | 96-00 | 97-00 |

# Unarmed - landed strikes (MMA): Elbow/knee

By situation. Roll d100 in the column that fits; 00 = 100.

| Location | Normal | Down |
|---|---|---|
| Skull (left) | 01-14 | 01-22 |
| Skull (right) | 15-25 | 23-38 |
| Face | 26-68 | 39-83 |
| Neck | 69-70 | 84-86 |
| Chest (left) | 71-77 | 87-89 |
| Chest (right) | 78-85 | 90-93 |
| Abdomen | 86-95 | 94-97 |
| Upper back | 96 | 98 |
| Thigh (left) | 97 | 99 |
| Thigh (right) | 98 | 00 |
| Lower leg (left) | 99 | - |
| Lower leg (right) | 00 | - |

# Wound Mechanism

| Weapon | Cut | Pierce | Crush | Ballistic | Threat | Notes |
|---|---|---|---|---|---|---|
| Sword (arming sword) | 01-70 | 71-00 | - | - | melee | Baseline weapon; the Visby tables are applied unchanged. |
| Axe (hand or long-hafted) | 01-85 | - | 86-00 | - | melee | Visby work describes axes taking the legs out; leg bias reflects that. Heavy cuts bite through mail (ignores 1 step vs cut). |
| Spiked club / morning star (knout) | - | 01-45 | 46-00 | - | melee | Square skull perforations at Visby match a spiked club (Randolph-Quinney et al., 2026); head bias reflects that. Spikes punch through mail (ignores 1 step vs pierce). |
| Spear / lance | 01-10 | 11-00 | - | - | melee | Thrusting weapon aimed at the centre mass and face over the shield rim. |
| Bill (English bill, hooked polearm) | 01-60 | 61-90 | 91-00 | - | melee | The common infantry polearm of the Wars of the Roses: a cleaving blade with a hook and spike on a long haft. On the Towton evidence table it rolls on the blade wounds. |
| Poleaxe (axe, hammer and spike) | 01-20 | 21-55 | 56-00 | - | melee | The man-at-arms' weapon against plate: hammer and spike to crack or pierce armour (ignores 1 step vs crush and pierce: design estimate). On the Towton evidence table it rolls on the blunt-force wounds, which were mostly to the head and face. |
| Bow (war arrows) | 01-05 | 06-00 | - | - | - | Its own table is the US Army's 1865-71 arrow-wound reports (Plains and Apache bows, iron and flint heads). At Visby, 126 arrow and crossbow-bolt wounds fell much like the blade wounds, mostly on the lower body, so on the Visby tables it uses their weights unchanged. Arrows are pierce wounds, not gunfire: bleeding, deep tracks and infection; a barbed head left inside causes trouble for months. |
| Longbow (English war bow, bodkin and broadhead arrows) | 01-05 | 06-00 | - | - | - | Heavy war bow of the Hundred Years' War and the Wars of the Roses. Arrows mostly left soft-tissue wounds, so bone evidence under-counts them; at Towton a few head wounds are arrowheads. Heavy arrows punch through mail and jacks (ignores 1 step vs pierce): design estimate; against good plate they mostly glanced. |
| Crossbow (bolts) | - | 01-00 | - | - | - | Visby's arrow and bolt wounds are counted together (126), mostly on the lower body; the Visby weights are used with a small head bias for aimed shots at the face and helmet. Heavy bolts punch through mail (ignores 1 step vs pierce): design estimate. |
| Sling (stones, lead bullets) | - | - | 01-00 | - | - | No battlefield tally separates sling wounds. Andean skeletons from the Wari era show cranial trauma in about a quarter to a third of adults, much of it consistent with sling stones, but the cause is not counted apart. Head bias is a design estimate. Blunt: fractures and shock; a helmet takes the edge off (crush steps). |
| Dagger / knife | 01-30 | 31-00 | - | - | melee | Grappling and finishing weapon. On medieval tables it seeks gaps at neck, face and armpit; elsewhere it uses the forensic knife-assault table as recorded. |
| Bayonet (musket or rifle) | 01-15 | 16-00 | - | - | melee | Rolls on the Peninsular War table, which records where 97 bayonet wounds fell (thighs most, then arms and hands, then chest). Rare in every war with records: 3% of French officers' wounds in Spain, and under 1% of Civil War wounds (922 sabre and bayonet wounds together). The zone bias is used only on tables with raw weapon biases. |
| Sharpened entrenching tool (spade) | 01-50 | - | 51-00 | - | melee | Edges sharpened from 1915 for trench raids; used at Stalingrad and taught to Soviet Spetsnaz. Short overhead chops in a confined space: head, neck and collarbone. No wound data exists; shaped on the Visby axe and club evidence. |
| Sabre or broadsword | 01-80 | 81-00 | - | - | melee | Rolls on the Peninsular War table, which records where 284 sword wounds fell: a third on the head and face, then arms, hands and shoulders. Cuts from horseback. The zone bias is used only on tables with raw weapon biases. |
| Cavalry lance | 01-10 | 11-00 | - | - | melee | Polish and French lancers, British lancers after 1816. Rolls on the Peninsular War table (43 lance wounds: arms and hands, chest, head). Long reach against men on foot or fleeing. |
| Clubbed musket (butt stroke) | - | - | 01-00 | - | melee | Swung like a club when there was no time to reload or no bayonet. Revolutionary War pensioners lost eyes and had skulls broken by it. Rolls on the Visby melee table with no shield; head bias is a design estimate. |
| Punch | - | - | 01-00 | - | melee | Punches land head and body; leg strikes are kicks. |
| Kick | - | - | 01-00 | - | melee | Most landed kicks are low kicks and body kicks. |
| Elbow or knee | 01-25 | - | 26-00 | - | melee | Clinch strikes; elbows split skin (cut). |

| Weapon | Cut | Pierce | Crush | Ballistic | Threat | Notes |
|---|---|---|---|---|---|---|
| Matchlock musket | - | - | - | 01-00 | rifle | Heavy infantry musket fired from a rest or the shoulder; a lead ball of about .70-.80 calibre. On the Lützen table it uses the 32 recorded gunshot injuries (mostly carbines and pistols; no source separates musket hits). |
| Wheellock pistol or carbine (cavalry) | - | - | - | 01-00 | pistol | Cuirassiers and harquebusiers fired pistols and carbines at close range, aimed at the head and the left side of the chest. On the Lützen table it uses the 32 recorded gunshot injuries, 22 of them to the skull. |
| Smoothbore musket (Brown Bess, Charleville) | - | - | - | 01-00 | rifle | Soft lead ball of about .69-.75 calibre. On the Peninsular table it uses the 3,331 recorded musket and rifle wounds; elsewhere the table's pooled odds. |
| Flintlock rifle (Pennsylvania, Baker, Jäger) | - | - | - | 01-00 | rifle | Riflemen and Jäger picked off officers and gunners at long range. No source records rifle wounds apart from musket wounds; the slight head bias is a design estimate and applies only on tables with raw weapon biases. |
| Flintlock pistol | - | - | - | 01-00 | pistol | Cavalry and officers' sidearm; one shot, then the sword. |
| Smoothbore artillery (round shot, canister, grape, shell) | - | - | 01-30 | 31-00 | fragment | Round shot took off limbs; canister and grape acted like giant shotgun loads; shells burst. On the Peninsular table it uses the 209 recorded ordnance wounds, which fell lower on the body than musket wounds (legs and feet). Three times as lethal as musketry there. |
| Rifle-musket (Minie ball) | - | - | - | 01-00 | rifle | Soft lead .58 ball; shatters rather than drills bone. |
| Artillery (canister, shell fragments) | - | - | 01-20 | 21-00 | fragment | Canister balls and jagged fragments; heavy fragments crush. |
| Revolver | - | - | - | 01-00 | pistol | Cavalry and officers' sidearm. |
| Service rifle (bolt or semi-auto, 1914-53) | - | - | - | 01-00 | rifle | Lee-Enfield, Mauser, Springfield, Garand, Mosin. Aimed fire at whatever shows. |
| Assault rifle (M16, AK) | - | - | - | 01-00 | rifle | Intermediate cartridge. Common steel-core AK rounds count as rifle, not armour-piercing. |
| Rifle, armour-piercing ammunition | - | - | - | 01-00 | rifle_ap | Hardened-core rounds that defeat SAPI plates; ESAPI is rated against them. |
| Machine gun | - | - | - | 01-00 | rifle | Grazing fire sweeps low across the ground. |
| Pistol or submachine gun | - | - | - | 01-00 | pistol | Handgun cartridges; soft armour stops them. |
| Hand grenade | - | - | 01-15 | 16-00 | fragment | Bursts on or near the ground: legs and lower body take most fragments. |
| Mortar, rocket or artillery | - | - | 01-15 | 16-00 | fragment | Air and ground bursts; fragments come down onto the head, shoulders and back of prone men. |
| RPG (rocket-propelled grenade) | - | - | 01-30 | 31-00 | fragment | Blast plus fragments; often aimed at vehicles and bunkers. |
| Mine, booby trap or IED | - | - | 01-40 | 41-00 | fragment | Buried and tripwire devices: feet, legs and groin. Crush covers blast and traumatic amputation. |
| Punji stake | - | 01-00 | - | - | - | Sharpened, often fouled bamboo in pits and trails; 2% of Vietnam wounds. Pierce wounds carry high infection risk. |
| Fragments, unspecified (shell, grenade, mine) | - | - | 01-15 | 16-00 | fragment | Used where a source only says 'fragments' or 'explosion'. |

**Cut** - Edge blow: opens soft tissue; can sever limbs.

**Pierce** - Point or spike: small entry, deep track; internal damage and infection.

**Crush** - Blunt impact: fractures and shock; less external bleeding.

**Ballistic** - Bullet, ball or shell fragment: shatters bone, carries cloth and dirt deep into the wound.

# Severity

| d100 | Severity | Use when |
|---|---|---|
| 01-55 | Light | Damage well under what your system treats as a serious hit (e.g. < 1/3 of a wound level / HP threshold). |
| 56-85 | Serious | A solid hit: about 1/3 to 2/3 of a threshold, or a hit that beat armour by a clear margin. |
| 86-00 | Critical | A crippling or killing blow: over 2/3 of a threshold, or a critical success. |

## Severity for ballistic hits

Use instead of the table above. Used for every hit from a firearm or explosive (any weapon with a threat level), including a blast's crush wounds. Calibrated to Bougainville 1944, where 38-42% of head, chest and abdomen hits killed but only 0.3% of arm and 3.4% of leg hits did. If your own system sets severity, give gunshots to the head and trunk its harshest damage.

| Hit zone | Light | Serious | Critical |
|---|---|---|---|
| head, torso | 01-25 | 26-62 | 63-00 |
| arms, legs | 01-55 | 56-90 | 91-00 |

# Fighting While Wounded

Optional: resolve the attack here so that wounds count. Skip this page if your own system rolls the attack; you can still use the penalty table for it.

1. Pick the **attacker** and the **defender**. Add up each one's wound penalties (table below), capped at the limit shown.

2. Attacker rolls d100 against **attack % minus penalties** (on the battle map, also minus any range penalty and men in the way). Equal or under hits; the **margin** is the effective chance minus the roll. A roll at or under a tenth of the effective chance is a **critical**: it cannot be defended.

3. If the defender may parry or dodge, they roll d100 against **defence % minus penalties**; equal or under avoids the blow.

4. On a hit, roll location and mechanism as usual. **Severity comes from the margin** (bands below) instead of the severity table.

5. Wounds switch situations on by themselves: a defender who cannot stand is **Target down**, a defender whose shield arm is useless has **No shield**, and an attacker on the ground fights as **Attacker lower**.

## Wound penalties (add up, each total capped at -60%)

Arm wounds count against the **weapon arm** (the side of the fighter's weapon hand, usually right) or the **shield arm**. Wounds elsewhere that impair an arm count against the weapon arm. Each impairment counts once however many wounds cause it. A fighter with neither hand usable cannot wield a weapon.

| Cause | Attack | Defence | Also |
|---|---|---|---|
| Pain (per step, max 3 steps) | -10% | -10% | |
| Blood at 75% or less | -10% | -10% | |
| Blood at 50% or less | -20% | -20% | |
| Blood at 25% or less | -40% | -40% | on the ground |
| Blood at 0% or less | - | - | incapacitated |
| dazed | -20% | -20% | |
| vision | -30% | -30% | |
| breath | -10% | -10% | |
| collapse | -20% | -30% | on the ground |
| stand | -20% | -30% | on the ground |
| leg | - | -20% | |
| move | - | -10% | |
| unconscious | - | - | incapacitated |
| arm (weapon arm) | -20% | - | |
| arm (shield arm) | - | -10% | |
| grip (weapon arm) | -30% | - | fights off-hand |
| grip (shield arm) | - | - | no shield |
| hand useless (weapon arm) | -30% | - | fights off-hand |
| hand useless (shield arm) | - | - | no shield |
| arm useless (weapon arm) | -40% | - | fights off-hand |
| arm useless (shield arm) | - | -10% | no shield |

## Severity from the margin of success

| Weapon / hit zone | Light | Serious | Critical |
|---|---|---|---|
| Default | margin 0-19 | 20-49 | 50+ or critical roll |
| Firearm or explosive: head, torso | margin 0-9 | 10-34 | 35+ or critical roll |
| Firearm or explosive: arms, legs | margin 0-29 | 30-59 | 60+ or critical roll |

Automatic situations: defender cannot stand = Target down (fallen, kneeling, helpless); defender's shield arm useless = Target has no shield; attacker on the ground = Attacker lower (in a ditch, kneeling, vs. rider) (where the table offers them).

# Graze, stop check and team morale

**Graze:** a hit by a margin of 2 or less (not a critical) only grazes: the wound's bleed and pain drop one step each and it calls for no stop check.

**Stop check:** a serious or critical wound marked for one (most are) calls for d100 against the fighter's **Nerve** (green 35, regular 50, veteran 65), minus pain and blood-loss penalties, and -10 more for a critical wound. Equal or under: he fights on. Failed by 1-20: **defend only** for 2 rounds; by 21-40: **stunned** for 2 rounds; by more than 40: **out of the fight** until helped or the fight ends. Defend only and stunned fighters cannot attack; stunned also -50% defence. Pain above 3 steps calls for a stop check every round.

**Team morale:** at the end of each round, a side with 25% or more of its fighters down rolls d100 against the best Nerve among those still up (its leader's, if he is up), +10 with a leader up, -20 if a leader is down, -20 once 50% are down. Over: the side breaks and runs or surrenders.

# Battle Map

**Scale:** one hex is 2.2 yd (2 m) and a round is 6 seconds. The map has pointy-top hexes; the roller's default is 24 x 16.

**A fighter's turn:** stay in place, **advance** up to 4 hexes and still attack, or **run** up to 12 hexes and not attack. He may pass through friends but not stop on them, and cannot enter an enemy's hex.

**Wounds and movement:** the leg and move impairments halve both distances (round down); the collapse and stand impairments leave only a crawl of 1 hex; the breath impairment stops him running. A fighter who is stunned or out of the fight cannot move. Defend only: he may move but not attack.

**Leaving contact:** a fighter who moves more than 1 hex and ends no longer next to an enemy he started beside gives that enemy one free attack, if the enemy is up and able to attack. Stepping back 1 hex is a careful withdrawal and draws none.

**Reach and range:** a close-combat weapon strikes enemies within its reach (1 is the next hex). A missile weapon has short, medium and long range in hexes, with attack modifiers short +0, medium -20%, long -40%; no shot past long range. Weapons marked off-map (artillery, mines, stakes) are not aimed on the map: the GM decides whom they hit.

## Weapon reach and range

Ranges are design estimates from each weapon's effective range in its period.

| Weapon | Reach or range (short / medium / long) |
|---|---|
| Sword (arming sword) | reach 1 |
| Axe (hand or long-hafted) | reach 1 |
| Spiked club / morning star (knout) | reach 1 |
| Spear / lance | reach 2 |
| Bill (English bill, hooked polearm) | reach 2 |
| Poleaxe (axe, hammer and spike) | reach 1 |
| Bow (war arrows) | 15 / 40 / 90 hexes (33 / 87 / 197 yd; 30 / 80 / 180 m) |
| Longbow (English war bow, bodkin and broadhead arrows) | 20 / 60 / 125 hexes (44 / 131 / 273 yd; 40 / 120 / 250 m) |
| Crossbow (bolts) | 15 / 50 / 100 hexes (33 / 109 / 219 yd; 30 / 100 / 200 m) |
| Sling (stones, lead bullets) | 10 / 30 / 60 hexes (22 / 66 / 131 yd; 20 / 60 / 120 m) |
| Dagger / knife | reach 1 |
| Bayonet (musket or rifle) | reach 1 |
| Sharpened entrenching tool (spade) | reach 1 |
| Sabre or broadsword | reach 1 |
| Cavalry lance | reach 2 |
| Clubbed musket (butt stroke) | reach 1 |
| Punch | reach 1 |
| Kick | reach 1 |
| Elbow or knee | reach 1 |
| Matchlock musket | 20 / 40 / 75 hexes (44 / 87 / 164 yd; 40 / 80 / 150 m) |
| Wheellock pistol or carbine (cavalry) | 3 / 8 / 15 hexes (7 / 17 / 33 yd; 6 / 16 / 30 m) |
| Smoothbore musket (Brown Bess, Charleville) | 25 / 50 / 100 hexes (55 / 109 / 219 yd; 50 / 100 / 200 m) |
| Flintlock rifle (Pennsylvania, Baker, Jäger) | 50 / 100 / 150 hexes (109 / 219 / 328 yd; 100 / 200 / 300 m) |
| Flintlock pistol | 3 / 8 / 15 hexes (7 / 17 / 33 yd; 6 / 16 / 30 m) |
| Smoothbore artillery (round shot, canister, grape, shell) | off-map (GM decides) |
| Rifle-musket (Minie ball) | 50 / 150 / 250 hexes (109 / 328 / 547 yd; 100 / 300 / 500 m) |
| Artillery (canister, shell fragments) | off-map (GM decides) |
| Revolver | 5 / 12 / 25 hexes (11 / 26 / 55 yd; 10 / 24 / 50 m) |
| Service rifle (bolt or semi-auto, 1914-53) | 100 / 200 / 300 hexes (219 / 437 / 656 yd; 200 / 400 / 600 m) |
| Assault rifle (M16, AK) | 50 / 150 / 250 hexes (109 / 328 / 547 yd; 100 / 300 / 500 m) |
| Rifle, armour-piercing ammunition | 100 / 200 / 300 hexes (219 / 437 / 656 yd; 200 / 400 / 600 m) |
| Machine gun | 100 / 300 / 500 hexes (219 / 656 / 1094 yd; 200 / 600 / 1000 m) |
| Pistol or submachine gun | 25 / 50 / 100 hexes (55 / 109 / 219 yd; 50 / 100 / 200 m) |

| Weapon | Reach or range (short / medium / long) |
|---|---|
| Hand grenade | 5 / 10 / 15 hexes (11 / 22 / 33 yd; 10 / 20 / 30 m) |
| Mortar, rocket or artillery | off-map (GM decides) |
| RPG (rocket-propelled grenade) | 50 / 100 / 150 hexes (109 / 219 / 328 yd; 100 / 200 / 300 m) |
| Mine, booby trap or IED | off-map (GM decides) |
| Punji stake | off-map (GM decides) |
| Fragments, unspecified (shell, grenade, mine) | off-map (GM decides) |

# Camps and Works

**Making camp:** a camp's works are those the battle map lays out for it: a ring of hexes big enough for the force (about 20 m2 a man on foot, 40 with horses), with the bank and palisade on its edge and the ditch one ring out. 75% of the men work at once; the rest guard and cook. Work uses what is left of the day's marching hours and then 4 evening hours, so a camp made after the march costs no marching time; beyond that it runs on into the next day. Timber for palisades and stakes takes longer to fetch when no woods are within a hex. A fight can be set up only once an enemy force has marched to within 1 hex of the camp. If it arrives before the camp is finished, the GM chooses: attack the works built so far, or wait. Works go up in the order listed, each kind all round from the gate before the next.

**Works on the battle map:** ditches, banks, palisades, gates and walls lie along hex edges; stakes, abatis, pavises and wagons fill a hex. Crossing costs extra movement; a palisade, a barred gate or a wall stops movement until breached. Missile attacks on a man right behind a work lose its cover. In close combat only reach-2 weapons (spears, bills) strike over a palisade, gate or wall, and a man attacking up at a defender on the high side of a bank, ditch or wall takes its height penalty.

**Breaching:** a fighter next to a work can spend his attack on it: one man-round. When the man-rounds reach the work's breach number it is open. A gate is opened or barred from inside.

**In the way:** each fighter standing on the line between a shooter and his target costs the shot 10% (at most 30%). A miss goes into the first man in the way at 50%: roll the hit on him with no attack roll and the severity table. Bows and crossbows loose high over the men in the way: the penalty, but never a stray.

## Camps (hours to make, on foot, timber hauled)

| Camp | What it is | Needs | 12 men | 100 men | 1,000 men |
|---|---|---|---|---|---|
| Bivouac | Sleep where you stop. Nothing built and no proper watch. | - | none | none | none |
| Watched camp | A cleared site with fires, the baggage drawn up and a sentry roster. | - | 1.0 h | 1.0 h | 1.0 h |
| Staked camp | A watched camp ringed with sharpened stakes, like archers' stakes set against horse. | tools | 6.8 h | 2.8 h | 1.5 h |
| Fortified camp | Ditch, bank and palisade all round, with a gate. | tools | 54.9 h | 17.4 h | 6.0 h |
| Quartered | Billeted in the houses of a village or town. | a village or town | 0.5 h | 0.5 h | 0.5 h |

## Medieval works

| Work | Lies on | Labour | Crossing | Cover | Close combat | Breach |
|---|---|---|---|---|---|---|
| Ditch | edge | 2.5 man-h/m | +2 | - | -10% attacking up | - |
| Earth bank | edge | 1 man-h/m | +1 | -20% (high side) | -10% attacking up | - |
| Palisade | edge | 1.5 man-h/m (+1 hauled) | blocks | -40% | reach 2 only | 12 man-rounds |
| Gate | edge | 6 man-h each | barred (open: free) | -40% | reach 2 only | 20 man-rounds |
| Stone wall | edge | not built in the field | blocks | -50% | -20% attacking up; reach 2 only | - |
| Stakes | hex | 1 man-h (+0.5 hauled) | +1 | - | no horses | 2 man-rounds |
| Abatis | hex | 4 man-h (+4 hauled) | +2 | -20% in it | no horses | 8 man-rounds |
| Pavise | hex | carried | free | -20% in it | - | - |
| Wagon | hex | 0.25 man-h | blocks | -40% behind it | no horses | 10 man-rounds |

# Weather

**Calendar and daylight:** each map has a start date (Julian or Gregorian), a latitude and a climate. The march starts 1 hour after sunrise and stops 1 before sunset, up to 8 hours.

**Weather:** each day is rolled from the climate's monthly normals, wet and dry spells tending to last (40% persistence). Wet days fall as snow when the high is 36 °F (2 °C) or less. The GM can change any day.

**Ground:** rain builds mud and snow lies, wearing off in dry or mild weather; heavy rain floods fords for two days. Mud and snow slow the march (see the table below); deep mud or snow costs +1 movement per hex in battle.

**Rest and fatigue:** the night's rest starts from the camp's and drops a step for each point of hardship (wet, cold) beyond its shelter. Fatigue changes each night: heat or cold on a march of 4 hours or more, 6 hours in deep mud or snow, and the night (bad +1, poor 0, fair -1, good -2).

## Conditions

| Weather | Missile sight | Bows, crossbows | Firearms | Night hardship |
|---|---|---|---|---|
| Fair | clear | - | - | 0 |
| Overcast | clear | - | - | 0 |
| Fog | 25 hexes, 55 yd (50 m) | - | - | 0 |
| Rain | 150 hexes, 328 yd (300 m) | -10% | may misfire | 1 |
| Heavy rain | 75 hexes, 164 yd (150 m) | -20% | may misfire | 2 |
| Storm | 50 hexes, 109 yd (100 m) | -20% | may misfire | 2 |
| Snow | 75 hexes, 164 yd (150 m) | -10% | may misfire | 1 |
| Heavy snow | 40 hexes, 87 yd (80 m) | -10% | may misfire | 2 |
| Blizzard | 15 hexes, 33 yd (30 m) | -20% | may misfire | 3 |

## Misfires in the wet

| Ignition | Dry | Wet | Very wet |
|---|---|---|---|
| Match | 0% | 30% | 60% |
| Wheel | 0% | 15% | 30% |
| Flint | 0% | 15% | 30% |
| Percussion | 0% | 5% | 10% |
| Cartridge | 0% | 0% | 0% |

## Fatigue

| Fatigue | Attack and defence | Marching speed |
|---|---|---|
| Fresh | - | 100% |
| Tired | -5% | 100% |
| Weary | -10% | 90% |
| Exhausted | -20% | 75% |
| Spent | -30% | cannot march |

## Mud and snow on the march (time x)

| Terrain | Mud | Snow |
|---|---|---|
| Road | x1.5 | x1.5 |
| Open ground | x1.3 | x1.5 |
| Farmland | x1.5 | x1.5 |
| Heath | x1.2 | x1.5 |
| Forest | x1.1 | x1.5 |
| Hills | x1.2 | x1.75 |
| Mountains | x1.1 | x2 |
| Marsh | x1.2 | x1.5 |
| Ford | x1 | x1.5 |
| Bridge | x1 | x1.25 |
| Village | x1.1 | x1.25 |
| Town | x1 | x1.25 |

## Climates (modern normals, °F with °C below, and days with rain or snow)

| Climate | | J | F | M | A | M | J | J | A | S | O | N | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| North-west Europe, maritime | high/low °F | 43/34 | 45/34 | 48/36 | 54/39 | 59/45 | 64/50 | 68/54 | 68/54 | 63/50 | 55/45 | 48/39 | 45/36 |
| | (°C) | 6/1 | 7/1 | 9/2 | 12/4 | 15/7 | 18/10 | 20/12 | 20/12 | 17/10 | 13/7 | 9/4 | 7/2 |
| | wet days | 13 | 10 | 11 | 10 | 10 | 9 | 10 | 10 | 10 | 12 | 12 | 12 |
| Central and eastern Europe, continental | high/low °F | 34/23 | 37/25 | 46/30 | 57/37 | 66/46 | 72/52 | 75/55 | 75/55 | 66/48 | 55/39 | 43/32 | 36/27 |
| | (°C) | 1/-5 | 3/-4 | 8/-1 | 14/3 | 19/8 | 22/11 | 24/13 | 24/13 | 19/9 | 13/4 | 6/0 | 2/-3 |
| | wet days | 9 | 8 | 8 | 8 | 9 | 10 | 10 | 9 | 8 | 7 | 8 | 9 |
| Mediterranean and Levant coast | high/low °F | 64/48 | 66/48 | 70/52 | 77/57 | 82/63 | 86/70 | 90/73 | 90/75 | 88/72 | 82/66 | 75/57 | 66/52 |
| | (°C) | 18/9 | 19/9 | 21/11 | 25/14 | 28/17 | 30/21 | 32/23 | 32/24 | 31/22 | 28/19 | 24/14 | 19/11 |
| | wet days | 10 | 9 | 6 | 3 | 1 | 0 | 0 | 0 | 1 | 3 | 6 | 9 |
| Hot desert and steppe | high/low °F | 61/39 | 66/43 | 75/50 | 86/59 | 99/70 | 108/75 | 111/79 | 111/77 | 106/70 | 93/61 | 75/48 | 64/41 |
| | (°C) | 16/4 | 19/6 | 24/10 | 30/15 | 37/21 | 42/24 | 44/26 | 44/25 | 41/21 | 34/16 | 24/9 | 18/5 |
| | wet days | 4 | 4 | 4 | 3 | 1 | 0 | 0 | 0 | 0 | 1 | 3 | 4 |
| Carolina piedmont, humid subtropical | high/low °F | 54/32 | 57/36 | 64/41 | 73/48 | 81/57 | 88/64 | 90/70 | 88/68 | 82/63 | 73/52 | 63/39 | 55/36 |
| | (°C) | 12/0 | 14/2 | 18/5 | 23/9 | 27/14 | 31/18 | 32/21 | 31/20 | 28/17 | 23/11 | 17/4 | 13/2 |
| | wet days | 9 | 8 | 9 | 8 | 9 | 9 | 10 | 9 | 6 | 6 | 7 | 9 |
| Virginia piedmont, humid | high/low °F | 43/25 | 46/27 | 55/32 | 68/41 | 75/52 | 84/61 | 88/64 | 86/63 | 79/55 | 68/45 | 57/34 | 46/28 |
| | (°C) | 6/-4 | 8/-3 | 13/0 | 20/5 | 24/11 | 29/16 | 31/18 | 30/17 | 26/13 | 20/7 | 14/1 | 8/-2 |
| | wet days | 9 | 8 | 10 | 10 | 11 | 10 | 9 | 8 | 8 | 7 | 8 | 9 |
| Great Lakes, humid continental | high/low °F | 30/18 | 34/19 | 43/27 | 55/36 | 66/46 | 75/57 | 81/63 | 79/61 | 72/54 | 59/43 | 46/34 | 36/25 |
| | (°C) | -1/-8 | 1/-7 | 6/-3 | 13/2 | 19/8 | 24/14 | 27/17 | 26/16 | 22/12 | 15/6 | 8/1 | 2/-4 |
| | wet days | 15 | 12 | 12 | 12 | 11 | 10 | 9 | 9 | 10 | 11 | 12 | 15 |

# Unit combat

**Unit combat (the rank and file, on the field map):** one exchange is about a minute. In melee the front rank of each hex touching the enemy strikes (10 men a hex face in close order; reach-2 weapons add the second rank). 9% of them make a telling attempt each exchange; archers loose their rate a minute, 3.3% of the shots aimed at a man. Each attempt is an individual attack against the unit's attack %, parried on its defence % (not from flank or rear, nor against missiles), then location, severity from the margin, and armour, as for one fighter.

Light wounds and grazes fight on. Serious and critical wounds call for a stop check against the unit's Nerve; a failure is down. A wound fatal within rounds kills. Flank +20, rear +30, a mounted charge +20 at x2 tempo, shaken -10.

**Unit morale:** after an exchange a unit checks if it lost the exchange, has 25% of its men down, lost 10% in the exchange, or was struck in flank or rear: d100 against Nerve, +20 while steady, +10 with a leader, -20 if he is down, -20 at half strength, -20 if flanked, -5 a man it lost more than it put down (at most -30). Steady fails: shaken; shaken fails: broken. A broken unit flees: every enemy in reach strikes it at +20, no parry, x2 tempo.

**Ground and works on the field map:** open ground (move 1); rough ground (move 2); woods (move 2, cover -20%); marsh (move 3, no horses); high ground (move 1, -10% striking up at it). Works are the battle map's, along the 11 yd (10 m) hex sides: only reach-2 weapons strike over a palisade, wall or barred gate, from the front rank; striking up a bank, ditch or wall costs its height; a volley takes the best cover where it lands. A unit ordered to breach puts half its front rank to hacking at the work in front (a field-map side takes 5 times a battle-map side's breach figure). The battle map's weather applies.

**Guns:** a battery is a unit armed with guns, in battery order; its men are the crews, 10 to a gun, and as crews fall fewer guns are served. Each served gun fires 1 round an exchange at the nearest enemy in front within 1094 yd (1000 m): canister to 328 yd (300 m), round shot beyond. Round shot strikes the unit (60% to 547 yd (500 m), 40% to 820 yd (750 m), 25% to 1094 yd (1000 m)), then each man in its path (the ranks deep, at most 12) is hit at 40%, so columns suffer most. Canister throws 8 chances a round (50% to 109 yd (100 m), 30% to 219 yd (200 m), 15% to 328 yd (300 m)). Cover, the weather and open order cut both. Each man hit takes a wound roll at 80% with no parry.

**Targets and line of fire:** the GM can name a battery's target; it fires at it when it can, otherwise at the nearest. No unit shoots through another unit, friend or enemy, except from high ground over a unit on lower ground (guns: round shot, never canister); bows and crossbows may loose high over units at 20% off. **Shell:** howitzers fire shell beyond canister range over anything between; a shell lands (45% to 547 yd (500 m), 30% to 820 yd (750 m), 20% to 1094 yd (1000 m)), then its fuse is rolled: 15% early, 25% air, 45% ground, 15% blind; a burst throws 12 chances at 35% in the air (cover counts half) or 25% on the ground. Canister gains 5% beyond 109 yd (100 m) on firm ground. **Ricochet:** on firm ground round shot bounds on, striking the first other unit within 328 yd (300 m) past the target (30% to 109 yd (100 m), 20% to 219 yd (200 m), 10% to 328 yd (300 m) past it) and going through its ranks; rough ground, woods, marsh, mud or snow stop it. A round-shot strike on the battery it is fired at knocks out a gun at 10%. **Ammunition:** 50 rounds a gun; resupplying takes an exchange out of contact, without moving or firing.

**Limbering:** a limbered battery moves but cannot fire; unlimbered it fires but can only be manhandled 2 hexes. Changing over takes an exchange, without moving or firing. Hand to hand the crews fight with sword (arming sword). A battery that breaks loses its guns.

**Heroes in units:** a named fighter in the front rank makes 3 times a ranker's telling attempts with his own attack, and blows on the front fall on him 2 times as often as on a ranker there; a shooting hero looses 3 times a ranker's aimed shots. His wounds use the full rules and his own stop check (a failure: down for the battle). A leader adds his unit's leader modifier while he is up.

**After the battle:** a side with no unit standing has lost the field, and its men down there are left behind, dead. A wound fatal in minutes kills unless he is bound in time (10%); one fatal in hours kills unless his camp treats it (no camp 0%, a camp 40%, quarters 60%). The rest are carried to camp and heal on the wound track (Camp disease).

## Unit quality

| Quality | Attack | Defence | Nerve |
|---|---|---|---|
| Green | 40% | 25% | 35 |
| Regular | 50% | 35% | 50 |
| Veteran | 60% | 45% | 65 |

# Missile rates

| Missile weapon | Shots a minute |
|---|---|
| Longbow (English war bow, bodkin and broadhead arrows) | 8 |
| Bow (war arrows) | 8 |
| Crossbow (bolts) | 2 |
| Sling (stones, lead bullets) | 6 |
| Matchlock musket | 1 |
| Smoothbore musket (Brown Bess, Charleville) | 2 |
| Flintlock rifle (Pennsylvania, Baker, Jäger) | 1 |
| Rifle-musket (Minie ball) | 2 |

# Team generation

**Medieval (c.1250-1460): a team in twenty minutes.** A player runs a team of four to six; the first member leads. Each character is made in five steps.

**1. Characteristics:** Strength, Agility, Endurance, Wits, Nerve, Presence: roll 3d6 x 5 for each in order (15-90), or take the array 70, 60, 55, 50, 45, 35 and place it, highest first, in the order his career prefers (the GM may let him swap two).

**2. Origin:** d100 on the origin table: a skill +15, a trait and its drawback (a characteristic +/-10 or a skill +/-15).

**3. Career terms:** leaders serve three terms, the rest one or two. Each term: his career's main skill +15 the first term and +10 after, its second skill +5, and d100 on the career's table: the skill named +15 and the event. An event he has already had moves to the next row down. Nobody dies in generation. He is 16 plus 4 a term.

**4. Old wounds:** a row marked * leaves an old wound, as does d100 at or under the career's old-wound %. Roll its location on the region's wound table for the weapon shown, then d100: 70 or less a healed scar; more, a lasting hurt: -5 to Wits (head), Presence (face), Strength (arms), Agility (legs) or Endurance (body).

**5. Skills and kit:** each skill starts at a characteristic divided as shown (rounded down), plus what he gained, at most 90. His attack is the skill his weapon is fought with; his defence the better of Parry and Dodge; his Nerve the characteristic. Kit and weapon come with his career and region (the table below).

## Skills

| Skill | Starts at | Weapons |
|---|---|---|
| Blade | Agility / 2 | sword, dagger |
| Axe and mace | Strength / 2 | axe, spiked club |
| Polearm | Strength / 2 | spear, bill, poleaxe |
| Bow | Agility / 2 | bow, longbow |
| Crossbow | Agility / 2 | crossbow |
| Sling | Agility / 3 | sling |
| Brawl | Strength / 2 | punch, kick, elbow/knee |
| Parry | Agility / 2 | |
| Dodge | Agility / 3 | |
| Ride | Agility / 3 | |
| Fieldcraft | Wits / 3 | |
| Watch | Wits / 2 | |
| Stealth | Agility / 3 | |
| Leechcraft | Wits / 4 | |
| Seamanship | Wits / 4 | |
| Craft | Wits / 3 | |
| Letters | Wits / 5 | |
| Persuade | Presence / 2 | |
| Command | Presence / 3 | |

## Origins

| d100 | Origin | Skill | Trait | Drawback |
|---|---|---|---|---|
| 01-05 | Ploughman's son | Fieldcraft | Strong back (Strength +10) | Cannot read a word (Letters -15) |
| 06-10 | Shepherd | Watch | Good with a sling (Sling +15) | Uneasy in crowds (Presence -10) |
| 11-15 | Fisherman's son | Seamanship | Swims like a seal (Endurance +10) | Smells of the catch (Persuade -15) |
| 16-20 | Smith's apprentice | Craft | Arms like hams (Strength +10) | Half deaf from the anvil (Watch -15) |
| 21-25 | Carter | Ride | Knows every road (Fieldcraft +15) | Bad back (Agility -10) |
| 26-30 | Townsman's apprentice | Craft | Quick tongue (Persuade +15) | Never slept under the sky (Fieldcraft -15) |
| 31-35 | Merchant's son | Persuade | Lettered (Letters +15) | Soft hands (Strength -10) |
| 36-40 | Huntsman's son | Bow | Moves quietly (Stealth +15) | Known as a poacher (Presence -10) |

| d100 | Origin | Skill | Trait | Drawback |
|---|---|---|---|---|
| 41-45 | Miller's son | Craft | Good at sums (Wits +10) | Mistrusted, as millers are (Persuade -15) |
| 46-50 | Younger son of a knight | Ride | Raised to arms (Blade +15) | Proud and touchy (Persuade -15) |
| 51-55 | A squire's bastard | Blade | Hard to frighten (Nerve +10) | No inheritance and no patience (Presence -10) |
| 56-60 | Novice who left the cloister | Letters | Knows his herbs (Leechcraft +15) | Faints at the sight of blood (Nerve -10) |
| 61-65 | Barber's boy | Leechcraft | Steady hands (Agility +10) | Never stops talking (Stealth -15) |
| 66-70 | Groom in a great stable | Ride | Horse sense (Fieldcraft +15) | Kicked by a horse as a boy (Endurance -10) |
| 71-75 | Sailor's orphan | Seamanship | Climbs anything (Agility +10) | Cannot sit a horse (Ride -15) |
| 76-80 | Woodcutter | Axe and mace | Tireless (Endurance +10) | Slow of speech (Wits -10) |
| 81-85 | Raised on the pilgrim roads | Fieldcraft | Speaks three tongues (Persuade +15) | Belongs nowhere (Presence -10) |
| 86-90 | Street brawler | Brawl | Takes a punch (Endurance +10) | Known to the watch (Persuade -15) |
| 91-95 | Servant in a great house | Watch | Knows how to please his betters (Persuade +15) | Never held a weapon (Parry -15) |
| 96-00 | Herdsman of the border | Polearm | Grew up among raids (Nerve +10) | Hot-tempered (Wits -10) |

## Careers

| Career | Main skill | Second | Old wound | Hurt by |
|---|---|---|---|---|
| Man-at-arms | Blade | Parry | 25% | sword |
| Archer | Bow | Dodge | 15% | sword |
| Crossbowman | Crossbow | Dodge | 15% | crossbow |
| Spearman or billman | Polearm | Parry | 20% | sword |
| Mounted sergeant | Ride | Parry | 20% | sword |
| Mercenary | Axe and mace | Parry | 25% | sword |
| Sailor | Seamanship | Dodge | 10% | axe |
| Clerk or priest | Letters | Persuade | 3% | dagger |
| Barber-surgeon | Leechcraft | Craft | 5% | dagger |
| Forester or huntsman | Fieldcraft | Stealth | 10% | bow |
| Servant or groom | Ride | Watch | 5% | dagger |
| Merchant | Persuade | Letters | 5% | dagger |

## Career terms (* leaves an old wound)

| Career | d100 | Skill | Event |
|---|---|---|---|
| Man-at-arms | 01-20 | Ride | Rode in a lord's household for a season. |
| Man-at-arms | 21-40 | Command | Led a handful of men on a raid. |
| Man-at-arms | 41-60 | Polearm | Fought on foot at a barrier. |
| Man-at-arms | 61-75 | Parry | Came through a tournament melee. |
| Man-at-arms | 76-90 | Watch | Held the gate of a besieged town. * |
| Man-at-arms | 91-00 | Persuade | Took a knight prisoner and haggled his ransom. |
| Archer | 01-20 | Watch | Stood sentry on a border tower. |
| Archer | 21-40 | Stealth | Scouted ahead of the army. |
| Archer | 41-55 | Fieldcraft | Foraged through a hungry winter. |
| Archer | 56-70 | Craft | Made arrows by the thousand in a long siege. |
| Archer | 71-85 | Blade | Fought hand to hand when the line broke. * |
| Archer | 86-00 | Dodge | Ran from a charge of horse and lived. * |
| Crossbowman | 01-20 | Craft | Kept the windlasses and bolts of a garrison. |
| Crossbowman | 21-40 | Watch | Shot from the walls through a siege. |
| Crossbowman | 41-55 | Seamanship | Served aboard a galley. |
| Crossbowman | 56-70 | Persuade | Bargained over pay for a free company. |
| Crossbowman | 71-85 | Parry | Held a pavise in the front of the line. * |
| Crossbowman | 86-00 | Dodge | Crawled back from a sally that failed. * |
| Spearman or billman | 01-20 | Fieldcraft | Marched through a long campaign. |

| Career | d100 | Skill | Event |
|---|---|---|---|
| Spearman or billman | 21-40 | Parry | Held his place in the line. * |
| Spearman or billman | 41-55 | Craft | Dug ditches and threw up banks for a camp. |
| Spearman or billman | 56-70 | Brawl | Won a brawl in camp. |
| Spearman or billman | 71-85 | Watch | Guarded the baggage carts. |
| Spearman or billman | 86-00 | Dodge | Was caught in a rout and lived. * |
| Mounted sergeant | 01-20 | Polearm | Rode in a charge with couched spears. |
| Mounted sergeant | 21-40 | Fieldcraft | Carried messages across hostile country. |
| Mounted sergeant | 41-55 | Command | Kept order among the grooms and the baggage. |
| Mounted sergeant | 56-70 | Watch | Patrolled a frontier road. |
| Mounted sergeant | 71-85 | Blade | Fought in a skirmish at a ford. * |
| Mounted sergeant | 86-00 | Persuade | Bought remounts at a horse fair. |
| Mercenary | 01-20 | Persuade | Changed masters for better pay. |
| Mercenary | 21-40 | Stealth | Looted a town and kept his share hidden. |
| Mercenary | 41-55 | Blade | Fought a duel over a debt. |
| Mercenary | 56-70 | Fieldcraft | Lived off the land between contracts. |
| Mercenary | 71-85 | Parry | Stormed a breach. * |
| Mercenary | 86-00 | Command | Was made corporal of a lance. |
| Sailor | 01-20 | Watch | Kept the night watch on a cog. |
| Sailor | 21-40 | Craft | Mended sail and rigging. |
| Sailor | 41-55 | Persuade | Traded on his own account in a foreign port. |
| Sailor | 56-70 | Brawl | Fought in a dockside tavern. |
| Sailor | 71-85 | Axe and mace | Beat off pirates. * |
| Sailor | 86-00 | Fieldcraft | Was wrecked and walked home. |
| Clerk or priest | 01-20 | Persuade | Wrote letters for a lord's steward. |
| Clerk or priest | 21-40 | Leechcraft | Kept the infirmary of a religious house. |
| Clerk or priest | 41-55 | Command | Kept the accounts of a garrison. |
| Clerk or priest | 56-70 | Watch | Took down the testimony of witnesses. |
| Clerk or priest | 71-85 | Ride | Travelled as an envoy. |
| Clerk or priest | 86-00 | Fieldcraft | Went on pilgrimage on foot. * |
| Barber-surgeon | 01-20 | Craft | Pulled teeth and let blood in a market town. |
| Barber-surgeon | 21-40 | Letters | Studied an anatomy with a master. |
| Barber-surgeon | 41-55 | Watch | Worked through a pestilence. |
| Barber-surgeon | 56-70 | Persuade | Was paid by a captain to follow the army. |
| Barber-surgeon | 71-85 | Leechcraft | Cut arrows out of men after a battle. |
| Barber-surgeon | 86-00 | Dodge | Was caught when the camp was overrun. * |
| Forester or huntsman | 01-20 | Bow | Hunted deer for a lord's table. |
| Forester or huntsman | 21-40 | Stealth | Hunted poachers in the forest. |
| Forester or huntsman | 41-55 | Watch | Tracked a band of outlaws. |
| Forester or huntsman | 56-70 | Ride | Rode with a great hunt. |
| Forester or huntsman | 71-85 | Polearm | Killed a boar with a spear. * |
| Forester or huntsman | 86-00 | Leechcraft | Kept the hounds and doctored them. |
| Servant or groom | 01-20 | Craft | Kept a lord's harness and arms. |
| Servant or groom | 21-40 | Watch | Waited at table and heard everything. |
| Servant or groom | 41-55 | Fieldcraft | Pitched tents and made camp on campaign. |
| Servant or groom | 56-70 | Persuade | Bought provisions for the household. |
| Servant or groom | 71-85 | Leechcraft | Nursed his master through a fever. |
| Servant or groom | 86-00 | Blade | Fought beside his master when the camp was raided. * |
| Merchant | 01-20 | Letters | Kept the books of a trading house. |
| Merchant | 21-40 | Seamanship | Sailed with his goods to a far port. |
| Merchant | 41-55 | Ride | Travelled the fairs with a pack train. |
| Merchant | 56-70 | Command | Hired and paid guards for a caravan. |
| Merchant | 71-85 | Watch | Was cheated once and learned from it. |
| Merchant | 86-00 | Blade | Fought off robbers on the road. * |

## Kit and weapon by region

| Career | Latin East | Gotland and the Baltic | England | France |
|---|---|---|---|---|
| Man-at-arms | Mail hauberk and helm; sword | Danish man-at-arms; sword | Brigandine, sallet and arm splints; sword | Full plate harness; sword |
| Archer | Gambeson and cap; bow | Gotland levy: mail coif, padded coat; bow | Jack and sallet; longbow | Jack and sallet; longbow |
| Crossbowman | Hardened leather; crossbow | Gotland, better equipped: coif, coat of plates; crossbow | Gambeson and cap; crossbow | Gambeson and cap; crossbow |
| Spearman or billman | Gambeson and cap; spear | Gotland levy: mail coif, padded coat; spear | Jack and sallet; bill | Jack and sallet; spear |
| Mounted sergeant | Mail hauberk and helm; spear | Gotland, better equipped: coif, coat of plates; spear | Brigandine, sallet and arm splints; spear | Coat of plates and mail; spear |
| Mercenary | Mail hauberk and helm; axe | Coat of plates and mail; axe | Brigandine, sallet and arm splints; axe | Brigandine, sallet and arm splints; axe |
| Sailor | Unarmoured; dagger | Hardened leather; axe | Unarmoured; axe | Unarmoured; axe |
| Clerk or priest | Unarmoured; dagger | Unarmoured; dagger | Unarmoured; dagger | Unarmoured; dagger |
| Barber-surgeon | Unarmoured; dagger | Unarmoured; dagger | Jack and sallet; dagger | Unarmoured; dagger |
| Forester or huntsman | Hardened leather; bow | Hardened leather; bow | Hardened leather; bow | Hardened leather; bow |
| Servant or groom | Unarmoured; dagger | Unarmoured; dagger | Unarmoured; dagger | Unarmoured; dagger |
| Merchant | Gambeson and cap; sword | Gambeson and cap; sword | Gambeson and cap; sword | Gambeson and cap; sword |

## Teams

| Team | Members (terms) |
|---|---|
| A lord's retinue | Man-at-arms (3), Archer (2), Archer (1), Spearman or billman (1), Barber-surgeon (1), Servant or groom (1) |
| A free company | Mercenary (3), Mercenary (2), Crossbowman (2), Archer (1), Spearman or billman (1) |
| A garrison patrol | Mounted sergeant (3), Crossbowman (2), Crossbowman (1), Spearman or billman (1), Clerk or priest (1) |
| A farmers' muster | Spearman or billman (3), Spearman or billman (1), Spearman or billman (1), Archer (1), Forester or huntsman (1) |
| A pilgrims' escort | Man-at-arms (3), Mounted sergeant (2), Crossbowman (1), Clerk or priest (1), Barber-surgeon (1) |
| A merchant's company | Merchant (3), Sailor (2), Crossbowman (1), Servant or groom (1) |

# Camp disease

**Camp disease (forces on the travel map):** each force counts its sick, by disease and by step on the track (mending, serious, grave, deadly). Once a week each well man may catch each disease: his chance is the outbreak chance below, with the week's modifiers, times the chance he fails to resist (half, at Endurance 50). A man carries one disease at a time.

A case shows after its incubation at mending, climbs a step a day to the peak rolled below, then makes a recovery check each day: d100 against Endurance, plus care (no camp -20, a camp +0, quarters +10), activity (marched -20, rested +10), a cold wet night beyond the shelter -10, poor hygiene -10, less the disease's virulence. Success by 30 or more: two steps better; success: one; failure: no change; failure by 30 or more: one worse. At deadly, any failure kills. Typhus and plague leave survivors immune; ague relapses in marsh country; flux can turn chronic.

Men at serious or worse do not dig or fight. Grave and deadly cases go on litters: the force marches at 90%, or 50% with fewer than 2 fit men a litter, and cannot march with none. Hygiene comes from the camp (bivouac poor, quarters good, others fair) unless the GM sets it. Plague needs the GM's "plague in the region".

**The wounded (from the field map):** a wounded man starts at mending (light), serious (serious) or grave (critical), and at grave if the wound would kill in hours. He makes the same recovery check, but once every 7 days, at virulence -20. 3-5 days after the battle the wound turns septic on d100: low infection risk 5%, medium 15%, high 30% (no camp +5, quarters -5). Wound fever is one step worse and checks every day at virulence 25.

| Disease | Caught from | Shows after | Outbreak chance a week | Peak (d100) | Deaths per case |
|---|---|---|---|---|---|
| Bloody flux | Dirty water or food, latrines too near the water | 1-3 days | 3%; poor +22, good -5, staying +4, marsh +5, upland -5, hot +10 | 01-03 deadly, 04-15 grave, 16-00 serious | 2.6% (virulence 5) |
| Camp fever (typhus) | Lice in crowded, unwashed camps | 7-14 days | 0%; poor +4, good -5, staying +0.3, crowded +14, cold wet +10 | 01-30 deadly, 31-70 grave, 71-00 serious | 40% (virulence 16) |
| Typhoid | Dirty water or food, often from a carrier in camp | 7-21 days | 0.3%; poor +5, good -5, town +5, hot +5 | 01-30 deadly, 31-60 grave, 61-00 serious | 36% (virulence 15) |
| Ague (malaria) | Mosquitoes in marshes and low wetlands | 7-28 days | 15% (only with marsh and warm) | 01-02 deadly, 03-30 grave, 31-00 serious | 0.8% (virulence -9) |
| Plague | Flea bites; worst in sieges and towns full of rats | 2-6 days | 5% (only with plague); staying +5, town +5 | 01-50 deadly, 51-80 grave, 81-00 serious | 45% (virulence 15) |

# Travel

**Travel map:** hexes are usually 0.6 mi (1 km). A force marches 8 hours a day from 08:00 at its type's road speed. Crossing from one hex to the next takes the average of the two terrains' time multipliers; rivers and lakes are crossed only at a ford or bridge.

A force stops for the night rather than start a hex it cannot finish that day. Wagons cannot enter forest, marsh, mountains or heath.

Speeds are for a small, fit band. A large army's column made less ground a day; cut its hours or speed to suit.

**Marching together:** forces on one side within a hex can march as a column behind a leader. The column keeps out of ground any of them cannot enter, moves at its slowest and most tired force's pace, keeps one clock, and makes one camp sized for all its men, with every man digging.

## Terrain

Km a day across whole hexes of each terrain; a road day is the type's speed times its marching hours.

| Terrain | Time | On foot<br>mi (km) a day | Mounted<br>mi (km) a day | Baggage wagons<br>mi (km) a day |
|---|---|---|---|---|
| Road | x1 | 20 (32) | 30 (48) | 15 (24) |
| Open ground | x1.5 | 13 (21) | 20 (32) | 10 (16) |
| Farmland | x1.5 | 13 (21) | 20 (32) | 10 (16) |
| Heath | x1.75 | 11 (18) | 17 (27) | - |
| Forest | x2.5 | 8 (13) | 12 (19) | - |
| Hills | x2 | 10 (16) | 15 (24) | 7 (12) |
| Mountains | x4 | 5 (8) | 7 (12) | - |
| Marsh | x3 | 7 (11) | 10 (16) | - |
| River | impassable | - | - | - |
| Ford | x2 | 10 (16) | 15 (24) | 7 (12) |
| Bridge | x1 | 20 (32) | 30 (48) | 15 (24) |
| Lake or sea | impassable | - | - | - |
| Village | x1 | 20 (32) | 30 (48) | 15 (24) |
| Town | x1 | 20 (32) | 30 (48) | 15 (24) |

# Armour

1. Use armour only on tables marked **armour allowed** (all-hits, baseline, knife and unarmed tables). The other tables already show the effect of what their soldiers wore.

2. Find the hit location's **armour slot** in the target's kit. Empty slot: no protection.

3. If the slot shows d100 bands (e.g. *Rifle plate 01-65 / Soft Kevlar 66-00*), roll d100 to see which layer the hit meets. A roll above the last band is a gap: no protection.

4. Protection = the material's steps against the wound mechanism. For gunfire use the weapon's **threat** column (fragment, pistol, rifle, AP rifle). Subtract the weapon's armour defeat, if any.

5. Lower the severity that many steps: Critical > Serious > Light > Stopped. Stopped is a bruise, no wound.

6. **Shields.** The Visby tables assume the target carries a shield. If the target's kit has none (every modern kit, and full plate), use the **No shield** column or situation.

## Protection (severity steps)

| Material | vs Cut | vs Pierce | vs Crush | Gunfire: fragment | Gunfire: pistol | Gunfire: rifle | Gunfire: rifle ap |
|---|---|---|---|---|---|---|---|
| Padded (gambeson) | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| Hardened leather | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| Mail | 2 | 1 | 0 | 1 | 0 | 0 | 0 |
| Mail over padding | 2 | 1 | 1 | 1 | 0 | 0 | 0 |
| Coat of plates / brigandine | 2 | 2 | 1 | 1 | 1 | 0 | 0 |
| Plate | 3 | 2 | 1 | 2 | 1 | 0 | 0 |
| Jack (many layers of linen) | 2 | 1 | 1 | 1 | 0 | 0 | 0 |
| Buff coat (thick oiled leather) | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| Steel cuirass (cuirassier, carabinier) | 3 | 2 | 1 | 1 | 2 | 1 | 0 |
| Cavalry helmet (brass or steel, with crest) | 2 | 1 | 1 | 1 | 1 | 0 | 0 |
| Thin steel vest (private purchase) | 2 | 1 | 1 | 1 | 1 | 0 | 0 |
| Heavy steel plate (Sappenpanzer) | 3 | 2 | 1 | 2 | 2 | 1 | 0 |
| Steel plates in fabric (aircrew flak vest) | 2 | 1 | 1 | 2 | 1 | 0 | 0 |
| Steel helmet (M1, Brodie) | 2 | 1 | 1 | 1 | 1 | 0 | 0 |
| Nylon flak vest (M1952, M69) | 1 | 0 | 0 | 2 | 1 | 0 | 0 |
| Doron fibreglass plates (M1955) | 2 | 1 | 1 | 2 | 2 | 0 | 0 |
| Soft Kevlar (PASGT, IIIA vest) | 1 | 0 | 0 | 2 | 2 | 0 | 0 |
| Kevlar helmet (PASGT, ACH) | 2 | 2 | 1 | 2 | 2 | 0 | 0 |
| Rifle plate (SAPI) | 3 | 3 | 1 | 3 | 3 | 2 | 0 |
| AP rifle plate (ESAPI) | 3 | 3 | 1 | 3 | 3 | 2 | 2 |

| Weapon | Armour defeat |
|---|---|
| Axe (hand or long-hafted) | ignores 1 step vs cut |
| Spiked club / morning star (knout) | ignores 1 step vs pierce |
| Poleaxe (axe, hammer and spike) | ignores 1 step vs crush, ignores 1 step vs pierce |
| Longbow (English war bow, bodkin and broadhead arrows) | ignores 1 step vs pierce |
| Crossbow (bolts) | ignores 1 step vs pierce |

**Sources.** Visby kit contents follow the finds summarised in Thordeman (1939): mail coifs, coats of plates, gauntlets. Not yet checked against the text. Vietnam vests: M1952/M69 nylon stopped fragments and low-velocity pistol rounds but not rifle rounds; M1955 Doron plates stopped .45 ACP at the muzzle (Wikipedia, 'Flak jacket'). Protection steps and coverage percentages are design estimates.

# Armour Kits: Visby 1361

| Slot (covers) | Gotland levy: mail coif, padded coat | Gotland, better equipped: coif, coat of plates | Danish man-at-arms (1361) |
|---|---|---|---|
| helm (Skull) | Mail | Mail over padding | Plate |
| visor (Face) | - | - | - |
| gorget (Neck) | Mail | Mail | Mail |
| torso_upper (Chest, upper back) | Padded | Coat of plates / brigandine | Coat of plates / brigandine |
| torso_lower (Abdomen, lower back) | Padded | Coat of plates / brigandine | Coat of plates / brigandine |
| skirt (Groin / pelvis) | Padded | Mail | Mail |
| pauldron (Shoulders) | Padded | Mail | Mail over padding |
| arm (Upper arms, forearms) | Padded 01-60 | Mail 01-70 | Mail over padding |
| gauntlet (Hands) | - | Coat of plates / brigandine 01-50 | Coat of plates / brigandine |
| leg (Thighs, knees) | - | - | Mail over padding |
| greave (Lower legs) | - | - | Plate 01-50 |
| sabaton (Feet) | - | - | Mail |
| shield (Visby tables) | Yes | Yes | Yes |

**Gotland levy: mail coif, padded coat:** Many Visby dead were buried wearing mail coifs.

**Gotland, better equipped: coif, coat of plates:** Coats of plates and plate gauntlets were recovered from the Visby graves.

**Danish man-at-arms (1361):** Professional kit of the period; face left open (visors uncertain for 1361).

# Armour Kits: Wars of the Roses (c.1460)

| Slot (covers) | Jack and sallet (archer or billman, c.1460) | Brigandine, sallet and arm splints (c.1460) |
|---|---|---|
| helm (Skull) | Plate | Plate |
| visor (Face) | - | - |
| gorget (Neck) | Mail 01-50 | Mail |
| torso_upper (Chest, upper back) | Jack | Coat of plates / brigandine |
| torso_lower (Abdomen, lower back) | Jack | Coat of plates / brigandine |
| skirt (Groin / pelvis) | Jack | Mail |
| pauldron (Shoulders) | Jack | Plate 01-60 |
| arm (Upper arms, forearms) | Jack 01-70 | Plate 01-50 |
| gauntlet (Hands) | - | Plate 01-50 |
| leg (Thighs, knees) | - | Mail over padding 01-40 |
| greave (Lower legs) | - | - |
| sabaton (Feet) | - | - |
| shield (Visby tables) | No | No |

**Jack and sallet (archer or billman, c.1460):** Padded jack of many linen layers and an open sallet; legs and hands mostly bare. Jack protection is a design estimate.

**Brigandine, sallet and arm splints (c.1460):** Better-equipped retainer: brigandine (plates riveted inside cloth), sallet, splints on the arms, some plate on the hands.

# Armour Kits: Medieval, generic

| Slot (covers) | Rawhide shield, no armour (Plains warrior) | Gambeson and cap | Hardened leather | Mail hauberk and helm (c.1100-1300) | Coat of plates and mail (c.1350) | Full plate harness (c.1450) |
|---|---|---|---|---|---|---|
| helm (Skull) | - | Padded | Hardened leather | Plate | Plate | Plate |
| visor (Face) | - | - | - | - | - | Plate |
| gorget (Neck) | - | - | - | Mail | Mail | Plate |
| torso_upper (Chest, upper back) | - | Padded | Hardened leather | Mail over padding | Coat of plates / brigandine | Plate |
| torso_lower (Abdomen, lower back) | - | Padded | Hardened leather | Mail over padding | Coat of plates / brigandine | Plate |
| skirt (Groin / pelvis) | - | Padded | - | Mail | Mail | Mail over padding |
| pauldron (Shoulders) | - | Padded | Hardened leather | Mail over padding | Mail over padding | Plate |
| arm (Upper arms, forearms) | - | Padded 01-70 | Hardened leather | Mail over padding | Mail over padding | Plate |
| gauntlet (Hands) | - | - | - | Mail 01-70 | Coat of plates / brigandine | Plate |
| leg (Thighs, knees) | - | Padded 01-40 | - | Mail 01-50 | Mail over padding | Plate |
| greave (Lower legs) | - | - | Hardened leather | - | Plate 01-50 | Plate |
| sabaton (Feet) | - | - | - | - | Mail | Plate |
| shield (Visby tables) | Yes | Yes | Yes | Yes | Yes | No |

**Rawhide shield, no armour (Plains warrior):** Round shield of thick, shrunken buffalo hide. It could turn arrows and glancing balls, but covered nothing by itself: use the shield situations.

# Armour Kits: Early modern (1600-1815)

| Slot (covers) | Cuirassier, three-quarter armour (1630s) | Harquebusier: buff coat, breastplate, pot helmet | Pikeman: morion and corslet | Cuirassier: steel cuirass and helmet | Dragoon or light dragoon helmet |
|---|---|---|---|---|---|
| helm (Skull) | Plate | Plate 01-80 | Plate 01-85 | Cavalry helmet | Cavalry helmet |
| visor (Face) | Plate 01-70 | - | - | - | - |
| gorget (Neck) | Plate | - | - | - | - |
| torso_upper (Chest, upper back) | Steel cuirass | Steel cuirass 01-80 / Buff coat 81-00 | Steel cuirass | Steel cuirass | - |
| torso_lower (Abdomen, lower back) | Steel cuirass | Buff coat | Steel cuirass 01-60 | Steel cuirass 01-70 | - |
| skirt (Groin / pelvis) | Plate | Buff coat | - | - | - |
| pauldron (Shoulders) | Plate | Buff coat | - | - | - |
| arm (Upper arms, forearms) | Plate | Buff coat | - | - | - |
| gauntlet (Hands) | Plate 01-50 | Plate 01-50 | - | - | - |
| leg (Thighs, knees) | Plate 01-70 | Buff coat 01-50 | Plate 01-35 | - | - |
| greave (Lower legs) | - | - | - | - | - |
| sabaton (Feet) | - | - | - | - | - |
| shield (Visby tables) | No | No | No | No | No |

**Cuirassier, three-quarter armour (1630s):** Closed helmet, proofed breastplate, arm defences and tassets to the knee; one gauntlet (the bridle hand). Heavy, and going out of use by the 1640s.

**Harquebusier: buff coat, breastplate, pot helmet:** Lighter cavalry of the Thirty Years' War and the English Civil War: 'lobster-tail' pot helmet, breast and back plate over a buff coat, a gauntlet on the bridle arm.

**Pikeman: morion and corslet:** Open helmet, breast and back plates and short tassets. Many pikemen left the armour behind on campaign.

**Cuirassier: steel cuirass and helmet:** French cuirassiers and carabiniers; breast and back plates were proofed against musket balls at range (1 step), better against pistols. The skirt of the cuirass left the belly partly open.

**Dragoon or light dragoon helmet:** Brass, steel or leather helmet with a crest, worn by dragoons on both sides; turned sabre cuts.

# Armour Kits: Modern (helmets, flak vests, Kevlar, plates)

| Slot (covers) | Private steel vest (Civil War, 1862) | Steel helmet and Sappenpanzer (WWI German sentry) | Aircrew flak vest and helmet (WWII bomber crew) | Steel helmet only (1915-1950s riflem an) | M1 helmet, nylon flak vest (US Army M 1952/M 69, 195 2-75) | M1 helmet, Doron vest (USMC M1951/ M1955, 1951-75 ) | PASGT vest and helmet (1980s-90s) | Ranger Body Armor, front plate only (M ogadis hu 1993) | Interceptor vest, SAPI plates, ACH (2000s) | Plate carrier, ESAPI plates, helmet (2010s-) | Concealable soft vest (police IIIA) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| helm (Skull) | - | Steel helmet | Steel helmet | Steel helmet | Steel helmet | Steel helmet | Kevlar helmet | Kevlar helmet | Kevlar helmet | Kevlar helmet | - |
| visor (Face) | - | - | - | - | - | - | - | - | - | - | - |
| gorget (Neck) | - | - | - | - | Nylon flak vest 01-30 | - | Soft Kevlar 01-50 | - | Soft Kevlar 01-60 | - | - |
| torso_upper (Chest, upper back) | Thin steel vest 01-85 | Heavy steel plate | Steel plates in fabric | - | Nylon flak vest | Doron fibreglass plates 01-75 / Nylon flak vest 76-00 | Soft Kevlar | Rifle plate 01-35 / Soft Kevlar 36-00 | Rifle plate 01-65 / Soft Kevlar 66-00 | AP rifle plate 01-65 | Soft Kevlar |
| torso_lower (Abdomen, lower back) | Thin steel vest 01-50 | Heavy steel plate 01-70 | Steel plates in fabric 01-70 | - | Nylon flak vest 01-70 | Doron fibreglass plates 01-70 | Soft Kevlar 01-80 | Soft Kevlar 01-70 | Soft Kevlar 01-70 | Soft Kevlar 01-25 | Soft Kevlar 01-70 |
| skirt (Groin / pelvis) | - | - | Steel plates in fabric 01-60 | - | - | - | - | - | Soft Kevlar 01-50 | - | - |
| pauldron (Shoulders) | - | - | - | - | - | - | Soft Kevlar 01-40 | - | Soft Kevlar 01-50 | - | - |
| arm (Upper arms, forearms) | - | - | - | - | - | - | - | - | - | - | - |
| gauntlet (Hands) | - | - | - | - | - | - | - | - | - | - | - |
| leg (Thighs, knees) | - | - | - | - | - | - | - | - | - | - | - |
| greave (Lower legs) | - | - | - | - | - | - | - | - | - | - | - |
| sabaton (Feet) | - | - | - | - | - | - | - | - | - | - | - |
| shield (Visby tables) | No | No | No | No | No | No | No | No | No | No | No |

**Private steel vest (Civil War, 1862):** Atwater and G&D; Cook vests, sold by the thousand in 1862; about 8 lb, most were thrown away on the march.

**Steel helmet and Sappenpanzer (WWI German sentry):** Heavy plate issued to sentries and machine gunners in static positions; too heavy for the attack.

**Aircrew flak vest and helmet (WWII bomber crew):** Steel plates sewn into fabric against flak fragments; worn by US bomber crews from 1943.

**Steel helmet only (1915-1950s rifleman):** Most infantry from 1915 to the early 1950s wore a steel helmet and no body armour.

**M1 helmet, nylon flak vest (US Army M1952/M69, 1952-75):** M69 nylon vest stopped fragments and low-velocity pistol rounds, not an AK-47 round.

**M1 helmet, Doron vest (USMC M1951/M1955, 1951-75):** Doron fibreglass plates over the chest, sides and back; nylon over the upper torso. Stopped .45 ACP at the muzzle, not rifle rounds.

**PASGT vest and helmet (1980s-90s):** Kevlar against fragments and handgun rounds; no rifle plates.

**Ranger Body Armor, front plate only (Mogadishu 1993):** IIIA soft vest with a single ceramic plate in front; the first version had no back plate, so only about half the usual plate coverage.

**Interceptor vest, SAPI plates, ACH (2000s):** Plates cover the centre of the chest and back (about 65% of upper-torso hits); soft Kevlar elsewhere on the vest. Neck, groin and shoulder pieces were not always worn.

**Plate carrier, ESAPI plates, helmet (2010s-):** Lighter than a full vest: plates over the vital area, soft side panels only. Outside the plates is a gap.

**Concealable soft vest (police IIIA):** Stops handgun rounds, not rifles. Soft Kevlar gives little protection against a knife (pierce 0).

# Wound Effects - Cut

Edge blow: opens soft tissue; can sever limbs. Left and right share a row. Effects are untreated.

| Location | Light | Serious | Critical | Infection |
|---|---|---|---|---|
| **Skull** | B1, P1 | B2, P2, Stop check, dazed | B3, P3, Stop check, unconscious, Lethal: hours | medium |
| **Face** | B1, P1 | B3, P2, vision | B3, P3, Stop check, vision, speech, Lethal: days | medium |
| **Neck** | B1, P1 | B3, P2, Stop check, Lethal: hours | B3, P3, Stop check, breath, Lethal: rounds | medium |
| **Chest** | B1, P1 | B3, P2, Stop check, breath, Lethal: days | B3, P3, Stop check, breath, collapse, Lethal: minutes | medium |
| **Abdomen** | B1, P1 | B2, P2, Lethal: days | B3, P3, Stop check, collapse, Lethal: hours | high |
| **Groin** | B1, P2 | B3, P3, Stop check, leg | B3, P3, Stop check, stand, Lethal: minutes | high |
| **Back Upper** | B1, P1 | B2, P2, arm | B3, P3, Stop check, breath, collapse, Lethal: hours | medium |
| **Back Lower** | B1, P1 | B2, P2, move | B3, P3, Stop check, stand, Lethal: days | medium |
| **Shoulder** | B1, P1 | B2, P2, arm | B3, P3, Stop check, arm useless, Lethal: days | low |
| **Upper Arm** | B1, P1 | B2, P2, arm | B3, P3, Stop check, SEVERED, arm useless, Lethal: minutes | low |
| **Forearm** | B1, P1 | B2, P2, grip | B3, P3, Stop check, SEVERED, arm useless, Lethal: minutes | low |
| **Hand** | B0, P1 | B2, P2, grip | B3, P3, SEVERED, hand useless, Lethal: minutes | low |
| **Thigh** | B1, P1 | B3, P2, leg | B3, P3, Stop check, SEVERED, stand, Lethal: minutes | medium |
| **Knee** | B0, P1 | B2, P2, leg | B2, P3, Stop check, stand | medium |
| **Lower Leg** | B1, P1 | B2, P2, leg | B3, P3, Stop check, SEVERED, stand, Lethal: minutes | medium |
| **Foot** | B0, P1 | B2, P2, leg | B3, P3, SEVERED, stand, Lethal: minutes | medium |

# Wound Effects - Pierce

Point or spike: small entry, deep track; internal damage and infection. Left and right share a row. Effects are untreated.

| Location | Light | Serious | Critical | Infection |
|---|---|---|---|---|
| **Skull** | B1, P1 | B1, P2, Stop check, dazed, Lethal: days | B2, P3, Stop check, unconscious, Lethal: minutes | high |
| **Face** | B1, P1 | B2, P2, vision, Lethal: days | B2, P3, Stop check, vision, speech, Lethal: hours | high |
| **Neck** | B1, P1 | B2, P2, Stop check, Lethal: minutes | B3, P3, Stop check, breath, Lethal: rounds | high |
| **Chest** | B1, P1 | B2, P2, Stop check, breath, Lethal: hours | B3, P3, Stop check, breath, collapse, Lethal: rounds | high |
| **Abdomen** | B1, P1 | B1, P2, Lethal: hours | B2, P3, Stop check, collapse, Lethal: rounds | high |
| **Groin** | B1, P2 | B2, P3, Stop check, leg, Lethal: days | B3, P3, Stop check, stand, Lethal: rounds | high |
| **Back Upper** | B1, P1 | B1, P2, arm, Lethal: days | B2, P3, Stop check, breath, collapse, Lethal: rounds | high |
| **Back Lower** | B1, P1 | B1, P2, move, Lethal: days | B2, P3, Stop check, stand, Lethal: minutes | high |
| **Shoulder** | B1, P1 | B1, P2, arm | B2, P3, Stop check, arm useless, Lethal: days | medium |
| **Upper Arm** | B1, P1 | B1, P2, arm | B3, P3, Stop check, arm useless, Lethal: hours | medium |
| **Forearm** | B1, P1 | B1, P2, grip | B2, P3, Stop check, arm useless, Lethal: hours | medium |
| **Hand** | B0, P1 | B1, P2, grip | B2, P3, hand useless | medium |
| **Thigh** | B1, P1 | B2, P2, leg | B3, P3, Stop check, stand, Lethal: minutes | high |
| **Knee** | B0, P1 | B1, P2, leg | B1, P3, Stop check, stand | high |
| **Lower Leg** | B1, P1 | B1, P2, leg | B2, P3, Stop check, stand, Lethal: hours | high |
| **Foot** | B0, P1 | B1, P2, leg | B1, P3, stand | high |

# Wound Effects - Crush

Blunt impact: fractures and shock; less external bleeding. Left and right share a row. Effects are untreated.

| Location | Light | Serious | Critical | Infection |
|---|---|---|---|---|
| **Skull** | B0, P1 | B0, P2, Stop check, Fracture, dazed | B1, P3, Stop check, Fracture, unconscious, Lethal: hours | medium |
| **Face** | B0, P1 | B1, P2, Stop check, Fracture, vision | B1, P3, Stop check, Fracture, vision, speech, Lethal: days | medium |
| **Neck** | B0, P1 | B1, P2, Stop check, Lethal: hours | B2, P3, Stop check, breath, Lethal: rounds | medium |
| **Chest** | B0, P1 | B1, P2, Stop check, Fracture, breath, Lethal: days | B2, P3, Stop check, Fracture, breath, collapse, Lethal: minutes | medium |
| **Abdomen** | B0, P1 | B0, P2, Stop check, Lethal: days | B1, P3, Stop check, collapse, Lethal: hours | high |
| **Groin** | B0, P2 | B1, P3, Stop check, Fracture, leg | B2, P3, Stop check, Fracture, stand, Lethal: minutes | high |
| **Back Upper** | B0, P1 | B0, P2, Stop check, Fracture, arm | B1, P3, Stop check, Fracture, breath, collapse, Lethal: hours | medium |
| **Back Lower** | B0, P1 | B0, P2, Stop check, Fracture, move | B1, P3, Stop check, Fracture, stand, Lethal: days | medium |
| **Shoulder** | B0, P1 | B0, P2, Stop check, Fracture, arm | B1, P3, Stop check, Fracture, arm useless, Lethal: days | low |
| **Upper Arm** | B0, P1 | B0, P2, Stop check, Fracture, arm | B2, P3, Stop check, Fracture, arm useless, Lethal: minutes | low |
| **Forearm** | B0, P1 | B0, P2, Stop check, Fracture, grip | B1, P3, Stop check, Fracture, arm useless, Lethal: hours | low |
| **Hand** | B0, P1 | B0, P2, Stop check, Fracture, grip | B1, P3, Stop check, Fracture, hand useless | low |
| **Thigh** | B0, P1 | B1, P2, Stop check, Fracture, leg | B2, P3, Stop check, Fracture, stand, Lethal: minutes | medium |
| **Knee** | B0, P1 | B0, P2, Stop check, Fracture, leg | B0, P3, Stop check, Fracture, stand | medium |
| **Lower Leg** | B0, P1 | B0, P2, Stop check, Fracture, leg | B1, P3, Stop check, Fracture, stand, Lethal: hours | medium |
| **Foot** | B0, P1 | B0, P2, Stop check, Fracture, leg | B0, P3, Stop check, Fracture, stand | medium |

# Wound Effects - Ballistic

Bullet, ball or shell fragment: shatters bone, carries cloth and dirt deep into the wound. Left and right share a row. Effects are untreated.

| Location | Light | Serious | Critical | Infection |
|---|---|---|---|---|
| **Skull** | B1, P1 | B2, P2, Stop check, Fracture, dazed, Lethal: days | B3, P3, Stop check, Fracture, unconscious, Lethal: minutes | high |
| **Face** | B1, P1 | B3, P2, Stop check, Fracture, vision, Lethal: days | B3, P3, Stop check, Fracture, vision, speech, Lethal: hours | high |
| **Neck** | B1, P1 | B3, P2, Stop check, Lethal: minutes | B3, P3, Stop check, breath, Lethal: rounds | high |
| **Chest** | B1, P1 | B3, P2, Stop check, Fracture, breath, Lethal: hours | B3, P3, Stop check, Fracture, breath, collapse, Lethal: rounds | high |
| **Abdomen** | B1, P1 | B2, P2, Stop check, Lethal: hours | B3, P3, Stop check, collapse, Lethal: rounds | high |
| **Groin** | B1, P2 | B3, P3, Stop check, Fracture, leg, Lethal: days | B3, P3, Stop check, Fracture, stand, Lethal: rounds | high |
| **Back Upper** | B1, P1 | B2, P2, Stop check, Fracture, arm, Lethal: days | B3, P3, Stop check, Fracture, breath, collapse, Lethal: rounds | high |
| **Back Lower** | B1, P1 | B2, P2, Stop check, Fracture, move, Lethal: days | B3, P3, Stop check, Fracture, stand, Lethal: minutes | high |
| **Shoulder** | B1, P1 | B2, P2, Stop check, Fracture, arm | B3, P3, Stop check, Fracture, arm useless, Lethal: days | medium |
| **Upper Arm** | B1, P1 | B2, P2, Stop check, Fracture, arm | B3, P3, Stop check, Fracture, arm useless, Lethal: hours | medium |
| **Forearm** | B1, P1 | B2, P2, Stop check, Fracture, grip | B3, P3, Stop check, Fracture, arm useless, Lethal: hours | medium |
| **Hand** | B0, P1 | B2, P2, Stop check, Fracture, grip | B3, P3, Stop check, Fracture, hand useless | medium |
| **Thigh** | B1, P1 | B3, P2, Stop check, Fracture, leg | B3, P3, Stop check, Fracture, stand, Lethal: minutes | high |
| **Knee** | B0, P1 | B2, P2, Stop check, Fracture, leg | B2, P3, Stop check, Fracture, stand | high |
| **Lower Leg** | B1, P1 | B2, P2, Stop check, Fracture, leg | B3, P3, Stop check, Fracture, stand, Lethal: hours | high |
| **Foot** | B0, P1 | B2, P2, Stop check, Fracture, leg | B2, P3, Stop check, Fracture, stand | high |

# Key

| Term | Meaning |
|---|---|
| B0 | None |
| B1 | Slow - seeping; lose 1 Blood per minute |
| B2 | Heavy - lose 1 Blood per round; bind within minutes |
| B3 | Arterial - lose 3 Blood per round; tourniquet or pressure now |
| P0 | None |
| P1 | -10% to all actions |
| P2 | -20% to all actions |
| P3 | -30% to all actions |
| dazed | Stunned: lose next action, then -20% for 1d6 rounds |
| unconscious | Knocked out (check each minute to wake) |
| vision | Vision impaired: -30% to sight-based actions; blinded if both eyes |
| speech | Cannot speak clearly |
| breath | Winded: no sustained exertion; fatigue doubled |
| collapse | Collapses; can only crawl or crouch |
| grip | Drops anything held in that hand; cannot grip with it |
| arm | That arm -20%; no two-handed use |
| arm useless | That arm useless |
| hand useless | That hand useless |
| leg | Movement halved; -20% to footwork and dodging |
| stand | Falls and cannot stand unaided |
| move | Movement halved |
| Stop check | Stop check: d100 against Nerve, or defend only, stunned or out of the fight |
| Fracture | Bone broken: no use of the part until set; pain +1 when used |
| Severed | Part severed: arterial bleeding until cauterised or tourniqueted |
| Lethal: none | Not life-threatening by itself |
| Lethal: days | Fatal in days without care (infection, internal damage) |
| Lethal: hours | Fatal in hours without care |
| Lethal: minutes | Fatal in minutes without care |
| Lethal: rounds | Fatal within a few rounds |
| Lethal: instant | Killed outright |

# Survival Tracking (optional)

Each character has **20 Blood**. Bleed drains it; total Pain above 3 calls for a stop check each round.

| Blood | Effect |
|---|---|
| 75% | Pale and weak: -10% to all actions |
| 50% | Faint check each minute; -20% to all actions |
| 25% | Barely conscious: -40% to all actions, can only crawl |
| 0% | Unconscious and dying |

| Treatment | Effect |
|---|---|
| Bind | Bandage / pressure: reduce bleed by 1 (min 0) |
| Tourniquet | Limb only: stops bleeding; limb lost if left on for hours |
| Cauterise | Stops bleeding; pain +1 |